CLEARED
For Open Publication

Jul 29, 2024

Department of Defense
OFFICE OF PREPUBLICATION AND SECURITY REVIEW

ANNUAL ENERGY PERFORMANCE, RESILIENCE, AND READINESS REPORT

Department of Defense

Fiscal Year 2023

COST ESTIMATE

The estimated cost of this report or study for the Department of Defense is approximately $336,000 in Fiscal Years 2023–2024. This includes $250,000 in expenses and $86,000 in DoD labor. Cost estimate generated on April 01, 2024.

RefID: C-94101F9

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY 1

2. DOD ENERGY DEMAND OVERVIEW 3

3. OPERATIONAL ENERGY 5

Operational Energy Demands and Expenditures 5
Implementing the Operational Energy Strategy 6
Operational Energy Initiatives 7
Conclusion 18

4. INSTALLATION ENERGY 19

Energy Resilience Policies and Guidance 19
Installation Energy Plans (IEPs): Title 10 U.S.C. § 2911 19
Black Start Exercises (BSEs) 20
Critical Energy Capacity and Availability: Title 10 U.S.C. § 2920 20
Utility Outages on Privatized Systems: Title 10 U.S.C. § 2688(g)(4) 22
The Energy Resilience and Conservation Investment Program (ERCIP) 22
Installation Energy Alternative Funding 23
Installation Renewable Energy Goals 24
Renewable Energy Project Deployment and Carbon Pollution-Free Electricity (CFE) 25
Availability and Use of Energy Cost Savings: Title 10 U.S.C. § 2912 27
Facility-Related Control Systems 28

APPENDIX A - LIST OF ACRONYMS 30

APPENDIX B - CONGRESSIONAL REPORTING REQUIREMENTS 37

APPENDIX C - COMPLIANCE MATRIX 39

APPENDIX D - ENERGY PERFORMANCE MASTER PLAN 42

APPENDIX E - DOD ENERGY PERFORMANCE SUMMARY 44

APPENDIX F - RECOMMENDED CHANGE IN ORGANIZATION OR AUTHORITY 47

APPENDIX G - INSTALLATION ENERGY PROGRAM MANAGEMENT 48

APPENDIX H - THIRD PARTY FINANCING AUTHORITIES 55

APPENDIX I - BRIEFING ON OVERSEAS EFFICIENCY STUDY 57

APPENDIX J – REPORT ON PILOT PROGRAM ON INSTALLATION ENERGY MANAGEMENT 69

APPENDIX K – REPORT ON PROPANE/ALTERNATIVE FUEL POWERED VEHICLES 72

APPENDIX L – BRIEFING ON HYDROGEN FUEL CELL RESEARCH FOR DEFENSE APPLICATIONS 75

APPENDIX M – BRIEFING ON THE BENEFITS AND USE OF ELECTROCHROMIC GLASS IN MILITARY BASES AND INSTALLATIONS 81

APPENDIX N – BRIEFING ON DOD GROUND SOURCE HEAT PUMPS 84

APPENDIX O – FY 2023 OPERATIONAL ENERGY INITIATIVES 88

APPENDIX P – REPORT ON INSTALLATION RESILIENCE PROJECTS 128

APPENDIX Q – POINTS OF CONTACT 150

1. Executive Summary

The Department of Defense (DoD) is required to submit the Annual Energy Performance, Resilience, and Readiness (AEPRR)[1] Report annually to detail the Department's energy consumption and performance toward achieving greater energy resilience for installations and warfighting platforms.[2] The 2022 *National Defense Strategy* (NDS) directs the Department to "adopt more efficient and clean-energy technologies that reduce logistics requirements in contested or austere environments."[3] These efforts are part of a larger Department shift to "reinforce our capability to quickly mobilize and deploy forces and to sustain high-intensity joint denial operations despite kinetic and non-kinetic attack and disruption." The DoD must ensure mission readiness is supported with resilient, reliable, clean, and efficient installation energy (IE) and operational energy (OE) capabilities.

In Fiscal Year (FY) 2023, the Department consumed 613,712 Billion British Thermal Units (BBtus) of total energy, costing a total of $16.5 billion, to support operations and training across a worldwide set of installations with thousands of energy consuming platforms. The DoD is the largest single energy consumer both in the United States (U.S.) and among the Federal Government. DoD OE and IE consumption represents approximately 75 percent of total Federal energy consumption, approximately 13 times the total energy consumption of the next closest Federal agency (the United States Postal Service)[4].

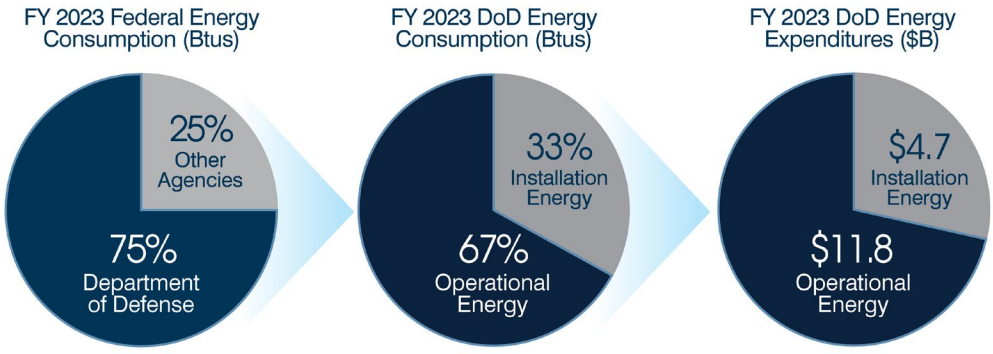


Figure 1. DoD Energy Consumption in FY 2023

The DoD expended most of its energy (67 percent) on **Operational Energy**, the vast majority of which came in the form of liquid fuel. Operational energy is defined as the "energy required for training, moving, and sustaining military forces and weapons platforms for military operations and includes energy used by ships, aircraft, combat vehicles, and tactical power generators."[5] In FY 2023, the DoD consumed

[1] The AEPRR Report was renamed from the Annual Energy Management and Resilience Report (AEMRR) in the FY 2023 National Defense Authorization Act and is required annually by Title 10 U.S.C. § 2925(a).

[2] "Energy Resilience" is defined as the "ability to avoid, prepare for, minimize, adapt to, and recover from anticipated and unanticipated energy disruptions in order to ensure energy availability and reliability sufficient to provide for mission assurance and readiness, including mission essential operations related to readiness, and to execute or rapidly reestablish mission essential requirements." Title 10 U.S.C. § 101(e)(6).

[3] Department of Defense, National Defense Strategy, 2022. p. 20

[4] FY 2022 energy data (Table A-1) collected annually by the Federal Energy Management Program (FEMP) and available here: https://ctsedwweb.ee.doe.gov/Annual/Report/TotalSiteDeliveredEnergyUseInAllEndUseSectorsByFederalAgencyBillionBtu.aspx

[5] Title 10 U.S.C. § 2924(4)

over 70 million barrels (408,830 BBtu) of fuel with a total cost of $11.8 billion to support worldwide operations and training, with 47 percent of this amount purchased outside of the U.S.

The DoD published a significantly revised *Operational Energy Strategy*, as required by title 10 United States Code (U.S.C.) § 2926, in May 2023.[6] The strategy identified initiatives across the four major lines of effort:

- Energy demand reduction;
- Energy substitution and diversification;
- Supply chain resilience; and
- Enterprise-wide energy visibility.

To meet the challenge of logistics in contested environments, the DoD invested over $2.8 billion in FY 2023 to improve propulsion, upgrade current equipment, revise operating procedures, and develop all new platform designs. Included in these investments is the Operational Energy Capability Improvement Fund (OECIF) and Operational Energy Prototyping Fund (OEPF) for advancing, demonstrating, and maturing advanced Joint technologies prior to transition to programs of record.

Installation Energy consists of the energy needed to power fixed installations and enduring locations as well as non-tactical vehicles (NTVs). In FY 2023, the DoD consumed 204,883 BBtus of energy and spent approximately $4.7 billion on installation energy, which included $4.5 billion to power, heat, and cool buildings; and $0.2 billion to supply fuel to the fleet of NTVs.

The DoD is working to publish an *Installation Energy Strategy* to ensure that our military installations continue to support the Joint Force's ability to fight and win in contested environments. The IE Strategy will support six (6) lines of effort (listed below), towards a strategic end state where secure, reliable, resilient, and efficient low-carbon energy for military installations provides increased readiness, increased options for sourcing and using energy, reduced fuel logistics tail, and command freedom of action.

- Ensure energy resilience for critical missions;
- Reduce demand and optimize installation energy efficiency;
- Support the transition to carbon-free energy;
- Data-driven decision making;
- Innovation; and
- Energy capable workforce.

In FY 2023, the DoD continued to invest heavily in installation energy, awarding just shy of $267M in third party financed contracts and $653M through Energy Resilience and Conservation Investment Program (ERCIP) project funding. Both programs provide more resilient and energy efficient installation energy infrastructure. These efforts focus on closing critical energy gaps at priority locations through energy solution sets that include application of cyber-security improvements, installation of field-flexible and expandable microgrids, deployment of energy storage technologies, and the leveraging of existing renewable energy generation resources. To support all its energy initiatives, the DoD is also strengthening its energy data collection and management systems, energy program analysis, and data-driven decision-making.

[6] DoD Operational Energy Strategy, released May 2023: https://www.acq.osd.mil/eie/Downloads/OE/2023%20Operational%20Energy%20Strategy.pdf

2. DoD Energy Demand Overview

The NDS set the policy direction for the Department's energy efforts by making "reducing energy demand a priority," and DoD's total energy consumption and cost has trended downward over time since FY12 (Figure 2).

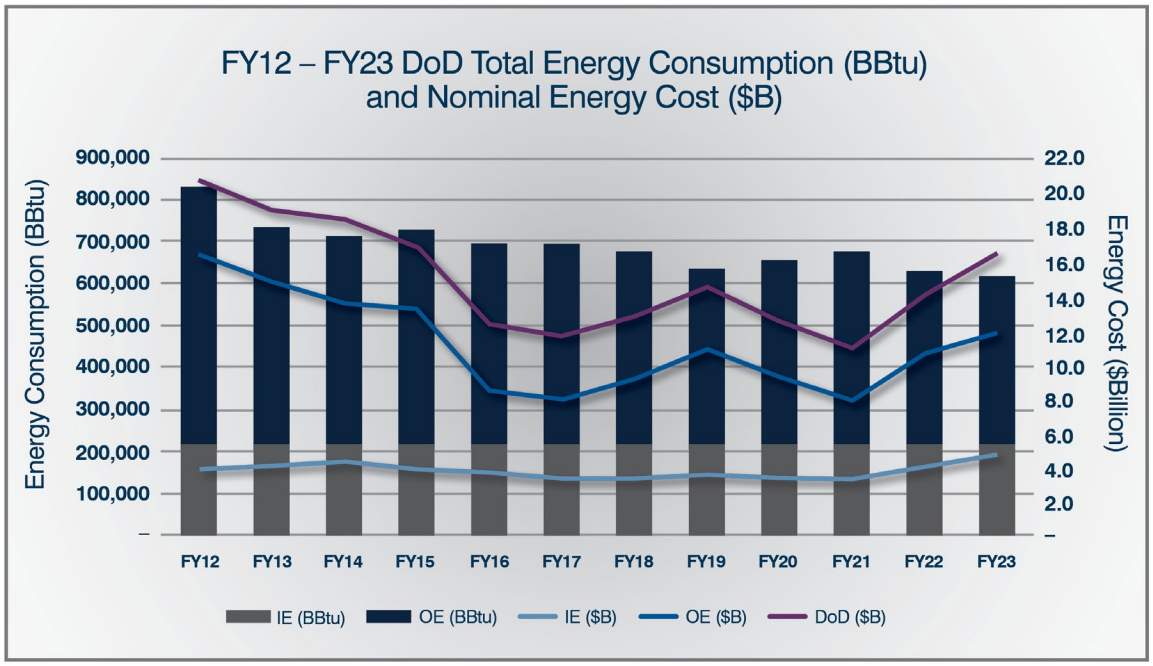


Figure 2. FY12 – FY23 DoD Total Energy Consumption (BBtu) and Nominal Energy Cost ($B)

Within DoD, the Department of the Air Force (DAF) was the largest consumer of operational energy in FY 2023, followed by the Navy and the U.S. Marine Corps (USMC), and Army (Left, Figure 3). The Army was the largest consumer of installation energy in FY 2023, followed by the Navy and USMC, DAF, and the Defense Agencies (Right, Figure 3).

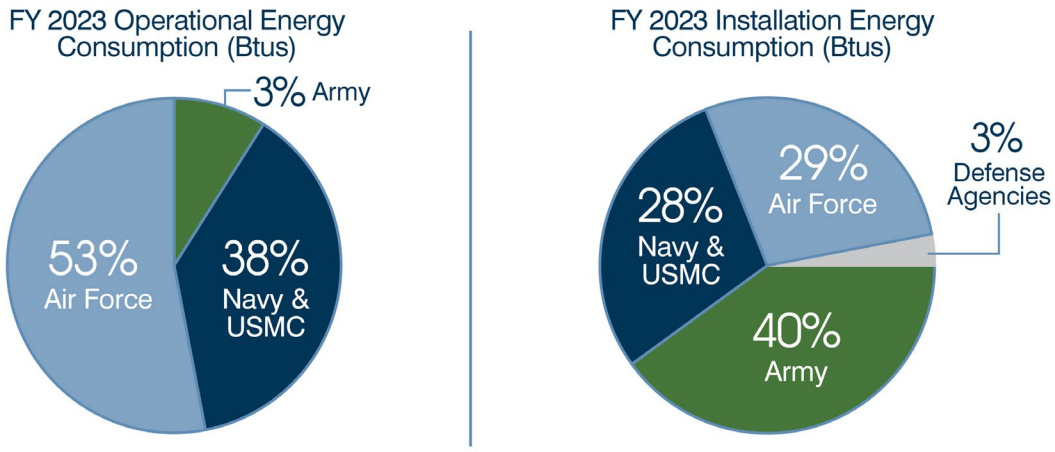


Figure 3. DoD Percentage of Energy Consumption for OE and IE by Service (FY 2023)

DoD's installation energy consumption mix mirrors that of the U.S. commercial sector, where natural gas and electricity dominate the supply mix (Figure 4).[7] The DoD operates over 500 installations all over the world, and is one of the largest customers for energy (particularly electricity and natural gas) within the U.S.

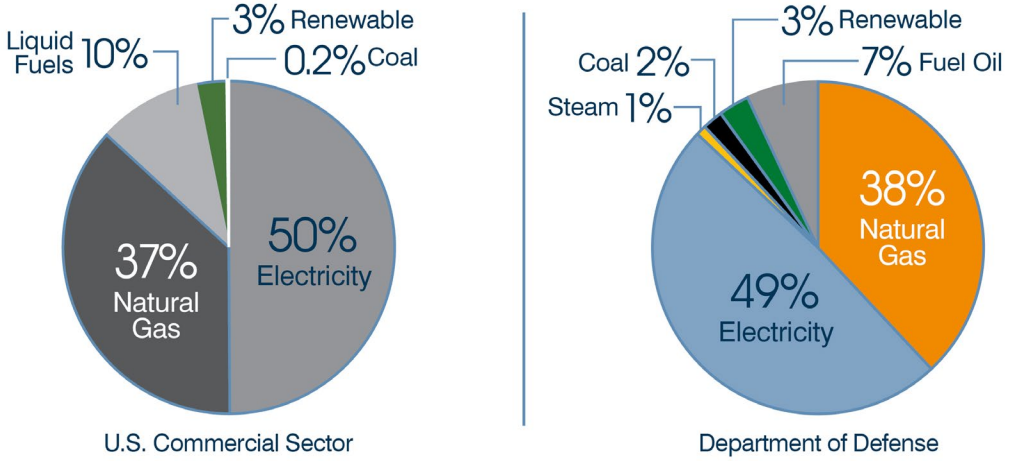


Figure 4. U.S. Commercial Sector vs. DoD Installation Energy Consumption (Btus), FY 2023

[7] U.S. Energy Information Administration, Monthly Energy Review, Table 2.3, as of March 26, 2024: https://www.eia.gov/totalenergy/data/monthly/

3. Operational Energy

By statute, operational energy is defined as "energy required for training, moving, and sustaining military forces and weapons platforms for military operations,"[8] and includes energy used by ships, aircraft, combat vehicles, and tactical power generators.

The significance of operational energy as an enabler of military capability is reflected in the 2022 *National Defense Strategy*. In alignment with the strategic environment and defense priorities, the Department will "make reducing energy demand a priority and seek to adopt more efficient clean-energy technologies that reduce logistics requirements in contested or austere environments."[9] The Department is also increasingly aware of the role of energy demand reduction in enabling operations in contested environments and the Deputy Secretary of Defense directed that the "Department's capability development activities, from requirements to acquisition to sustainment, must increase energy supportability and must reduce energy demand across all capability solutions."[10]

As described in this report, the Department's initiatives and programs increase resilience and support enhanced range, reach, time-on-station, and performance of the Joint Force. In FY 2023, the Department invested over $2.8B in initiatives to procure new or upgrade existing vehicles and aircraft, increase the range and endurance of platforms, enhance energy resiliency at contingency bases, and plan and execute war-games to account for increasing risks to logistics and sustainment. As the Department prepares to operate in contested environments around the globe, these investments increase range, endurance, and lethality while decreasing risks to warfighters.

Operational Energy Demands and Expenditures

In FY 2023, the Department consumed over 70 million barrels of fuel to power ships, aircraft, combat vehicles, and contingency bases with a total cost of $11.8 billion. To respond to the needs of a global force, the Department purchased 47 percent of this fuel outside of the U.S.

[8] Title 10 U.S.C. § 2924(4).

[9] Department of Defense, National Defense Strategy, 2022. p. 20.

[10] Kathleen Hicks, Deputy Secretary of Defense, Energy Supportability and Demand Reduction in Capability Development, April 21, 2022.

Table 1. DoD Operational Energy Demand by Service[11]

		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24e
Operational Energy Demand, Million Barrels	Army	7.3	7.1	7.6	9.2	9.0	8.1	7.3	6.1	6.0	6.2
	Navy	28.5	28.5	28.4	26.0	28.1	27.9	27.0	27.5	26.3	27.7
	Air Force	52.0	49.6	49	51.9	45.3	41.2	43.0	38.9	37.3	37.6
	USMC	0.2	0.2	0.2	0.5	0.4	0.4	0.3	0.3	0.3	0.4
	Other DoD	0.5	0.4	0.3	0.9	0.8	0.3	0.7	0.5	0.6	1.2
	Total Demand	**88.6**	**85.7**	**85.5**	**88.5**	**83.6**	**77.6**	**78.4**	**73.2**	**70.8**	**73.0**
	Expenditures ($ Billions)	**$14.1**	**$8.7**	**$8.2**	**$9.1**	**$11.0**	**$9.20**	**$7.91**	**$10.30**	**$11.8**	**$11.4**

Historical operational energy demand is based on net sales of selected liquid fuels by Defense Logistics Agency (DLA) Energy to the Services, while future operational energy demand estimates are based on the FY 2024 President's Budget. Expenditures for operational energy are estimated using the average fuel sales price for the specific fuel provided to the customer at the point of sale and include procurement and overhead costs. This price does not reflect additional costs imposed on the Department for force protection, storage, and transportation beyond the point of sale. As a purchaser of fuel on the open market, the Department is subject to the same price volatility experienced by commercial customers.[12]

Implementing the Operational Energy Strategy

Section 2926(c) of title 10, U.S.C., assigns the Assistant Secretary of Defense for Energy, Installations and Environment (ASD(EI&E)) with the responsibility to establish and maintain a department-wide transformational strategy for operational energy. In FY 2023, the Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization, ODASD(ER&O), issued a new *Operational Energy Strategy* to meet the requirements of section 2926(e) of title 10, U.S.C. To ensure the Joint Force can fight and win in contested environments, the *Operational Energy Strategy* includes four lines of effort:

- Energy Demand Reduction;
- Energy Substitution and Diversification;
- Supply Chain Resilience; and
- Enterprise-wide Energy Visibility.

To assist with the implementation of this new strategy, the ASD(EI&E) leads the Contested Logistics Operational Energy Working Group (CLOE WG), which was established in FY 2022 in accordance with section 2926(d) of title 10, U.S.C. With senior leader representatives from the armed Services, Joint Staff, and Combatant Commands, the working group is shaping the direction and implementation of the *Operational Energy Strategy* and inform future planning and programming guidance. Operational energy issues span requirements, acquisition, planning, programming, budgeting, execution, and sustainment activities across the Department. To ensure the successful implementation of the strategy, the CLOE WG will elevate issues and challenges to other governance forums including the Deputy's Management Action

[11] Updated analysis of expenditures may lead to different results from previous Operational Energy Annual Reports. Expenditures are not adjusted for inflation; data on historical demand may not capture final end use nor account for fuel transfers between the Services; Historical and Estimated Demand include Base and Overseas Contingency Operations (OCO) funding and purchases using Transportation Working Capital Fund (TWCF). Note: while operational energy includes more sources other than liquid fuels, DoD lacks enterprise-wide visibility regarding demand for power in warfighting platforms; this gap is being addressed in the Department's recently revised OE Strategy.

[12] Standard DLA Energy fuel prices can be found at http://www.dla.mil/Energy/Business/StandardPrices.aspx

Group (DMAG), the Climate Working Group, and the Service Acquisition Executive Meetings, to address specific decision-making processes in support of the *Operational Energy Strategy*.

The CLOE WG met in FY 2023 to align on the approach to implement the *Operational Energy Strategy* and begin execution of the near-term (2023-2024) and mid-term (2024-2028) initiatives. In FY 2024, the CLOE WG will meet on a quarterly basis as required by section 343 of the FY 2024 National Defense Authorization Act (NDAA) and subsequent annual energy reports will use the *Operational Energy Strategy* as the baseline for evaluating initiatives and measuring progress in the implementation of the strategy.

Operational Energy Initiatives

The Department's activities in operational energy are carried out by the Office of the Secretary of Defense and its defense agencies, the Military Departments, and the Joint Staff and the Combatant Commands. Highlights of these activities are provided below and are detailed in Appendix O.

Office of the Assistant Secretary of Defense for Energy, Installations, and Environment

In accordance with section 2926(c) of title 10, U.S.C., the ASD(EI&E) "shall oversee the operational energy activities of the Department of Defense," "oversee the investments of the Department in such activities," and "make recommendations to the Secretary [of Defense] regarding the policies and investments that affect the use of operational energy across the Department of Defense." Through ODASD(ER&O), the notable operational energy initiatives in FY 2023 included:

- **Wargame Support and Analytics**: ODASD(ER&O) continues to exercise oversight of operational energy support to the warfighter through participation in Joint Staff, Service Title 10, and Combatant Commands (CCMDs) sponsored wargames, with a focus on demand reduction and increased energy supportability as outlined in the Deputy Secretary of Defense memo. During the Globally Integrated Wargame '23, ODASD(ER&O) assessed fuel supply and distribution within the U.S. Indo-Pacific Command (USINDOPACOM) area of responsibility (AOR), examining alternative approaches to meeting warfighter needs. ODASD(ER&O) also participated in the Joint Staff sponsored effort Determining Climate Challenges for Defense Logistics (DC2DL) and various Service wargames (e.g., USMC Contested Logistics Wargame), with the goal of ensuring supply chain resilience for both climate and contested logistics challenges. Engagement with allies and partners is another critical area, and ODASD(ER&O) participated in the U.S. European Command (USEUCOM) sponsored Nordic Fuel Game, the first look at assessing fuel demand and distribution in that region.

- **Energy in Requirements and Acquisition**: Reflecting the priority on capability development, ODASD(ER&O) is focused on ensuring that requirements and acquisition decision-making reflects challenges of contested environments and the benefits of advancing energy-related technologies and practices. Through the Climate Working Group, the Military Departments adapted policy, processes, and procedures to ensure consideration of energy supportability analyses and evolving operational requirements. The Office of Under Secretary of Defense for Acquisition and Sustainment (OUSD(A&S)) continued to prepare a recurring data collection framework to provide higher quality, more frequent updates on Department-wide programs and inform policy and oversight. Finally, OUSD(A&S) initiated a pilot program for the application of energy supportability analyses in Integrated Acquisition Portfolio Reviews (IAPRs). Focused on specific groups of platforms, energy demand and supportability analyses will better assess overall Joint capabilities, assess any risks related to the provision of energy, and enhance the identification

of interdependencies outside of any single portfolio. Working with OASD(Acquisition), the Office of the ASD(EI&E) (OASD(EI&E)) will focus on the Defense of Guam as the pilot effort and will finish the initial analyses in FY 2024.

- **Bulk Fuel Governance**: In collaboration with the Military Departments, Joint Staff, CCMDs, and DLA, ODASD(ER&O) ensured successful completion of several actions to enhance globally integrated operations and energy supportability. Supporting United States Transportation Command's (USTRANSCOM) transition as Single Manager for Global Bulk Fuel Management and Delivery (GBFMD), ODASD(ER&O) remained a key stakeholder organization for the GBFMD Operational Planning Team (OPT) to establish and modify policy that enhances global integration and governance of bulk fuel. ODASD(ER&O) developed the implementation memorandum, signed by the Secretary of Defense in May 2023, that designated USTRANSCOM as the Single Manager and incorporated non-Integrated Material Management (IMM) responsibilities into the Single Manager mission; DLA retained all previous IMM and IMM-related responsibilities. This guidance sets the framework to codify policy and responsibilities in the revision of DoD Instruction 4140.25, DoD Management Policy for Energy Commodities and Related Services and the corresponding DoD manuals. ODASD(ER&O) then spearheaded working group meetings to align and revise DoD Instruction 4140.25, which is projected to complete by end of FY 2024.

Supporting GBFMD efforts, ODASD(ER&O) participated in multiple USTRANSCOM Single Manager Table-Top Exercises executed at the O6/GS15 and Flag Officer/SES levels to review and confirm bulk fuel responsibilities and decision-making across CCMDs, DLA, Joint Staff, and the Office of the Secretary of Defense (OSD) within contested environments. The incorporation of USTRANSCOM into decision-making processes enabled improvements at echelon across the Department. ODASD(ER&O) coordinated CCMD requests regarding changes to Preposition War Reserve Stock (PWRS) and establishing bulk fuel storage in support of USEUCOM and USINDOPACOM, consistent with the criteria outlined in the Department's "Adapting Bulk Fuel Planning and Operations," policy dated 3 September 2020. Additionally, ODASD(ER&O) finalized Department guidance for the FY 2024 Petroleum War Reserve Requirement (PWRR) regarding sizing, acquisition, management, and positioning of PWRS within the Inventory Management Plan for approved Operational Plans.

- **Operational Energy Innovation:** The Department utilizes the OECIF to advance technology development and the OEPF to validate more mature technologies to accelerate transition. OECIF and the OEPF are investing to develop new technologies and practices in four main thrust areas:

    - Contested Logistics and Battlespace Awareness – focuses investments on technologies that enable operations in contested environments and bring energy awareness into the DOD command and control architectures.
    - Operational Energy Resilience – a critical enabler for operational flexibility, freedom of maneuver, and mission assurance. Technical areas of this project include energy diversification, standardization, interoperability, and energy components advancing next generation warfighting capabilities.
    - Operational Energy Demand Reduction – drives innovations that reduce operational energy demand, increase efficiencies for aviation and space domains, and reduce risks by providing next generation power and clean energy delivery options (power beaming) for military effectiveness in contested environments.
    - Nuclear Power – leads DOD innovations enabling nuclear power across the battlespace. Technical areas of this project include nuclear fuel fabrication processes, energy conversion methods, novel power generation and energy storage technologies, and analytical study and strategy development.

The OECIF program continues achieving significant results for the warfighter. OECIF showed success in FY 2023, by: addressing key operational energy gaps for battlespace awareness; evaluating energy impacts of joint logistics resilience and sustainability in ongoing operations; advancing power and thermal management and energy storage to advance high power weapons; maturing advanced hydrogen and synthetic/alternative fuel production; increasing power and energy solutions for Arctic environments; and emphasized ground vehicle hybridization and electrification. OECIF also focused on longer-term, higher-risk investments such as advancing nuclear fuel options, successfully increasing distance and power-levels of wireless power transmissions for terrestrial and space applications, and advanced aircraft development to increase operational reach and speed. Finally, OEPF culminated their inaugural projects with a joint warfighting demonstration at the Marine Corps Tactical Systems Support Activity (MCTSSA) in Camp Pendleton, CA in late 2023. The demonstration showcased advances in energy storage, tactical microgrids, anti-idle kits for ground vehicles, vehicle centric microgrid capabilities, improved range/endurance of Unmanned Aerial Vehicles (UAVs), and advanced fuel monitoring, reporting and predictive logistics capabilities. A warfighter touchpoint database was piloted at the event to collect and aggregate feedback further accelerating transition of emerging operational energy capabilities.

Strategic Capabilities Office

The Strategic Capabilities Office (SCO) continued to execute Project Pele, a transportable microreactor which is on pace to be the first-generation IV nuclear reactor operated outside of China. The reactor is designed to fit inside four 20-foot shipping containers and can be transported by existing military transportation infrastructure. The Pele reactor will generate 1-5 Megawatts of electrical power for three years without refueling. In FY 2023, the Pele team worked toward finalizing the design of the reactor while also initiating the ordering of long lead hardware. This includes beginning to manufacture the enriched fuel for the Pele core, as well as beginning forgings and the manufacture of core moderator blocks. The SCO team is working closely with the Department of Energy regulators toward approval of the engineering design in FY 2024, which will allow the final assembly of the prototype reactor.

Office of the Under Secretary of Defense for Research and Engineering

The Office of the Under Secretary of Defense for Research and Engineering, OUSD(R&E), identified Renewable Energy and Storage (REGS) as one of 14 critical technology areas. This new critical technology area includes increasing energy efficiency; utilizing solar wind, bio-based and geothermal technologies; advanced energy storage; small modular reactors; and power grid integration. This portfolio includes both operational and installation energy renewable energy technologies to decrease warfighter vulnerability and deliver new operational capabilities for the Department. From more efficient batteries to diversifying energy sources and reduced fuel transportation risks, renewable energy generation and storage will add resilience and flexibility in a contested logistics environment. In support of these goals, OUSD(R&E) developed a DoD REGS technology strategy which will be followed in FY 2024 by a technical roadmap. OUSD(R&E) will continue to focus on prioritizing Department-wide Research, Development, Test, and Evaluation (RDT&E) resources toward these areas, overseeing U.S. Military Department Science and Technology to provide a consistent technical direction, supporting transition of these technologies into programs of record, and set the foundation to attract and build a talented future technical workforce. As promising technologies mature, REGS coordinates with OECIF and OEPF to provide funds for further development and transition. In addition, REGS participates with efforts led by the Assistant Secretary of Defense to coordinate battery communities within DoD.

Defense Logistics Agency

The DLA enables mission readiness by providing globally resilient energy solutions to the Warfighter and the Federal Government. Highlights for FY 2023 include:

- **Integrated Material Manager**: As the IMM for Bulk Petroleum, DLA focused on end-to-end supply chain management and oversight for all DoD bulk petroleum products and systems in coordination with the DoD Components. Based on the changes in the FY 2022 NDAA, DLA Energy supported implementation of USTRANSCOM to become the single element responsible for bulk fuel management and delivery. DLA participated in the OPT that developed the GBFMD memorandum required by Congress which confirms USTRANSCOM as the Single Manager for bulk petroleum and outlines DLA's retained IMM responsibilities. Additionally, DLA supported USTRANSCOM in the development and coordination of the Joint Petroleum Executive Council (JPEC) and Joint Petroleum Steering Group (JPSG) charters. The JPSG capitalizes upon the governance process previously led by the former Executive Agent Component Steering Group (CSG). DLA collaborated with USTC to begin transitioning CSG processes and actions to the new JPSG construct. Additionally, DLA established the DLA Energy GBFMD cell, which is co-located at the USTRANSCOM headquarters to enable a direct synchronization function between the USTRANSCOM single manager role and DLA Energy's IMM role.

- **Additization**: DLA continued to implement a comprehensive policy for additizing commercial jet fuel, including the establishment of expeditionary additization capability and enhancements to the resilience of the additive supply chain. DLA continues to collaborate with USINDOPACOM and Pacific Air Forces to preposition individual additive components (Fuel System Icing Inhibitor, Corrosion Inhibitor/Lubricity Improver, Static Dissipator Additive) at three DLA Distribution Centers in the Pacific theater. Storing the additive individually prevents issues with shelf life of the product. DLA has also established the capability to build additive kits at two of these three locations. The fuel additive kitting can provide the warfighter with a Medium kit (additizes 60,000 gallons of neat jet fuel) or a Large kit (additizes 250,000 gallons of neat jet fuel). If requested the warfighter would be sent one complete kit or the required number of individual items to support their fuel requirement. This process simplifies the additive process for the warfighter, improves DLA's flexibility to support specific fuel requirements, mitigates shelf-life concerns, and allows DLA to support all customers in theater. Currently DLA is posturing individual fuel additive components in USEUCOM and U.S. Central Command (USCENTCOM) distribution centers and working with the combatant commands to review their requirements.

- **Energy Readiness Research and Development (R&D)**: The Energy Readiness R&D Program is a designated Defense Operational Energy Program that directly supports the DoD Class III Bulk fuel and energy supply system through a multi-faceted approach to addressing issues connected to the entire spans of the system, in coordination with the military services. The program addresses issues connected with ensuring operation fuel integrity, supply chain infrastructure maintenance, and alternative (non-petroleum based) drop-in replacement fuel certifications. Current ongoing projects include:

    - **Woody Biomass Conversion to Liquid Hydrocarbon Fuels (Congressional Add Project) –** will result in viable fuel production alternative to petroleum using available forestry material.
    - **Evaluation of Hypersonic Fuel Conditions –** will develop test protocol for determining thermal stability of endothermic (high-heat absorption) fuels used for hypersonic (greater than MACH 5) flight requirements.
    - **Test and Evaluation of Alternative Rocket Fuels (TEARF)** – entails the selection, analysis, and experimental evaluation of existing or readily available alternative rocket kerosene formulations to broaden the supplier base.

- **Method Comparison for Quantification of Nitrogen Content in Mobility Fuel –** will consolidate DoD and industry analytical test methods for nitrogen content in jet fuels to enable improved fuel quality monitoring across the DOD.
- **Method Optimization for Quantification of Polar Heteroatomic Species in Mobility Fuel –** will optimize previously developed analytical methods that detect and quantify deleterious polar heteroatomic species in mobility fuels.
- **Using a Field Portable Liquid Chromatography (LC) Device for Detecting Additives in Fuel –** evaluating the capability of a field-portable liquid chromatography system to replicate the capabilities of a laboratory benchtop instrument with respect to current LC-based methods for analysis of metal deactivator and corrosion inhibitor/lubricity improver additives.
- **Trace Metals and their Prevalence, Removal, and Impacts on Thermal Stability** – will develop DOD analytical methods for trace metals harmful to fuel stability and evaluate the metal content of petroleum fuels with various available instruments and novel techniques.
- **Arctic Fuel System Supply Point –** supports the DoD's Arctic Strategy to enhance cold climate capabilities necessary to provide the Army with the ability to operate in extreme cold-weather environments.
- **High Performance Green Aerospace Propellant Composition Characterization –** investigates and further develops liquid monopropellant-103S (LMP-103S) aerospace propellant quality check processes and methodology in order to create an industry standard that can be used for increased source procurement and sampling verification.
- **Determination of Aviation Fuel Thermal Stability Breakpoint Via Single Run** – will streamline and shorten the test method process for determining thermal (high-heat) operational stability properties of jet fuels.
- **Development of Blend Rules for Thermal Stability** – will establish protocols for reconstituting non-compliant in meeting thermal (high-heat) operational stability properties through blending with compliant fuels.
- **Low Cetane Fuel Durability Study** – will study the impacts of using jet fuels that have low ignition quality in compression ignition (diesel cycle) automotive engine vehicles in adherence to the U.S. DOD Single Fuel Usage on the Battlefield Policy.

- **Battery Network**: The Battery Network manufacturing technology R&D program, along with Small Business Innovation Research (SBIR) investments, supported multiple efforts to improve supply and performance, address obsolescence, and reduce costs for batteries used in fielded weapons or soldier systems.

    - DLA established the manufacturing design and processes for significantly lighter, bi-polar lead-acid batteries and plans to deliver Military Performance Specifications, MIL-PRF-32143 and MIL-PRF-8565/11 battery sets for testing and qualification, respectively, to the Army Ground Vehicle Systems Center (GVSC) in 2024 and Navy Surface Warfare Center (NSWC) in 2025. Lighter batteries will lower fuel use in assigned systems and distribution logistics.
    - DLA also delivered design analyses of lithium-ion alternatives for multiple nickel-cadmium batteries which can improve supply chain viability, lower battery weight, increase power and energy, and significantly extend battery life (which lowers replenishment requirements). The Air Force is now prototyping C-17A lithium titanate batteries and considering the technology for other nickel-cadmium replacements. DLA is prototyping nickel-cadmium replacements for Naval Sea Systems Command (NAVSEA) MK-45 Gun Systems and Naval Air Systems Command (NAVAIR) is reviewing proposals for new lithium-ion alternatives.

- DLA also continued work on a range of technical enhancements to battery systems. These improvements include lower cost, high-quality production of carbon monofluoride used in superior non-rechargeable batteries (BA-5790), leveraging solid-state electrolytes to improve small system battery safety (BB-2590), and leveraging battery deactivation and materials separation/reclamation processes for more efficient, non-hazardous, disposition of batteries around the world.

Department of the Army

The Department of the Army consumed 9 percent of the Department's operational fuel in FY 2023, which compares to 6 percent in FY 2022. Total Army flight hours for FY 2023 have yet to be fully reported, but in FY 2022, 12,404 BBtus (47.7 percent) of Army operational fuel consumption was for air operations and 13,619 BBtus (52.3 percent) was for ground operations. Overall, the Army consumed fuel to train, deploy, conduct contingency operations worldwide, and sustain those operations.

- In support of the *National Defense Strategy* and the Army's ability to conduct logistics in contested, multi-domain environments, the Army is developing energy technologies that will provide warfighters tactical advantage through an organized campaign of rapid experimentation, prototyping and Soldier evaluations. By minimizing the demand for supplies and services, including fuel, operational forces are capable of greater range and endurance, giving Commander's greater freedom of action and reducing risk to combat and sustainment forces. These modernization and demand reduction efforts are reflected in the Army's investments in energy-enabled capabilities.

- The Army continued to test and develop anti-idle capabilities for tactical wheeled vehicles throughout FY 2023 in preparation of retrofitting the current fleet beginning in FY 2025. Anti-idle kits are expected to reduce the fuel consumption of a vehicle by at least 20 percent and engine runtime by up to 50 percent, improving fuel efficiency, extending operational endurance, and reducing maintenance requirements.

- The Army is also developing technologies for hybrid-electric drive vehicles. In FY 2023, the Army completed testing on two Bradley hybrid-electric technology demonstrators, and is acquiring two hybrid-electric High Mobility Multipurpose Wheeled Vehicles (HMMWVs), and one hybrid-electric Joint Light Tactical Vehicle (JLTV) for experimentation. Hybrid-electric vehicles are expected to reduce fuel consumption by up to 35 percent, provide extended silent watch and silent mobility capabilities, and reduce thermal and audible signatures to increase survivability. The advanced electrical systems will also provide more power to support emerging capabilities such as active protection systems, directed energy weapons, or power import/export capabilities. The advancements being made in hybrid-electric technologies are informing the development of future tactical and combat vehicle platforms.

- In 2023, the Army began planning to deliver experimental battlefield recharging technologies to Army units to test these concepts in austere environments under simulated combat conditions. The recharging technologies will power both Soldier systems and Tactical and Combat Vehicle-Electrification (TaCV-E) prototype platforms, enabling the Army to test future hybrid-electric and all-electric vehicles alongside legacy internal combustion engine fleets. The Soldier touchpoint opportunities provided by these proofs of concept will not only assist the Army's modernization efforts with refining the development of future materiel capabilities to improve warfighter protection, maneuverability, and lethality, but will inform the development of modern Army doctrine, training, and strategy. The lessons learned from these training events will hone tactics, techniques, and procedures (TTPs) at the tactical edge of the battlefield, ensuring Army forces

most efficiently and effectively employ the most modern weapon and vehicle systems to their maximum capability.

- Fast-forming, cyber-secure tactical microgrids can provide resilient and efficient ground-based electrical power for contingency base applications, air and missile defense systems, and tactical command posts. In January 2023, the Army-developed Tactical Microgrid Standard was approved as MIL-STD-3071. The integration of more efficient Advanced Medium Mobile Power Source (AMMPS) generators with a microgrid can decrease fuel consumption by 45 percent, while the addition of tactical storage to the microgrid could decrease fuel consumption by 65 percent. In August 2023, the Army demonstrated powering elements of a Terminal High Altitude Area Defense (THAAD) missile battery using an advanced TMS-compliant microgrid at McGregor Range on Fort Bliss, Texas. Tactical microgrid controls enable autonomous start-up and shutdown of generators based on demand to maximize fuel efficiency and improve power redundancy and resilience by employing multiple generators capable of sharing loads.

- Additional Army RDT&E efforts in FY 2023 to advance Army operational energy capabilities include the completion of a series of functional, safety, and performance tests for the T901 Improved Turbine Engine. The Army received the first T901 flight test engine in September 2023. The T901 replaces the 1970's era T700 design and will retrofit the UH-60 Blackhawk and AH-64 Apache fleets. The improved turbine engine is expected to reduce fuel consumption by 13 percent to 25 percent, provide 50 percent increased shaft horsepower, decrease maintenance time and costs by 30 percent, and provide the ability to operate in high/hot conditions at full payload.

- The Army is working with the other Services and DLA to qualify sustainable fuels from alternative pathways and certify aircraft for their use as drop-in replacements for conventional petroleum. To date, the Army has qualified alternative JP8 fuels from two pathways for use in all Army equipment, and certified Army aircraft to use alternative fuels from an additional two pathways. During FY 2023, the Services completed testing fuels from four additional JP8 pathways which are awaiting final Department approval for use and inclusion in standards and contracts.

To support both the development of the future force and the readiness of the current force, the Army invested significantly in science and technology, the development of modernized capabilities, and upgrades to the legacy fleet. See Appendix O for details on FY 2023 investments.

Department of the Navy

The Department of the Navy (DoN) remains steadfast in its commitment to advancing energy initiatives in line with the National Defense Strategy and the evolving landscape of Great Power Competition. Building upon the foundations laid in FY 2022, the DoN has further intensified its focus in FY 2023 on Distributed Maritime Operations (DMO) and Expeditionary Advanced Base Operations (EABO). The integration of energy objectives into Design 2.0 continues to be a priority, emphasizing resupply capability, operational reach of weapons systems, and energy command and control for the forward-deployed integrated force in contested environments.

- **Summary of DoN Fuel Use**: Including both Navy and Marine Corps, the DoN consumed 37 percent of the Department's operational energy use in FY 2023. Overall, the Department of Navy consumed energy across these broad mission areas (37 percent Navy aircraft, 33 percent surface ships, 17 percent Military Sealift Command ships, 12 percent Marine Corps aircraft, and 1 percent Marine Corps ground tactical platforms). In FY 2023, the DoN's operational energy program focused on power and fuel in support of platforms, advanced sensors, and directed energy weapon

systems, fuel distribution in contested environments, as well as developing systems that provide commanders information about their energy use to achieve a more lethal force with increased operational reach and time-on-station.

- **Alternative Fuel Qualifications**: Investments in mobility tactical fuels and alternative fuels continued, with the completion of three tri-service assessments for synthetic fuels to fully align with commercially approved synthetic aviation turbine fuels (SATF). Additionally, testing and analysis of two synthetic ship fuels was completed. Additional property data for 1,743 fuel samples (both within and outside of the continental U.S.) was added to the NFACT Data analytics tool to develop hardware data modules to reduce cost (~$1M+), time and testing required to evaluate new fuels. Models were developed to predict fuel bulk properties from fuel composition that reduce the volume of fuel required for analysis by greater than 99 percent. Technical data and recommendations were updated and provided to NAVSEA in FY 2023 for the use of F-75 as an alternative fuel in areas where F-76 is limited. These efforts aim to enhance mission capability, reduce Total Ownership Cost (TOC), and improve worldwide fuel availability and operability.

- **Improvements to Current Equipment**: The Navy has prioritized the reduction of fuel consumption through the implementation of energy efficiency measures in existing and future systems. Continued development of the Integrated Power Systems (FY 2023, $146.7M) will provide improved platform flexibility by allowing selection of the most efficient combination of engines to supply the total electric power required for the combined propulsion and ship service loads, which provides for greater fuel efficiency and enhanced warfighting in comparison to a mechanically driven ship propulsion system. This will enable current and future weapon systems balance of power between propulsion and system loads resulting in significant efficiencies in fuel consumption. Current efforts are underway to improve energy storage systems on ships by consolidating uninterrupted power supplies (UPSs) and leveraging lithium-ion battery technology. Enhancing maintenance and modernization of ship systems through metering and monitoring efforts will allow for data-driven support of optimal operations of existing, and integration of future, energy storage and weapons systems.

- **New Capabilities**: The Navy continued investment in refueling infrastructure, energy storage, and research and development for energy command and control tools, such as the Global Energy Information System (GENISYS) and Replenishment at Sea Planner (RASP). This demonstrates the Navy's commitment to optimizing energy and logistics supply and demand for distributed operations. There were 12 completed GENISYS installations on DDG-51 class ships in FY 2023, with 30 planned installations for FY 2024, and expansion plans in place across various ship classes. The Navy continues to develop the first CVN-based organic mission and recovery tanker. The MQ-25 Program (FY 2023, $239M) will extend the range and increase lethality of the Carrier Aircraft Wing and will contribute to F/A-18EF shortfall by relieving tanker duties and returning aircraft to the strike fighter role. MQ-25 will focus on improving the range and endurance and have a secondary ISR capability.

The Marine Corps continued to invest in advanced power and mobility capabilities that use less fuel than previous technologies. Mobile electric hybrid power systems, small unit power, and energy storage units give expeditionary units access to renewables and microgrids to reduce fuel demand. Another major focus is the use of standard lithium batteries to offset disposable battery use and improve the mobility and lethality of dismounted warfighters. In FY 2023, the Marine Corps used an array of funding sources (e.g. OECIF, Sec 2912, etc.) to continue investing in hybrid tactical vehicles and combat systems research with the prototype Advanced Reconnaissance Vehicle (ARV) (FY 2023, $3.7M), High Power and Hybrid Electric ULTV (FY 2023, $7.6M), Medium Tactical Vehicle Replacements (MTVR) (FY 2023, $5M),

Next Generation-Medium Tactical Truck (NG-MTT) (FY 2023, $6M), and Logistics Vehicle System Replacement (LVSR) (FY 2023, $1M).

- **Advanced Batteries**: The Battery Safety Community made significant progress this year in battery containment and sensing as well as completing safety certification on a Lithium 6T vehicle battery that will enable DOD ground vehicles to improve energy performance. The certification also allows those batteries to be transported on most Navy ships. The containment system designs that Navy Laboratories are developing are available to share with other agencies and industry.

- **Energy in Requirements and Acquisition:** In FY 2023, DoN drafted a memorandum to provide guidance on how the DoN will reinvigorate current practices to accurately account for energy and intentionally adopt more efficient technologies and practices. DoN's approach describes a systematic, full lifecycle solution that links energy across wargaming, requirements, acquisition, operational and workforce behaviors to address challenges in each area. The DoN conducted a comprehensive evaluation of acquisition policy to identify areas for improvement; produced up-to-date and consistent energy supportability analysis for five acquisition programs to include Landing Ship Medium (LSM), Joint Petroleum Over the Shore (JPOTS), Next Generation Logistics Ship (NGLS), DDG(X) Next Generation Destroyer, and Expeditionary Medical (EXMED). In addition, ESA leads conducted additional analysis to evaluate fleet energy sensitivities to further understand and improve ESA capabilities. The results of the analysis will lead to refinement and development of enhanced energy specific acquisition milestone / gate guidance, best practices, and templates.

- **Operational Energy Coast Savings**: In FY 2023, DoN Operational Energy retained and executed $27.68M of the identified $171M of validated operational energy cost savings to invest in additional operational energy efficiencies in accordance with title 10 U.S.C. § 2912 in support of DOD and DoN operational energy strategy and goals.

- **Training and Education**: In FY 2023, DoN initiated an Operational Energy Certificate Program at Naval Post Graduate School in the areas of Unmanned Systems Persistence, Refueling Logistics and Directed Energy Weapons in Asymmetric Warfare and launched an online Operational Energy General Military Training (GMT) course for which 2974 military personnel completed and passed training.

In conclusion, the DoN's energy initiatives in FY 2023 demonstrate a continued commitment to enhancing operational effectiveness, resilience, and readiness across the Navy and Marine Corps, aligning with the dynamic challenges presented by the evolving global security landscape. See Appendix O for details on FY 2023 investments.

Department of the Air Force

The DAF consumed 54 percent of the Department's operational energy in FY 2023 and remains consistent with previous years' consumption numbers. Overall, the Air Force consumed energy across these broad mission areas: 53 percent mobility, 33 percent fighter/attack, 6 percent bomber, and 4 percent training. Through the rapid deployment of 21st century technology and a data-driven approach to optimizing energy use in operations, the Air Force continues to enhance combat capability and readiness.

- **Data Collection and Analysis**: In 2023, the DAF continued the partnership with the 618th Air Operations Center (AOC) and MIT Lincoln Laboratory (MIT LL) to further develop the Next Generation Information Technology for Mobility Readiness Enhancement (NITMRE) concept to digitally transform the Mobility Air Forces (MAF) C2 architecture. This architecture optimizes

the wartime needs of multiple combatant commanders by employing ChatOps mobile collaboration software as a primary method of communication. Additionally, the effort-built NLP-enabled technologies to extract and provide insights from the ChatOps to aid warfighters by accelerating information flow and reducing uncertainty in planning through more timely and precise cargo, fuel, weather, and routing information.

- **Cargo Optimization**: The DAF also continued to partner with the Defense Innovation Unit (DIU) and Air Mobility Command (AMC) on Air Logistics Optimization (ALO), which leverages proven operations research methodologies to optimize airlift cargo and aircraft scheduling through efficient cargo routing, ultimately reducing empty and partially empty legs. It also provides a more comprehensive operational picture for disruption management and dynamic scheduling. Additionally, the solution will ingest and aggregate requests for movement of cargo for processing, including a modified user interface that enables users to oversee the ALO's generated schedule and quickly identify potential issues.

- **Aviation Alternative Fuels**: Research and analysis of conventional and alternative fuel certification processes, fuel additives, and fuel transfer and storage equipment help to identify and address problem areas resulting in improved resilience of the jet fuel supply chain while conducting logistics while under attack.

The Logistics and Sustainment division maintained its ongoing engagement with commercial stakeholders such as the Commercial Aviation Alternative Fuels Initiative (CAAFI), International Civil Aviation Organization (ICAO), our OEM manufacturers and other government agencies to maintain awareness of the research, development, testing, certification, and commercialization of "drop-in" alternative aviation fuel in the ASTM International approval process. Air Force Operational Energy continued to team with DOD and DAF stakeholders, such as DLA Energy, Air Force Petroleum Office, Air Force Research Laboratory, and Air Staff, as well as international allies and partners, to monitor alternative fuel production, pricing, and commercial use. These partnerships position the DAF to support changes in the defense alternative fuels strategy and provide operational flexibility based on fuel availability in specific regions/circumstances.

- **Legacy Aircraft Sustainment**: The DAF liaised with multiple defense industry and commercial aviation entities to identify industry best practices and technologies in aircraft sustainment that could potentially translate to the Air Force. By leveraging innovative material for coatings of engine compressor blades, the DAF is looking to ensure overhauled legacy engines deliver optimized engine performance. The commercial aviation industry is realizing fuel savings of 1.5 percent for turbine compressor bladed coatings, which also have the potential of extending blade life 10x. The Air Force performed certification of coatings for the F117 engine (C-17) and will integrate them at the Depot starting in 2024. The Air Force is also performing certifications for coated blades to be incorporated into the F108 (KC-135), which is projected to reduce fuel demand by $25 million per year if fully implemented across the C-17 and KC-135 fleets.

- **Mission Execution Excellence Program (MEEP)**: As part of DAF efforts to streamline operations and increase aircraft range and capability, Air Force Operational Energy, in partnership with Air Mobility Command, launched the MEEP in FY 2022 and substantially expanded the program in FY 2023. MEEP is a fuel savings incentivization program promoting optimized flying on the largest fuel consumers in the Air Force, heavy mobility aircraft. MEEP's overarching goal is to drive cultural change among Airmen to optimize use of aviation fuel in preparation for future conflict in a fuel-constrained environment (i.e., USINDOPACOM) with the largest point of emphasis on precise fuel planning. Additionally, MEEP encourages use of efficient flying 'best

practices' through education and incentivization without negatively impacting mission and training requirements.

In 2023, MEEP expanded to a total of 15 Squadrons including both the C-17 and the C-5 MDS with both active-duty and AFRC participating. MEEP saved 4.8M gallons of aviation fuel or $18.6M in cost avoidance in 2023. As a result of these accomplishments, Air Force Operational Energy provided over $7M of incentive funds to the Wings participating in the program. These funds will be executed on unit-level operational energy initiatives and to procure resources that improve DAF's worldwide combat capability. MEEP is one of the most cost-effective programs in SAF/IEN's portfolio and is a significant contributor to the overall climate energy intensity goal, representing approximately 1.5 percent of total AF energy intensity progress.

- **Capability Development and Acquisitions**: The Future Operations division of Air Force Operational Energy guides acquisition policy and influences plans and programs to address operational energy requirements associated with new and major modification programs through the eKPP and ESA. Through research, development, prototyping, and demonstration efforts, the Air Force works closely with AFRL to push the technology envelope and help advance key disruptive technologies to maximize operational energy efficacy and outpace our competitors.

The DAF has invested in and advocated for several technologies and platforms that increase range or endurance and optimize fuel use for numerous critical missions. The DAF also plays an active role in the Capability Development process, participating in Capability Development Working Groups. Through these engagements, the DAF helps ensure future systems will not only be optimized for energy considerations but will be sustainable in their intended operational environments.

The DAF works closely with the office of Future Force Energy and Power (F2EP), the Air Vehicles and Turbine Engines Divisions, and the Center for Rapid Innovation at AFRL, as well as Air Force Futures Disruptive Technologies Division and DIU to transition new technology to the Air Force and Combatant Commands.

- **Drag Reduction**: AFRL Future Force Energy and Power efforts that were new or continuing in 2023 span a wide range of capability improvements on legacy Air Force platforms. The DAF continues to support and advocate for C-17 and C-130 Drag Reduction programs, which smooth aft body turbulence and will save the Air Force $10 million per year in fuel costs, with a return on investment of less than six months. Reorientation of KC-135 wipers from horizontal to vertical will increase efficiency by 1 percent, saving an estimated $7 million per year in fuel. Additionally, promising research into KC-135 aft body drag reduction continues. The DAF conducted initial investigations in coordination with industry and AFRL partners for KC-135 aft body drag reduction devices, which are expected to save more than $7 million per year. The DAF secured $15.5 million in funding to implement these technologies on 300-395 KC-135s through FY 2023. Similarly, the DAF secured $20.5 million in funding to complete development and install C-130 aft body drag reduction devices through 2023, and $4 million in funding from FY 2022 to FY 2023 to complete prior development efforts for C-17 engine pylon fairings. Finally, in FY 2023, the DAF began a 15-month prototyping effort to optimize mobility aircraft flight control surface rigging and identify potential fuel savings using industry partner proprietary data analytics. All together, these initiatives offer and estimated annual savings of more than $20 million.

- **Blended Wing Body Aircraft**: The DAF awarded a $245M contract to design and build a detailed prototype Blended Wing Body (BWB) aircraft in August 2023. BWB is expected to increase fuel efficiency of large aircraft by 30 percent (with today's engines) significantly impacting both mission and climate. This prototype project aims to accelerate future optionality

for tanker, cargo, and bomber fleets while leveraging significant private capital to maximize DoD return on investment. The project aims to complete the first flight of a full-scale BWB aircraft by December 2026 and initial flight testing by September 2027.

- **DAF Climate Action Plan**: In 2022, Air Force Operational Energy finalized climate objectives and key results as part of the Department of the Air Force Climate Action Plan (CAP), as directed by Executive Order (EO) 14008, *Tackling the Climate Crisis at Home and Abroad*, EO 14057 *Clean Energy Industries and Jobs through Federal Sustainability*, and the Department of Defense Directive (DoDD) 4715.21 *Climate Change, Adaptation, and Resilience.* The DAF CAP is a living document that will be modified periodically to address new policies and guidance, technology innovation, and evolving missions. The DAF Senior Leader Climate Forum, comprised of members from the Offices of the Secretariat, Deputy Chiefs of Staff, Deputy Chiefs of Space Operations, and chaired by the ASD(EI&E), will provide oversight, monitoring, and implementation direction through quarterly status briefings, while DAF Climate Action Team members will develop more specific Plan of Action and Milestones, convene monthly, and execute goal activities. Air Force Operational Energy's climate mitigation and fuel efficiency efforts within the DAF CAP are detailed below:

    - Objective 3.1: Improve Operational Energy Intensity: Optimized aircraft energy use increases the readiness and operational capability of the department, decreases risk to logistics supply chains, and informs operational plans while reducing greenhouse gas emissions. Optimization includes the adoption of ultra-fuel-efficient aircraft designs, which provide an opportunity to significantly improve combat capability.

        - **Key Result 3.1.1**: Operational energy intensity of Air Force flying missions increased 5 percent by FY 2027 and 7.5 percent by FY 2032 through standardized use of aircraft drag reduction technologies, modern software scheduling tools, and enhanced engine sustainment practices.
        - **Key Result 3.1.2**: Development and testing of a full-scale blended wing body prototype completed by FY 2027.

Conclusion

In summary, the Department is leading multiple efforts to reduce energy demand, mitigate risk, and enhance operations in contested environments through the adoption of new weapons systems and technologies and improved operational concepts, training, and command and control. To ensure that Joint Forces have access to the energy needed to fight and win while operating within contested environments, the Department will reduce operational energy demand, diversify our energy sources, improve supply chain resilience, and enhance the enterprise-wide visibility of energy supply and demand. The Department investments and initiatives reviewed in this report will increase the resilience of the force against emerging threats and enable the successful implementation of the *National Defense Strategy*.

4. Installation Energy

Energy Resilience Policies and Guidance

In managing an energy resilient enterprise, the DoD must be prepared for and be able to recover from utility disruptions that impact mission assurance on its installations. DoD relies on commercial power to conduct missions from its installations, and these commercial power supplies can be threatened by physical attacks, cyberattacks, climate change, and other environmental factors. Climate change is expected to increase risks of wildfires, winter storms, hurricanes, storm surges, rain and heat hazards, and other extreme weather events, which are threat multipliers and degrade installation functions and infrastructure. The DoD recognizes that such events could result in power outages affecting critical DoD missions and continues to work to understand and address the vulnerabilities that could lead to such events and impact mission readiness.

As defined by Title 10 U.S.C. 101(e)(6), energy resilience is

"the ability to avoid, prepare for, minimize, adapt to, and recover from anticipated and unanticipated energy disruptions in order to ensure energy disruptions in order to ensure energy availability and reliability sufficient to provide for mission assurance and readiness, including mission essential operations related to readiness, and to execute or rapidly reestablish mission essential requirements."

Energy resilience can be enhanced in a variety of ways, including the use of technologies, policies, readiness exercises, and energy resilience management best-practices. Mission capability concerns override preferences for specific technology implementation goals. An important aspect of energy resilience is to establish an iterative planning and implementation cycle in which mission owners conduct risk analyses and specify requirements, infrastructure stakeholders solve for the specified requirements, and the process repeats itself as needed to meet changing mission parameters.

DoD's primary policy for energy strategy and guidance is DoD Instruction (DoDI) 4170.11, Installation Energy Management which provides a framework for reducing energy consumption and investing in energy reduction measures, with the goal of providing resilient energy to support the warfighter. A significant update to DoDI 4170.11 began in 2022 and is expected to be completed in 2024, to incorporate energy resilience strategies and climate resilience goals for the DoD.

DoD publishes policies, guidance, and the status of its energy resilience program on this website: OUSD(A&S) - ASD(EI&E) (osd.mil).

Installation Energy Plans (IEPs): Title 10 U.S.C. § 2911

Title 10 U.S.C. § 2911(b)(1) and (2) requires the DoD and Components to develop comprehensive master plans for the achievement of DoD energy goals. The DoD continues to ensure the use of structured, rigorous evaluations of critical energy requirements, risks, and mitigations by completing Installation Energy Plans (IEPs). IEPs drive a more integrated and systematic approach to energy and water management through informed energy and water resilience planning and support a more holistic energy project implementation with input from all stakeholders. Installations are responsible for conducting gap analysis and developing appropriate options for risk mitigation, while developing energy projects that optimize energy efficiency, reliability, and resiliency. In FY 2023, the DoD continued completion of IEPs across the enterprise, focusing installation resilience requirements on the most advantageous

technology and funding execution paths. To date, the DoD has completed IEPs at 73 percent of installations.

Black Start Exercises (BSEs)

BSEs are a mission-assurance focused exercise mandated by 10 U.S.C. § 2920 in which Military Departments must carry out no fewer than five (5) planned BSEs per year through FY 2027. As a planned, risk-mitigated exercise, BSEs emulate a loss of utility power by disconnecting an installation from off-installation energy sources. These exercises identify gaps in critical mission requirements and in conjunction with IEPs assist installation personnel in identifying and executing energy resilience solutions to support critical missions.

While each exercise has been different, general findings and resilience gaps identified from prior BSEs include:

- Backup generators alone can be insufficient to provide energy resilience with challenges that include being severely under-loaded, electrical misconfiguration, and improper prioritization of critical loads;
- Lack of resilient information technology (IT) architecture for cyber-security; and
- Unknown energy and control system interdependencies that lead to unanticipated failures.

In FY 2023, the DoD completed 23 BSEs across the Services identifying energy vulnerabilities and solutions to improve installation energy resilience.

Table 2. BSEs for Determining Energy Vulnerabilities

Service	FY 2023 BSE SNAPSHOT
ARMY	Six (6) BSEs in FY 2023: Fort Campbell, Fort Carson, Fort Drum, and U.S. Army Garrison Bavaria (which included BSEs conducted at the main campus and geographically separated sub-Installations - Grafenwoehr Training Area and Hohenfels Training Area).
NAVY	Ten (10) BSEs in FY 2023: Naval Air Station (NAS) Meridian, Naval Support Activity (NSA) Naples, NAS Jacksonville, NAS Key West, NAS Joint Reserve Base New Orleans, Naval Submarine Base (NSB) Kings Bay, NSA Panama City, NSA Hampton Roads, Naval Station Everett, and NSA Mid-South.
USMC	Two (2) BSEs in FY 2023: Marine Corps Air Station (MCAS) Yuma and Marine Corps Air Ground Combat Center (MCAGCC).
DAF	Five (5) BSEs in FY 2023: Beale Air Force Base, Robins Air Force Base, Creech Air Force Base, Otis Air National Guard, and Cape Cod Space Force Station.

Critical Energy Capacity and Availability: Title 10 U.S.C. § 2920

Critical missions, as defined in title 10 U.S.C. § 2920, refer to those missions of an installation, including mission-essential operations, that are critical to successful performance of the strategic national defense mission. Energy systems used to support critical missions are developed and managed at the installation level by each of the Components and expected to meet the requirements outlined in title 10 U.S.C. § 2920. Figure 5 below provides details on critical energy system capacity and progress toward achieving a minimum level of 99.9 percent energy system reliability for critical missions by 2030. The average system reliability, defined as the percentage of time the energy system is functional, is approximately 99.6 percent for the entire DoD.

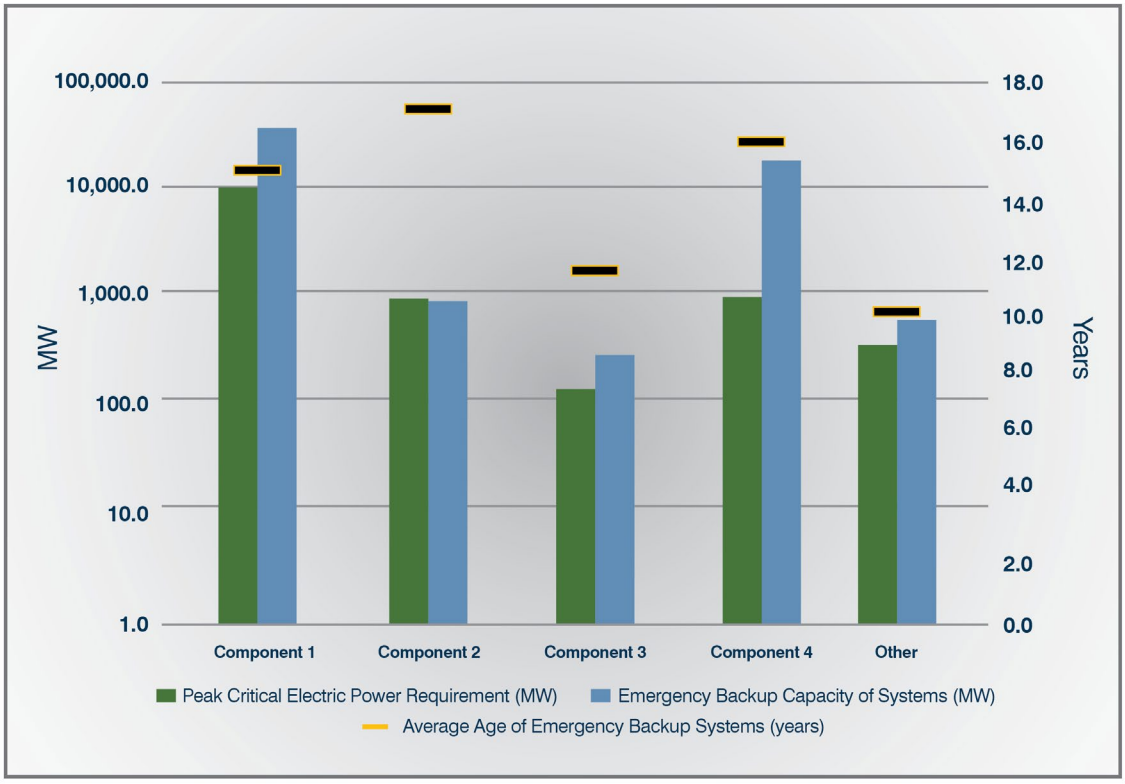


Figure 5: Critical Energy Requirements and Emergency Backup System Capacities and Average Age by DoD Component

Figure 6 below provides details on average system downtimes and average availability over the course of the year of these critical energy systems for the DoD Components.

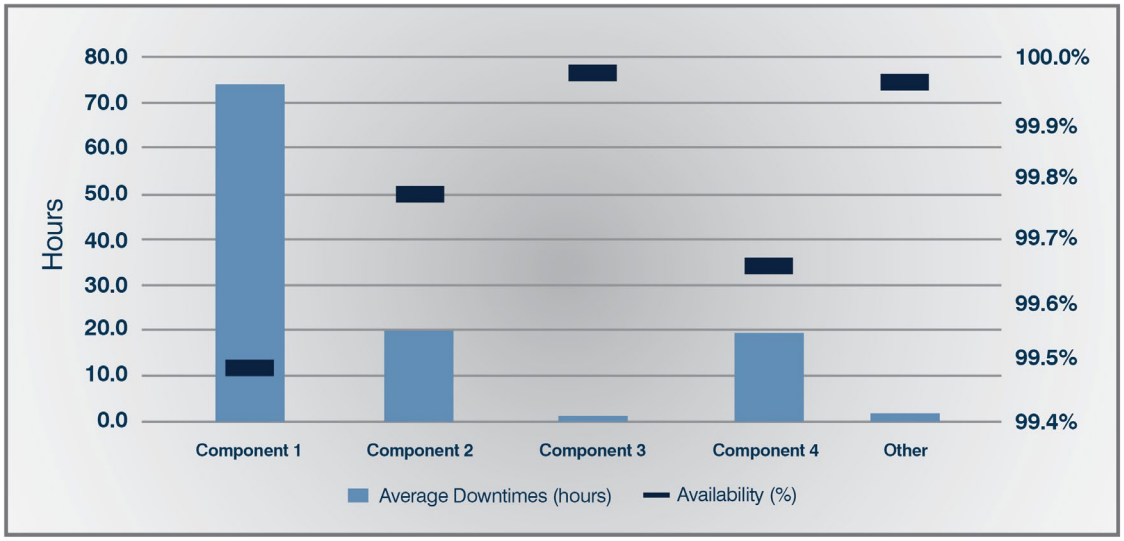


Figure 6: Average downtimes for critical energy system and average energy system availability by DoD Component

Utility Outages on Privatized Systems: Title 10 U.S.C. § 2688(g)(4)

To meet statutory requirements, DoD began collecting outage data to measure progress on privatized systems in FY19. Table 3 below provides details of FY 2023 outage data on privatized systems to help meet this requirement. The average cost of outages has increased over the past three years due to extreme weather events that affected energy systems, including Winter Storm Uri in FY 2021, heat waves across the U.S. in summer of FY 2022, wildfires on Maui Island and Typhoon Mawar in Guam in FY 2023, and other external factors affecting the cost of energy.

Table 3. Privatized System Outage Summary

PRIVATIZED SYSTEM OUTAGE SUMMARY					
	FY 2019	FY 2020	FY 2021	FY 2022	FY 2023
Percent of outages on privatized systems	19%	21%	15%	15%	19%
Percent of privatized system outages (8+ hours)	13%	17%	14%	16%	16%
NOMINAL COSTS ($)					
Total cost of outages on privatized systems	$ 404,071	$ 190,725	$76,507,606	$ 2,975,694	$ 8,620,333
Average cost per day of outages	$ 981	$ 540	$ 62,755	$ 2,190	$ 8,449

The Energy Resilience and Conservation Investment Program (ERCIP)

ERCIP, which is authorized by title 10 U.S.C. § 2914, is a critical element of the DoD's strategy to improve the energy resilience, energy security, and energy conservation of its fixed installations. ERCIP is the principal mechanism applied to execute DoD high-priority, high-value projects that improve energy resilience and mission readiness. The FY 2021 NDAA implemented significant 10 U.S.C. § 2914 modifications, transitioning ERCIP to a project/line-item authorized Military Construction (MILCON) program. ERCIP improves readiness by focusing on the closure of critical energy gaps at priority locations, increasingly through the installation of flexible and expandable microgrids, deployment of energy storage technologies, the leveraging of existing renewable energy generation resources, and application of cyber-security improvements. As ERCIP projects employ increasingly complex technologies, the cost to design and construct ERCIP projects also increases. In FY 2023, Congress added $100M to ERCIP, increasing the FY 2023 appropriations to just over $653M. In FY 2024, just over $634M was requested and appropriated for ERCIP.

Table 4. Energy Resilience Project Snapshots

Service	FY 2023 ERCIP SNAPSHOT
ARMY	In FY 2023, the Army received $119M for five power generation and microgrid projects. Additionally, the Army received $36.8M in cost to complete funds for four FY 2022 projects and began the detailed planning and design work for 14 projects.
NAVY	The Navy received $132.72M for eight projects at seven locations. This included critical electrical distribution infrastructure upgrades at Naval Base (NAVBASE) Guam and Joint Base Pearl Harbor-Hickam, Hawaii. The Navy also received approximately $32M in cost to complete funds for two projects in Guam and began design efforts for seven projects.
USMC	In FY 2023, the USMC received $25.5M for Marine Corps Mountain Warfare Training Center Bridgeport, California. $608K in cost to complete funds were provided for MCAS Miramar to complete construction on the landfill gas (LFG) Power Meter Station. An additional three projects received funding to begin design efforts.
DAF	The Department of Air Force received $15.7M for two projects at Patrick Space Force Base (SFB), Florida. The DAF received an additional $30M in cost to complete funds for two FY 2022 projects and began design efforts for nine projects at eight locations.

Installation Energy Alternative Funding

The DoD seeks to maximize the use of authorities to pursue third-party financing of installation efficiency, renewable, other carbon pollution-free (CFE) energy generation, and resilience projects. These authorities are Utility Energy Service Contracts (UESCs) and Energy Savings Performance Contracts (ESPCs) (title 10 U.S.C. § 2913), power-purchase agreements (PPAs) (title 10 U.S.C. § 2922a), and outgrants (title 10 U.S.C. § 2667 and 10 U.S.C. § 2668)(7) (see Appendix H).

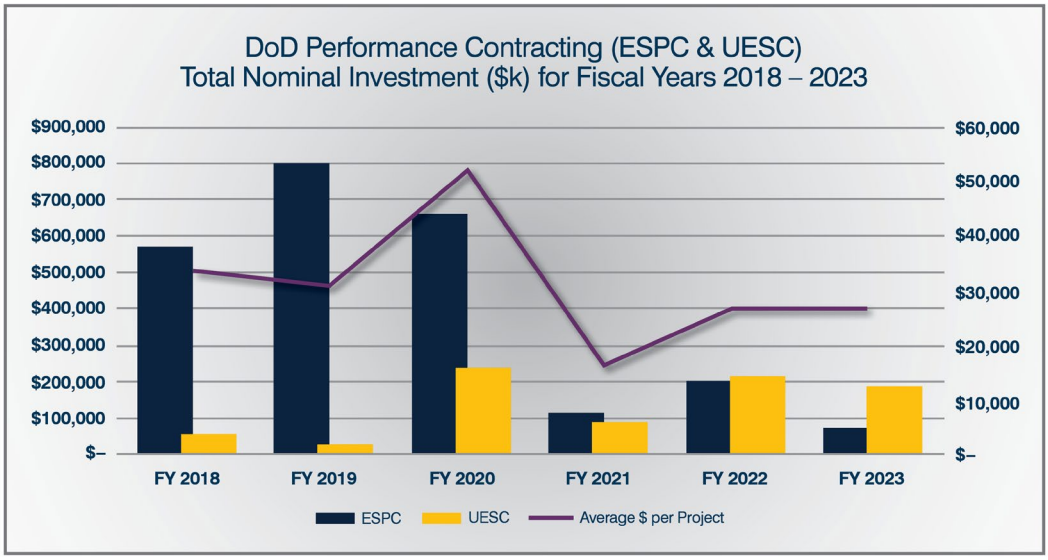


Figure 7: DoD Performance Contracting (ESPC and UESC) Total Nominal Investment Value ($k) for Fiscal Years 2018 – 2023

The DoD has the largest performance contracting portfolio in the Federal Government (over $5B). As enabled by the Energy Act of 2020, Energy Independence and Security Act (EISA) 2007, and Energy Policy Act (EPAct) 2005, DoD continues to use performance contracts to bolster resilience and reduce energy and water intensity. By statute, all such projects must pay for themselves through energy savings. The average size of DoD contracts is comparatively large in comparison to other Federal agencies ($32M

average per project for FY 2018 – FY 2023, Figure 7 above). From FY 2018 – FY 2023 the Department awarded just over $3.2 billion through 101 performance contracts. In FY 2023, the DoD awarded three ESPCs with a total investment value of $74.4M, and seven UESCs with a total investment value of $192.6M across all the Components. In FY 2023, ODASD(ER&O) began evaluating strategies to align third-party financing with building level efforts (i.e., Deep Energy Retrofit (DER), Electrification, Audits, etc.), and coordinating with the Military Departments to increase the third-party financing pipeline.

Table 5. Alternative Financing Project Snapshots

Service	FY 2023 ALTERNATIVE FINANCING SNAPSHOT
ARMY	In FY 2023, the Army awarded $20.4M across two ESPC projects and $135.8M across four UESC projects. These projects will reduce greenhouse gas (GHG) emissions by 63,386 tons carbon dioxide (CO_2) equivalent, include 3.5 Megawatt (one million watts) (MW) of CFE generation, and 7.5 megawatt hours (MWh) of battery energy storage. The Army partnered with the Department of Energy (DOE) to provide Army-specific ESPC and UESC training to new project staff to increase execution capacity.
NAVY	The Navy was granted approval by OSD to utilize 10 U.S.C. 2913(c) to allow Naval Base San Diego to accept incentives in the form of goods and services. On 20 September 2022, Navy Region Southwest entered into an Intergovernmental Support Agreement (IGSA) with the Port of San Diego to register and generate Low Carbon Fuel Standard (LCFS) credits based on the Navy vessels' use of shore power in port that meets the requirements of the LCFS regulation.
USMC	In FY 2023, MCAS Beaufort awarded a $23.7 million UESC to Dominion Energy for electrical distribution and lighting improvements. The UESC will contribute to the resilience of MCAS Beaufort by providing funding to upgrade and reconfigure over 15 miles of distribution lines from overhead to underground and complete lighting upgrades. The contract leverages $20.2M ERCIP cost avoidance funding leaving a capital cost of $3.5M, before financing. The contract will be financed over 22.5 years, including a 29-month construction period.
DAF	Joint Base San Antonio reported decreased consumption by 18.6 percent from the previous year. This decrease is a direct result of the fully commissioned energy conservation measures from the last ESPC contract.

Installation Renewable Energy Goals

The DoD currently works to achieve two annual Federally mandated renewable energy goals: Section 203 of the EPAct 2005 (title 42 U.S.C. § 15852(a)) and title 10 U.S.C. § 2911(g). Section 319(c)(2) of the NDAA for FY 2022, (Public Law 117-81) requires that the DoD provide an annual briefing on the progress made toward meeting the goal described in title 10 U.S.C. § 2911(g)(1)(A), with respect to FY 2025. This section and the following section (Renewable Energy Project Deployment and Carbon Pollution-Free Electricity (CFE)) satisfy that briefing request.

The goal outlined in title 10 U.S.C. § 2911(g) is specific to DoD, and progress toward the goal cannot include the purchase of unbundled renewable energy credits (RECs). DoD progress toward both installation renewable energy goals for FY 2023 is shown in Table 6.

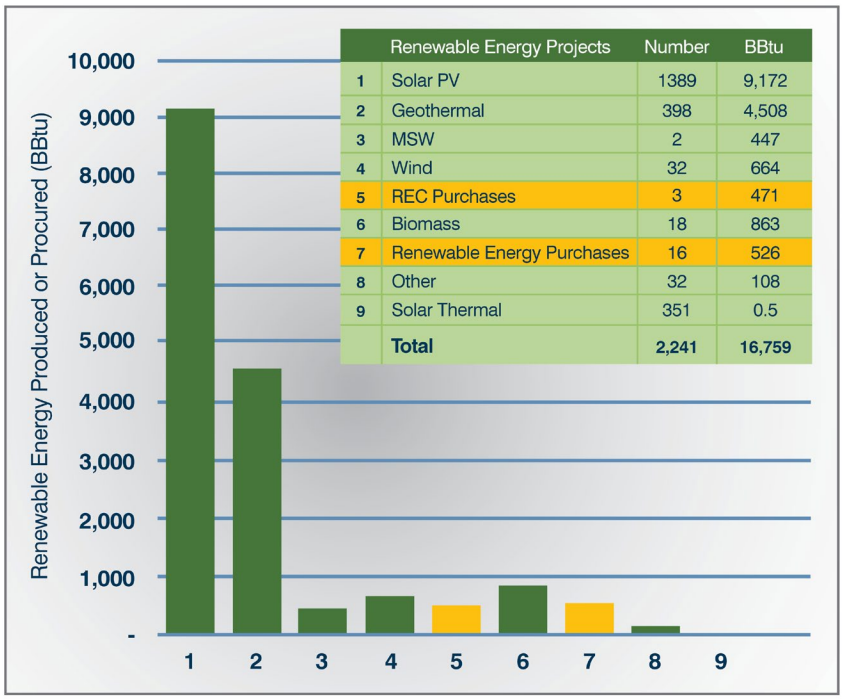
Table 6. FY 2023 Progress Toward Installation Energy Goals

GOALS AND OBJECTIVES	METRIC	COMPONENT	FY 2023	GOAL
Consume More Electric Energy from Renewable Sources Title 42 U.S.C. § 15852(a) – EPAct 2005	Total renewable electricity consumption as a percentage of total facility electricity consumption.	DoD	7.5%	7.5%
		ARMY	10.6%	
		NAVY	5.9%	
		USMC	11.8%	
		USAF	6.6%	
Produce or Procure More Energy from Renewable Sources Title 10 U.S.C. § 2911(g)	Total renewable energy (electric and non-electric) produced or procured as a percentage of total facility energy consumption.	DoD	17.9%	25% by 2025
		ARMY	12.9%	
		NAVY	33.6%	
		USMC	9.4%	
		USAF	17.7%	

Renewable Energy Project Deployment and Carbon Pollution-Free Electricity (CFE)

DoD pursues renewable energy and other CFE to support and advance installation energy and climate resilience, while also complying with legal requirements to increase its renewable energy supply. DoD continues to invest in an array of cost-effective renewable and distributed energy solutions such as solar photovoltaic (PV), wind, and geothermal technologies, as well as deploying on-site energy storage solutions. The DoD focuses renewable energy investments using IEPs and Service installation-level assessments to identify locations and facilities where the greatest benefit in energy resilience improvement and greenhouse gas emissions reduction can be achieved. Figure 8 below demonstrates DoD's commitment and progress towards renewable energy goals, DoD renewable energy generation increased by roughly 25 percent in FY 2023 (16,759 BBtu) as compared to FY 2022 (13,401 BBtu)[13].

[13] See FY 2022 AEPRR Report, page 30 (Figure 8). Available here: https://www.acq.osd.mil/eie/Downloads/IE/FY22%20AEPRR%20Report.pdf



	Renewable Energy Projects	Number	BBtu
1	Solar PV	1389	9,172
2	Geothermal	398	4,508
3	MSW	2	447
4	Wind	32	664
5	REC Purchases	3	471
6	Biomass	18	863
7	Renewable Energy Purchases	16	526
8	Other	32	108
9	Solar Thermal	351	0.5
	Total	**2,241**	**16,759**

Figure 8. FY 2023 Total Renewable Energy Projects and Energy Produced or Procured in BBtu by Type

As detailed in the *Department of Defense Carbon Pollution-Free Electricity Strategic Plan for FY 2024*, the Department obtained 46 percent of its electricity from CFE sources in FY 2023. DoD aims to transition to 100 percent CFE on an annual basis by FY 2030, with at least 50 percent of demand matched to CFE regional supply on an hourly basis in accordance with EO 14057.[14] To achieve this goal, in FY 2022 the DoD began an evaluation of changes to organizational structures and acquisition methods. In FY 2023, DoD has coordinated closely with other government agencies, especially the General Services Administration (GSA) and the DOE, in a "whole-of-government" approach that leverages the purchasing power of the U.S. Government, as well as combining Service-level contracts to increase purchasing power and encourage utilities to source additional CFE.

The DoD uses various authorities to increase the supply of distributed (on-site) and renewable energy sources on its installations. DoD uses both appropriated funds and non-Governmental ("third-party" or "alternative") financing to pursue renewable energy projects. Typically, DoD contracts with non-Federal entities to enable the development of large-scale renewable (or other distributed) energy projects, using third-party financing to fund the initial project costs. DoD also relies on Congressional appropriations to fund cost-effective, small-scale distributed generation projects that do not require third-party financing.

In FY 2023, the DoD made progress on multiple fronts with respect to CFE:

- Launched Pathfinders and other engagements covering ~40 percent of CFE need (up to 183 MW in Carolinas, ~75 MW in Florida, and ~36 MW in Arkansas);
- Continued to advance onsite CFE development (~300 MW in development); and
- Developed a CFE Policy to define roles and responsibilities for CFE (to be issued FY 2024).

[14] E.O 14057: https://www.whitehouse.gov/briefing-room/presidential-actions/2021/12/08/executive-order-on-catalyzing-clean-energy-industries-and-jobs-through-federal-sustainability/

Table 7. Renewable Energy Projects

Service	FY 2023 Renewable Energy Project SNAPSHOT
ARMY	In FY 2023, the Army added eight new renewable energy projects to its portfolio for a combined total of 1.3 MW of rated capacity. Additionally, a new 28 MW solar microgrid project, completed in FY 2023 at the Joint Forces Training Base in Los Alamitos, will begin generating power in FY 2024.
NAVY	The Navy executed two Enhanced Use Leases (EULs) to develop renewable energy power generation projects on Joint Base Pearl Harbor-Hickam (JBPHH). These projects will provide 250+ MW of renewable energy to the local utility, Hawaiian Electric Company (HECO), in support of island grid reliability, energy gap mitigation, and mission continuance in the event of grid outage.
USMC	MCAS Beaufort was competitively selected for a DOE Assisting Federal Facilities with Energy Conservation Technologies (AFFECT) grant for $7.8M. The AFFECT funding supports the installation of solar PV carport over the parking lot adjacent to the gymnasium and dining facility.
DAF	The major element to DAF renewable energy performance improvement was the installation of a 460 MW solar EUL located on Edwards Air Force Base (AFB) which all the power was sold into the state grid and a percentage of the rent revenues may be used to fund energy-related projects at Edwards AFB.

Availability and Use of Energy Cost Savings: Title 10 U.S.C. § 2912

Title 10 U.S.C. § 2912 allows the DoD Components to reinvest energy cost savings. In FY 2023, the Components awarded over 150 projects totaling $53.1M in reinvested energy cost savings. Table 8 includes highlighted FY 2023 projects and activities from the Services.

Table 8. Energy Cost Savings Projects

Service	FY 2023 2912 Project SNAPSHOT
ARMY	In FY 2023, the Army captured more than $49 million of validated installation energy cost savings. Over $44 million of those savings were reinvested into qualified projects before the end of FY 2023.
NAVY	In FY 2023, the Navy continued its implementation of the 10 U.S.C. § 2912 Pilot Program through the Navy Energy Headquarters Program Director that was launched in FY 2021. This program incentivizes and rewards installations for implementing and leading energy consumption reduction efforts by re-investing savings back into all Navy installations. Over the last three years, the Navy's 2912 program has earned $100+M and funded a total of 290 energy projects, advancing the Navy's energy goals and objectives.
USMC	In FY 2023, the USMC recovered $1.2 million. The amount recovered funds was limited by the Assistant Secretary of the Navy (Financial Management and Comptroller).
DAF	The Air Force did not harvest energy cost savings or award any projects in FY 2023, but they continued to award and execute projects from the existing "no-year" funds captured before the statute was amended in the FY 2022 NDAA[15].

The authorized uses of harvested energy cost savings for reinvestment are defined in 10 U.S.C. § 2912(b)(1), § 2912(b)(2), and § 2912(c) and listed in Table 9 below.

[15] Section 315 of the FY 2022 NDAA changed the language of 10 U.S.C. § 2912 in subsection (a) by striking "until expended" and inserting "for that fiscal year and the succeeding fiscal year." Funds previously transferred remain available for obligation for section 10 U.S.C. § 2912 purposes and retain their "no-year" period of availability.

Table 9. Authorized Uses for Energy Cost Savings Reinvestment

Installation Energy		Operational Energy
Enterprise Funds - 10 U.S.C. § 2912(b)(1)	Installation Funds - 10 U.S.C. § 2912(b)(2)	Enterprise Funds - 10 U.S.C. § 2912(c)
Installation energy resilience	Improvements to existing military family housing units	Operational energy resilience
Mission assurance	Unspecified minor construction project that will enhance the quality of life of personnel	Efficiencies
Weather damage repair and prevention		Mission assurance
Energy conservation	Morale, welfare, or recreation facility or service	Energy conservation
Energy security measures		Energy security.

In accordance with 10 U.S.C. § 2912(e)(3), DoD Components are also required to provide a detailed description of the funds are transferred between appropriations/authorizations (10 U.S.C. § 2912(e)(1)) and the purpose for which such amounts have been used (as shown in Table 10).

Table 10. FY 2023 Transfers Between Appropriations/Authorizations (10 U.S.C. 2912(e)(1))

Service	Funds Transferred	Transferred From	Transferred To	Funds Use Description
ARMY	$857,000	206520D22	204020D22	All-terrain electric Mission Module trailers
	$13,666,000	202020D22	204020D22	All-electric Infantry Squad Vehicles and an H2-fuel cell powered mobile charging station
	$426,000	202020D22	204020D22	All-terrain electric Mission Module trailers
USMC	$600,000	1711062222D	1718042223D	Update the main controllers at Marine Corps Air Ground Combat Center (MCAGCC) Energy Program in all the buildings required to get these systems back on the network and operational. Support security and energy savings.

Facility-Related Control Systems

In FY 2023, Deputy ASD(EI&E) provided funding for the planning and site visits needed to conduct Cybersecurity Resilience Readiness Exercises (CRREs) at four mission critical installations in FY 2024. These CRREs build on a successful live cyber resilience readiness exercise, under sponsorship of ODASD(ER&O), that was performed on an active DoD installation over a 12-hour period in FY 2021, in conjunction with a BSE. The privatized utility power was disconnected via cyber means and demonstrated two different cyber effects on both mission-owned and real-property control systems. The findings from the planning, execution, analysis, and reporting from this live-fire cyber exercise were impactful and actionable for the installation, critical mission partner, and military department that hosted the exercise.

The success of this initial exercise demonstrated the broad range of control systems on an installation that can impact mission assurance; the variability in understanding, coordination, and ownership of the relevant systems; and the limited guidance, tools and resources that are available for installations to identify the unique cyber-physical vulnerabilities of their control systems. Analogous to BSEs, CRREs will be conducted with each Service to inform the development of Control Systems Cybersecurity policy, Unified Facilities Criteria and a CRRE playbook outlining execution of the exercises as a mandatory effort whose outputs inform mission impact, risk mitigation and investment.



https://media.careersinthemilitary.com/citm-images/0040.02.jpg

Appendix A - List of Acronyms

Acronym	Definition
AEMRR	Annual Energy Management and Resilience Report
AEPRR	Annual Energy Performance, Resilience, and Readiness Report
AETP	Advanced Engine Transition Program
AF/A4	HQ USAF Deputy Chief of Staff for Logistics, Engineering, and Force Protection
AFB	Air Force Base
AFFECT	Assisting Federal Facilities with Energy Conservation Technologies
AFSOC	Air Force Special Operations Command
AMI	Advanced meter infrastructure
AMMPS	Advanced Medium Mobile Power Source
AOR	Area of responsibility
APS	Advanced power strips
ARC	Army Reserve Center
ARCENT	Army Central Command
ARNG	Army National Guard
ASA(IE&E)	Assistant Secretary of the Army for Installations, Energy and Environment
ASD(EI&E)	Assistant Secretary of Defense for Energy, Installations and Environment
ASN(EI&E)	Assistant Secretary of the Navy for Energy, Installations and Environment
ATO	Air Tasking Order
ATS	Automatic Transfer Switch
BBtu	Billion British thermal units
BESS	Battery Energy Storage System
BOS	Base operations support
BSE	Black Start Exercise
Btu	British thermal unit
CAAFI	Commercial Aviation Alternative Fuels Initiative
CAP	Climate Action Plan
CO_2	Carbon dioxide
CCMD	Combat Commands
CFE	Carbon pollution-free electricity
CAP	Climate Action Plan
CLOE WG	Contested Logistics Operational Energy Working Group

Acronym	Definition
CNIC	Commander, Navy Installations Command
CNO	Office of the Chief of Naval Operations
COOP	Continuity of Operations
CRRE	Control System Resilience Readiness Exercise
CSG	Executive Agent Component Steering Group
DAF	Department of the Air Force
DAFI	Department of the Air Force Instruction
DASA(E&S)	Deputy Assistant Secretary of the Army for Energy and Sustainability
DASN(E)	Deputy Assistant Secretary of the Navy for Energy
DC2DL	Determining Climate Challenges for Defense Logistics
DCA	Defense Critical Asset
DC I&L	USMC Deputy Commandant for Installations and Logistics
DCMA	Defense Contract Management Agency
DCS	Deputy Chief of Staff
DeCA	Defense Commissary Agency
DER	Deep energy retrofit
DFARS	Defense Federal Acquisition Regulation Supplement
DFAS	Defense Finance and Accounting Service
DIA	Defense Intelligence Agency
DLA	Defense Logistics Agency
DMAG	Deputy's Management Action Group
DMO	Distributed Maritime Operations
DoD	Department of Defense
DoDD	DoD Directive
DoDI	DoD Instruction
DOE	Department of Energy
DoN	Department of the Navy
EABO	Expeditionary Advanced Base Operations
EISA 2007	Energy Independence and Security Act 2007
eKPP	Energy Key Performance Parameter
eLRV	Electric light reconnaissance vehicle (eLRV)
Energy HPD	Energy Headquarters Program Director
EO	Executive Order

Acronym	Definition
EPAct 2005	Energy Policy Act of 2005
ERCIP	Energy Resilience Conservation and Investment Program
ERP	Energy Readiness Program
ESA	Energy Supportability Analysis
ESCO	Energy service contractor
ESPC	Energy Savings Performance Contract
ESTCP	Environmental Security Technology Certification Program
EUL	Enhanced use lease
F2EP	Office of Future Force Energy and Power
FCBs	Functional Capability Boards
FEB	Facility Energy Board
FEC	Navy Facility Engineering Command
FEG	Facility Energy Group
FEMP	Federal Energy Management Program
FEP	Facility Energy Panel
FRCS	Facility-related control systems
FEOC	Facility Energy Operations Center
FY	Fiscal year
FYDP	Future Years Defense Program
GBFMD	Global Bulk Fuel Management and Delivery
GENISYS	Navy Global Energy Information System
GSA	General Services Administration
GVSC	Army Ground Vehicle Systems Center
HAF	Headquarters Air Force
HASC	House Armed Services Committee
HECO	Hawaiian Electric Company
HMMWV	High Mobility Multipurpose Wheeled Vehicle
HQ	Headquarters
HVAC	Heating, ventilation, and air conditioning
IAPR	Integrated Acquisition Portfolio Reviews (IAPRs)
ICAO	International Civil Aviation Organization
IE	Installation energy
IEP	Installation Energy Plan

Acronym	Definition
IGSA	Intergovernmental Services Agreement
IMM	Integrated Material Management
IMM	Integrated Materiel Manager
IoT	Internet of things
IRP	Installation Resilience Plan
IT	Information technology
JB	Joint Base
JBPHH	Joint Base Pearl Harbor-Hickam
JLENT	Joint Logistics Enterprise
JLTV	Joint Light Tactical Vehicles
JPEC	Joint Petroleum Executive Council
JPSG	Joint Petroleum Steering Group
JSOC	Joint Special Operations Command
KW	Kilowatt, 1 thousand Watts
kWh	Kilowatt-hours
LC	Liquid chromatography
LCFS	Low Carbon Fuel Standard
LED	Light-emitting diode
LFG	Landfill gas
LMP	Liquid monopropellant
MAF	Mobility Air Forces
MCAGCC	Marine Corps Air Ground Combat Center
MCAS	Marine Corps Air Station
MCICOM	Marine Corps Installations Command
MCICOM GF	MCICOM, Facilities Directorate
MCICOM GF-PW	MCICOM, Public Works
MCTSSA	Marine Corps Tactical Systems Support Activity
MEEP	Air Force Mission Execution Excellence Program
MDA	Missile Defense Agency
MILCON	Military Construction
MMBtu	Million British thermal units
MOSAICS	More Situational Awareness for Industrial Control Systems
MOTCO	Military Ocean Terminal Concord

Acronym	Definition
MSIC	Missile and Space Intelligence Center
MW	Megawatt, 1 million watts
MWh	Megawatt-Hour, 1 million watt-hours
NAS	Naval Air Station
NAVAIR	Naval Air Systems Command
NAVBASE	Naval Base
NAVFAC	Naval Facilities Engineering Command
NAVSEA	Naval Sea Systems Command
NDAA	National Defense Authorization Act
NDS	National Defense Strategy
NGA	National Geospatial-Intelligence Agency
NGAD	Next Generation Air Dominance
NGAP	Next Generation Adaptive Propulsion Program
NH139	Joint Use Intelligence Center
NITMRE	Next Generation Information Technology for Mobility Readiness Enhancement
NREL	National Renewable Energy Laboratory
NRO	National Reconnaissance Office
NSA	National Security Agency
NSA [*base name*]	Naval Supply Activity
NSB	Naval Submarine Base
NSWC	Navy Surface Warfare Center
NTV	Non-tactical vehicle
OASD(Acquisition)	Office of the Assistant Secretary of Defense for Acquisition
OASD(EI&E)	Office of the Assistant Secretary of Defense for Energy, Installations and Environment
OCO	Overseas Contingency Organization
ODASD(ER&O)	Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization
OE	Operational energy
OECIF	Operational Energy Capability Improvement Fund
OEPF	Operational Energy Prototyping Fund
OPNAV N4I	CNO Shore Installations Division
OPT	Operational Planning Team
OSD	Office of the Secretary of Defense
OUSD(A&S)	Office of the Under Secretary of Defense for Acquisition and Sustainment

Acronym	Definition
OIUSD(R&E)	Office of the Under Secretary of Defense for Research and Engineering
PDM	Program Decision Memorandum
PLC	Programmable logic controller
PPA	Power purchase agreement
PREPA	Puerto Rico Electric Power Authority
PWRR	Petroleum War Reserve Requirement
PWRS	Preposition War Reserve Stock
PV	Photovoltaic
RASP	Navy Replenishment at Sea Planner
RASP-CASA	Refueling and Support Package to Enable Communications and Situational Awareness
RDT&E	Research, Development, Test and Evaluation
REC	Renewable Energy Credit
REGS	Renewable energy and storage
RICE	Reciprocating internal combustion engines
SAF/IE	Assistant Secretary of the Air Force (Installations, Environment and Energy)
SAF/IEE	Deputy Assistant Secretary of the Air Force Environment, Safety and Infrastructure
SASC	Senate Armed Services Committee
SATF	Synthetic Aviation Turbine Fuels
SBIR	AFWERX Small Business Innovation Research Program
SCADA	Supervisory Control and Data Acquisition
SCO	Strategic Capabilities Office
SOTF	Security Operations Training Facility
SFB	Space Force Base
TaCV-E	Tactical and combat vehicle - electrification
THAAD	Terminal High Altitude Area Defense
TTPs	Tactics, techniques, and procedures
TEARF	Test and Evaluation of Alternative Rocket Fuels
TOC	Total Ownership Cost
TWCF	Transportation Working Capital Fund
UAS	Unmanned aerial system
UAVs	Unmanned aerial vehicles
UESC	Utility Energy Service Contract
UMCS	Utility monitoring and control system

Acronym	Definition
UP	Utilities privatization
UPS	Uninterruptible power supply
U.S.	United States
USAF	U.S. Air Force
USAR	U.S. Army Reserve
USASOCOM	U.S. Army Special Operations Command
U.S.C.	United States Code
USCENTCOM	U.S. Central Command
USCG	U.S. Coast Guard
USEUCOM	U.S. European Command
USINDOPACOM	U.S. Indo-Pacific Command
USMC	U.S. Marine Corps
USSF	U.S. Space Force
USSOCOM	U.S. Special Operations Command
USTRANSCOM	U.S. Transportation Command
WHS	Washington Headquarters Services

Appendix B - Congressional Reporting Requirements

Section 2925(a) of title 10, United States Code (U.S.C.) requires the Secretary of Defense to provide "a report detailing the fulfillment during that fiscal year of the authorities and requirements under Sections 2688, 2911, 2912, 2920, and 2926 of this title, including progress on energy resilience at military installations and the use of operational energy in combat platforms and at contingency locations."

The FY 2023 Annual Energy Performance, Resilience, and Readiness (AEPRR) Report satisfies the DoD's statutory reporting requirements pursuant to the following:

- Title 42 U.S.C. § 8258, which requires Federal agencies to describe their energy management activities in a report to the Secretary of Energy;
- Title 10 U.S.C. § 2925(a), which requires DoD to submit to Congress an AEPRR describing its installation energy activities;
- Title 10 U.S.C. § 2688 (g)(4), which requires DoD to report progress in meeting energy resilience metrics for all utility conveyance contracts entered into;
- Title 10 U.S.C. § 2911(c), which requires DoD to establish and report energy performance goals for transportation systems, support systems, utilities, and infrastructure and facilities;
- Title 10 U.S.C. § 2912 (e)(3), which requires DoD to provide a detailed description of the use of funds attained from energy cost savings realized by the Department; and
- Title 10 U.S.C. § 2926 (e)(4), which requires DoD to provide proposed operational energy budgets for each of the Services as well as an operational energy strategy and updates to the strategy every 5 years.

The Compliance Matrix in Appendix C provides all reporting requirements and requests satisfied by this report, including the following additional report requests:

- Public Law 114-328, S.2943, the National Defense Authorization Act for Fiscal Year 2017, 10 U.S.C. § 2815(f).
    - Report on Military Installation Resilience Projects, requires an annual report to the congressional defense committees 90 days after the end of each fiscal year until December 31, 2025
- Public Law 117-81, S.1605, the National Defense Authorization Act for Fiscal Year 2022:
    - Energy, Water, and Waste Net-zero Requirement for Major Military Installations. - Status Report and Briefing on Progress (Section 319(c)(2))
- Senate Report 117–130, accompanying SR 7776, the National Defense Authorization Act for Fiscal Year 2023:
    - Briefing on Overseas Energy Efficiency Study
- Senate Report 116–236, accompanying SR 4049, the National Defense Authorization Act for Fiscal Year 2021:

- Report on Installation Energy - Pilot Program on Energy Management

- House Report 118–125, accompanying HR 2670, the National Defense Authorization Act for Fiscal Year 2024:
  - Report on Propane/Alternative Fuel Powered Vehicles

- Senate Report 118–58, accompanying SR 2226, the NDAA for Fiscal Year 2024:
  - Briefing on Hydrogen Fuel Cell Research for Defense Applications
  - Briefing on Benefits and Use of Electrochromic Glass in Military Bases and Installations
  - Briefing on Ground Source Heat Pumps

Appendix C - Compliance Matrix

	Subsection	Description	FY 2023 AEPRR Chapter/Appendix	Page Number
Title 10 U.S.C. § 2925	(a)	Annual Report Related to Installations Energy Performance, Resilience, and Readiness — Not later than 240 days after the end of each fiscal year, the Secretary of Defense shall submit to the congressional defense committees a report detailing the fulfillment during that fiscal year of the authorities and requirements under sections 2688, 2911, 2912, 2920, and 2926 of this title, including progress on energy resilience at military installations and the use of operational energy in combat platforms and at contingency locations.		
	(b)(1)(A)	A description of the progress made to achieve the goals of the Energy Policy Act of 2005 (Public Law 109-58), section 2911(g) of this title, and the Energy Independence and Security Act of 2007 (Public Law 110-140).	**Chapter 4, Appendix D**	**19– 29, 42 - 43**
	(b)(1)(B)	A description of the energy savings, return on investment, and enhancements to installation mission assurance realized by the fulfillment of the goals described in subparagraph (A).	**Chapter 4**	**19– 29**
	(b)(1)(C)	A description of and progress toward the energy security, resilience, and performance goals and master planning for the Department of Defense, including associated metrics pursuant to subsections (c) and (d) of section 2911 of this title and requirements under section 2688(g) of this title.	**Appendix D**	**42- 43**
	(b)(1)(D)	An evaluation of progress made by the Department in implementing the operational energy strategy of the Department, including the progress of key initiatives and technology investments related to operational energy demand and management.	**Chapter 3**	**5 – 18**
	(b)(1)(E)	Details of the amounts of any funds transferred by the Secretary of Defense pursuant to section 2912 of this title, including a detailed description of the purpose for which such amounts have been used.	**Chapter 3**	**5 – 18**
	(b)(2)	Statistical information on operational energy demands of the Department, in terms of expenditures and consumption, for the preceding five fiscal years, including information on funding made available in regular defense appropriations Acts and any supplemental appropriations Acts.	**Chapter 3**	**5 – 18**
	(b)(3)	A description of each initiative related to the operational energy strategy of the Department and a summary of funds appropriated for each initiative in the previous fiscal year and current fiscal year and requested for each initiative for the next five fiscal years.	**Chapter 3, Appendix O**	**5 – 18, 88 – 127**
	(b)(4)	Such recommendations as the Secretary considers appropriate for additional changes in organization or authority within the Department to enable further implementation of the energy strategy and such other comments and recommendations as the Secretary considers appropriate.	**Appendix F**	**47**

	Subsection	Description	FY 2023 AEPRR Chapter/Appendix	Page Number
Title 10 U.S.C. § 2911	(c)(1)	The Secretary of Defense shall submit to the Congressional defense committees the energy performance goals for the Department of Defense regarding transportation systems, support systems, utilities, and infrastructure and facilities.	Appendix D	42 – 43
	(c)(3)	The Secretary of Defense shall include the energy security and resilience goals of the Department of Defense in the installation energy report submitted under Section 2925(a) of this title for Fiscal Year 2018 and every fiscal year thereafter. In the development of energy security and resilience goals, the Department of Defense shall conform with the definitions of energy security and resilience under this title. The report shall include the amount of critical energy load, together with the level of availability and reliability by fiscal year the Department of Defense deems necessary to achieve energy security and resilience.	Chapter 4	19– 29
	(d)(1)	The Secretary of Defense shall develop a comprehensive master plan for the achievement of the energy performance goals of the Department of Defense, as set forth in laws, executive orders, and Department of Defense policies.	Appendix D	42 – 43
Title 10 U.S.C. § 2688	(g)(4)	The Secretary of Defense, in consultation with the Secretaries of the military departments, shall include in the installation energy report submitted under Section 2925(a) of this title a description of progress in meeting energy resilience metrics for all conveyance contracts entered into pursuant to this section.	Chapter 4	19– 29
Title 10 U.S.C. § 2912	(e)(3)	At the end of each fiscal year, the Secretary of Defense shall submit to Congress a report detailing any funds transferred pursuant to paragraph (1) during that fiscal year, including a detailed description of the purpose for which such amounts have been used.	Chapter 3	5 – 18
Title 10 U.S.C. § 2920	(g)(1)	If by the end of fiscal year 2029, the Secretary determines that the Department will be unable to meet the requirements under subsection (a), not later than 90 days after the end of such fiscal year, the Secretary shall submit to the Committees on Armed Services of the Senate and House of Representatives a report detailing the projected shortfall;	Chapter 4	19– 29
	(g)(2)	reasons for the projected shortfall;	Chapter 4	19– 29
	(g)(3)	any statutory, technological, or monetary impediments to achieving such requirements;	Chapter 4	19– 29
	(g)(4)	any impact to readiness or ability to meet the national defense posture; and	Chapter 4	19– 29
	(g)(5)	any other relevant information as the Secretary considers appropriate.	Chapter 4	19– 29

	Subsection	Description	FY 2023 AEPRR Chapter/Appendix	Page Number
Title 10 U.S.C. § 2926	(e)(4)	Not later than 30 days after the date on which the budget for fiscal year 2024 is submitted to Congress pursuant to section 1105 of title 31, and every five years thereafter, the Assistant Secretary shall submit to the congressional defense committees the strategy required under paragraph (1).	**Chapter 3**	**5 – 18**
Public Law 117-81, S.1605, the National Defense Authorization Act for Fiscal Year 2022	Section 319(c)(2)	During fiscal year 2022 and each succeeding fiscal year through fiscal year 2025, the Secretary shall provide to the Committees on Armed Services of the House of Representatives and Senate a briefing on the progress the Secretary has made toward meeting the goal described in section 2911(g)(1)(A) of title 10, U.S.C., with respect to fiscal year 2025.	**Chapter 4**	**Renewable Energy Goals / Project Deployment**
10 U.S.C. § 2815. FY17 NDAA	(f)	Military installation resilience projects, requires an annual report to the congressional defense committees 90 days after the end of each fiscal year until December 31, 2025	**Appendix P**	**128 - 149**
Report 117-130: Senate Armed Services Committee (SASC) FY23 Committee Report	Page 353	Overseas Energy Efficiency Study	**Appendix I**	**57 – 68**
Report 116-236: SASC FY21 Committee Report	Page 180	Installation Energy - Pilot program on Energy management	**Appendix J**	**69 – 71**
Report 118-125: HASC FY24 Committee Report	Page 100	Propane/Alternative Fuel Powered Vehicles	**Appendix K**	**72 – 74**
Report 118-58: SASC FY24 Committee Report	Page 365	Hydrogen fuel cell research for defense applications	**Appendix L**	**75 – 80**
	Page 358	Benefits and use of electrochromic glass in military bases and installations	**Appendix M**	**81 – 83**
	Page 363	Ground source heat pumps	**Appendix N**	**84 – 87**

Appendix D - Energy Performance Master Plan

D.1. Energy Performance Master Plans

Title 10 U.S.C § 2911(d)(1) and (2) requires the Secretary of Defense and each military department and defense agency, or DoD Component, to develop comprehensive master plans for the achievement of DoD energy performance goals. This has been accomplished through the development of comprehensive installation and energy master plans by each Component.

Installation Energy Plans (IEPs) incorporate installation, Component, and higher-level guidance into a holistic roadmap that enables the installation to work toward its energy efficiency, renewable energy, energy resilience, and climate resilience goals. Energy resilience exercise outputs are combined with analyses from the Defense Climate Assessment Tool and data from the DoD Regional Sea Level Database to inform the IEPs. The IEPs, in turn, are used to inform the installation resilience plans (IRPs), required by title 10 U.S.C. § 2864 (a)(2), which then are incorporated into the installation master planning process and feed into the installation master plan.

Overall, an IEP directs a structured and effective approach to selecting, prioritizing, sequencing, and implementing, energy projects and programs that ultimately result in better long-term installation energy performance. IEPs should be developed in coordination with other installation plans. Every effort should be made to integrate the IEP development process for developing master plans. While the IEP must be compatible with the installation-level master plan, the IEP should be either a stand-alone document or a severable product within the master plan. IEPs should be continuously maintained, incorporating projects' lessons learned and synchronizing with the annual energy reporting process.

D.2. Energy Performance Goals

The DoD energy goals in Tables C-1 and C-2 are set forth by title 42 U.S.C. § 15852(a) and title 10 U.S.C. § 2911(g). These goals focus on renewable energy use. The Department remains committed to maximizing the efficient use of energy to free up resources for higher priorities. With respect to renewable energy, the DoD strives to optimize the use of on-site distributed energy resources from all sources of energy generation to directly improve mission assurance. The type of generation source is determined by local availability, market conditions, a business case, or mission requirements. As such, the Department is committed to optimizing the effective and efficient use of generating sources.

Section 2911(c)(3) of title 10 U.S.C. requires DoD to include installation energy security and resilience goals in this report and subsequent AEPRRs. In FY 2022, DoD components continued use of BSEs and other energy assessments to conduct full-scale and routine testing on the infrastructure and equipment supporting all critical energy requirements and determine energy availability at the mission level. In FY 2023, the Services successfully conducted 19 independent, full-scale exercises. Planning is in progress for additional BSE frameworks, such as cybersecurity modules.

Table D-1. DoD Renewable Energy Goal

Goal	Description	Uniform Measure	Method of Measurement	Metric
Consume More Electric Energy from Renewable Sources Title 42 U.S.C. § 15852(a)	Increase consumption of renewable energy	Installation renewable energy consumption	Total renewable electricity consumption as a percentage of total facility electricity consumption	MWh
Produce or Procure More Energy from Renewable Sources Title 10 U.S.C. § 2911(g)	Increase production or procurement of renewable energy	Electric and non-electric renewable energy production and procurement	Electric and non-electric renewable energy produced or procured compared to total facility electricity consumption	MWh

Table D-2. Renewable Energy Performance Targets

Target	FY11	FY12	FY13	FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY25
Consume More Renewable Energy	+5%	+5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%
Produce/Procure More Renewable Energy [1]	-	-	-	-	-	-	-	+15%	-	-	+25%
[1]FY18 interim target required by Title 10 U.S.C. § 2911(g)(2).											

Table D-3. Critical Energy Availability

Target	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27	FY28	FY29	FY30
100% of critical mission energy load has a minimum of at least 99.9% or higher availability by FY30 10 U.S.C. § 2920	-	-	-*	-	-	-	-	-	-	-	99.9+%
*Specific availability standards between 99.9% and 99.9999% availability per fiscal year (depending on criticality of mission), and interim goals prior to FY25, may be established by the Secretary of Defense pursuant to Title 10 U.S.C. § 2920.											

Table D-4. Black Start Exercises

Target	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27
Each military department will conduct not fewer than 5 BSEs per year through FY27 Title 10 U.S.C. § 2920	-	-	15	15	15	15	15	15

Appendix E - DoD Energy Performance Summary

Table E-1. Renewable Electric Energy

Renewable Electric Energy Requirement per Title 42 U.S.C. 15852(a)	Renewable Electricity Use (MWh)	Total Electricity Use (MWh)	Percentage of Facility Electric Use	EPAct 2005 Requirement
Eligible renewable electricity use as a percentage of total electricity use	2,175,921.0	28,901,230.9	7.5%	7.5%

Table E-2. Procuring Renewable Energy Sources

Produce or Procure More Energy from Renewable Sources per Title 10 U.S.C. 2911(g)	Renewable Energy Produced/Procured (MWh)	Total Electricity Use (MWh)	Percentage of Facility Electric Use	Compliance Target by 2025
Total renewable energy (electric and non-electric) produced or procured as a percentage of total facility electricity consumption	5,177,422	28,901,230.9	17.9%	25.0%

Table E-3. Goals for Buildings Metered for Electricity

Meter Goals	Cumulative # of Buildings Metered for Electricity	Cumulative % of Appropriate Buildings Metered for Electricity	Cumulative # of Buildings Metered for Natural Gas	Cumulative % of Appropriate Buildings Metered for Natural Gas	Cumulative # of Buildings Metered for Steam	Cumulative % of Appropriate Buildings Metered for Steam
Standard Meters in FY23	49,791	44.6%	24,879	38.4%	2,173	6.0%
Advanced Meters in FY23	26,325	23.6%	8,488	13.1%	2,462	6.8%
Total Meters in FY23	76,116	68.2%	33,367	51.6%	4,635	12.8%

Table E-1. Energy Efficiency Standards for Federal Buildings

Federal Building Energy Efficiency Standards	Percentage of New Building Designs	Compliance Target
Percentage of new building designs started since beginning in FY 2007 that are 30% more energy efficient than relevant code, where lifecycle cost-effective (including 8/2012 standards)	76.5%	100.0%

E.1. Investments in Energy Management

Table E-2. Investment Sources for Energy

Sources of Investment	Investment Value (Thou. $)	Anticipated Annual Savings (MMBtu)
Direct obligations for facility energy efficiency improvements	$28,993.0	5,055.2
Investment value of ESPC Task/Delivery Orders awarded in fiscal year	$74,365.3	67,901.0
Investment value of UESC Task/Delivery Orders awarded in fiscal year	$192,563.5	574,533.0
TOTAL	**$295,921.8**	**647,489.2**

Table E-3. Funding for ESPC/UESC

Investment	Percentage
Total Investment as a percentage of total facility energy cost	6.3%
Financed (ESPC/UESC) investment as a percentage of total facility energy costs	5.7%

Table E-4. Installation Energy Type and Cost

Total Installation Energy Consumption and Cost[16]			
Energy Type	BBtu	Cost (thou.)	% of Consumption
Electricity	95,157.45	$3,167,345.78	48.1%
Fuel Oil	13,731.78	$293,101.57	6.9%
Natural Gas	74,725.81	$760,238.74	37.8%
Liquified Petroleum Gas	904.74	$18,292.75	0.5%
Coal	3,957.48	$24,237.73	2.0%
Steam	2,938.34	$117,187.34	1.5%
Other	871.29	$22,882.23	0.4%
Renewable Electric, On-site	3,089.69	$89,681.96	1.6%
Renewable Electric Off-Site	987.93	$5,510.32	0.5%
Renewable, Other, On-Site	1,178.28	$2,969.53	0.6%
Renewable, Off-Site Green Energy Purchases	232.62	$16,906.99	0.1%
Renewable Energy On-site, Replacement REC Adjustment	763.71	$0.00	0.4%
TOTAL BUILDING	**197,775.4**	**$4,518.354.93**	**100%**
Non-Tactical Vehicles	7,107.34	$213,841.88	
TOTAL Building + NTVs	**204,882.75**	**$4,732,196.81**	

[16] Total energy consumption and cost does not include on-site produced, renewable energy replacement REC adjustment.

Appendix F - Recommended Change in Organization or Authority

The Department has no recommendations for changes in organization or authority.

Appendix G - Installation Energy Program Management

G.1. Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization (ODASD(E&RO))

The Assistant Secretary of Defense for Energy, Installations, and Environment (ASD (EI&E)) is within the Office of the Under Secretary of Defense for Acquisition and Sustainment. The Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization (ODASD(E&RO) is within ASD (EI&E) and prescribes policies and procedures, provides guidance, and monitors and reviews programs related to operational energy (OE), installation energy (IE), climate resilience, innovation, and the siting clearinghouse (see Figure G-1).

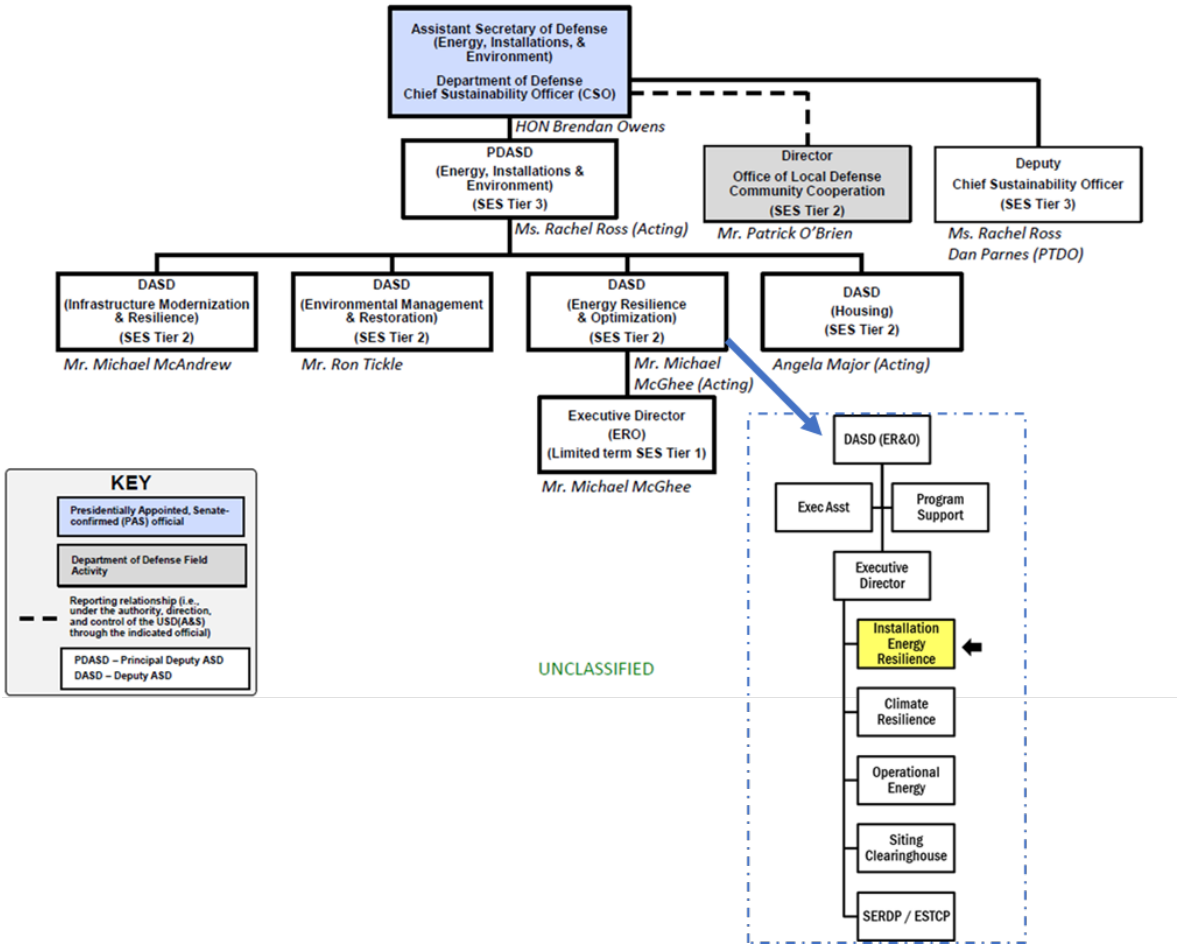


Figure G-1. Department of Defense Energy Management Organizational Structure

The mission of ODASD(ER&O) is to sustain warfighting readiness and lethality by providing all energy-related policy and governance for programs and activities that enable resilient, efficient, and cyber-secure energy for Joint forces, weapon systems, and installations. To accomplish this mission, ODASD(ER&O) supports initiatives across three primary areas:

- **Energy Resilience:** Enhancing the military capability, readiness, and resilience of DoD installations and forces through assured access to resilient and cyber-secure fuel and power;

- **Energy Risk:** Identifying, assessing, and integrating energy-related analyses and risks into Department decision-making associated with requirements, deliberate planning, wargames and exercises, installation master planning, Energy Resilience and Conservation Investment Program (ERCIP), and investments in forces and installations; and

- **Energy Performance:** Ensuring energy efficiency and lower costs at DoD installations through reliable, efficient use of power and alternative financing mechanisms.

G.2. Department of the Army—Assistant Secretary of the Army for Installations, Energy, and Environment (ASA (IE&E))

The Assistant Secretary of the Army for Installations, Energy and Environment (ASA(IE&E)) oversees the Army's energy, water, and sustainability programs, see Figure G-2. The ASA(IE&E) establishes policy, oversees programs, and sets the strategic direction for installations, housing, and partnerships; energy and sustainability; environment, safety, and occupational health; and installation modernization. The Army's senior energy official, the Deputy Assistant Secretary of the Army for Energy and Sustainability (DASA(E&S)), is responsible for overall program direction, establishing policies, developing and refining strategies, and oversight of all programs and initiatives related to energy and water resilience and sustainability within the Army. The Deputy Chief of Staff (DCS), G-9, is the Army's lead integrator for installation matters, including installation readiness and operations, military facilities investment requirements, installation services, privatization and partnerships, energy and water resilience, and sustainability. The G-9 provides installation, energy, and environmental program policy implementation guidance to Landholding Organizations: Army Materiel Command (AMC), Army Central (ARCENT), Army National Guard (ARNG), and the U.S. Army Reserve (USAR). Each Army Landholding Organization implements Army policy and guidance in support of the Army goals and strategy.

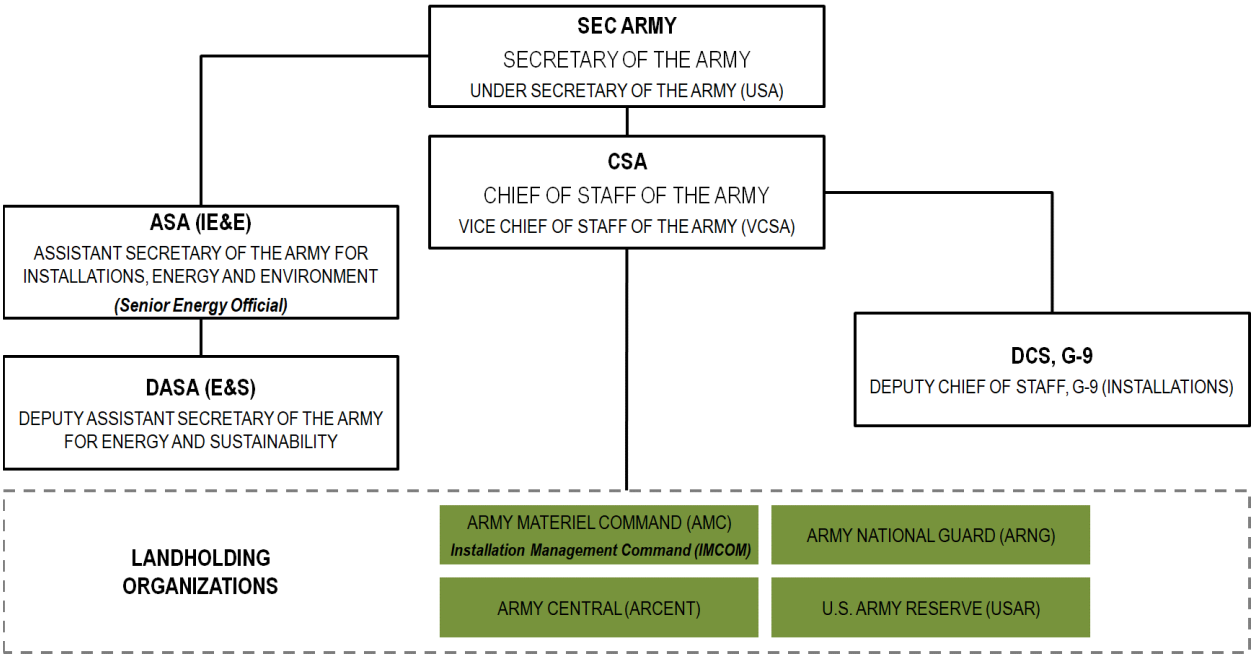


Figure G-2. Department of Army Energy Management Organizational Structure

G.3. Department of the Navy (DoN)—Assistant Secretary of the Navy for Energy, Installations, and Environment (ASN(EI&E))

The Assistant Secretary of the Navy for Energy, Installations and Environment (ASN(EI&E)) serves the Department of Navy (DoN) by enhancing combat capabilities for the warfighter through greater energy security; the acquisition and disposal of real property; construction and maintenance of installations; environmental protection, planning and restoration ashore and afloat; and conservation of natural and cultural resources. The Deputy Assistant Secretary of the Navy for Energy (DASN(E)) is the principal energy advisor to ASN(EI&E). Within the Secretariat, the Director for Installation Resilience is responsible for military installation readiness through mission assurance and resilience. The DoN's program for energy security promotes innovation and optimizes the use of congressional authorities to deliver energy and water resilience on and off DoN military installations. The Navy Shore Energy Management Enterprise Organizational Structure is shown in Figure G-3 below.

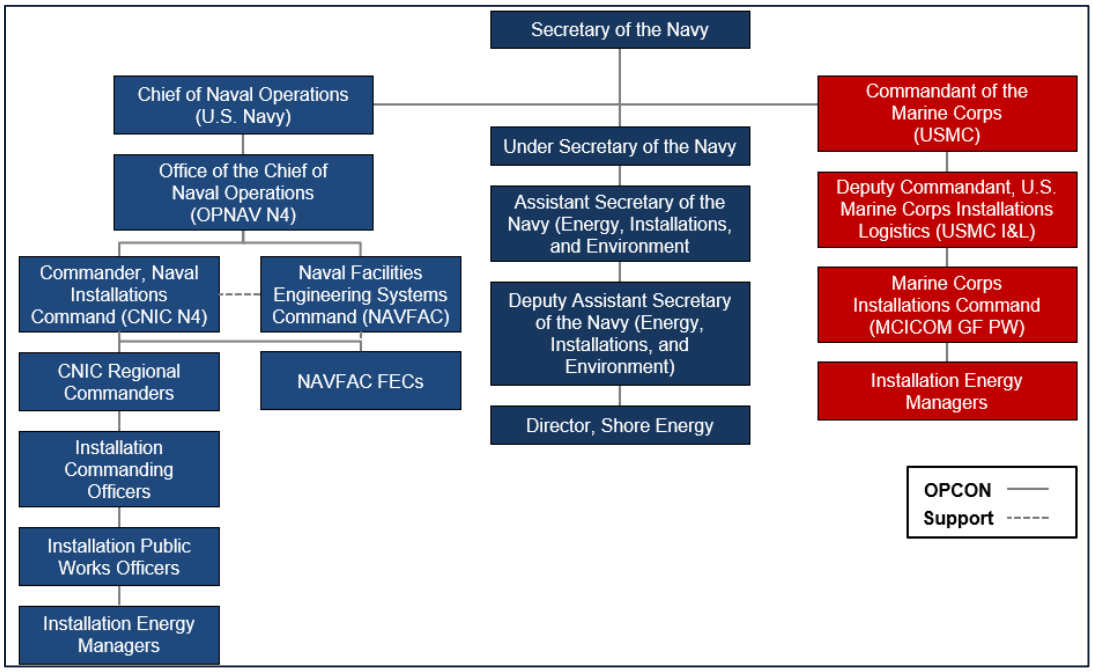


Figure G-3. Department of Navy Energy Management Organizational Structure

The Office of the Chief of Naval Operations (CNO) Installations Division (OPNAV N4I) is responsible for developing policy and programming resources for the DoN's Facility Energy and Mission Assurance Programs. OPNAV N4I also ensures compliance with DoN Shore Energy Goals through OPNAV Instruction 4100.5E (Shore Energy Management). Commander, Navy Installations Command (CNIC) is responsible for leading, managing and integrating the Navy's Budget, Policy, and Requirements across the Navy Shore Energy Enterprise to include current and future shore energy, water, cybersecurity, and mission requirements across warfighter enterprises. The DoN's energy program aligns under CNIC N4 (Facilities and Environmental Department) and N42 (Energy and Base Operations Support (BOS) Division), with the Energy Headquarters Program Director (Energy HPD) responsible for leading and managing the Shore Energy Program across all 10 Regions and 70 Installations. This includes providing budgetary, policy, personnel, and management oversight of diverse programs, projects, and initiatives that directly enhance the Navy's Shore Energy Program.

The Deputy Commandant for Installations and Logistics (DC I&L) is responsible for establishing energy and water management policy for Marine Corps installations in accordance with the Commandant's direction. The Assistant Deputy Commandant for Installations and Logistics (Facilities) serves as the single point of contact responsible for program management and resourcing. The Commander, Marine Corps Installations Command (MCICOM) oversees program planning and execution with direct support provided by the Director, Facilities (MCICOM GF). The Public Works Section (MCICOM GF- PW) includes program management for the Installation Energy, Utilities, Facilities Maintenance, Real Property Services, and Facility-Related Control Systems (FRCS) programs for the Marine Corps.

G.4. Department of the Air Force—Assistant Secretary of the Air Force for Energy, Installations, and Environment (SAF/IE)

Headquarters, United States Air Force Deputy Chief of Staff for Logistics, Engineering, and Force Protection (AF/A4) provides information required to support governance and oversight of energy management activities across DAF installations. The AF/A4 develops guidance, procedures, and practices to enhance DAF energy assurance with the goal of energy resilience and ensures energy security to meet mission-essential requirements. The DAF's organizational structure for energy oversight is shown in Figure G-4.1.

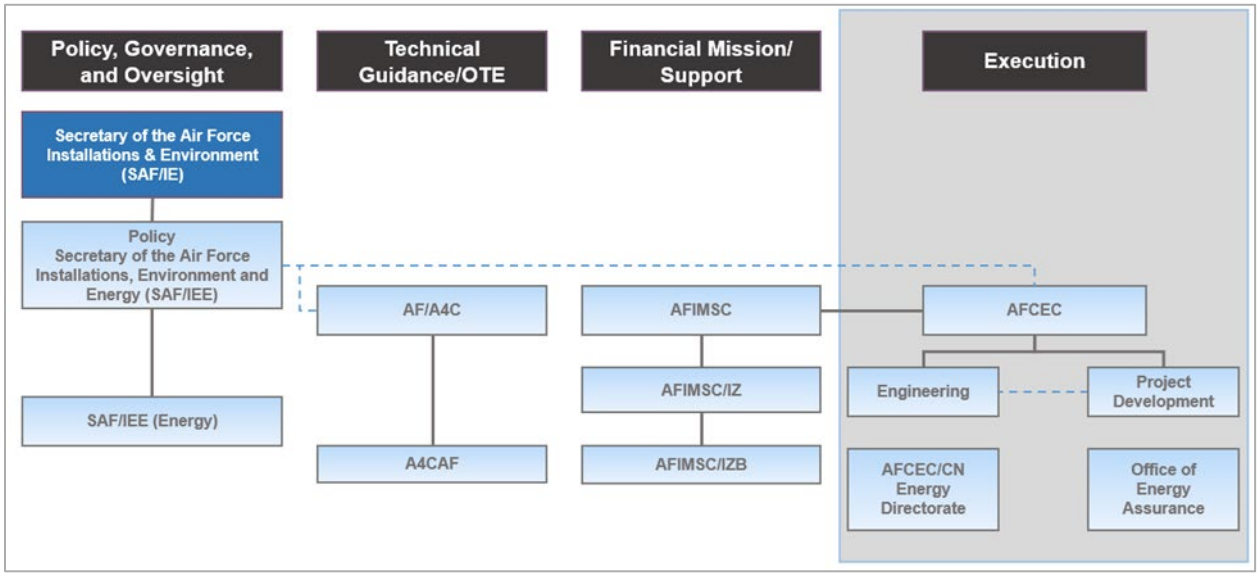


Figure G-4.1. Department of Air Force Energy Management Organizational Structure

Policy: DAF installation energy management is spelled out in Department of the Air Force Instruction (DAFI) 90-1701, Installation Energy and Water Management, last updated in December 2020. DAFI 90-1701 provides guidance on energy and water resource management and resilience, as well as project development and funding. The instruction outlines the specific roles and responsibilities for all individuals and organizations involved in the management of facilities energy and water.

The publication of EO 14057 in December 2021 has led DAF to consider how to incorporate multiple elements of sustainability into its mission-focused approach to installation energy management. As part of the approach to address the targets specified in the EO, DAF published the Climate Action Plan (CAP) in October 2022. The DAF CAP spells out three climate priorities for the Department: 1) Maintain air

and space dominance in the face of climate risks, 2) Make climate-informed decisions, and 3) Optimize energy use and pursue alternative energy sources. In July 2023, DAF published the Climate Campaign Plan, which outlines specific objectives and key results to address the DAF CAP priorities.

DAF issued a memo on Advancing Utilities Privatization Investments in August 2023, and a memo on Achieving Carbon Pollution-Free Electricity (CFE) Targets in November 2023. The Deputy Assistant Secretary of the Air Force Environment, Safety and Infrastructure (SAF/IEE) will be issuing several more memos related to the implementation of EO 14057 in late 2023 or early 2024.

Governance: The DAF Energy Program governance structure provides strategic guidance for development, review, validation, and approval for near- and long-term energy project direction.

Facility Energy Board (FEB): The FEB is the strategic forum used to ensure that the Secretariat and Air Staff vision and priorities are fully incorporated into the DAF Facility Energy Program's decisions and processes. This body ensures that Headquarters Air Force (HAF), DAF Facility Energy Program, and Major Command strategic goals and priorities are considered when balancing risk to mission and risk to Airmen and Guardians within available resources, as well as whether the DAF Corporate Structure should increase energy resiliency efforts. Where practical, the FEB also serves as a forum to discuss industry initiatives and/or best practices in achieving DAF energy goals.

Facility Energy Group (FEG): The FEG serves as the mid-level decision forum to ensure appropriate oversight of facility energy projects. This governance board is focused on ranking projects based on HAF priorities, identifying risks and mitigation plans to be briefed at the FEB, and discussing opportunities for collaboration across organizations to increase efficiencies for project transition and execution.

Facility Energy Panel (FEP): The FEP serves as an action officer-level advisory committee for all DAF energy and water resilience projects. Members of the FEP determine if energy, utility, or water resilience requirements will be transitioned to their appropriate execution partner. The FEP is also responsible for validating the prioritization of the resilient energy and water project portfolio.

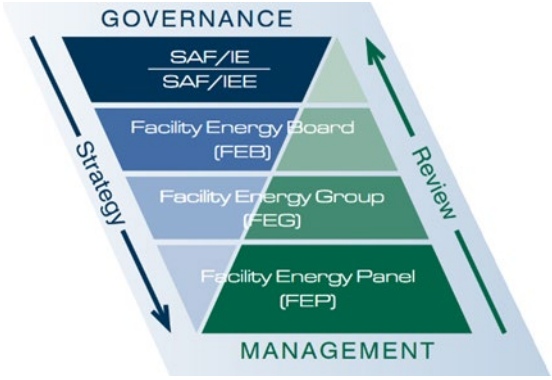


Figure G-4.2. Department of Air Force Energy Management Governance/Management Structure

G.5. Defense Agencies

The Defense Agencies continue to enhance their installation energy management programs and coordinate with stakeholders to address energy and water resilience issues. Where agencies exist as tenants on installations, they work to participate in host-installation energy conservation and resilience

programs. Each has a designated senior energy official to administer their respective program (Table G-1).

Table G-1. Defense Agencies Senior Energy Officials

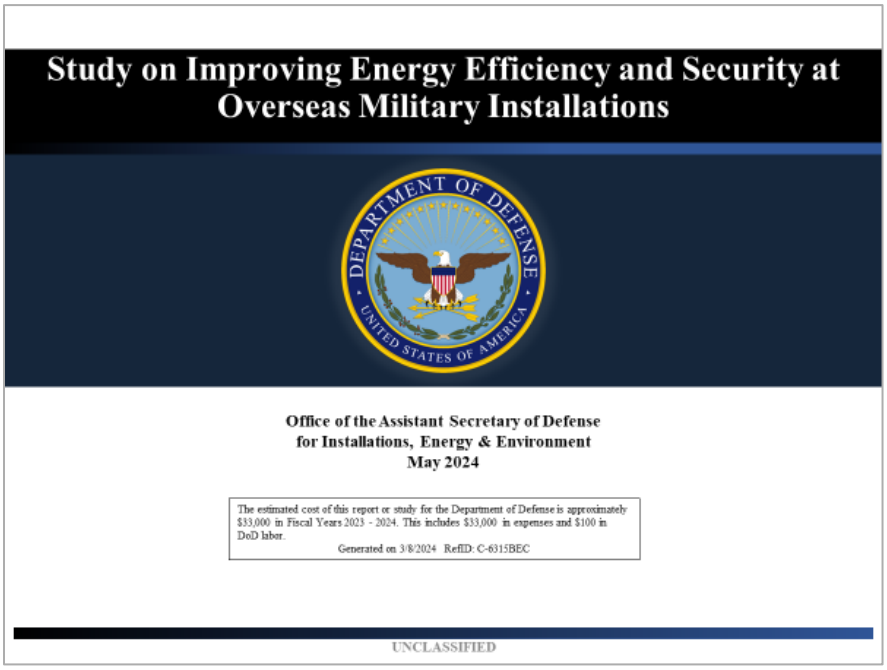
DoD Component	Senior Energy Official
Defense Contract Management Agency (DCMA)	Energy Program Manager
Defense Commissary Agency (DeCA)	Energy Program Manager
Defense Finance and Accounting Service (DFAS)	Director, Enterprise Management Service
Defense Intelligence Agency (DIA)	Director, Office of Facilities and Services
Defense Logistics Agency (DLA)	Installation Management Director
Missile Defense Agency (MDA)	Environmental Executive
National Geospatial-Intelligence Agency (NGA)	Director, Installation Operations Office
National Reconnaissance Office (NRO)	Chief, Facilities Support Group
National Security Agency (NSA)	Chief of Installations and Logistics
Washington Headquarters Services (WHS)	Pentagon Energy Program Manager

Appendix H - Third Party Financing Authorities

Funding Mechanism	Authority	Definition
Energy Savings Performance Contracts (ESPC)	Title 42 U.S.C. § 8287	Title 42 U.S.C. § 8287 c(3) defines an ESPC as "a contract that provides for the performance of services for the design, acquisition, installation, testing, and, where appropriate, operation, maintenance, and repair, of an identified energy or water conservation measure or series of measures at 1 or more locations." Title 42 U.S.C. § 8287 provides that ESPCs may be entered into "solely for the purpose of achieving energy savings and benefits ancillary to that purpose," the term of which may be up to 25 years. The Energy Service Contractor (ESCO) conducts a comprehensive energy audit for the Federal facility and identifies improvements to save energy. In consultation with the Federal agency, the ESCO designs and constructs a project that meets the agency's needs. The ESCO must incur the costs of implementing energy savings measures, in exchange for a share of any energy savings directly resulting from implementation of such measures during the term of the contract. The ESPC must guarantee savings. Aggregate annual payments by the agency to both utilities and ESCOs may not exceed the amount that the agency would have paid for utilities without an ESPC during contract years. After the contract ends, all additional cost savings accrue to the agency.
Utility Energy Service Contracts (UESC)	Title 10 U.S.C. § 2913(d)	Title 10 U.S.C. § 2913 (d) authorizes agreements "with gas or electric utilities to design and implement cost-effective demand and conservation incentive programs (including energy management services, facilities alterations, and the installation and maintenance of energy saving devices and technologies by the utilities) to address the requirements and circumstances of the installation." These agreements are referred to as UESCs. While the maximum term of a UESC is not set in statute, Defense Federal Acquisition Regulation Supplement (DFARS) 241.103(2) authorizes UESCs for a period not to exceed 25 years. A UESC may be structured to provide that the utility advances the initial capital costs of the project, which are repaid over the contract term from the cost savings generated by the energy efficiency measures from funds available to a military department for the purchase of utility services. Title to energy-saving devices or technologies installed pursuant to the UESC must vest in the United States. There is no statutory requirement that the utility guarantee savings.
Power Purchase Agreement (PPA)	Title 10 U.S.C. § 2922a	Title 10 U.S.C. § 2922a authorizes contracts for periods of up to 30 years for (1) development of geothermal energy resources on military land, and (2) for the provision and operation of energy production facilities (located on property under the jurisdiction of the Secretary concerned or on private property) and the purchase of energy produced from such facilities. The costs of the project may be paid from annual appropriations each year. Contracts must be approved by the Secretary of Defense.
Enhanced Use Leases (EUL)	Title 10 U.S.C. § 2667	Title 10 U.S.C. § 2667 authorizes the Secretaries of military departments to lease property under the Secretary's jurisdiction if certain conditions are met and accept cash or in-kind consideration in an amount not less than the fair market value of the lease interest. This authority is used in a wide array of contexts. When Title 10 U.S.C. § 2667 is used as the primary authority for an energy project, land is typically leased to a third party for the purpose of operating and maintaining an energy generation and/or storage facility. The fair market value of the lease interest is provided to the installation in the form of energy services – such as the right to use the facility during grid outages, or, if the lease interest is more valuable, the provision of related equipment to enhance installation use of the facility, such as the provision of equipment to form a functioning microgrid. Initial investment using appropriated

Funding Mechanism	Authority	Definition
		funds may also be required to enable the project to provide energy resilience to the installation if the fair market value of the lease interest is low.
Easements for rights-of-way	Title 10 U.S.C. § 2668	Title 10 U.S.C. § 2668 authorizes the Secretaries of military departments to grant easements for rights-of-way over lands under the Secretary's control if certain conditions are met. Title 10 U.S.C. § 2668 specifically authorizes twelve easements for rights-of-way, and provides that the Secretary concerned may grant easements for rights-of-way for "any other purpose that the Secretary considers advisable." Such purposes could include purposes relating to energy use and transmission, if appropriate for a right-of-way, such as inverters, transfer switches, uninterruptible power supplies (UPS), and depending on the project, batteries and generation.
Utilities Privatization (UP)	Title 10 U.S.C. § 2688	Title 10 U.S.C. § 2688 authorizes the conveyance of utility systems owned by a military department to a third party, along with a contract for utility services to operate and maintain the utility system on behalf of the installation, which may have a term of up to 50 years. Improvements to the privatized utility system, including those that that enhance energy resilience, may be financed through the contract for utility services.

Appendix I - Briefing on Overseas Efficiency Study

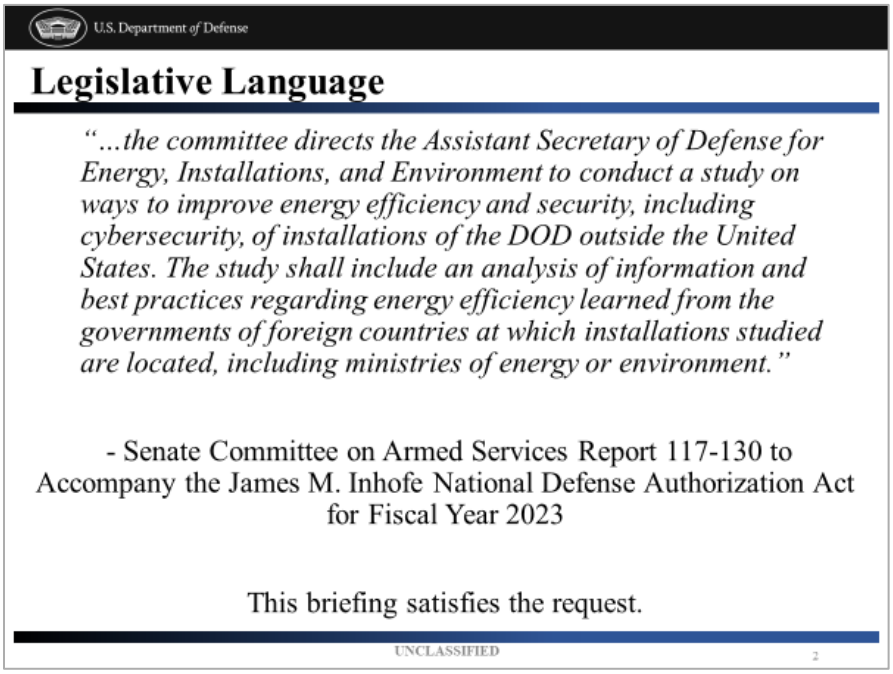


Study on Improving Energy Efficiency and Security at Overseas Military Installations

Office of the Assistant Secretary of Defense
for Installations, Energy & Environment
May 2024

The estimated cost of this report or study for the Department of Defense is approximately $33,000 in Fiscal Years 2023 - 2024. This includes $33,000 in expenses and $100 in DoD labor.
Generated on 3/8/2024 RefID: C-6315BEC

UNCLASSIFIED



U.S. Department of Defense

Legislative Language

"...the committee directs the Assistant Secretary of Defense for Energy, Installations, and Environment to conduct a study on ways to improve energy efficiency and security, including cybersecurity, of installations of the DOD outside the United States. The study shall include an analysis of information and best practices regarding energy efficiency learned from the governments of foreign countries at which installations studied are located, including ministries of energy or environment."

- Senate Committee on Armed Services Report 117-130 to Accompany the James M. Inhofe National Defense Authorization Act for Fiscal Year 2023

This briefing satisfies the request.

UNCLASSIFIED

2

U.S. Department of Defense

Summary

- Foreign countries are addressing energy security, efficiency, renewable technologies in existing building portfolios that can be applied to U.S. installations overseas
- Most host nations have plans to reach carbon neutrality by 2050
  - Short-term and long-term strategic plans related to energy and emissions
  - DoD alignment with host nation governments' energy and climate goals
- Technology approaches and future fuel mixes vary by country, location
- Drivers of approach vary (e.g., energy availability, vulnerability to natural disasters, reliance on Russian energy sources)
- Building-related activities include:
  - Building evaluations, audits, and retrofits
  - Increased performance and/or efficiency standards (e.g., envelope, equipment)
  - Net or near-zero energy and emission building requirements
  - Deployment of renewable energy technologies

Approaches for U.S. installations located overseas may vary slightly based on location, but overall energy efficiency, energy security, and carbon neutrality driving building policies and best practices.

UNCLASSIFIED 3

U.S. Department of Defense

Contents

- DoD Overseas Footprint
- Methodology for conducting study
- Evaluating potential impacts by DoD locations
- Energy and security policies and best practices of foreign countries
- Cybersecurity
- Key challenges, activities and opportunities
- DoD strategies and example projects
- Major outcomes

UNCLASSIFIED 4

U.S. Department of Defense

Department of Defense Overseas Footprint

DoD operates over 560,000 buildings and structures at over 500 military installations in the United States and overseas*

- Of the total DoD installation footprint, overseas installations comprise approximately:
  - 17% of gross square feet
  - 14% of building energy use
- Across all DoD installations (FY21)**
  - Building energy usage: 198,674 billion Btus
  - Building utility costs: $3.28 billion

From: *https://www.acq.osd.mil/eie/IE/FEP_index.html
** https://www.acq.osd.mil/eie/Downloads/IE/FY%202021%20AEMRR.pdf

UNCLASSIFIED 5

U.S. Department of Defense

Methodology

- Compile list of locations of DoD installations, by Command region
- Organize by energy usage intensity, gross square footage (GSF)
- Research individual countries, international organizations
- Document key ministries, agencies that set and enforce policy
- Analyze energy efficiency best practices
- Research country energy security and cybersecurity
- Identify challenges and outcomes related to energy efficiency at overseas installations

UNCLASSIFIED 6

U.S. Department of Defense

Foreign Countries Included in Evaluation



Highlighted countries were included in study and are where DoD installations consume energy.
(from AEMRR 2021 dataset)

UNCLASSIFIED 7

U.S. Department of Defense

Evaluating Potential Impacts by Foreign Country

Countries with Largest Footprint, by gross square foot (GSF)

Country	Total Gross Square Footage (Thou. SF)	Number of Installations	Total Energy Consumption (BBtu)	Energy Use Intensity (kBtu/GSF)	CO COM
Japan	107,841	32	10,665	98.9	INDOPACOM
Germany	100,235	25	5,602	55.9	EUCOM
Republic of Korea	54,257	20	3,712	68.4	INDOPACOM
Italy	20,903	9	1,490	71.3	EUCOM
United Kingdom	13,093	10	1,076	82.2	EUCOM
Cuba	6,506	1	1,010	155.2	SOUTHCOM
Belgium	5,895	2	172	29.1	EUCOM
Turkey	5,531	3	244	44.1	EUCOM
Spain	4,629	3	319	68.9	EUCOM
Bahrain	2,948	1	281	95.3	CENTCOM

Average EUI across all DoD installations	116.1 kBtu/GSF
Average EUI across DoD overseas installations	118.3 kBtu/GSF

Energy use intensity = total annual energy consumption in British thermal units (Btu) divided by facility square footage (GSF)

UNCLASSIFIED

U.S. Department of Defense

Evaluating Potential Impacts by Foreign Country

Countries with Highest Energy Use Intensity (kBtu/GSF)

Country	Energy Use Intensity (kBtu/GSF)	Number of Installations	Total Gross Square Footage (Thou. SF)	Total Energy Consumption (BBtu)	CO COM
Djibouti	400.2	1	2,164	866	AFRICOM
Diego Garcia	336.3	1	2,441	821	INDOPACOM
Romania	267.3	1	169	45	EUCOM
Poland	234.0	1	128	30	EUCOM
Honduras	196.6	1	1,088	214	SOUTHCOM
Cuba	155.2	1	6,506	1,010	SOUTHCOM
Netherlands	136.5	1	24	3	EUCOM
Japan	98.9	32	107,841	10,665	INDOPACOM
Bahrain	95.3	1	2,948	281	CENTCOM
United Kingdom	82.2	10	13,093	1,076	EUCOM

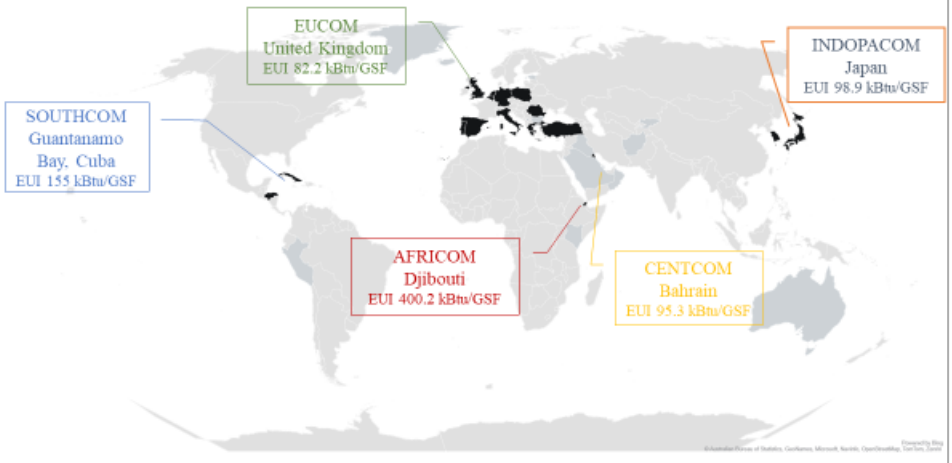
Average EUI across all DOD installations 116.1 kBtu/GSF
Average EUI across DOD overseas installations 118.3 kBtu/GSF

Energy use intensity = total annual energy consumption in British thermal units (Btu) divided by facility square footage (GSF)

UNCLASSIFIED 9

U.S. Department of Defense

Countries with Highest Footprint, EUI by Command



UNCLASSIFIED 10

U.S. Department of Defense

Energy and Energy Security Policies

European Command

- European Union
  - Carbon neutrality by 2050, 55% reduction in GHG emissions by 2030
  - Energy-related directives set targets for efficiency, emissions, and renewables
    - 10-year National Energy and Climate Plan (NECP) required of each Member State
  - European Energy Security Strategy
    - Resource diversification, cross-border interconnections, independence from Russian energy sources
- Italy, Germany, United Kingdom
  - Key energy policies: Eliminate Russian gas imports, phase out coal, increase renewables and advanced clean energy technologies
  - Key challenges: Unique, lengthy host nation approval process for energy efficiency project implementation at DoD installations; working with local utilities; building envelope considerations

UNCLASSIFIED 11

U.S. Department of Defense

Energy and Energy Security Policies

Indo-Pacific Command

- Japan's Strategic Energy Plan
  - 36% energy from renewable by 2030, carbon neutrality by 2050
- Korea's Third Energy Master Plan
  - Deploy zero-carbon energy generation technologies – nuclear and hydrogen
- Key challenges: limited domestic energy sources, relying on importing energy sources; space-constraints for on-site clean energy generation

Southern Command

- Cuba's External Investment Law
  - Goal to increase renewable energy generation and independence from fossil fuels
- Key challenges: non-transparent, slow approval process for energy projects; heavily subsidized electricity prices

UNCLASSIFIED 12

U.S. Department of Defense

Energy and Energy Security Policies

Central Command

- Bahrain energy efficiency and renewable energy action plans
  - 6% reduction in energy consumption by 2025 across all sectors
  - 5% renewable energy by 2025 and 20% by 2035
  - Initiatives to encourage energy efficiency implementation
  - Key Challenges: current high dependency on fossil fuels

Africa Command

- Vision 2035 - 100% renewable energy power production by 2035
  - Geothermal generation, solar and wind farms
  - Address longstanding problem of electricity access
- Energy security is supported by foreign investment
  - G7's Partnership for Global Infrastructure and Investment
  - Europe's recent actions to secure African oil/gas supplies (replacing Russia)
  - Key Challenges: limited internal energy resources, interconnections, developing framework for public-private partnerships

UNCLASSIFIED 13

U.S. Department of Defense

Foreign Country Buildings Best Practices

- Building Requirements
  - Energy codes, net-zero buildings, building rating systems, thermal efficiency requirements, energy use disclosure, shift to non-fossil fuel energy sources, optimizing electricity demand (load shifting, energy storage), interim energy reduction milestones
- Equipment Standards
  - Minimum energy performance standards for systems and components, appliance labeling, phaseouts of inefficient technologies (e.g., incandescent bulbs, R22 refrigerant
- Energy Efficiency Programs
  - Demand, load management
  - Incentives, Tax Credits, Grants/Loans targeting energy conservation measures (e.g., building envelope retrofits, heat pumps, rooftop solar, combined heat and power, electric vehicles)

Common objective: reduce building heating and cooling loads through envelope retrofits, then apply efficiency improvements to energy using equipment (e.g., HVAC with minimum efficiency requirements)

UNCLASSIFIED 14

U.S. Department of Defense

Energy Security and Cybersecurity

- Due to critical nature of energy sector infrastructure, energy security and cybersecurity are linked.
  - Critical energy infrastructure systems at greater risk of increased cyberattacks and impacts energy security
  - Digitization of controls increases exposure to cyberattacks
  - Level of utility interconnections between countries lead to different approaches and impacts to surrounding countries, particularly in Europe
- The proliferation of smart metering, microgrids, and facility-related control systems presents a growing cybersecurity threat for countries and consumers.
- Energy sector critical infrastructure is a main driver for cybersecurity strategies for foreign countries, with planned periodic updates to policies to account for changing technologies and risks.

UNCLASSIFIED 15

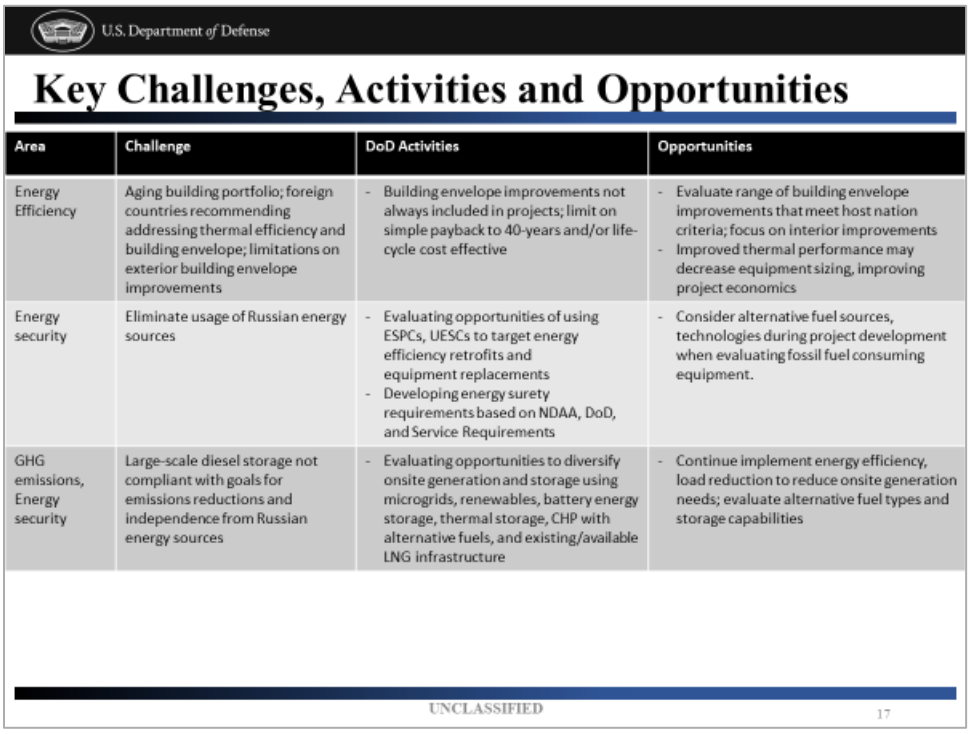
U.S. Department of Defense

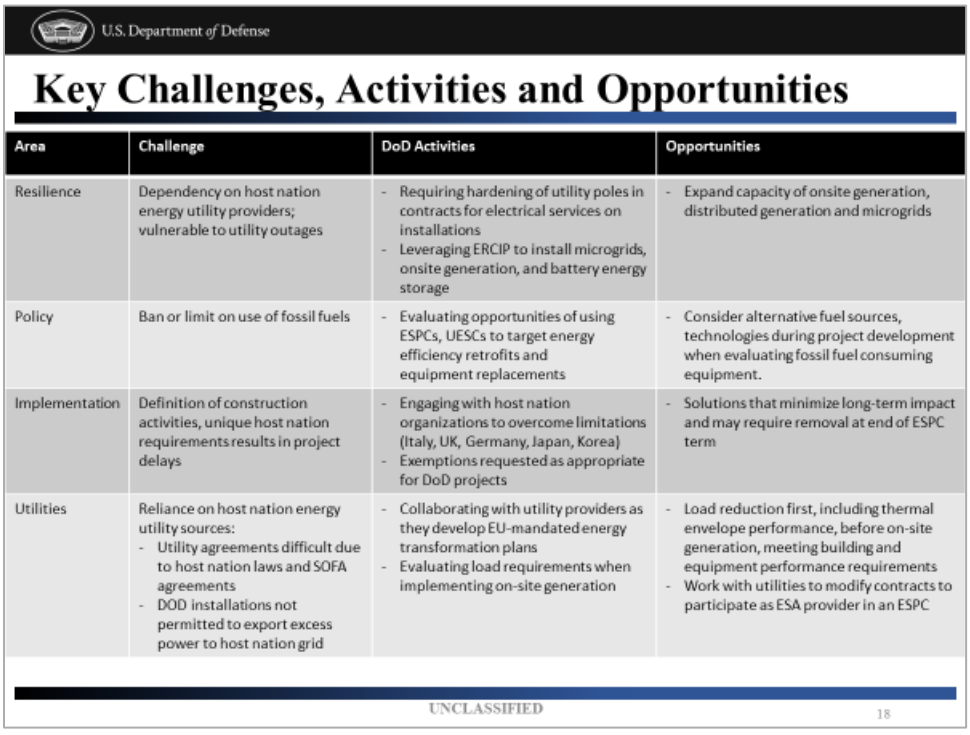
Cybersecurity

- Increasing prominence of cybersecurity, with some nations establishing committees and developing policies for minimum security standards, cyber-defense mechanisms, and collaboration across all critical sectors, including energy and information technology services.
  - EU's *Network and Information Systems (NIS) Directive on Security*
  - Bahrain's *National Cyber Security Strategy*
  - Japan's *Cybersecurity Policy for Critical Infrastructure Protection*

Country	Authority for cybersecurity
Bahrain	Ministry of the Interior
Belgium	Prime Ministry, Center for Cybersecurity Belgium
Cuba	Ministry of Communications
Germany	Federal Office for Information Security
Greece	Ministry of Digital Governance
Italy	Prime Ministry, National Cybersecurity Agency
Japan	Cabinet Secretary, Cybersecurity Strategic Headquarters
Netherlands	Ministry of Justice and Security
Poland	Ministry of Digital Affairs
Portugal	Security Information Service
Republic of Korea	National Intelligence Service
Romania	National Cybersecurity Directorate
Singapore	Ministry of Communication and Information
Spain	Prime Ministry, National Security Council
Turkey	Ministry of Transport and Infrastructure
United Kingdom	Government Communications Headquarters
Djibouti	Ministry of Digital Economy and Innovation
Honduras	National Telecommunications Commission

UNCLASSIFIED 16


U.S. Department of Defense
Key Challenges, Activities and Opportunities
Area
Challenge
DoD Activities
Opportunities
Energy Efficiency
Aging building portfolio; foreign countries recommending addressing thermal efficiency and building envelope; limitations on exterior building envelope improvements
- Building envelope improvements not always included in projects; limit on simple payback to 40-years and/or life-cycle cost effective
- Evaluate range of building envelope improvements that meet host nation criteria; focus on interior improvements
- Improved thermal performance may decrease equipment sizing, improving project economics
Energy security
Eliminate usage of Russian energy sources
- Evaluating opportunities of using ESPCs, UESCs to target energy efficiency retrofits and equipment replacements
- Developing energy surety requirements based on NDAA, DoD, and Service Requirements
- Consider alternative fuel sources, technologies during project development when evaluating fossil fuel consuming equipment.
GHG emissions, Energy security
Large-scale diesel storage not compliant with goals for emissions reductions and independence from Russian energy sources
- Evaluating opportunities to diversify onsite generation and storage using microgrids, renewables, battery energy storage, thermal storage, CHP with alternative fuels, and existing/available LNG infrastructure
- Continue implement energy efficiency, load reduction to reduce onsite generation needs; evaluate alternative fuel types and storage capabilities
UNCLASSIFIED
17


U.S. Department of Defense
Key Challenges, Activities and Opportunities
Area
Challenge
DoD Activities
Opportunities
Resilience
Dependency on host nation energy utility providers; vulnerable to utility outages
- Requiring hardening of utility poles in contracts for electrical services on installations
- Leveraging ERCIP to install microgrids, onsite generation, and battery energy storage
- Expand capacity of onsite generation, distributed generation and microgrids
Policy
Ban or limit on use of fossil fuels
- Evaluating opportunities of using ESPCs, UESCs to target energy efficiency retrofits and equipment replacements
- Consider alternative fuel sources, technologies during project development when evaluating fossil fuel consuming equipment.
Implementation
Definition of construction activities, unique host nation requirements results in project delays
- Engaging with host nation organizations to overcome limitations (Italy, UK, Germany, Japan, Korea)
- Exemptions requested as appropriate for DoD projects
- Solutions that minimize long-term impact and may require removal at end of ESPC term
Utilities
Reliance on host nation energy utility sources:
- Utility agreements difficult due to host nation laws and SOFA agreements
- DOD installations not permitted to export excess power to host nation grid
- Collaborating with utility providers as they develop EU-mandated energy transformation plans
- Evaluating load requirements when implementing on-site generation
- Load reduction first, including thermal envelope performance, before on-site generation, meeting building and equipment performance requirements
- Work with utilities to modify contracts to participate as ESA provider in an ESPC
UNCLASSIFIED
18

U.S. Department of Defense

On-Going DoD Energy Efficiency Strategies

- Installation Energy Plans
  - Utilize installation- and facility-level data to develop implementation roadmap
  - Audit facilities to identify opportunities for energy and water demand reduction
  - Implement solutions to address energy and water support for critical missions
- Energy Efficiency Sample Projects
  - Demand management, HVAC retuning, re/retro-commissioning, building automation and data analytics
  - Energy Conservation Measures: building envelope retrofits, heat pumps, solar PV, battery storage, combined heat and power, heat recovery
  - Clean energy technology strategies: carbon pollution-free electricity, nuclear, carbon capture
  - Leverage ESPCs, UESCs and Department of Energy's AFFECT grant funding
    - ESPCs, UESCs capitalize on energy savings and provide revenue source to achieve resilience
    - 2022 AFFECT selected projects include U.S. Army Garrison Ansbach, Germany and Marine Corps Installation Pacific, Okinawa, Japan

UNCLASSIFIED 19

U.S. Department of Defense

Examples – Past Projects

DoD has awarded over $4 billion in performance contracts (FY2016 – FY 2022) to enhance resilience, reduce energy intensity and address decarbonization, incorporating more efficient and clean-energy technologies*

Department of Air Force
Yokota Air Base, Japan
$167M ESPC project

Impacts:
Reduce energy consumption by 24,245 MMBTU/yr
Decrease water consumption by 25,188,000 gallons
19 ECMs across 1,100 bldgs, 14.3M sq.ft.

Efficiency:
Optimization of central plants, building automation system expansion, lighting upgrades, and plumbing fixture replacement, reducing demand

Resilience:
10.72MW Combined Heat and Power Plant , with microgrid controls

Department of Army
US Army Garrison Ansbach, Germany
$24.8M ESPC project

Impacts:
Reduce energy consumption by 22,857 MMBTU/yr
Over $1.5M cost avoidance
Five installations, 6.9M sq. ft.
Received $2M FY22 AFFECT grant

Efficiency:
Utility monitoring and control systems; building automation system improvements; interior, exterior, and high-bay LED lighting upgrades; and domestic hot water decentralization

Resilience:
Carbon-free electricity generation with 568kW solar PV; Combined heat and power plant upgrades

Department of Navy
Naval Station Guantanamo Bay, Cuba
$368.8M ESPC project

Impacts:
27.4% reduction in fuel consumption
18 percent power generated from renewable sources

Efficiency:
Interior, exterior lighting upgrades
Upgrades to water fixtures
Heating, ventilation, air conditioning improvements
Commercial refrigeration improvements
Water and sewer improvements

Resilience:
12 MW of solar photovoltaic energy generation; battery energy storage
20 MW Liquefied Natural Gas Plant

*From DOE Comprehensive Annual Energy Data and Sustainability Performance, G-2 Agency Investment in Efficiency and Conservation Projects by Funding Type. https://ctsedwweb.ee.doe.gov/Annual/Report/Report.aspx

UNCLASSIFIED 20

U.S. Department of Defense

Summary of Findings

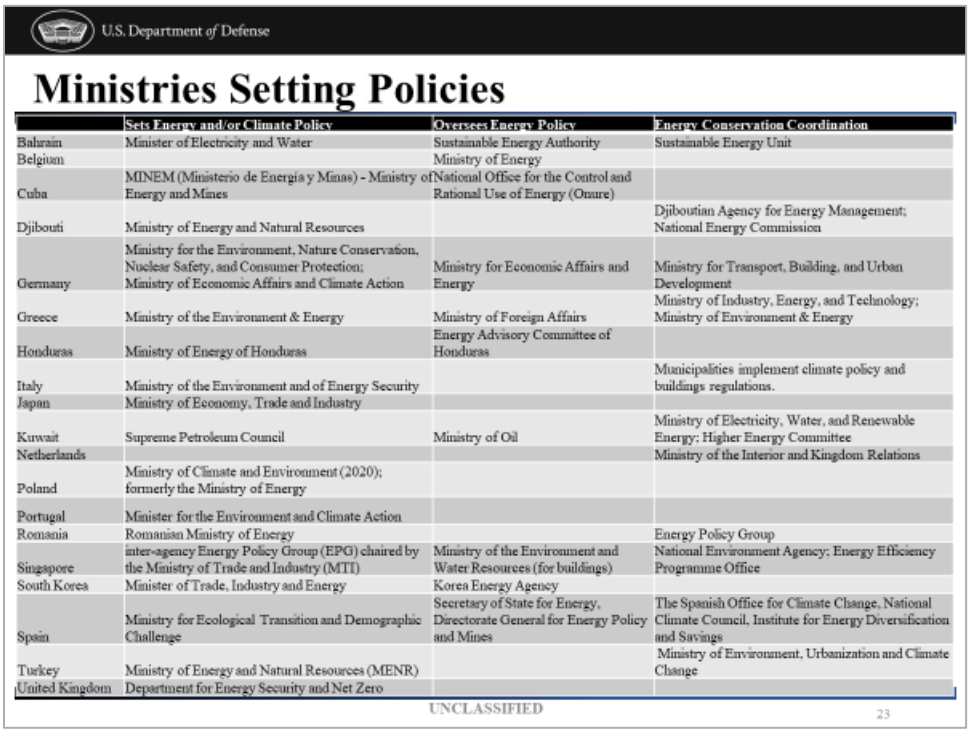
- Efficiency is driving force for several outcomes, including resilience, energy security and decarbonization, augmented by clean energy generation
  - DoD is identifying efficiency opportunities through Comprehensive Energy and Water Evaluations
  - DoD is capitalizing energy savings stream from ESPC, UESC projects to address energy efficiency and resilience
- Overarching goals for energy security and improved resilience
  - Many DoD projects include resilience measures (e.g., microgrids, renewable energy)
  - Energy independence from Russia
  - FY23 NDAA, Section 1086: DoD to implement plans to reduce reliance on Russian energy source for all main operating bases
- Modernization and environmental considerations
  - Shift away from fossil fuel generation; coal power plant phaseouts
  - Foreign country efforts align with targets for GHG emission reductions, decreased energy consumption, and renewable energy production
- Heightened cybersecurity awareness
  - DoD is conducting cyber and control system resilience readiness exercises

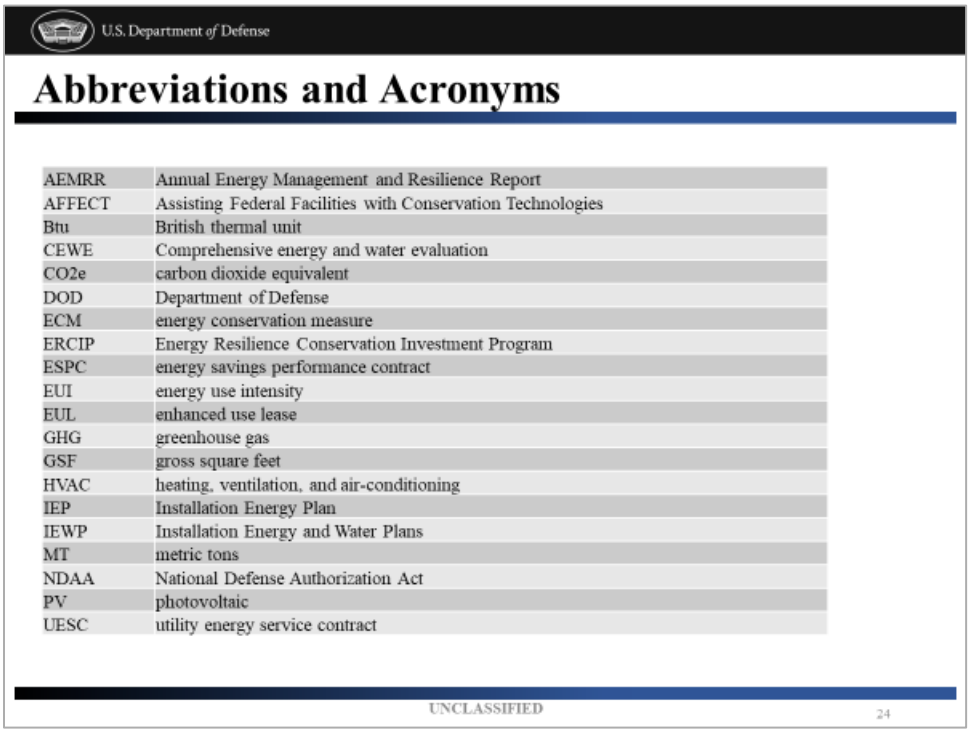
UNCLASSIFIED 21

U.S. Department of Defense

Backup Slides

UNCLASSIFIED 22


U.S. Department of Defense
Ministries Setting Policies
Sets Energy and/or Climate Policy
Oversees Energy Policy
Energy Conservation Coordination
Bahrain
Minister of Electricity and Water
Sustainable Energy Authority
Sustainable Energy Unit
Belgium
Ministry of Energy
Cuba
MINEM (Ministerio de Energia y Minas) - Ministry of Energy and Mines
National Office for the Control and Rational Use of Energy (Onure)
Djibouti
Ministry of Energy and Natural Resources
Djiboutian Agency for Energy Management; National Energy Commission
Germany
Ministry for the Environment, Nature Conservation, Nuclear Safety, and Consumer Protection; Ministry of Economic Affairs and Climate Action
Ministry for Economic Affairs and Energy
Ministry for Transport, Building, and Urban Development
Greece
Ministry of the Environment & Energy
Ministry of Foreign Affairs
Ministry of Industry, Energy, and Technology; Ministry of Environment & Energy
Honduras
Ministry of Energy of Honduras
Energy Advisory Committee of Honduras
Italy
Ministry of the Environment and of Energy Security
Municipalities implement climate policy and buildings regulations.
Japan
Ministry of Economy, Trade and Industry
Kuwait
Supreme Petroleum Council
Ministry of Oil
Ministry of Electricity, Water, and Renewable Energy; Higher Energy Committee
Netherlands
Ministry of the Interior and Kingdom Relations
Poland
Ministry of Climate and Environment (2020); formerly the Ministry of Energy
Portugal
Minister for the Environment and Climate Action
Romania
Romanian Ministry of Energy
Energy Policy Group
Singapore
inter-agency Energy Policy Group (EPG) chaired by the Ministry of Trade and Industry (MTI)
Ministry of the Environment and Water Resources (for buildings)
National Environment Agency; Energy Efficiency Programme Office
South Korea
Minister of Trade, Industry and Energy
Korea Energy Agency
Spain
Ministry for Ecological Transition and Demographic Challenge
Secretary of State for Energy, Directorate General for Energy Policy and Mines
The Spanish Office for Climate Change, National Climate Council, Institute for Energy Diversification and Savings
Turkey
Ministry of Energy and Natural Resources (MENR)
Ministry of Environment, Urbanization and Climate Change
United Kingdom
Department for Energy Security and Net Zero
UNCLASSIFIED
23


U.S. Department of Defense
Abbreviations and Acronyms
AEMRR Annual Energy Management and Resilience Report
AFFECT Assisting Federal Facilities with Conservation Technologies
Btu British thermal unit
CEWE Comprehensive energy and water evaluation
CO2e carbon dioxide equivalent
DOD Department of Defense
ECM energy conservation measure
ERCIP Energy Resilience Conservation Investment Program
ESPC energy savings performance contract
EUI energy use intensity
EUL enhanced use lease
GHG greenhouse gas
GSF gross square feet
HVAC heating, ventilation, and air-conditioning
IEP Installation Energy Plan
IEWP Installation Energy and Water Plans
MT metric tons
NDAA National Defense Authorization Act
PV photovoltaic
UESC utility energy service contract
UNCLASSIFIED
24

Appendix J – Report on Pilot Program on Installation Energy Management

The Office of the Secretary of Defense has prepared this report in response to Senate Report 116-236, page 180, accompanying S.4049 the National Defense Authorization Act for Fiscal Year 2021.

Installation Energy

The committee recognizes that the energy consumption of large, energy-intensive systems, such as heating, ventilation, and air conditioning (HVAC) systems of the Department of Defense (DOD), are monitored and managed by industrial control systems (ICS) to maximize their efficiency and cost savings. Unfortunately, computers, printers, and other smaller items plugged into an electrical system are not. While the More Situational Awareness for Industrial Control Systems (MOSAICS) program is focused on the cybersecurity of ICS platforms, it is not aimed at providing energy cost savings. Recent technologies have emerged on the market to provide comprehensive plug-load energy savings with cybersecurity protection.

Therefore, to further a DOD priority to leverage building control systems to achieve substantial energy savings in a highly secure architecture, the committee authorizes and encourages the Secretary of Defense to establish a pilot program that incorporates technologies relating to energy management on an installation that is not reliant on a single telecommunications provider or energy provider. This approach will assure that, if successful, the technology suite and architecture will be easily implemented across the DOD and will be able to generate the maximum level of energy savings in a secure environment.

The pilot should: (1) Include at least three installations; (2) Incorporate energy efficiency technology for an entire plug-load (large and small systems); (3) Incorporate comprehensive cybersecurity technology; (4) Integrate with current and future architectures; (5) Allow scalability and flexibility; and (6) Avoid single points of failure. The pilot program shall sunset on September 30, 2023. Within 90 days of completion of the pilot, the DOD shall brief the Committees on Armed Services of the Senate and the House Representatives on the results of the pilot, whether or not it should be expanded as appropriate, and explanation as to the outcome of that decision.

Background

The DoD has sought to research and implement emerging technologies to achieve its energy savings and resilience requirements. To make informed decisions on what measures and technologies the Department should pursue, the Department has leveraged the knowledge and experience of other Federal Agencies as well as their findings as it relates to emerging technologies.

The Department did not run its own pilot study on the uses of plug load management given our awareness of others in agencies work and research into this subject. Instead, the Department relies on results from a study conducted by GSA in partnership with the National Renewable Energy Laboratory into plug-load management and the energy savings which would result.[17] This study satisfies the parameters enumerated by Congress in their request for a study on the use of plug-load energy savings technology.

The Study

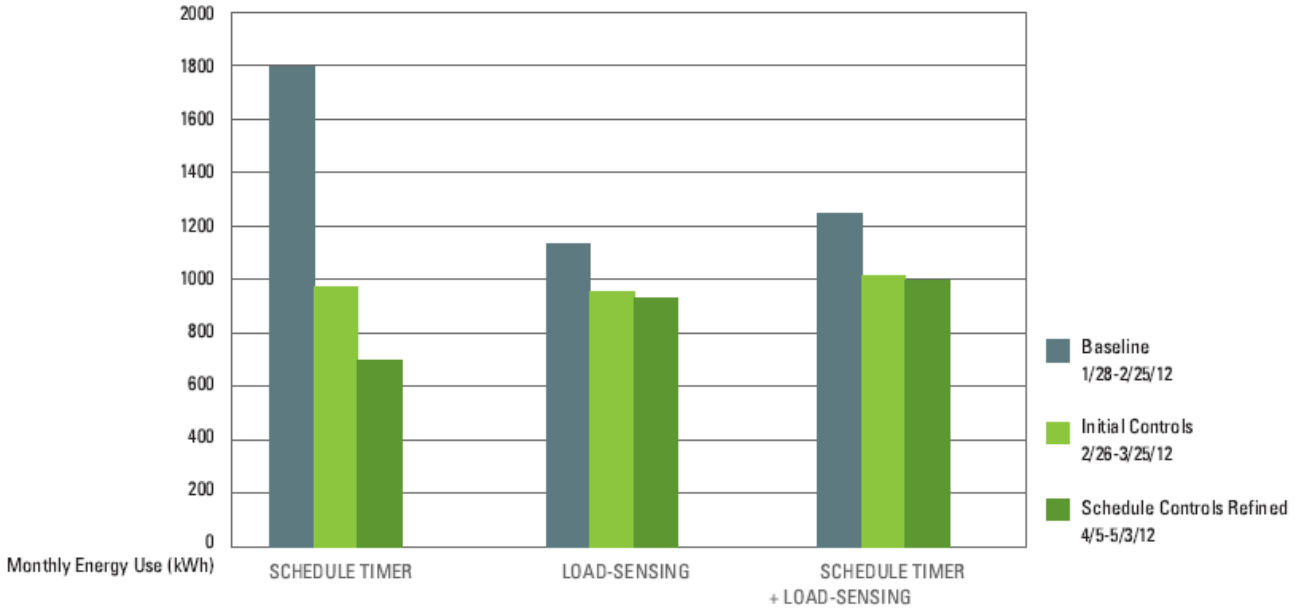
GSA's Green Proving Ground Program worked with a team from the National Renewable Energy Laboratory to identify offices that were representative of those in the wider GSA portfolio and selected eight from the Mid-Atlantic Region. In each building, approximately 12 standard power strips with no control capability were replaced with advanced power strips (APS) which monitored and provided power to an array of devices. Three control strategies were used: a schedule timer, load-sensing, and a combination of the two. Over 295 devices were monitored during the study which consisted of three separate test periods, each four weeks in length.

The APS in the evaluation provided web-based control, monitoring, and data collection. With this, the National Renewable Energy Laboratory (NREL) gathered three months of energy-usage data for each control strategy. Researchers used this information to determine overall performance at the end of each test period, compare that performance to the baseline energy consumption, and adjust control strategy parameters in an effort to achieve maximum energy savings.

Findings

Of the control strategies that were implemented, the schedule timer control was the most effective with an average energy savings of 48 percent. The largest savings were achieved when the schedule timer controls were applied to devices that were powered 24/7. Printers and copiers were among these devices as well as kitchen appliances such as coffee makers and water coolers.

[17] Full report conducted by GSA that was relied on for this response can be found here: Plug-Load Control and Behavioral Change Research in GSA Office Buildings



For the APS installed as part of the study the payback for the schedule timer was less than 8 years in all applications. Occupant surveys reveal that most of the users did not want more control over their individual APS but they were willing to program them in accordance with their own personal work schedules.

Conclusion

Desk-based technologies and other electronics in an office setting consume significant amounts of electricity that are often non-metered nor managed with energy reduction strategies. Several technologies that meter and control office equipment are available and those that employ strategies to match energy use per user work schedules are particularly effective.
The Department is committed to continually evaluate new technologies to reduce its energy use and improve reliance across all its installations. The results of this study show that load sensing controls can provide limited energy savings and a high, simple payback. The Department acknowledges that there are opportunities to deploy simple and low-cost schedule timer power strips to address most office plug loads, and this technology can play a role in the Department's efforts to meet its energy goals. Therefore, the Department recommends plug load management technology to be considered at its installations, as a measure to reduce its energy consumption and increase its resiliency.

Appendix K – Report on Propane/Alternative Fuel Powered Vehicles

May 2024

Office of the Under Secretary of Defense
for Acquisition and Sustainment

Pursuant to House Report 118-125, page 100, accompanying H.R. 2670, the National Defense Authorization Act for Fiscal Year 2024

The estimated cost of this report for the Department of Defense is approximately $9,000 in Fiscal Year 2024. This includes $0 in expenses and $9,000 in DoD Labor
Cost estimate generated on March 2, 2023, RefID: 4-42D3CA8

Report to Congress on Conducting Pilots to Study Replacement of Medium- and Heavy-Duty Non-Tactical Vehicles with Alternative-Fuel Vehicles

Requirement:

This report is submitted as requested by House Report 118-125, page 100, accompanying H.R. 2670, the National Defense Authorization Act for Fiscal Year 2024.

Page 100: ***Propane/Alternative Fuel Powered Vehicles***

"The committee notes numerous actions taken by the U.S. Department of Defense, and its various components, to invest significantly in the electric vehicle market. The committee notes there are other affordable alternative fuels that provide the environmental benefits the Department seeks. Additionally, the committee notes that as the Department seeks to reduce emissions, consideration should also be given to alternative fuel vehicles where fuel sources or key components produced in the United States. The committee notes the U.S. Department of Energy previously determined that while the initial cost for an alternatively fueled vehicle, such as propane, can be between 5–15 percent higher compared to a traditional vehicle, propane is typically cheaper creating a faster return on investment. The committee also notes that propane-fueled vehicles often have lower maintenance costs than other light-, medium-, and heavy-duty vehicles."

"Accordingly, the committee directs the Secretary of Defense to provide a report to the House Committee on Armed Services by no later than January 1, 2024, on the feasibility to implement a pilot program at three domestic facilities to analyze the benefits of alternative fuels for medium- and heavy-duty vehicles. The report shall include amongst other things a cost-benefit analysis that includes the cost of the vehicles, the cost of fueling infrastructure, the cost of fuel, affordability and availability of replacement parts, and an evaluation of life cycle, safety, and well-to-wheel emissions.

Response:

The Department shares the committee's interest in sustaining and developing domestic supply chains that underpin our Defense Industrial Base and ultimately improve the mission readiness of our forces. The Department also shares the committee's interest in mitigating the effects of climate change by reducing greenhouse gas emissions from transportation.

The DoD non-tactical vehicle fleet, while large, is a small fraction of America's overall vehicle market, so future costs and the economic viability of alternative-fueled vehicles will primarily be determined by market forces in the private sector and not by DoD's demand signal.

Any decision to transition a DoD vehicle to alternative fuel begins with a mission assessment. Available alternative-fuel vehicles must be able to perform the mission, or they are not considered. A secondary factor in the transition decision is cost-effectiveness.

There is a growing number of light-duty battery-electric vehicle models that already meet mission requirements at a lower total cost of ownership than comparable internal combustion engine models[18,19].

In accordance with Executive Order 14057, the Department's near-term focus is on transitioning light-duty vehicles to battery electric where possible. Transitioning Medium- and Heavy-Duty vehicles to zero-emission drivetrains will be more challenging due to their more demanding energy requirements for power and range. Because mission compatibility is the primary factor when the Department evaluates alternative fueled vehicles, and because it is unclear if battery-electric vehicles will be able to meet the mission needs of medium and heavy-duty vehicles, the Department is actively studying all types of potential alternative fuel drivetrains.

In FY 2023 the Environmental Security Technology Certification Program (ESTCP) funded several projects to study various aspects of transitioning medium and heavy-duty vehicles, including performance, range, cost-benefit analysis, fueling/charging infrastructure requirements, supply chain, safety, and greenhouse gas emissions. The initial phase of the ESTCP work consists of a feasibility analysis that will narrow the list of viable technologies. At the conclusion of the first phase, depending on the availability of funding, ESTCP plans to execute pilot studies of the most viable alternative fuel technologies.

[18] Consumer Reports. Electric Vehicle Ownership Costs: Chapter 2 – Maintenance. https://advocacy.consumerreports.org/wpcontent/uploads/2020/09/Maintenance-Cost-White-Paper-9.24.20-1.pdf

[19] Argonne National Laboratory. "U.S. DOE VTO/HFTO R&D Benefits." https://vms.taps.anl.gov/research-highlights/u-s-doe-vto-hfto-r-d-benefits/

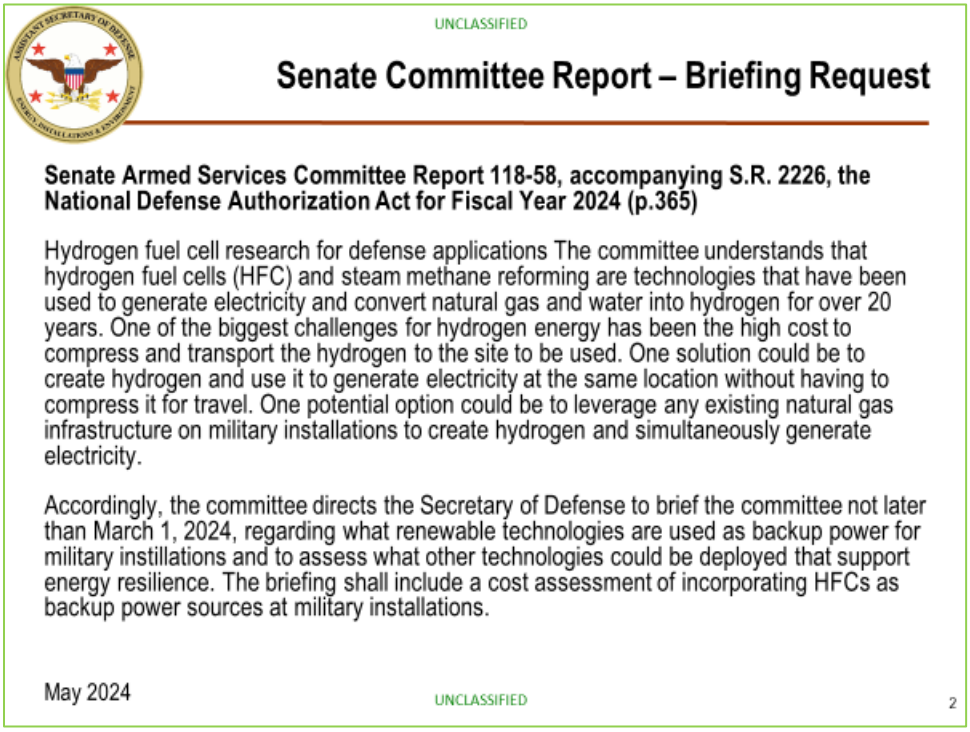
Appendix L – Briefing on Hydrogen Fuel Cell Research for Defense Applications


UNCLASSIFIED

Hydrogen Fuel Cell Research for Defense Applications

MAY 2024

UNCLASSIFIED


UNCLASSIFIED

Senate Committee Report – Briefing Request

Senate Armed Services Committee Report 118-58, accompanying S.R. 2226, the National Defense Authorization Act for Fiscal Year 2024 (p.365)

Hydrogen fuel cell research for defense applications The committee understands that hydrogen fuel cells (HFC) and steam methane reforming are technologies that have been used to generate electricity and convert natural gas and water into hydrogen for over 20 years. One of the biggest challenges for hydrogen energy has been the high cost to compress and transport the hydrogen to the site to be used. One solution could be to create hydrogen and use it to generate electricity at the same location without having to compress it for travel. One potential option could be to leverage any existing natural gas infrastructure on military installations to create hydrogen and simultaneously generate electricity.

Accordingly, the committee directs the Secretary of Defense to brief the committee not later than March 1, 2024, regarding what renewable technologies are used as backup power for military instillations and to assess what other technologies could be deployed that support energy resilience. The briefing shall include a cost assessment of incorporating HFCs as backup power sources at military installations.

May 2024

UNCLASSIFIED

2

UNCLASSIFIED

DoD Energy Resilience

From 10 U.S.C. § 101:

"Energy resilience" means the ability to avoid, prepare for, minimize, adapt to, and recover from anticipated and unanticipated energy disruptions in order to ensure energy availability and reliability sufficient to provide for mission assurance and readiness, including mission essential operations related to readiness, and to execute or rapidly reestablish mission essential requirements.

From 10 U.S.C. § 2920:

The Secretary of Defense shall, by the end of fiscal year 2030, provide that 100 percent of the energy load required to maintain the critical missions of each installation have a minimum level of availability of 99.9 percent per fiscal year.

DoD strives to optimize the use of on-site energy resources to directly improve mission assurance.



Schofield Generating Station, Hawaii

May 2024

UNCLASSIFIED

3

UNCLASSIFIED

DoD Backup Power

DoD Instruction 4170.11, Installation Energy Management, identifies the need for emergency backup generators among a diversified portfolio of backup power technologies, and points to additional guidance to support procurement of backup power, including generators and renewable power sources.

The table below provides a summary of DoD guidance and tools related to the procurement of backup generators for military installations, as part of a standalone procurement or as part of an Energy Savings Performance Contract (ESPC).

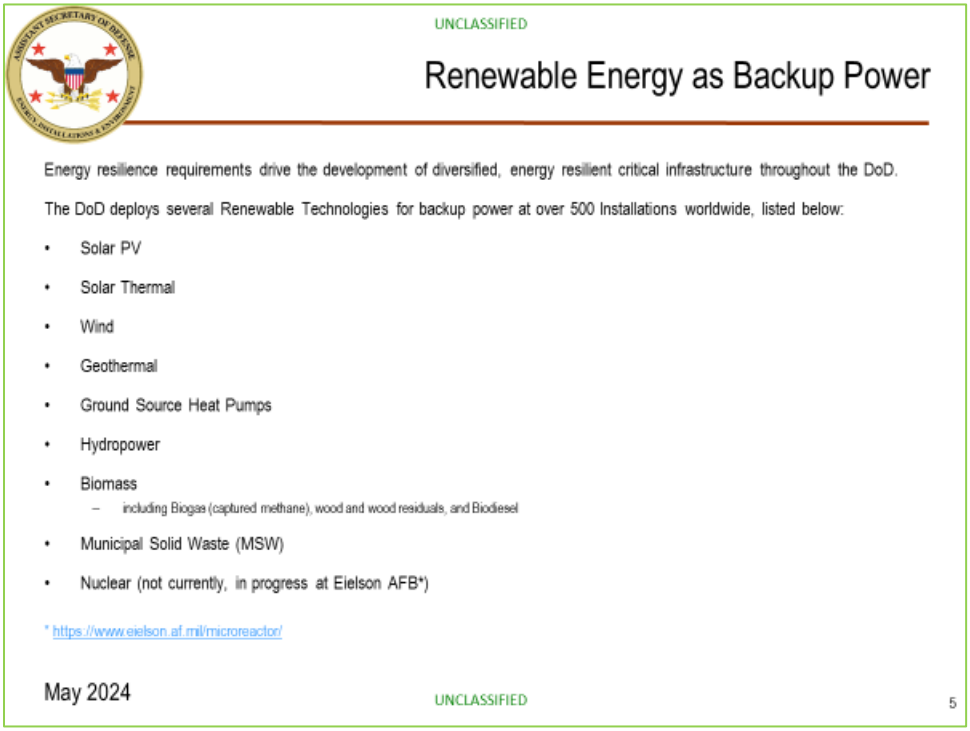
Document	Date of Issuance	Summary
DoD Instruction 4170.11 – Installation Energy Management	March 2016	The Department of Defense shall strive to modernize infrastructure, increase utility and energy conservation, enhance demand reduction, and improve energy flexibility resilience, thereby saving taxpayer dollars and reducing emissions that contribute to air pollution and global climate change. This document identifies the need for emergency backup generators, and also recognizes the need to diversify into integrated, distributed, or renewable energy sources.
Engine- Driven Generator Systems for Prime and Standby Power Applications (UFC 3-540-01)	November 2019	This UFC provides criteria for the design and installation of engine generator systems for use as standby and prime power systems.
Energy Resilience Assessment Tool	Initial release 2015; tool is updated regularly	The ERA Tool analyzes the energy resilience baseline for military installations in terms of the lifecycle cost and amount of unserved load associated with the current design of the utility system. It provides alternative resilient energy technology combinations capable of meeting the mission required electrical loads, that includes both centralized and distributed energy solutions, diesel and natural gas generation, solar photovoltaics, energy storage, fuel cells, and more.
Installation Energy Plan Guidance	March 2016; updated March 2018	Provides guidance to develop a holistic roadmap that enables the installation to work toward energy efficiency, renewable energy, and energy resilience goals. Through this process the installation identifies energy resilience gaps and can determine backup power requirements.

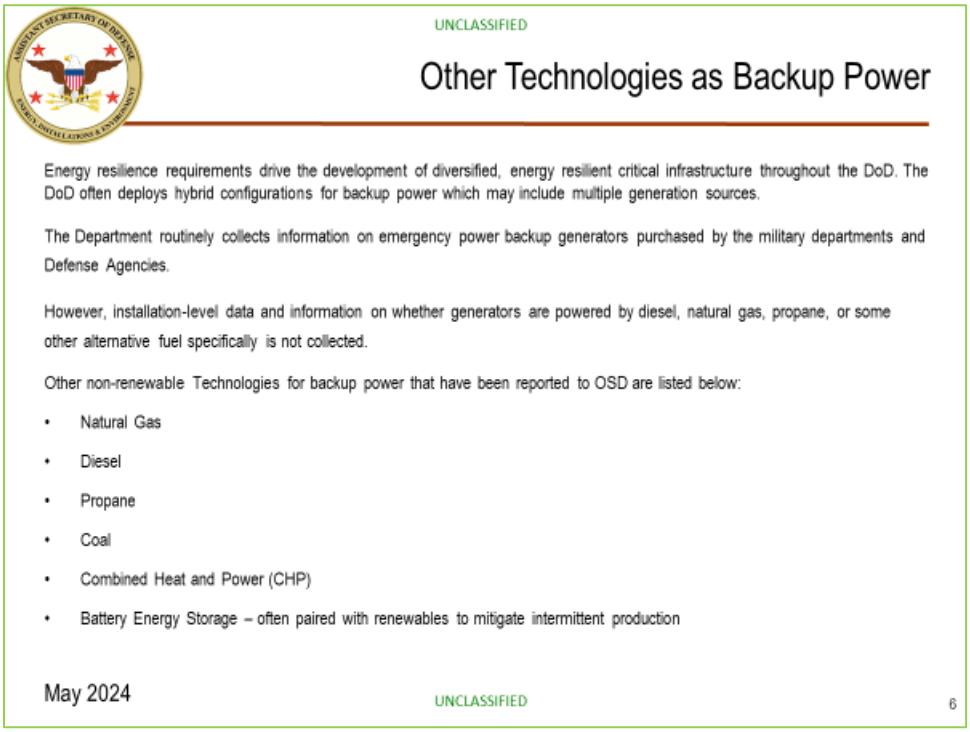
[1] https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc/ufc-3-540-01
[2] https://www.acq.osd.mil/eie/Downloads/IE/Signed%20Installation%20Energy%20Plan.pdf

May 2024

UNCLASSIFIED

4


UNCLASSIFIED
Renewable Energy as Backup Power
Energy resilience requirements drive the development of diversified, energy resilient critical infrastructure throughout the DoD.
The DoD deploys several Renewable Technologies for backup power at over 500 Installations worldwide, listed below:
Solar PV
Solar Thermal
Wind
Geothermal
Ground Source Heat Pumps
Hydropower
Biomass
including Biogas (captured methane), wood and wood residuals, and Biodiesel
Municipal Solid Waste (MSW)
Nuclear (not currently, in progress at Eielson AFB*)
* https://www.eielson.af.mil/microreactor/
May 2024
UNCLASSIFIED
5


UNCLASSIFIED
Other Technologies as Backup Power
Energy resilience requirements drive the development of diversified, energy resilient critical infrastructure throughout the DoD. The DoD often deploys hybrid configurations for backup power which may include multiple generation sources.
The Department routinely collects information on emergency power backup generators purchased by the military departments and Defense Agencies.
However, installation-level data and information on whether generators are powered by diesel, natural gas, propane, or some other alternative fuel specifically is not collected.
Other non-renewable Technologies for backup power that have been reported to OSD are listed below:
Natural Gas
Diesel
Propane
Coal
Combined Heat and Power (CHP)
Battery Energy Storage – often paired with renewables to mitigate intermittent production
May 2024
UNCLASSIFIED
6

UNCLASSIFIED

Hydrogen Fuel Cells & the DoD

Hydrogen offers a potential zero carbon source of energy. Its costs and benefits are under evaluation by the DOE and DoD.

The **DOE** has a large investment in hydrogen*, which DoD follows to help understand its role for DoD installations. The U.S. Department of Energy Hydrogen Program, led by the **Hydrogen and Fuel Cell Technologies Office (HFTO)** within the Office of Energy Efficiency and Renewable Energy (EERE), conducts research and development in hydrogen production, delivery, infrastructure, storage, fuel cells, and multiple end uses across transportation, industrial, and stationary power applications.

In FY 2023, **DoD's Environmental Security Technology Certification Program (ESTCP)**** has also initiated an assessment for using green hydrogen to provide energy resilience. This project aims to perform a feasibility study (FS) to design and demonstrate a Green Hydrogen System (GHS) for improved energy resilience across Department of Defense installations.

* https://www.hydrogen.energy.gov/
** https://serdp-estcp.mil/projects/details/ea8aa530-7252-4ce2-b95f-811ca2d42970

May 2024

UNCLASSIFIED

7

UNCLASSIFIED

Hydrogen and Natural Gas Infrastructure

In June 2023, the DOE HFTO published the *U.S. National Clean Hydrogen Strategy and Roadmap**, Figure 23 from page 43 of the Strategy appears to support the committee's suggested option to leverage existing natural gas infrastructure on military installations to create hydrogen and simultaneously generate electricity.

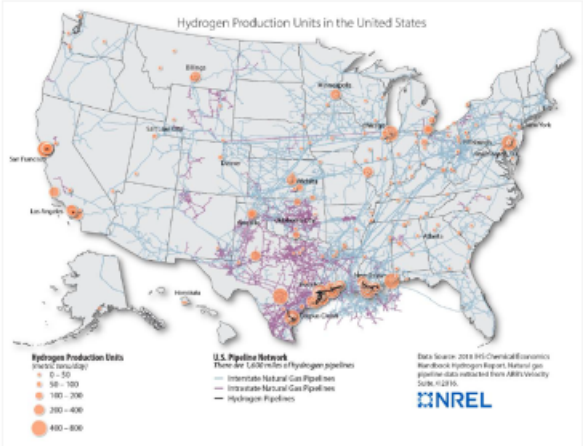


Figure 23: Hydrogen production units and pipelines for hydrogen and natural gas in the United States.

* https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/us-national-clean-hydrogen-strategy-roadmap.pdf

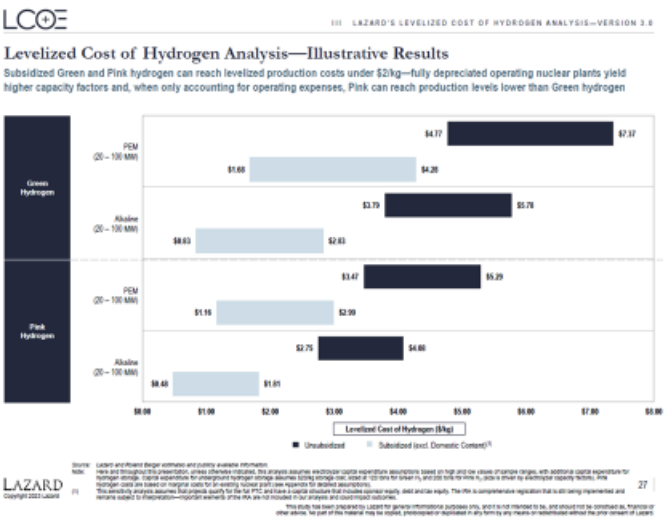
May 2024

UNCLASSIFIED

8

UNCLASSIFIED

HFC Cost Assessment



*Lazard's Levelized Cost of Hydrogen Analysis—Version 3.0** was published in April 2023 and included a Levelized Cost of Hydrogen for large scale (20 – 100 MW) Green and Pink Hydrogen.

*https://www.lazard.com/research-insights/levelized-cost-of-energyplus/

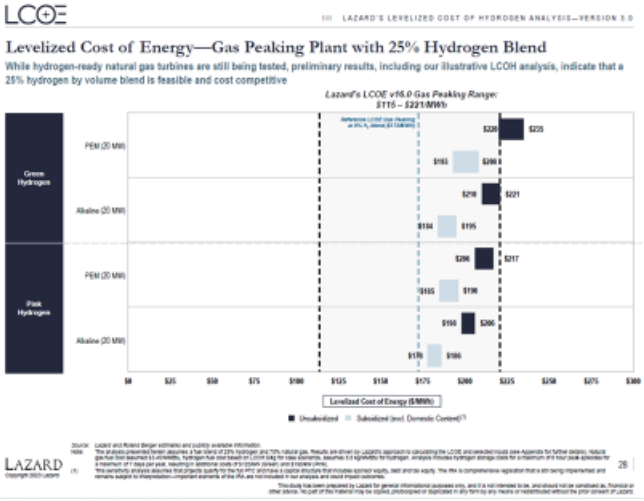
May 2024

UNCLASSIFIED

9

UNCLASSIFIED

HFC Cost Assessment



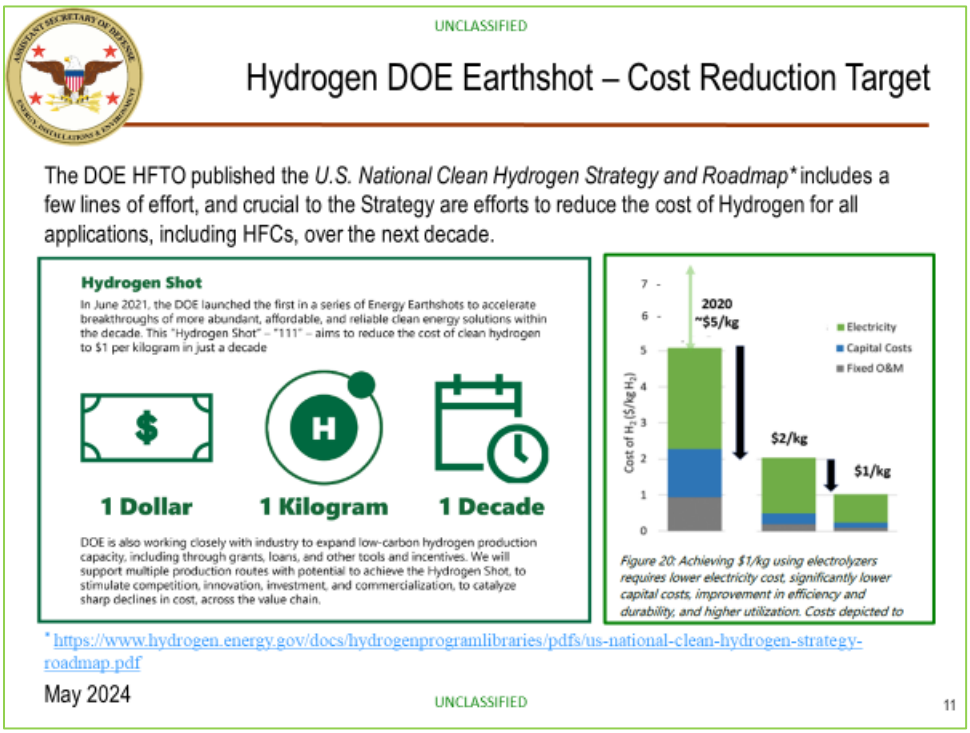
*https://www.lazard.com/research-insights/levelized-cost-of-energyplus/

*Lazard's Levelized Cost of Hydrogen Analysis—Version 3.0** also includes an evaluation of a 25% clean Hydrogen (Green, Pink) by volume blend – while these results are preliminary, Lazard asserts that this is feasible and cost competitive with Gas Peaker plants.

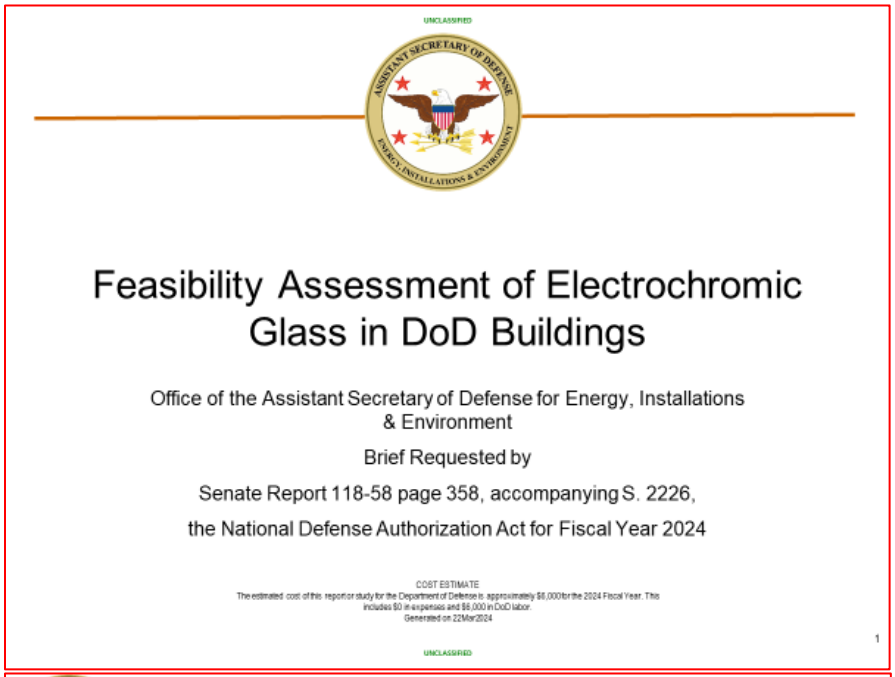
May 2024

UNCLASSIFIED

10


UNCLASSIFIED
Hydrogen DOE Earthshot – Cost Reduction Target
The DOE HFTO published the U.S. National Clean Hydrogen Strategy and Roadmap* includes a few lines of effort, and crucial to the Strategy are efforts to reduce the cost of Hydrogen for all applications, including HFCs, over the next decade.
Hydrogen Shot
In June 2021, the DOE launched the first in a series of Energy Earthshots to accelerate breakthroughs of more abundant, affordable, and reliable clean energy solutions within the decade. This "Hydrogen Shot" – "111" – aims to reduce the cost of clean hydrogen to $1 per kilogram in just a decade
1 Dollar
1 Kilogram
1 Decade
DOE is also working closely with industry to expand low-carbon hydrogen production capacity, including through grants, loans, and other tools and incentives. We will support multiple production routes with potential to achieve the Hydrogen Shot, to stimulate competition, innovation, investment, and commercialization, to catalyze sharp declines in cost, across the value chain.
2020
~$5/kg
$2/kg
$1/kg
Electricity
Capital Costs
Fixed O&M
Cost of H2 ($/kg H2)
Figure 20: Achieving $1/kg using electrolyzers requires lower electricity cost, significantly lower capital costs, improvement in efficiency and durability, and higher utilization. Costs depicted to
* https://www.hydrogen.energy.gov/docs/hydrogenprogramlibraries/pdfs/us-national-clean-hydrogen-strategy-roadmap.pdf
May 2024
UNCLASSIFIED
11

Appendix M – Briefing on the Benefits and Use of Electrochromic Glass in Military Bases and Installations



UNCLASSIFIED

Feasibility Assessment of Electrochromic Glass in DoD Buildings

Office of the Assistant Secretary of Defense for Energy, Installations & Environment

Brief Requested by

Senate Report 118-58 page 358, accompanying S. 2226,

the National Defense Authorization Act for Fiscal Year 2024

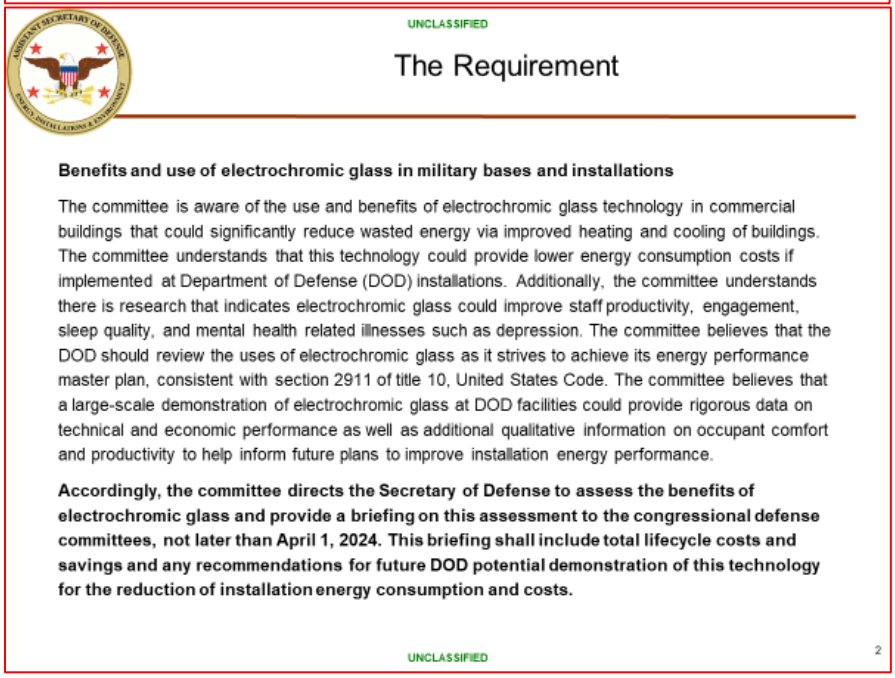
COST ESTIMATE

The estimated cost of this report or study for the Department of Defense is approximately $6,000 for the 2024 Fiscal Year. This includes $0 in expenses and $6,000 in DoD labor.

Generated on 22Mar2024

1

UNCLASSIFIED



UNCLASSIFIED

The Requirement

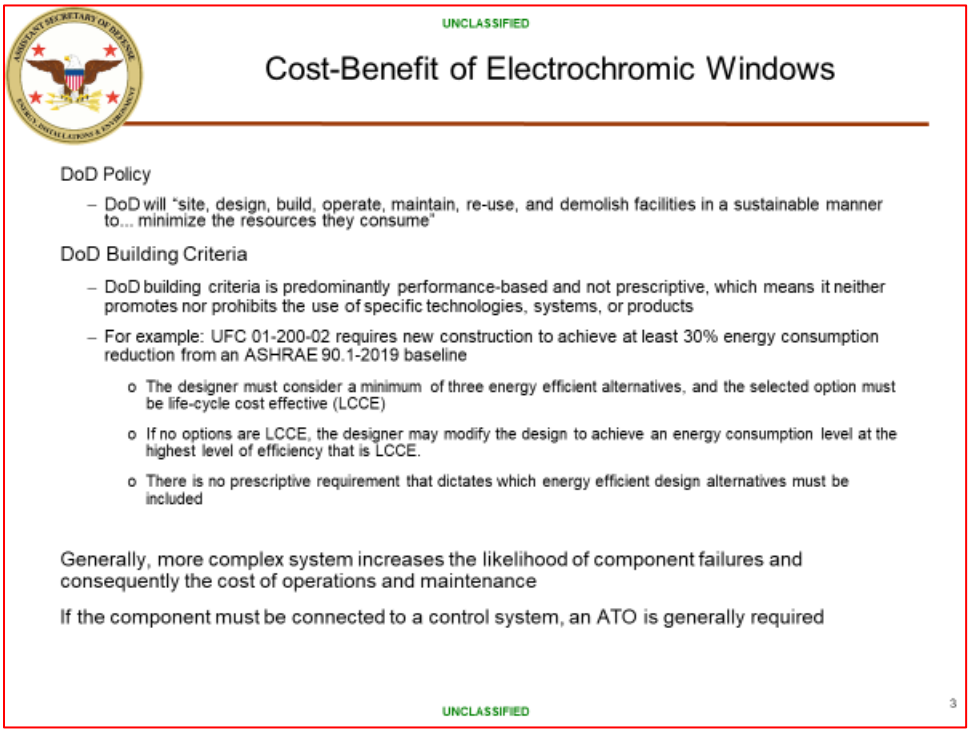
Benefits and use of electrochromic glass in military bases and installations

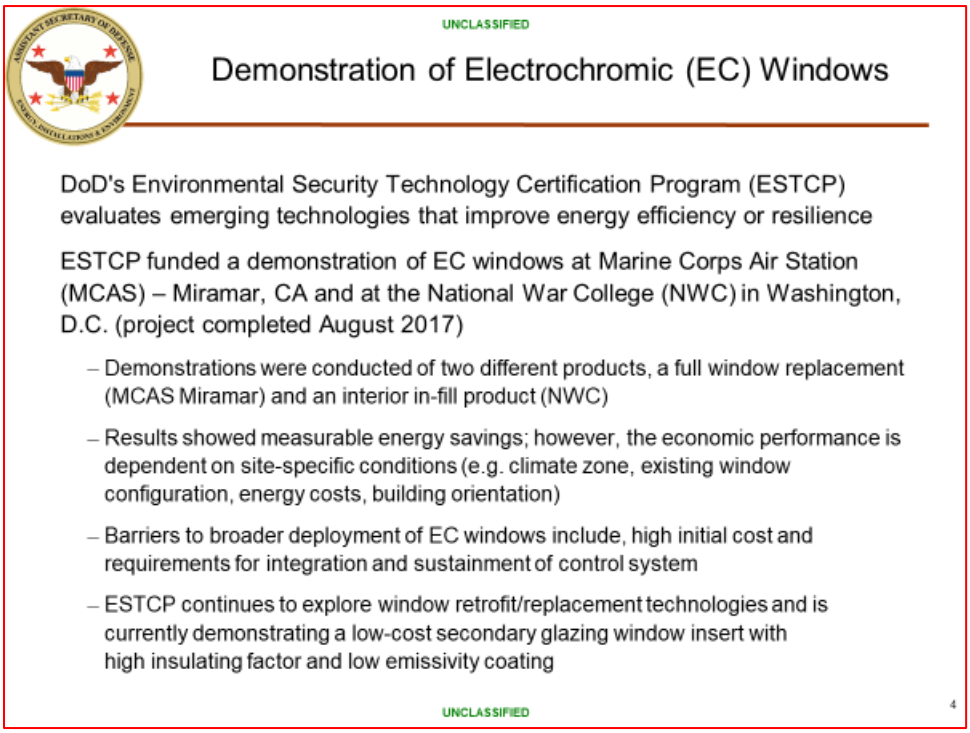
The committee is aware of the use and benefits of electrochromic glass technology in commercial buildings that could significantly reduce wasted energy via improved heating and cooling of buildings. The committee understands that this technology could provide lower energy consumption costs if implemented at Department of Defense (DOD) installations. Additionally, the committee understands there is research that indicates electrochromic glass could improve staff productivity, engagement, sleep quality, and mental health related illnesses such as depression. The committee believes that the DOD should review the uses of electrochromic glass as it strives to achieve its energy performance master plan, consistent with section 2911 of title 10, United States Code. The committee believes that a large-scale demonstration of electrochromic glass at DOD facilities could provide rigorous data on technical and economic performance as well as additional qualitative information on occupant comfort and productivity to help inform future plans to improve installation energy performance.

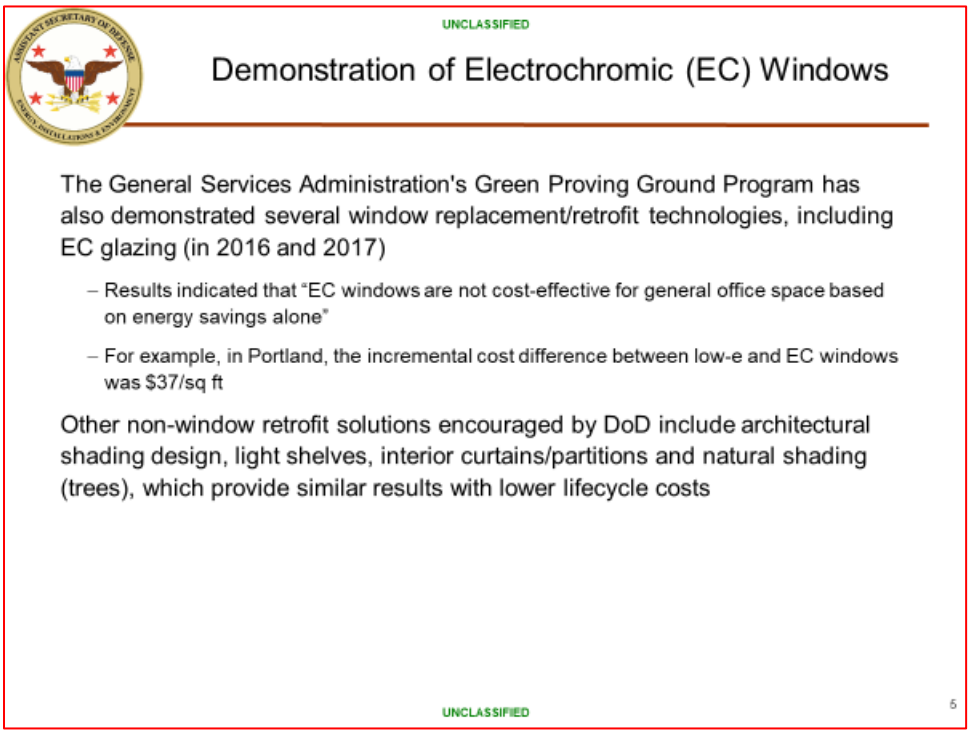
Accordingly, the committee directs the Secretary of Defense to assess the benefits of electrochromic glass and provide a briefing on this assessment to the congressional defense committees, not later than April 1, 2024. This briefing shall include total lifecycle costs and savings and any recommendations for future DOD potential demonstration of this technology for the reduction of installation energy consumption and costs.

UNCLASSIFIED

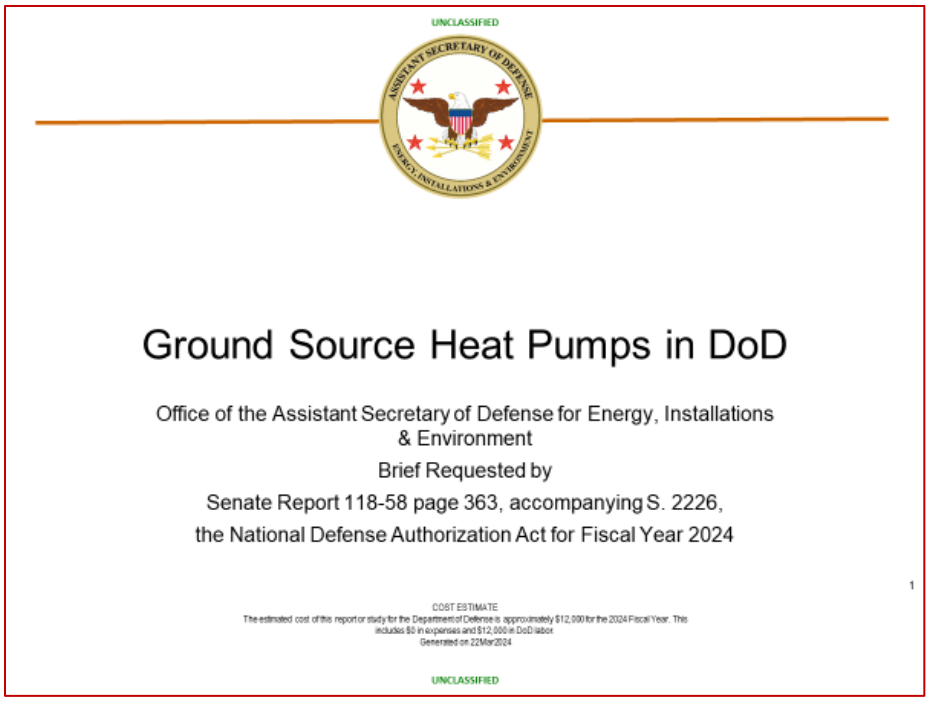
2


UNCLASSIFIED
Cost-Benefit of Electrochromic Windows
DoD Policy
– DoD will "site, design, build, operate, maintain, re-use, and demolish facilities in a sustainable manner to... minimize the resources they consume"
DoD Building Criteria
– DoD building criteria is predominantly performance-based and not prescriptive, which means it neither promotes nor prohibits the use of specific technologies, systems, or products
– For example: UFC 01-200-02 requires new construction to achieve at least 30% energy consumption reduction from an ASHRAE 90.1-2019 baseline
o The designer must consider a minimum of three energy efficient alternatives, and the selected option must be life-cycle cost effective (LCCE)
o If no options are LCCE, the designer may modify the design to achieve an energy consumption level at the highest level of efficiency that is LCCE.
o There is no prescriptive requirement that dictates which energy efficient design alternatives must be included
Generally, more complex system increases the likelihood of component failures and consequently the cost of operations and maintenance
If the component must be connected to a control system, an ATO is generally required
UNCLASSIFIED
3


UNCLASSIFIED
Demonstration of Electrochromic (EC) Windows
DoD's Environmental Security Technology Certification Program (ESTCP) evaluates emerging technologies that improve energy efficiency or resilience
ESTCP funded a demonstration of EC windows at Marine Corps Air Station (MCAS) – Miramar, CA and at the National War College (NWC) in Washington, D.C. (project completed August 2017)
– Demonstrations were conducted of two different products, a full window replacement (MCAS Miramar) and an interior in-fill product (NWC)
– Results showed measurable energy savings; however, the economic performance is dependent on site-specific conditions (e.g. climate zone, existing window configuration, energy costs, building orientation)
– Barriers to broader deployment of EC windows include, high initial cost and requirements for integration and sustainment of control system
– ESTCP continues to explore window retrofit/replacement technologies and is currently demonstrating a low-cost secondary glazing window insert with high insulating factor and low emissivity coating
UNCLASSIFIED
4


UNCLASSIFIED
Demonstration of Electrochromic (EC) Windows
The General Services Administration's Green Proving Ground Program has also demonstrated several window replacement/retrofit technologies, including EC glazing (in 2016 and 2017)
– Results indicated that "EC windows are not cost-effective for general office space based on energy savings alone"
– For example, in Portland, the incremental cost difference between low-e and EC windows was $37/sq ft
Other non-window retrofit solutions encouraged by DoD include architectural shading design, light shelves, interior curtains/partitions and natural shading (trees), which provide similar results with lower lifecycle costs
UNCLASSIFIED
5

Appendix N – Briefing on DoD Ground Source Heat Pumps



UNCLASSIFIED

Ground Source Heat Pumps in DoD

Office of the Assistant Secretary of Defense for Energy, Installations & Environment

Brief Requested by

Senate Report 118-58 page 363, accompanying S. 2226,

the National Defense Authorization Act for Fiscal Year 2024

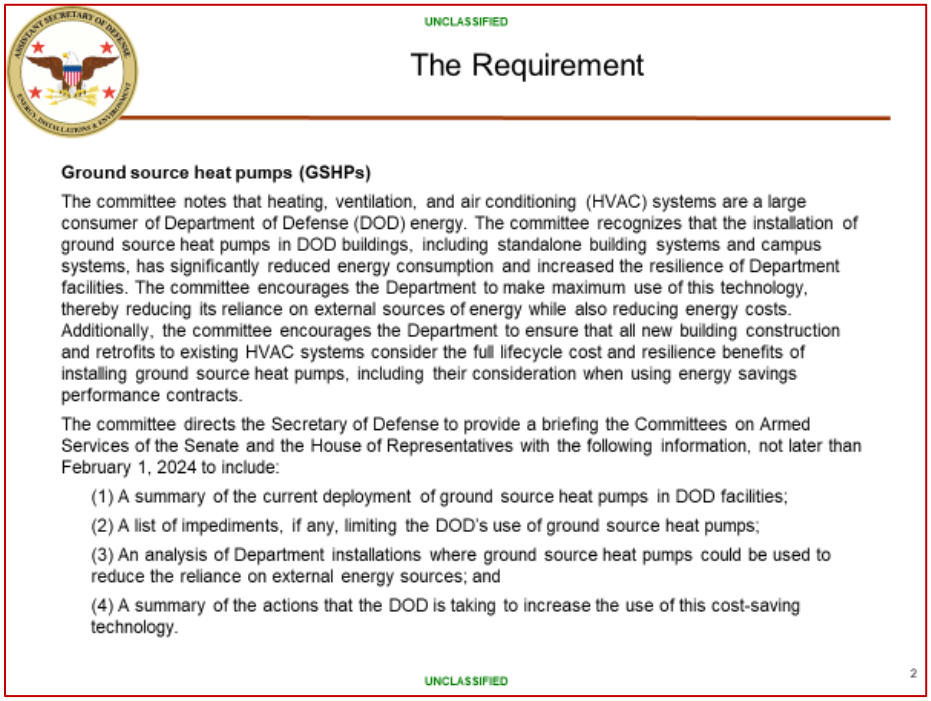
1

COST ESTIMATE

The estimated cost of this report or study for the Department of Defense is approximately $12,000 for the 2024 Fiscal Year. This includes $0 in expenses and $12,000 in DoD labor.

Generated on 22Mar2024

UNCLASSIFIED



UNCLASSIFIED

The Requirement

Ground source heat pumps (GSHPs)

The committee notes that heating, ventilation, and air conditioning (HVAC) systems are a large consumer of Department of Defense (DOD) energy. The committee recognizes that the installation of ground source heat pumps in DOD buildings, including standalone building systems and campus systems, has significantly reduced energy consumption and increased the resilience of Department facilities. The committee encourages the Department to make maximum use of this technology, thereby reducing its reliance on external sources of energy while also reducing energy costs. Additionally, the committee encourages the Department to ensure that all new building construction and retrofits to existing HVAC systems consider the full lifecycle cost and resilience benefits of installing ground source heat pumps, including their consideration when using energy savings performance contracts.

The committee directs the Secretary of Defense to provide a briefing the Committees on Armed Services of the Senate and the House of Representatives with the following information, not later than February 1, 2024 to include:

(1) A summary of the current deployment of ground source heat pumps in DOD facilities;

(2) A list of impediments, if any, limiting the DOD's use of ground source heat pumps;

(3) An analysis of Department installations where ground source heat pumps could be used to reduce the reliance on external energy sources; and

(4) A summary of the actions that the DOD is taking to increase the use of this cost-saving technology.

UNCLASSIFIED

2

UNCLASSIFIED

(1) Current deployment of GSHPs

- DoD Facility and Equipment management systems are not set up to track GSHPs specifically
- The BUILDER Sustainment Management System contains an inventory of heat pumps, but they are not differentiated as GSHPs
- DoD-wide inventory of heat pumps:

Heat Pumps, Total	65,875
Heat Pump, Air Source	
Heat Pump, Water Source	

UNCLASSIFIED

3

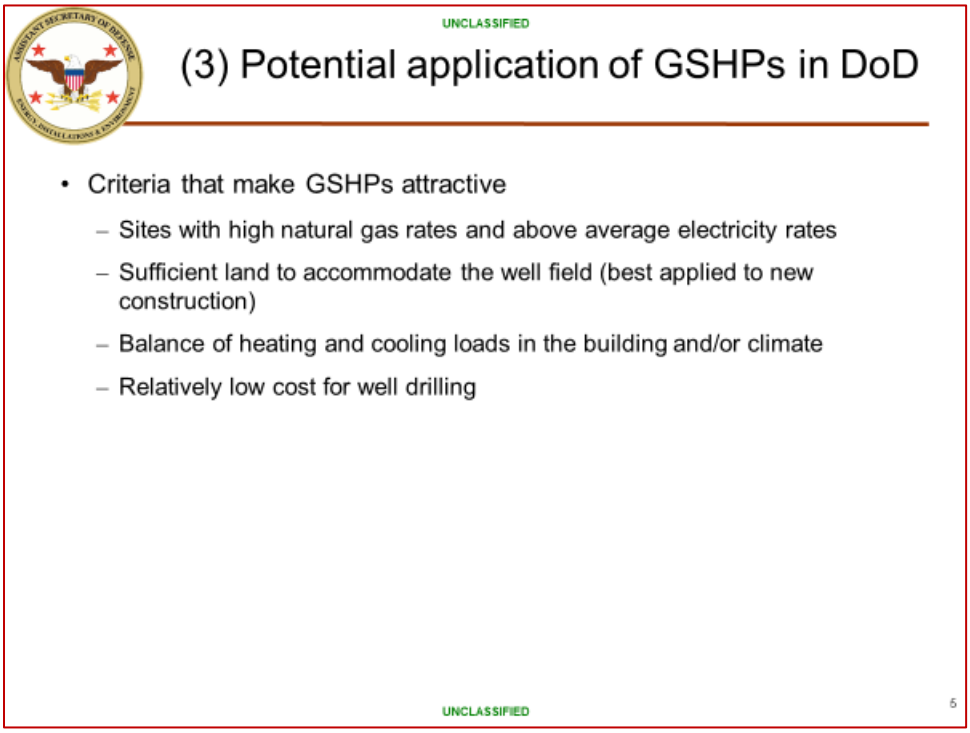
UNCLASSIFIED

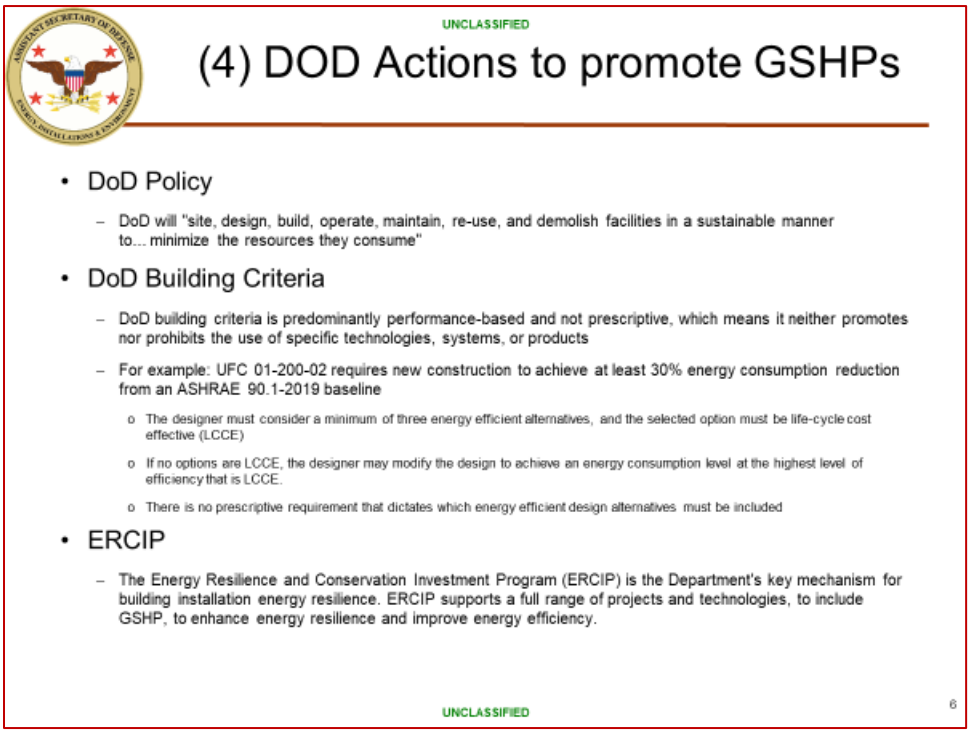
(2) Impediments limiting use of GSHPs

- High initial cost compared to alternative HVAC systems
- Lack of mission owner knowledge of or trust in GSHP benefits
- Limited in-house GSHP design and life-cycle cost analysis/modeling expertise
- Limited expertise in GSHP installation
- Lack of new technologies and techniques to improve GSHP cost/performance
- Variable state and local permitting requirements for wells
- System repair and maintenance challenges

UNCLASSIFIED

4


UNCLASSIFIED
(3) Potential application of GSHPs in DoD
• Criteria that make GSHPs attractive
– Sites with high natural gas rates and above average electricity rates
– Sufficient land to accommodate the well field (best applied to new construction)
– Balance of heating and cooling loads in the building and/or climate
– Relatively low cost for well drilling
UNCLASSIFIED
5


UNCLASSIFIED
(4) DOD Actions to promote GSHPs
• DoD Policy
– DoD will "site, design, build, operate, maintain, re-use, and demolish facilities in a sustainable manner to... minimize the resources they consume"
• DoD Building Criteria
– DoD building criteria is predominantly performance-based and not prescriptive, which means it neither promotes nor prohibits the use of specific technologies, systems, or products
– For example: UFC 01-200-02 requires new construction to achieve at least 30% energy consumption reduction from an ASHRAE 90.1-2019 baseline
o The designer must consider a minimum of three energy efficient alternatives, and the selected option must be life-cycle cost effective (LCCE)
o If no options are LCCE, the designer may modify the design to achieve an energy consumption level at the highest level of efficiency that is LCCE.
o There is no prescriptive requirement that dictates which energy efficient design alternatives must be included
• ERCIP
– The Energy Resilience and Conservation Investment Program (ERCIP) is the Department's key mechanism for building installation energy resilience. ERCIP supports a full range of projects and technologies, to include GSHP, to enhance energy resilience and improve energy efficiency.
UNCLASSIFIED
6

UNCLASSIFIED

(4) DOD Actions to promote GSHPs

- DoD's premier technology demonstration-validation program, the Environmental Security Technology Certification Program (ESTCP), has funded demonstrations of GSHP technologies and activities to inform the DoD community on best practices for GSHP project implementation
  - Sponsoring workshop on opportunities to implement Thermal Networks (which include GSHPs) on DoD installations
  - Next slide provides a summary of related projects and activities

UNCLASSIFIED

7

UNCLASSIFIED

Summary of ESTCP GSHP Projects

- Demonstrated GSHP Coupled with Underground Thermal Energy Storage (Completed 2018)
  - Marine Corps Logistics Base – Albany GA: Borehole Thermal Energy Storage (BTES)
  - Fort Moore, GA: Aquifer Thermal Energy Storage (ATES)
- Delivered training on GSHP system planning and design (Completed 2021)
  - Develop GSHP-BTES Design Guidance
  - Develop training webinars
    - Borehole Thermal Energy Storage
    - Commissioning of GSHP Systems
- Studying feasibility of thermal networks at three military installations (Report due Dec. 2024)
- Evaluating GSHP application in cold climates (Report due Aug. 2024)

UNCLASSIFIED

8

Appendix O – FY 2023 Operational Energy Initiatives

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
USAF	B-52H Re-Engine	B-52 Commercial Engine Replacement Program (CERP)	This sustainment program will replace the current unsustainable TF33 engine with engines of similar size, weight, and thrust characteristics. The development, production and installation of new engines and related subsystems will replace legacy equipment on 76 B- 52H a/c.	Energy Substitution and Diversification	Propulsion Upgrades_Air	3600	07	0101113F	$379,400	$4,198,700
USAF	Adaptive Engine Transition Program	Adaptive Engine Transition Program (AETP)	The AETP 3-stream adaptive cycle engine is not transitioning to a Program of Record. Testing of the full-size prototype engines has validated the revolutionary performance gains, engineering models, maturity of enabling technologies, transition readiness, as well as commensurate reduction in greenhouse gas emissions of the engines. AETP's advanced materials, technologies, manufacturing, assembly, and test results are informing design activities and prototype planning for the Next Generation Adaptive Propulsion (NGAP) program.	Energy Demand Reduction	Propulsion Upgrades_Air	3600	04	0604534F	$286,096	$0
USAF	Advanced Engine Development	Next Generation Adaptive Propulsion (NGAP)	The NGAP program builds upon the prototyping and technology maturation accomplished under the Adaptive Engine Transition Program (AETP) to provide enhanced engine performance key to enabling future Next Generation Air Dominance (NGAD) capabilities. Operational benefits include reduced energy demand and aerial tanker dependency, increased platform range and or time on station, added thermal management capacity to support platform lethality and survivability upgrades, and a choice between increased peacetime aircrew readiness or cost savings due to the less fuel needed.	Energy Substitution and Diversification	Propulsion Upgrades_Air	3600	04	0604004F	$212,586	$1,885,121
USAF	Aerospace Propulsion and Power Technology (Corey - RQ)	Power and Thermal for Omniscient Autonomous Platforms (PTOAP) (formerly Megawatt Aircraft Power and Thermal)	Integrating new developments in Power and Thermal components to demonstrate advanced architectures to enable high powered mission systems for future Air Superiority platforms while maintaining energy efficiency. Technology maturation in advanced power and thermal architecture, modeling and simulation, and integration. Developing flexible and adaptive Power and Thermal components that allow for synergetic architectures that leverage advanced engine capabilities and energy storage.	Energy Demand Reduction	Platform Thermal Management	3600	02	0602203F	$3,353	$1,648
USAF	Aerospace Propulsion and Power Technology (Corey - RQ)	Power and Thermal for Omniscient Autonomous Platforms (PTOAP) (formerly	Integrating new developments in Power and Thermal components to demonstrate advanced architectures to enable high powered mission systems for future Air Superiority	Energy Demand Reduction	Platform Thermal Management	3600	03	0603211F	$0	$5,236

[20] FY2024 – FY2028

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
		Megawatt Aircraft Power and Thermal)	platforms while maintaining energy efficiency. Technology maturation in advanced power and thermal architecture, modeling and simulation, and integration. Developing flexible and adaptive Power and Thermal components that allow for synergetic architectures that leverage advanced engine capabilities and energy storage.							
USAF	Aerospace Propulsion and Power Technology (Corey - RQ)	Power and Thermal for Omniscient Autonomous Platforms (PTOAP) (formerly Megawatt Aircraft Power and Thermal)	Integrating new developments in Power and Thermal components to demonstrate advanced architectures to enable high powered mission systems for future Air Superiority platforms while maintaining energy efficiency. Technology maturation in advanced power and thermal architecture, modeling and simulation, and integration. Developing flexible and adaptive Power and Thermal components that allow for synergetic architectures that leverage advanced engine capabilities and energy storage.	Energy Demand Reduction	Platform Thermal Management	3600	03	0603216F	$8,838	$10,067
USAF	Aerospace Vehicle Technologies	Legacy aircraft drag reduction (aggregate)	Development and implementation of various drag red initiatives (e.g., HF-antenna, other conformal antennas, C-5 anti-collision light, others)	Supply Chain Resilience	Platform Upgrades_Air	3600	05	0603211F	$0	$60,000
USAF	Aerospace Vehicle Technologies (Corey - RQ)	Attritable Structures (formerly Low Cost Attritable Aircraft Technology)	Develop, prototype and flight-demonstrate a series of low-cost attritable aircraft that can be rapidly manufactured, in large or small numbers and on-demand, as-needed, for a base platform cost NTE $3M, plus the cost of the mission system/payload. The capability will enable the Air Force to be able to levy a cost-imposing strategy on potential adversaries, near-peer or otherwise, with a focus on the A2AD threat. The capability will be able to operate as needed with fractionated capabilities, limited training, minimal maintenance, certified analytically, and economically produced at very low production quantities.	Supply Chain Resilience	Platform Upgrades_Air	3600	02	0602201F	$1,467	$0
USAF	Aerospace Vehicle Technologies (Corey - RQ)	Off Board Sensing Station (OBSS) Flight Demonstration (Formerly "Off-Borad Sensing Systems"; and "Low Cost Attritable Aircraft Platform Sharing")	Develop 2 variants of a long range attritable Unmanned Aerial System (UAS) for $3M AUFC. Define aircraft open architecture concept. The primary program objective is to build and fly an off-board sensing station (OBSS) variant consistent with the philosophy and approach of the overarching LCAAT initiative and the platform-sharing concept. This approach is predicated on the cost and time benefit inherent in deriving multiple vehicle variant instantiations from a common shared platform. The program will include the design, fabrication, and flight test an OBSS variant in a relevant environment.	Supply Chain Resilience	Platform Upgrades_Air	3600	02	0602201F	$1,472	$2,906
USAF	Aerospace Vehicle Technologies (Corey - RQ)	Off Board Sensing Station (OBSS) Flight Demonstration (Formerly "Off-Borad Sensing Systems"; and "Low Cost Attritable Aircraft Platform Sharing")	Develop 2 variants of a long range attritable UAS for $3M AUFC. Define aircraft open architecture concept. The primary program objective is to build and fly an off-board sensing station (OBSS) variant consistent with the philosophy and approach of the overarching LCAAT initiative and the platform-sharing concept. This approach is predicated on the cost and time benefit inherent in deriving multiple vehicle variant instantiations from a common shared platform. The program	Supply Chain Resilience	Platform Upgrades_Air	3600	02	0602203F	$0	$11,185

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
			will include the design, fabrication, and flight test an OBSS variant in a relevant environment.							
USAF	Aerospace Vehicle Technologies (Corey - RQ)	Off Board Sensing Station (OBSS) Flight Demonstration (Formerly "Off-Borad Sensing Systems"; and "Low Cost Attritable Aircraft Platform Sharing")	Develop 2 variants of a long range attritable UAS for $3M AUFC. Define aircraft open architecture concept. The primary program objective is to build and fly an off-board sensing station (OBSS) variant consistent with the philosophy and approach of the overarching LCAAT initiative and the platform-sharing concept. This approach is predicated on the cost and time benefit inherent in deriving multiple vehicle variant instantiations from a common shared platform. The program will include the design, fabrication, and flight test an OBSS variant in a relevant environment.	Supply Chain Resilience	Platform Upgrades_Air	3600	03	0603211F	$25,682	$45,242
USAF	AF Energy Program	Manpower for Op Energy (two ppl at HQ/energy office)	Provide technical and data analysis relating to aviation fuel and capabilities; develop, track and report on operational energy aviation metrics	Enterprise-wide Energy Visibility	Policy and Oversight	3400	04	0905015F	$0	$2,500
USAF	AF Energy Program	Mission Execution Excellence Program (MEEP)	The MEEP program aims to incentivize operational units toward fuel-efficient operations such as: precision fuel planning, engine start technique, cruise altitude selection, descent profile, and reduced engine taxi. Based on historical data analysis, AF planners and operators require incentives to use fuel more efficiently and become an energy-optimized force. SAF/IEN, the AF Operational Energy Office, created a Mission Execution Incentive Strategy to re-align incentives and remove barriers. This strategy includes a competitive, voluntary program to incentivize Airmen to operate with fuel-use optimization in mind. Financial incentives will aim to create future savings and reinvestment in fuel-efficient education, training, and programs.	Energy Demand Reduction	Policy and Oversight	3400	04	0905015F	$10,000	$82,500
USAF	AF Energy Program	Vertical Pallet Stacker - Manufacturing	Addresses wasted vertical storage space in C-17/C-5/C-130 aircraft - stacks a 463L pallet atop an already loaded 463L pallet; enable Agile Combat Employment (ACE); pallet is certified for cargo aircraft; funds will build 50 units for operational use	Energy Demand Reduction	Current Operations Tools	3400	04	0905015F	$0	$10,000
USAF	AF Energy Program	Operational Energy Education	Efforts to educate the Air Force and Joint Force regarding operational energy in DOD warfighting. Includes coordination with the AF Institute of Technology, Naval Post-Graduate School, and Defense Acquisition University. Includes initiatives such as curriculum development, force-wide education and strategic communications campaigns, and university or FFRDC-led studies to improve understanding of operational energy problem sets.	Enterprise-wide Energy Visibility	Policy and Oversight	3400	04	0905015F	$2,000	$10,000
USAF	AF Energy Program	Engine Detergent Wash (KC-135, C-5, B-52, E-3)	Jet engines ingest debris and contaminants during operations that reduces their time-on-wing and decreases engine efficiency and power. This causes increased fuel burn and exhaust gas temperature, leading to higher maintenance costs and decreased aircraft availability. Detergent engine washes	Energy Demand Reduction	Propulsion Upgrades_Air	3400	04	0905015F	$600	$3,000

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
			reach deep into the engine core and apply full coverage to engine parts, removing buildup of contaminants. Detergent washing is expected to improve performance, reduce engine temperatures, and decrease fuel burn. This program focuses on the F108 engine, which is the engine used on the KC-135.							
USAF	Aircraft Engine Component Improvement Program	Mobility Aircraft Engine Compressor Blade Scanning/Coating Technology for Increased Fuel Efficiency	Develops technologies, processes, procedures to leverage commercial airline methods for improving fuel efficiency of legacy aircraft propulsion systems through sustainment innovation.	Energy Demand Reduction	Propulsion Upgrades_Air	3600	07	0207268F	$1,000	$9,000
USAF	AOC WS Block 20 Optimization	AOC WS Block 20 - KRADOS Optimization	This effort optimizes for operational energy (aviation fuel) use and increases combat capability through software development activities within the Air Operations Center (AOC) Weapon System (WS) Modifications program. Funds support optimization for the Kessel Run All Domain Operations Suite (KRADOS) software capability to provide efficiencies for planning and executing the Air Tasking Order (ATO), Air Control Order (ACO), and Master Air Attack Plan (MAAP). Enhancements include MAAP process improvements, autoplanner service and algorithm for improving tanker sortie planning, integration to data exchange to ISR Dynamic Execution (IDEX) for Reconnaissance, Surveillance, and Target Acquisition (RSTA) Annex and ISR Sync, and an integrated route, fuel, and transit time calculation service for KRADOS.	Supply Chain Resilience	Current Operations Tools	3600	07	0207410F	$6,000	$18,000
USAF	C-130	Legacy Aircraft Drag Reduction - C-130 Aft Body Drag Reduction Devices	Procures and installs aft body drag reduction devices for C-130 fleet. Expected to result in an average 4.5 percent fuel efficiency improvement across the C-130 fleet (range: 3-6 percent), which results in approximately $10M/year fuel savings (based on FY 2020 fuel prices).	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401115F	$17,500	$0
USAF	C-17	C-17 Databus Collection and Analytics (DBCA)	Procure C-17 DBCA - Previously Databus Reporting System (DBRS), DBCA has been renamed to highlight both the hardware and analytics capabilities instead of focusing primarily on hardware. DBCA will capture and apply analytics to all data present across all databuses on the C-17 aircraft to provide and drive efficiencies in predictive maintenance, operations, training, safety considerations and cybersecurity concerns.	Enterprise-wide Energy Visibility	Current Operations Tools	3010	05	0401130F	$0	$20,000
USAF	C-17	C-17 Databus Collection and Analytics (DBCA) Development	Research and Development project - Data software to record, analyze, and provide and display aircraft system data, provide troubleshooting and engineering diagnoses, allow in-flight analysis and display of trends, faults and anomalies.	Enterprise-wide Energy Visibility	Current Operations Tools	3010	05	0401130F	$0	$3,000
USAF	C-17	C-17 Microvanes	Microvanes are drag reduction devices mounted permanently via adhesive on the aft fuselage to provide an approximate 1 percent drag reduction	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401130F	$3,000	$132
USAF	C-17	C-17 Engine Pylon Fairings	Engine pylon fairings are drag reduction devices mounted on the inboard and outboard pylon on each wing. The engine pylon fairings in conjunction with Microvanes were Flight Tested at Yuma in 2019 demonstrating an improved drag	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401130F	$2,500	$0

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
			reduction (1.4 percent) as opposed to solely Microvanes(1 percent)							
USAF	C-17	C-17 Advanced Winglets (proc)	Procures and installs winglets on C-17 platform that will improve aerodynamic efficiency and reduce fuel consumption	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401130F	$0	$80,000
USAF	Engine Component Improvement Program	Mobility Aircraft Engine Compressor Blade Scanning/Coating Technology for Increased Fuel Efficiency	Develops technologies, processes, procedures to leverage commercial airline methods for improving fuel efficiency of legacy aircraft propulsion systems through sustainment innovation.	Energy Demand Reduction	Propulsion Upgrades_Air	3600	07	0207268F	$96,538	$287,579
USAF	KC-135	KC-135 Winglets	KC-135 Winglets	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401218F	$0	$155,000
USAF	KC-135	KC-135 Vertical Windshield Wipers and Aft Body Drag Reduction Devices	Procures and installs vertical windshield wipers and aft body drag reduction devices for KC-135 fleet. Expected to result in a total 2 percent fuel efficiency improvement for KC-135, which results in approximately $14M/year fuel savings (based on FY 2020 fuel prices).	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401218F	$8,000	$0
USAF	KC-46A	RG	Funds development of Winglets for the KC-46 that will improve aerodynamic efficiency and reduce fuel consumption	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0401221F	$0	$60,000
USAF	Materials Development (Maria Murphy)	Low-Cost Attritable Structures	This program will provide manufacturing support to the Low Cost Attritable Aircraft Technology (LCAAT) Initiative by investigating and validating low-cost manufacturing materials and processes that can be introduced to reduce aerospace vehicle costs and minimize vehicle manufacturing time.	Supply Chain Resilience	Materials and Design	3600	03	0603680F	$2,768	$25,231
USAF	Operational Energy and Installation Resilience	Aviato (CAF Scheduling)	The CAF scheduling platform and related ecosystem extends the training requirements packages developed by HAF/A3 into squadron-level operations and integrates maintenance and other requirements, including deployments, with optimization algorithms that ensure a ready force of combat aircraft and aircrew.	Enterprise-wide Energy Visibility	Current Operations Tools	3600	04	0604860F	$0	$10,000
USAF	Operational Energy and Installation Resilience	KC-46 Winglet Development	Research and development project on installing advanced winglets on KC-46 aircraft fleet	Energy Demand Reduction	Platform Upgrades_Air	3010	05	0604860F	$0	$5,000
USAF	Operational Energy and Installation Resilience	Augmented Reality Adversary Air	Enable MAJCOMs to train forces more efficiently and realistically - increase training opportunities and enable aircrew, Special Tactics and other ground force mission training to occur almost anywhere, at any time, with any training partner; provide both in-flight and in the field training, to include creating immersive, Blue/Red-Force and threat laden environments, using both DoD and civilian airspace	Enterprise-wide Energy Visibility	Policy and Oversight	3600	04	0604860F	$0	$6,000
USAF	Operational Energy and Installation Resilience	KC-46 Winglets (RDT&E)	Research and development project on installing advanced winglets on KC-46 aircraft fleet	Energy Demand Reduction	Platform Upgrades_Air	3600	04	0604860F	$0	$15,000

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
USAF	Operational Energy and Installation Resilience	C-17 Advanced Winglets (RDT&E)	Research and development project on installing advanced winglets on C-17 aircraft fleet	Energy Demand Reduction	Platform Upgrades_Air	3600	04	0604860F	$0	$17,000
USAF	Operational Energy and Installation Resilience	Aircraft Engine Wash (nucleated foam)	Expansion of the nucleated foam wash technology to additional platforms outside of Air Force Special Operations Command's (AFSOC's) CV-22 and AC-130J platforms	Energy Demand Reduction	Propulsion Upgrades_Air	3600	04	0604860F	$0	$3,000
USAF	Operational Energy and Installation Resilience	Air Refueling Optimization	A comprehensive end-to-end requirements repository, validation, allocation, mission planning and execution tool for the entire air refueling enterprise	Enterprise-wide Energy Visibility	Current Operations Tools	3600	04	0604860F	$0	$15,000
USAF	Operational Energy and Installation Resilience	Aviato (CAF Scheduling)	The CAF scheduling platform and related ecosystem extends the training requirements packages developed by HAF/A3 into squadron-level operations and integrates maintenance and other requirements, including deployments, with optimization algorithms that ensure a ready force of combat aircraft and aircrew.	Enterprise-wide Energy Visibility	Current Operations Tools	3600	04	0604860F	$0	$10,000
USAF	Operational Energy and Installation Resilience	C-17 Mobile Mission Computer Application for EFB	The C--17 mission computer application will allow aircrew to run calculations and conduct other planning tasks without powering up the aircraft, which uses fuel. The application will also allow fuel planning optimization by streamlining calculations.	Enterprise-wide Energy Visibility	Current Operations Tools	3600	04	0604860F	$0	$2,500
USAF	Operational Energy and Installation Resilience	KC-135 and C-17 Warfighter Posture Predictive Logistics	KC-135 and C--17 data in an AI Predictive Logistics framework will allow the Air Force to improve the reliability of "last-mile" fuel logistics in aerial refueling and increase the performance of C-17s and KC-135s, enhancing efficiency. The project will focus on warfighter postures in the Pacific Theater to enable Agile Combat Employment concepts with minimal intratheater lift requirements.	Enterprise-wide Energy Visibility	Current Operations Tools	3600	04	0604860F	$0	$6,600
USAF	Operational Energy and Installation Resilience	Pilot Training Next	Pilot Training Next infuses a variety of advanced techniques that can streamline and evolve pilot training to reduce time in the pipeline and the need for aircraft and fuel to produce pilots ready for Formal Training Units. Leveraging VR platforms and extensive instructor interaction, the program can reduce dependence on T-6 availability by 33 percent.	Enterprise-wide Energy Visibility	Policy and Oversight	3600	04	0604860F	$0	$14,000
USAF	Operational Energy and Installation Resilience	Training Ops Modeling	SAF/IEN has worked with AETC on a predictive analytics tool that simulates flight training operations to inform leadership decision making with respect to aircrew training throughput, aircraft readiness, and energy/fuel optimization; increases decision speed and accuracy in complex training and ops environments	Enterprise-wide Energy Visibility	Policy and Oversight	3600	04	0604860F	$0	$1,000
USAF	Operational Energy and Installation Resilience	Advanced Distributed Operations	Mobility Air Forces (MAF) require extensive training for combined or joint operations, but platforms and crew are not always available in real-time to conduct training. Advanced Distributed Operations continues to sustain drop-in real--time virtual training capability while preparing the technology and warfighter for distributed operations in contingencies in the future.	Enterprise-wide Energy Visibility	Current Operations Tools	3600	04	0604860F	$0	$18,000

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
USAF	Operational Energy and Installation Resilience	Advanced Distributed Operations	Mobility Air Forces require extensive training for combined or joint operations, but platforms and crew are not always available in real-time to conduct training. Advanced Distributed Operations continues to sustain drop-in real--time virtual training capability while preparing the technology and warfighter for distributed operations in contingencies in the future.	Enterprise-wide Energy Visibility	Current Operations Tools	3400	08	0604860F	$0	$18,000
USAF	Space Solar Power Incremental Demonstrations and Research Project (SSPIDR)	Space Solar Power Incremental Demonstrations and Research Project (SSPIDR)	SSPIDR is a portfolio of Integrated Demonstrations and Research efforts addressing space-based power collection and transmission capabilities. SSPIDR's Component/Subsystem on-orbit demonstrations will help inform the Operational Spacecraft Prototype Design (~ 1MW Power Beaming).	Enterprise-wide Energy Visibility	Alternative Power Sources	3600	04	1206616SF	$34,500	$202,700
USAF	Tech Transition Program	Mobility Asset Optimization	This program will enhance the efficiency of Mobility Air Fleet and Combat Air Forces operations through tools that optimize processes such as cargo loading, munitions configurations, aircrew scheduling, and aircraft allocation. Optimized cargo load plans yield more efficient flight operations, and the possibility to decrease fuel use by maximizing loads and utilizing fewer aircraft. Improved load planning also requires a reassessment of under-loaded long-distance flights. Loading techniques range from bi-level aircraft loading system to dynamic re-planning software, to dual-rail swap and aggregation. For aircrew scheduling, Puckboard is a data-informed software application that plans aircrew flight qualification and ground training events. The tool, developed by and for Airmen, allows schedulers to rapidly match qualified aircrew with available flights that complete currency requirements. Digital interface enables planners to visualize flight schedules and generate schedules for aircrew while considering required qualifications, crew rest, and conflicts. Overall increases in operational efficiency are expected. For aircraft allocation, Magellan is Air Mobility Command's readiness-driven allocation process software program to synchronize aircraft and crew data, coordinate allocation cycles, enable real-time changes. These tools allow the force to fly more missions with the same number of aircraft.	Supply Chain Resilience	Current Operations Tools	3600	04	0604858F	$13,000	$49,000
USAF	Tech Transition Program	Blended Wing Body Aircraft Prototype	Cargo, tanker, and non-stealth bomber aircraft account for approximately 40 percent of the DOD's total annual operational energy consumption: about 1.2B gallons per year. Reducing this number reduces energy logistics risks, a significant challenge in vast theaters such as the Pacific. Beyond legacy aircraft drag reduction initiatives and improved propulsion sustainment technologies, significant efficiency potential lies within revolutionary improvements in aircraft design and systems. Shifting cargo, tanker, and non-stealth bomber aircraft from tube-and-wing designs to blended wing body (BWB) designs, via a demonstrator aircraft, would yield a	Supply Chain Resilience	Platform Upgrades_Air	3600	04	0604858F	$42,000	$197,000

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[20] ($K)
			minimum 30 percent increase in range and payload capabilities from current Air Force capabilities, and a corresponding 30 percent reduction in emissions. Much like the development of the KC-135 and Boeing 707, advancing BWB designs at present will have a synergistic effect on the U.S. commercial aerospace sector, including both passenger and cargo aircraft. This project, in partnership with industry and other U.S. Government agencies, accelerates large aircraft performance improvement for the DOD by providing an alternative to traditional 20th century tube-and-wing aircraft for aerial tankers.							
USSF	Space Solar Power Incremental Demonstrations and Research Project (SSPIDR)	Space Solar Power Incremental Demonstrations and Research Project (SSPIDR)	SSPIDR is a portfolio of Integrated Demonstrations and Research efforts addressing space-based power collection and transmission capabilities. SSPIDR's Component/Subsystem on-orbit demonstrations will help inform the Operational Spacecraft Prototype Design (~1MW Power Beaming).	Energy Substitution and Diversification	Alternative Power Sources	3620	04	1206616SF	$34,500	0
								Air and Space Forces Total	**$1,192,800**	**$7,778,547**

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Integrated Soldier Power Data System - Core	Small Unit Tactical Power (ISPDS-C)	ISPDS-C RDT&E	Supply Chain Resilience	Individual/Warfighter Power	2040	05	0604827A	$12,559	$23,565
ARMY	Universal Battery Charger	Small Unit Tactical Power (ISPDS-C)	UBC RDT&E	Supply Chain Resilience	Individual/Warfighter Power	2040	05	0604827A	$1,022	$5,058
ARMY	Mobile Soldier Power	Integrated Soldier Power Data System - Core (ISPDS-C)	ISPDS-C OPA	Supply Chain Resilience	Individual/Warfighter Power	2035	03	0211700A	$6,725	$34,075
ARMY	Mobile Soldier Power	Universal Battery Charger	UBC OPA	Supply Chain Resilience	Individual/Warfighter Power	2035	03	0211700A	$9,208	$39,572
ARMY	Ground Soldier System	Small Unit Tactical Power (Nett Warrior)	Nett Warrior CWB+Next Gen HUB OPA	Supply Chain Resilience	Individual/Warfighter Power	2035	03	0211700A	$25,113	$122,396
ARMY	Bradley Program (MOD)	Improved Bradley	Increases mobility with extended life track, and improved shocks, road arms and torsion bars	Energy Demand Reduction	Platform Upgrades_Land	2033	01	0211702A		$0

[21] FY2024 – FY2028

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Bradley Program (MOD)	Improved Bradley	More efficient Bradley - The Bradley improved engine and transmission generate an overall fuel reduction of 3 percent	Energy Demand Reduction	Propulsion Upgrades_Land	2033	01	0211702A	$12,070	$0
ARMY	Abrams Upgrade Program	Improved Abrams	Power Initiative for Abrams M1A2SEPv3 (Power ECP) vehicle - Auxiliary Power Unit (APU) for fuel efficiency	Energy Demand Reduction	Platform Upgrades_Land	2033	01	0211702A	$3,542	$35,261
ARMY	Abrams Upgrade Program	Improved Abrams	Advance Reliability and Cost Savings (ARCS) for the AGT-1500 Turbine Engine - Fuel Efficiency ECP	Energy Demand Reduction	Propulsion Upgrades_Land	2033	01	0211702A	$0	$0
ARMY	Improved Recovery Vehicle (M88A2 HERCULES)	Improved Recovery Vehicle (M88A2 HERCULES)	Engine Reliability Improvement (Engine Electronic Fuel Injection (EFI) AVDS 1790); To increase mean time between overhauls (MTBO) from 300 hours to 2500 hours	Supply Chain Resilience	Propulsion Upgrades_Land	2033	01	0211700A	$0	$51,190
ARMY	Hercules M88A2 Upgrade Recapitalization	Improved Recovery Vehicle (M88A2 HERCULES)	Engine Reliability Improvement (Engine Electronic Fuel Injection (EFI) AVDS 1790); To increase mean time between overhauls (MTBO) from 300 hours to 2500 hours	Supply Chain Resilience	Propulsion Upgrades_Land	2033	01	0211700A	$0	$51,190
ARMY	Defense Research Sciences	CHEMICAL SYNTHESIS and POWER - 66	Basic research to achieve advanced energy control. Research efforts will lead to: light-weight, reliable, compact power sources.	Energy Substitution and Diversification	Alternative Power Sources	2040	01	0601102A	$5,882	$27,912
ARMY	Defense Research Sciences	PROPULSION ENERGETICS FLIGHT - 97	Basic research for improved understanding of propulsion and combustion for improved efficiency and fuel flexibility and fluid dynamics for rotorcraft.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	01	0601102A	$3,094	$15,911
ARMY	Defense Research Sciences	Air Vehicle Structures and Dynamics Research - 118	Establish fundamental understanding in structural damage tracking methods, novel material/structures, and prognostic and diagnostic techniques to improve vehicle performance and capability. This includes the advancement of machine learning algorithms for deep learning, and the exploration of novel lightweight, durable, self-sensing structures for improved maneuver and reduced maintenance.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	01	0601102A	$0	$7,653
ARMY	Defense Research Sciences	Autonomous Vehicle Research - 120	Basic research focused on enabling robust autonomous mobility for small and human-scale robotic systems, including autonomous teaming behavior with hybrid human-robotic teams. Enablers for robust autonomous mobility include planning, behaviors, energy efficient maneuver, and the interface of manipulation technologies to support manned-unmanned teaming constructs.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	01	0601102A	$1,844	$9,393
ARMY	Defense Research Sciences	Fundamentals for Alternative Energy Applied Physics Research - 143	Explore novel concepts in energy generation and capture in technologies for efficient conversion of ambient energy to electrical energy for use and storage. Design novel structures to include microscale power devices for multimodal harvesting and efficient distributed power conversion.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	01	0601102A	$991	$5,025

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Defense Research Sciences	Intelligent Systems - 48	Research in autonomous systems that supports and unburdens Soldiers in a flexible, robust, survivable and comprehensive manner. This work addresses the cognitive requirements of humans and (non-human) agents, both hardware and software based, operating individually or in collaboration, on the battlefield.	Energy Substitution and Diversification	Platform Upgrades_Land	2040	01	0601102A	$6,443	$33,812
ARMY	Defense Research Sciences	Novel multi-fuel tolerant small vehicle power - 92	Basic research to enable highly efficient, multi-fuel conversion in small engines with reduced sensitivity to fuel property variation and extreme ambient conditions.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	01	0601102A	$3,177	$15,817
ARMY	Defense Research Sciences	Reconfigurable Platform Mechanics Propulsion - 146	Reconfigurable platform mechanics and propulsion science investigating technologies to enable subsystem configuration concepts for efficient hover and high-speed/range Vertical Take-Off and Landing (VTOL) aircraft.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	01	0601102A	$1,046	$5,305
ARMY	Defense Research Sciences	Research In Vehicle Mobility - 151	Research in support of advanced military mobility technologies with emphasis on Terramechanics (vehicle-terrain interaction), and complex vehicle dynamics and simulation.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	01	0601102A	$791	$4,181
ARMY	Defense Research Sciences	Sol Struct Mech - AMRDEC-AV - 101	Create robust experimental and computational approaches for understanding, modeling, and predicting the complex fluid flow and aerodynamics of next generation rotorcraft concepts.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	01	0601102A	$2,731	$13,855
ARMY	Defense Research Sciences	UAS Vehicle Research - 93	Basic research focused on topics that contribute to the body of knowledge required to create future intelligent, unmanned air systems that can effectively team with manned and unmanned aircraft, ground platforms, and human teammates.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	01	0601102A	$3,199	$16,238
ARMY	Defense Research Sciences	Vehicle Propulsion Power Research - 128	Investigate concepts and theories to provide enhanced tools, methods, and innovative concepts to enable improvements in propulsion power density, energy efficiency, reliability, and lifecycle costs for increased performance and capabilities in future Army systems.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	01	0601102A	$1,633	$8,537
ARMY	Soldier Lethality Technology	Efficient Compact Portable Power - 959	Develop more efficient power and thermal management for small systems and alternative energy technologies to substantially reduce the number of batteries required to accomplish dismounted Soldier/Squad mission objectives	Energy Substitution and Diversification	Individual/Warfighter Power	2040	02	0602143A	$950	$4,837
ARMY	Soldier Lethality Technology	Tactical Power for Soldier Lethality - 938	Designs, and develops innovative power generation and energy storage technologies that support next generation Soldier systems to decrease Soldier load and power burden and increase power capabilities by providing more energy to prolong mission run-time.	Energy Substitution and Diversification	Individual/Warfighter Power	2040	02	0602143A	$5,154	$40,365
ARMY	Ground Technology	Advanced Distributed Power for Autonomous Systems	Research power distribution concepts and architectures for autonomous systems.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	02	0602144A	$976	$8,620

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Ground Technology	Power Electronic Components and Materials	Research Power Electronic Components and Materials to enable high power electrification of manned and unmanned combat vehicles and other Army systems.	Energy Substitution and Diversification	Platform Upgrades_Land	2040	02	0602144A	$0	$0
ARMY	Next Generation Combat Vehicle Technology	Beyond Lithium-Ion Energy Storage 16	Research Lithium-Ion and other battery technologies to increase energy storage density.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	02	0602145A	$1,195	$9,314
ARMY	Next Generation Combat Vehicle Technology	Diesel Electric Power Generator 15	Research Diesel Electric Power Generator to enable hybrid powertrains for combat vehicles and silent mobility and watch.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	02	0602145A	$0	$0
ARMY	Next Generation Combat Vehicle Technology	Electric Drive Motors/Power Controllers/Conv 18	Research Electric Drive Motors, Power Controllers, and Converters to enable hybrid powertrains with electric drive for silent mobility and watch and to improve fuel efficiency of manned and unmanned combat vehicles.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	02	0602145A	$2,019	$0
ARMY	Next Generation Combat Vehicle Technology	High Voltage Modular Li-Ion Battery 17	Research High Voltage Modular Li-Ion Battery technologies to increase energy storage density and enable electrification of manned and unmanned combat vehicles.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	02	0602145A	$1,234	$1,167
ARMY	Next Generation Combat Vehicle Technology	Combat Electric Power Architecture - 1061	Researches combat vehicle power control architecture for the heavy combat vehicles	Supply Chain Resilience	Platform Upgrades_Land	2040	02	0602145A	$0	$7,072
ARMY	Next Generation Combat Vehicle Technology	Next Generation Tank Mobility System (NGTMS) - Power Dense Propulsion System - 1063	Researches power dense propulsion system for Heavy Combat Vehicles	Energy Demand Reduction	Propulsion Upgrades_Land	2040	02	0602145A	$0	$24,057
ARMY	Next Generation Combat Vehicle Technology	OMT Traction Turret Drive Motors Electric Cooling	Researches electric traction drive motors for turret operation.	Energy Substitution and Diversification	Platform Upgrades_Land	2040	02	0602145A	$1,415	$1,018
ARMY	Next Generation Combat Vehicle Technology	Power Dense Range Extender Tech	Researches range extending fuel cell design that uses solid-state energy storage.	Energy Demand Reduction	Storage, Power Controls and Distribution	2040	02	0602145A	$0	$4,956
ARMY	Next Generation Combat Vehicle Technology	RCV Silent Watch and Mobility Range Extension	Researches JP8 reformer based silent watch and mobility extension subsystem.	Energy Demand Reduction	Storage, Power Controls and Distribution	2040	02	0602145A	$1,714	$1,525
ARMY	Next Generation Combat Vehicle Technology	Scalable Electrification and Control Architecture 19	Research saleable Electrification and Control Architectures to enable electrified and energy efficient manned and unmanned combat vehicles.	Energy Demand Reduction	Platform Upgrades_Land	2040	02	0602145A	$1,981	$2,009
ARMY	Network C3I Technology	Energy Efficient Devices Technology 84	Develop supply and demand electronics for energy-constrained platforms that will extend mission duration (dismounted Soldier), reduce frequency of battery replacement (unattended Sensors) and increase endurance (Unmanned Aerial Vehicles (UAVs)).	Supply Chain Resilience	Individual/Warfighter Power	2040	02	0602146A	$5,480	$28,101
ARMY	Future Vertical Lift Technology	Future UAS Engine Technology 01	Research to enable intelligent and robust propulsion performance and noise signature reduction via multi-fuel and optimized hybrid electric capability for small	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	02	0602148A	$0	$0

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
			engines (20-150kW) powering future aerial and ground systems.							
ARMY	Future Vertical Lift Technology	High Reduction-Ratio Transmission (HRT) 63	Project develops FVL advanced drive train technologies that increase performance and double current drivetrain life cycles while improving their reliability and maintainability	Supply Chain Resilience	Propulsion Upgrades_Air	2040	02	0602148A	$0	$0
ARMY	Future Vertical Lift Technology	Adaptive Power Component Technologies	Project develops power and thermal management technologies to provide significantly higher electrical power capability to FVL aircraft while addressing consequential size, weight and thermal issues. Provides power capability for advanced electric aeromechanical effectors, advanced mission systems algorithms for route planning and teaming, and advanced electronic warfare devices	Energy Substitution and Diversification	Platform Upgrades_Air	2040	02	0602148A	$0	$12,434
ARMY	Future Vertical Lift Technology	Optimized Energy for C4ISR Platforms	Researches power components and control for high power aviation mission systems.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	02	0602148A	$5,185	$26,460
ARMY	Future Vertical Lift Technology	Power and Thermal Management Components	Project develops power and thermal management technologies to provide significantly higher electrical power capability to FVL aircraft while addressing consequential size, weight and thermal issues. Provides power capability for advanced electric aeromechanical effectors, advanced mission systems algorithms for route planning and teaming, and advanced electronic warfare devices	Energy Substitution and Diversification	Platform Upgrades_Air	2040	02	0602148A	$2,428	$1,200
ARMY	Air Platform Applied Research	Future UAS Engine Technology 01	Research to enable intelligent and robust propulsion performance and noise signature reduction via multi-fuel and optimized hybrid electric capability for small engines (20-150kW) powering future aerial and ground systems.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	02	0602183A	$3,414	$17,894
ARMY	Air Platform Applied Research	High Reduction-Ratio Transmission (HRT) 63	Project develops FVL advanced drive train technologies that increase performance and double current drivetrain life cycles while improving their reliability and maintainability	Supply Chain Resilience	Propulsion Upgrades_Air	2040	02	0602183A	$1,482	$1,511
ARMY	Soldier Applied Research	Technologies for Alternative Energy	Conducts applied research and development on materials and component level power and energy technologies in the areas of energy storage, power generation, alternative energy, and intelligent power distribution and thermal management designs that support Command, Control, Communications, Computers, Intelligence, Surveillance and Reconnaissance (C4ISR) and Soldier power needs to include next generation squad weapons and advanced optical devices and sensors. ?	Energy Substitution and Diversification	Individual/Warfighter Power	2040	02	0602184A	$2,387	$13,048

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Soldier Lethality Advanced Technology	Soldier Power and Energy Technology Demonstration - 936	Matures, integrates, and demonstrates advanced Soldier Power and Energy (P&E) technologies to power the dismounted Soldier and small unit's command and control, communications, computers, and sensor devices during tactical operations.	Energy Substitution and Diversification	Individual/Warfighter Power	2040	03	0603118A	$4,189	$22,405
ARMY	Ground Advanced Technology	Advanced Tribology Research 58	Mature tribology techniques to evaluate fluids and fuels.	Energy Substitution and Diversification	Conventional Fuels Testing	2040	03	0603119A	$480	$0
ARMY	Ground Advanced Technology	Enhanced Jet Fuel for Ground System Durability 56	Evaluate Enhanced Jet Fuel to determine its impact to Ground System Durability.	Energy Substitution and Diversification	Alternative Fuels Certification and Testing	2040	03	0603119A	$539	$3,187
ARMY	Ground Advanced Technology	Fuel Contamination Limits for Ground Systems 54	Evaluate Fuel Contamination Limits to establish standards for military Ground Systems based on emerging commercial fluids and standards.	Enterprise-wide Energy Visibility	Conventional Fuels Testing	2040	03	0603119A	$0	$0
ARMY	Next Generation Combat Vehicle Advanced Technology	Advanced Mobility Experimental Prototype - 728	Develops advanced powertrain, track and running gear, and unmanned robotic technologies for integration into a ground combat vehicle.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	03	0603462A	$0	$0
ARMY	Next Generation Combat Vehicle Advanced Technology	Electrification Materials Technology 11	Develop Electrification Technology in collaboration with the Department of Energy through the Advanced Vehicle Power Technology Alliance.	Energy Substitution and Diversification	Platform Upgrades_Land	2040	03	0603462A	$2,207	$4,966
ARMY	Next Generation Combat Vehicle Advanced Technology	Diesel Electric Power Generator 09	Develop Diesel Electric Power Generator to enable hybrid powertrains for combat vehicles and silent mobility and watch.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	03	0603462A	$1,646	$2,005
ARMY	Next Generation Combat Vehicle Advanced Technology	Electric Drive Motors/Power Controllers/Conv 07	Develop Electric Drive Motors, Power Controllers and Converters to enable hybrid powertrains for combat vehicles and silent mobility and watch.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	03	0603462A	$1,210	$1,618
ARMY	Next Generation Combat Vehicle Advanced Technology	Extreme Energy Density Energy Storage	Mature extremely high energy density energy storage for all-electric combat vehicles with advanced suspension designs.	Energy Demand Reduction	Storage, Power Controls and Distribution	2040	03	0603462A	$0	$10,336
ARMY	Next Generation Combat Vehicle Advanced Technology	High Density In-Sprocket Drive Advanced Technology	Develops components for hybrid-electric track drive combat vehicles.	Supply Chain Resilience	Propulsion Upgrades_Land	2040	03	0603462A	$0	$6,528
ARMY	Next Generation Combat Vehicle Advanced Technology	High Voltage Modular Li-Ion Battery 10	Develop High Voltage Modular Li-Ion Battery technologies to increase energy storage density and enable electrification of manned and unmanned combat vehicles.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	03	0603462A	$3,031	$3,057
ARMY	Next Generation Combat Vehicle Advanced Technology	Highly Electrified and Autonomous Platforms - GVSC/AAL 15	Develop electrification components to enable high power electrification of manned and unmanned combat vehicles.	Supply Chain Resilience	Storage, Power Controls and Distribution	2040	03	0603462A	$0	$0

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Next Generation Combat Vehicle Advanced Technology	Highly Electrified and Autonomous Platforms - GVSC/AAL 16	Demonstrate Highly Electrified Autonomous Platform components and architectures.	Supply Chain Resilience	Storage, Power Controls and Distribution	2040	03	0603462A	$4,469	$4,914
ARMY	Next Generation Combat Vehicle Advanced Technology	Next Generation Tank Mobility System (NGTMS) - Power Dense Propulsion System - 1069	Develop power dense propulsion system for the Next Generation Tank	Energy Demand Reduction	Propulsion Upgrades_Land	2040	03	0603462A	$0	$34,446
ARMY	Next Generation Combat Vehicle Advanced Technology	OMT Traction Turret Drive Motors Electric Cooling Advanced Technology	Researches electric traction drive motors for turret operation.	Energy Demand Reduction	Platform Upgrades_Land	2040	03	0603462A	$1,515	$6,174
ARMY	Next Generation Combat Vehicle Advanced Technology	Tactical and Wheeled Vehicles Hybrid Electric System	Develop integration software for anti-idle, high voltage energy storage, and hybrid functions of regenerative braking, electric launch assist, multi-vehicle power networking node, and mobility optimization for hybrid-electric tactical vehicles.	Supply Chain Resilience	Propulsion Upgrades_Land	2040	03	0603462A	$0	$18,221
ARMY	Next Generation Combat Vehicle Advanced Technology	Power Dense Range Extender Adv Technology	Develops range extending fuel cell design that uses solid-state energy storage.	Energy Demand Reduction	Storage, Power Controls and Distribution	2040	03	0603462A	$0	$7,032
ARMY	Next Generation Combat Vehicle Advanced Technology	RCV Silent Watch and Mobility Range Extension Advanced Technology	Matures and integrates JP8 reformer components and sub-systems in order to demonstrate extended silent watch and mobility as part of a modular electrification architecture supporting robotic combat vehicles.	Energy Demand Reduction	Storage, Power Controls and Distribution	2040	03	0603462A	$2,021	$3,356
ARMY	Next Generation Combat Vehicle Advanced Technology	Scalable Electrification and Control Architecture - 856	Validates component-level performance and integrates the power distribution and control components to implement a common, scalable, electrified vehicle power architecture on combat platforms from 15 to 50 tons.	Energy Substitution and Diversification	Platform Upgrades_Land	2040	03	0603462A	$3,536	$4,075
ARMY	Next Generation Combat Vehicle Advanced Technology	SiC Gen 4 Electronics Improvement - 1071	Mature Silicon Carbide-based High Power Electronic Subsystems for Combat Vehicles.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	03	0603462A	$0	$24,227
ARMY	Next Generation Combat Vehicle Advanced Technology	Battery Technologies for Supply Chain Security	Evaluate technologies that mitigate battery supply chain security issues	Supply Chain Resilience	Storage, Power Controls and Distribution	2040	03	0603462A	$8,800	$33,744
ARMY	Next Generation Combat Vehicle Advanced Technology	System Vehicle Integration and Test - 857	Mature Silicon Carbide-based High Power Electronic Subsystems for Combat Vehicles.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	03	0603462A	$3,983	$27,805
ARMY	Next Generation Combat Vehicle Advanced Technology	Platform Electrification E-Vectronics	Demonstrate Tactical Vehicle Electronics Architecture	Energy Substitution and Diversification	Platform Upgrades_Land	2040	03	0603462A	$0	$0

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Future Vertical Lift Advanced Technology	Alternative Concept Engine Advanced Technology 07	Project develops FVL engine technologies that could significantly improve platform performance (range, speed, payload), durability, and operational capability	Supply Chain Resilience	Propulsion Upgrades_Air	2040	03	0603465A	$0	$0
ARMY	Future Vertical Lift Advanced Technology	Next Generation Aviation Transmission Adv Tech	Project develops FVL advanced drive train technologies that increase performance and double current drivetrain life cycles while improving their reliability and maintainability	Supply Chain Resilience	Propulsion Upgrades_Air	2040	03	0603465A	$0	$0
ARMY	Future Vertical Lift Advanced Technology	Optimized Energy for C4ISR Platforms Adv Tech	Develops power components and control for high power aviation mission systems.	Energy Demand Reduction	Propulsion Upgrades_Air	2040	03	0603465A	$2,043	$10,299
ARMY	Future Vertical Lift Advanced Technology	Power and Thermal Management Tech Demo	Project effort will develop and demonstrate integrated power and thermal management technologies to provide significantly higher electrical power capability for FVL aircraft	Energy Demand Reduction	Propulsion Upgrades_Air	2040	03	0603465A	$2,353	$14,456
ARMY	Next Generation Combat Vehicle Technology	Battlefield Electric Vehicle Recharge Technology	Research highly mobile electric vehicle rechargers to enable highly electrified tactical and combat platforms	Energy Substitution and Diversification	Platform Upgrades_Land	2040	02	0602145A	$0	$10,109
ARMY	Soldier Applied Research	Energy Storage for Sustained Operations (ESSO)	alternative energy storage to Lithium-ion batteries and alternatives to storing energy from harvestable sources	Energy Substitution and Diversification	Individual/Warfighter Power	2040	02	0602184A	$0	$11,300
ARMY	Soldier Applied Research	Reformed Ethanol Fuel Cell Catalysts for Small Unit Power	Provide the dismounted Soldier with a small, lightweight, compact fuel cell to charge batteries on the battlefield.	Energy Substitution and Diversification	Individual/Warfighter Power	2040	02	0602184A	$0	$12,009
ARMY	Ground Advanced Technology	Fluids for Electrified Tactical Vehicles	Conduct bench top testing of thermal management fluids for vehicle electrification to evaluate and down-select fluid candidates.	Energy Substitution and Diversification	Alternative Fuels Certification and Testing	2040	03	0603119A	$0	$8,647
ARMY	Ground Advanced Technology	Smart Fuel Metering	Mature smart meter design based on baseline evaluation, add tank level monitoring, and assess fuel dashboard and data transfer performance.	Enterprise-wide Energy Visibility	Conventional Fuels Testing	2040	03	0603119A	$0	$12,419
ARMY	Future Vertical Lift Advanced Technology	Hybrid Propulsion Technology Demonstrator	Demonstrate hybrid-electric aircraft propulsion technologies.	Energy Substitution and Diversification	Propulsion Upgrades_Air	2040	03	0603465A	$0	$4,956
ARMY	Next Generation Combat Vehicle Technology	Extreme Energy Density Energy Storage	Mature extremely high energy density energy storage for combat vehicles.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	02	0602145A	$1,185	$7,069
ARMY	Next Generation Combat Vehicle Technology	High Density In-Sprocket Drive Applied Technology	Researches components for hybrid-electric track drive combat vehicles.	Supply Chain Resilience	Propulsion Upgrades_Land	2040	02	0602145A	$1,534	$1,928
ARMY	UH-60 Black Hawk M Models	Black Hawk Aircrew Simulators	Black Hawk Aircrew Simulators	Enterprise-wide Energy Visibility	Simulators Air	2031	01	0210101A	$4,182	$38,869
ARMY	UH-60 Black Hawk A and L Models	Black Hawk Aircrew Simulators and Concurrency	Black Hawk Aircrew Simulators	Enterprise-wide Energy Visibility	Simulators Air	2031	01	0210101A	$28,848	$13,655

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Chinook Product Improvement Program	Chinook Transportable Flight Proficiency Simulator (TFPS)	Chinook Transportable Flight Proficiency Simulator (TFPS)	Enterprise-wide Energy Visibility	Training and Education	2040	07	0607137A	$0	$0
ARMY	Chinook Product Improvement Program	CH-47F Block II rotor, engine, fuel and electrical system upgrades, and lightweighting efforts	Updates the CH-47F with more powerful engines, more efficient rotors, simplified fuel system with greater capacity, more powerful electrical system, and light weight components for greater payload and range	Energy Demand Reduction	Platform Upgrades_Air	2040	07	0607137A	$52,510	$44,781
ARMY	AH-64 Apache Attack Helicopter D Model	AH-64 Apache Attack Helicopter Simulators and Concurrency	AH-64 Apache Attack Helicopter Simulators	Enterprise-wide Energy Visibility	Simulators Air	2031	01	0210100A	$0	$0
ARMY	AH-64 Apache Attack Helicopter E Model	AH-64 Apache Attack Helicopter Simulators	AH-64 Apache Attack Helicopter Simulators	Enterprise-wide Energy Visibility	Simulators Air	2031	01	0210100A	$0	$0
ARMY	Improved Turbine Engine	Aviation - Improved Turbine Engine Program (ITEP)	Aviation - Improved Turbine Engine Program (ITEP)	Energy Demand Reduction	Propulsion Upgrades_Air	2040	07	0607139A	$228,036	$558,419
ARMY	Improved Turbine Engine	Aviation - Improved Turbine Engine Program (ITEP)	Aviation - Improved Turbine Engine Program (ITEP)	Energy Demand Reduction	Propulsion Upgrades_Air	2031	01	0216300A		$1,823,094
ARMY	Armored Multi-Purpose Vehicle (AMPV)	Armored Multi-Purpose Vehicle (AMPV)	Powertrain and electrical systems for the M113-family replacement: Supports mobility, protection, and mission command, and positions the vehicle for future upgrades	Energy Demand Reduction	Propulsion Upgrades_Land	2033	01	0211702A	$297,624	$2,310,385
ARMY	Distribution Systems, Petroleum and Water	Early Entry Fluid Distribution System (E2FDS)	Early Entry Fluid Distribution System (E2FDS)	Supply Chain Resilience	Fuel Infrastructure	2035	03	0216300A	$236	$47,608
ARMY	Semitrailers and Tankers	Bulk Fuel Distribution System (BFDS)	BFDS - Bulk Fuel Distribution System	Supply Chain Resilience	Fuel Infrastructure	2035	03	0216300A	$14,816	$153,224
ARMY	Semitrailers and Tankers	Tactical Fuel Distribution System (TFDS)	Tactical Fuel Distribution System (TFDS)	Supply Chain Resilience	Fuel Infrastructure	2035	03	0216300A	$4,553	$275,411
ARMY	Water and Petroleum Distribution - ED	Tactical Fuel Distribution System (TFDS)	Tactical Fuel Distribution System (TFDS)	Supply Chain Resilience	Fuel Infrastructure	2040	05	0604804A	$370	$3,480
ARMY	Modification Of In Svc Equip	HTV Anti-Idle Retrofit Kit	Anti-idle retrofit kit, ECP for existing HTV platforms (HEMTT, PLS, HET, M915A5) that enables HVAC operation and onboard comms / accessories to run on battery power for 1-4 hours with the engine off.	Energy Demand Reduction	Propulsion Upgrades_Land	2035	01	0216300A		$85,190
ARMY	Modification Of In Svc Equip	MTV/HTV On Board Vehicle Power (OBVP)	A highly mobile, lightweight fast-forming vehicle hybrid power system with AC / DC microgrid capability – ECP for MTV/HTV platforms.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2035	01	0216300A		$100,000
ARMY	Modification Of In Svc Equip	Fuel Sense 2.0	Transmission software upgrade / ECP on existing platforms designed to improve electronic shift strategy for Allison 5th and 6th generation tactical vehicle transmission assemblies.	Energy Demand Reduction	Propulsion Upgrades_Land	2035	01	0216300A		$7,500

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
ARMY	Water and Petroleum Distribution - ED	Early Entry Fluid Distribution System (E2FDS)	Early Entry Fluid Distribution System (E2FDS)	Supply Chain Resilience	Fuel Infrastructure	2040	05	0604804A	$359	$0
ARMY	Water and Petroleum Distribution - ED	Bulk Fuel Distribution System (BFDS)	BFDS - Bulk Fuel Distribution System	Supply Chain Resilience	Fuel Infrastructure	2040	05	0604804A	$1,409	$150
ARMY	Medium Tactical Vehicles	FMTV Anti-Idle Retrofit Kit	Anti-idle retrofit kit, ECP for the FMTV that enables HVAC operation and onboard comms / accessories to run on battery power for 1-4 hours with the engine off.	Energy Demand Reduction	Propulsion Upgrades_Land	2040	05	0604604A	$1,500	$500
ARMY	Medium Tactical Vehicles	HTV Anti-Idle Retrofit Kit	Anti-idle retrofit kit, ECP for existing HTV platforms (HEMTT, PLS, HET, M915A5) that enables HVAC operation and onboard comms / accessories to run on battery power for 1-4 hours with the engine off.	Energy Demand Reduction	Propulsion Upgrades_Land	2040	05	0604604A	$3,750	$30,500
ARMY	Medium Tactical Vehicles	MTV/HTV On Board Vehicle Power (OBVP)	A highly mobile, lightweight fast-forming vehicle hybrid power system with AC / DC microgrid capability – ECP for MTV/HTV platforms.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	2040	05	0604604A	$5,750	$8,500
ARMY	Medium Tactical Vehicles	MTV/HTV Advanced / Hybrid Propulsion with OBVP	By adding a specialized clutch, 200kW Inverter, and Li-ion battery pack, can turn the Transmission Integral Generator into an electric motor.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	05	0604604A	$3,000	$3,000
ARMY	Medium Tactical Vehicles	Fuel Sense 2.0	Transmission software upgrade / ECP on existing platforms designed to improve electronic shift strategy for Allison 5th and 6th generation tactical vehicle transmission assemblies.	Energy Demand Reduction	Propulsion Upgrades_Land	2040	05	0604604A	$550	$0
ARMY	Medium Tactical Vehicles	Commercial Class 8 Electrification Study / Line Haul	Commercial Class 8 Electrification Study / Line Haul	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	05	0604604A	$450	$0
ARMY	Joint Light Tactical Vehicle	Electrical System Energy Related Improvements - Engine Upgrade	Electrical System Energy Related Improvements - Engine Upgrade	Energy Demand Reduction	Propulsion Upgrades_Land	2035	01	0216300A	$86,145	$527,089
ARMY	Joint Light Tactical Vehicle	Electrical System Energy Related Improvements - Alternator and Li Battery	Electrical System Energy Related Improvements - Alternator and Li Battery	Energy Demand Reduction	Platform Upgrades_Land	2035	01	0216300A	$8,462	$47,734
ARMY	Joint Light Tactical Vehicle	Anti-Idle Kits (TVEK) (Procurement Effort)	Anti-idle kit for legacy JLTVs to reduce overall fuel consumption up to 20 percent (duty cycle dependent) (Production Kits)	Energy Demand Reduction	Propulsion Upgrades_Land	2035	01	0216300A		$184,417
ARMY	Joint Light Tactical Vehicle	Hybrid and Electric System Modeling	Modeling efforts to optimize the hybrid and full electric propulsion systems	Energy Demand Reduction	Propulsion Upgrades_Land	2040	05	0605812A		$0
ARMY	Joint Light Tactical Vehicle	Anti-Idle Kits (TVEK) (RDT&E Development Engineering Effort)	Anti-idle kit for legacy JLTVs to reduce overall fuel consumption up to 20 percent (duty cycle dependent) (A1)	Energy Demand Reduction	Propulsion Upgrades_Land	2040	05	0605812A	$7,060	$7,500
ARMY	Generators and Associated Equipment	Advanced Medium Mobile Power Source	Advanced Medium Mobile Power Sources (AMMPS) units power or are incorporated into digital systems across the Army Portfolio to include Network/C3I, Air and Missile Defense, Long Range Precision Fires, Command Post and Combat Support/Combat	Energy Demand Reduction	Contingency Basing	2035	03	0216300A	$110,000	$377,373

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
			Service Support Systems. AMMPS is a modernization program to provide 5kW, 10kW, 15kW, 30kW, or 60kW generators in either a skid, trailer mounted, or microgrid configuration to replace legacy generators and provide an ability to interface with newer technologies. AMMPS enhances operational effectiveness by increasing availability of power, while reducing noise, weight, and sustainment burdens for units. AMMPS Microgrid Configurations meet power needs while reducing the fuel consumption on the battlefield. The reduced logistics footprint significantly reduces the cost of operation.							
ARMY	Generators and Associated Equipment	Power Distribution Illumination Systems Electrical (PDISE)	Power Distribution Illumination Systems Electrical (PDISE) provides the linkage between the generators and the Network/C3I, Air and Missile Defense, Long Range Precision Fires, Command Post and Combat Support/Combat Service Support systems. Power distribution systems are the circuit breaker panel for communication and electronic systems. This program transmits electrical power efficiently from the point of generation to the point of need. PDISE components are man-portable and safe for all-weather operations to protect Soldiers from electric shock. PDISE is a reliable, quick to assemble, modular designed power distribution equipment that is critical to deploying power networks. PDISE is compatible with DoD generator sets from 5kW to 60kW. PDISE as a program is committed solely to the warfighter. Medium Power Distribution Systems (MPDS) provides the linkage between the 5-60kW generators and includes the current PDISE family, M200, M100, M40, M60, M46 and the multiple input microgrid feeder box.	Energy Demand Reduction	Contingency Basing	2035	03	0216300A	$1,076	$30,267
ARMY	LOGISTICS AND ENGINEER EQUIPMENT ENGINEERING DEV	Power Distribution Illumination Systems Electrical (PDISE) Expansion	Prepare PDISE- Prime effort by awarding the Prime Power Distribution System (PPDS) contract, developing Prime Power Connection Kit (PPCK) first article units and start developmental testing inclusive of the Improved Primary Switching Center (iPSC) and Improved Secondary Distribution Center (iSDC) The PPCK enables distribution of power from prime power sources which use medium voltages or higher. The system will transform medium or higher voltages down to standard 120/208 V, 3-phase power. Elements of the PPCK will enhance the existing Secondary Distribution Center (SDC) by incorporating advanced capabilities to accept either 4160 Volts Alternating Current (VAC) primary input power from a	Energy Demand Reduction	Contingency Basing	2040	05	0604804A	$2,500	$1,500

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
			USA Deployable Power Generation and Distribution System (DPGDS) or a USAF Basic Expeditionary Airfield Resources (BEAR) power source or 13,800 VAC from contracted and commercial power sources or host nation/existing distribution systems.							
ARMY	Generators and Associated Equipment	Small Tactical Electric Power (STEP)	STEP is a modernization program that will provide a family of systems of improved tactical electric mobile power sources for replacement of the currently fielded 2kW MTG and 3kW TQG.	Energy Demand Reduction	Contingency Basing	2035	03	0216300A	$1,613	$54,588
ARMY	LOGISTICS AND ENGINEER EQUIPMENT ENGINEERING DEV	STEP	RDT&E investment in STEP will further improve mobile Tactical Electric Power by providing lightweight, modular, reliable and more logistically supportable power sources.	Energy Demand Reduction	Contingency Basing	2040	05	0604804A	$11,487	$34,385
ARMY	LOGISTICS AND ENGINEER EQUIPMENT ENGINEERING DEV	Integrated Fire Control Network (IFCN) Relay	The effort will develop and integrate an advanced hybrid power solution for the AMMPS generators to initially support operation of the Integrated Fire Control Network (IFCN) Relay.	Energy Substitution and Diversification	Contingency Basing	2040	05	0604804A	$1,036	$1,079
ARMY	LOGISTICS AND ENGINEER EQUIPMENT ENGINEERING DEV	Field Hospital Microgrid Systems	RDT&E investment in Field Hospital Microgrid Systems will support medical operations in large scale combat operations.	Energy Substitution and Diversification	Contingency Basing	2040	05	0604804A		$1,500
ARMY	LOGISTICS AND ENGINEER EQUIPMENT ENGINEERING DEV	Lightweight Portable Power	RDT&E investment into Lightweight Portable Power supports the prototyping and test and evaluation of lightweight power systems in the 2-5kW range.	Energy Demand Reduction	Contingency Basing	2040	05	0604804A	$10,000	$0
ARMY	Mobile Soldier Power	Platoon Power Generator (PPG)	PPG provides small units with portable power. It reduces dependency on disposable batteries and the need for operational energy resupply for long duration missions.	Supply Chain Resilience	Contingency Basing	2035	03	0211700A		$26,369
ARMY	Soldier Systems - Warrior Dem/Val	Platoon Power Generator (PPG)	RDT&E investment in PPG will further reduce the logistical burden associated with fuel and primary (disposable) batteries.	Supply Chain Resilience	Individual/Warfighter Power	2040	05	0604827A	$4,145	$4,075
ARMY	Force Provider	Force Provider (FP)	The FPE 150-person module is 30-50 percent more energy efficient than the equivalent legacy FP (Rev B) module.	Energy Demand Reduction	Contingency Basing	2035	03	0216300A	$12,000	$34,200
ARMY	Heaters and ECU's	Improved Environmental Control Unit (IECU)	Family of IECUs (4 variants) are 20-30 percent more energy efficient than legacy ECUs. More energy efficient ECUs decrease electrical power needed to cool mission platforms, command posts and bases; and, thus, reduce overall operational energy usage.	Energy Demand Reduction	Contingency Basing	2035	03	0216300A	$7,672	$37,270
ARMY	LOGISTICS AND ENGINEER EQUIPMENT ENGINEERING DEV	Improved Environmental Control Unit (IECU)	RDT&E investment in family of IECU will further improve energy efficiency and enable compatibility with more environmentally friendly refrigerants.	Energy Demand Reduction	Contingency Basing	2040	05	0604804A	$1,529	$5,959
ARMY	LOGISTICS AND ENGINEER	Army Standard Family-Rigid Wall Shelters (ASF-RWS)	ASF-RWS shelters have 20-40 percent better system-level insulating capability (heat transfer coefficient) than applicable legacy shelters. This improvement in	Energy Demand Reduction	Contingency Basing	2040	05	0604804A	$3,376	$7,998

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
	EQUIPMENT ENGINEERING DEV		insulating capability results in a significant reduction (25-30 percent) in environmental control energy consumption and, thus, overall operational energy usage.							
ARMY	Logistics and Engineer Equipment - Eng Dev	194 / Engine Driven Gen Ed	Large Tactical Power System (LTPS) is a modernization program for existing legacy large power generation systems that will provide mobile, durable and reliable tactical electric power capabilities above 60kW, to support Multi Domain Operations in the austere environments of today's battlefield.	Energy Demand Reduction	Contingency Basing	2040	05	0604804A		$0
ARMY	LOGISTICS AND ENGINEER EQUIPMENT- ADV DEV	Adv Elec Energy Con Ad/ G11	RDT&E investment supports the modernization of the current Tactical Electric Power capability with technology insertions including, but not limited to hybrid capabilities, lightweight power solutions, vehicle/tactical microgrid interoperability and Tactical Microgrid Standards (TMS). Funding also supports developing initial prototypes to enable refinement of Operational Requirements and early user feedback to support future sustainment and operational energy concepts. Congressionally provided funds will support analysis and planning for potential transition to the Army of the mobile micro-reactor prototype and capability.	Energy Demand Reduction	Contingency Basing	2040	04	0603804A	$15,000	$0
ARMY	Modification Of In Svc Equip	FMTV Anti-Idle Retrofit Kit	Anti-idle retrofit kit, ECP for the FMTV that enables HVAC operation and onboard comms / accessories to run on battery power for 1-4 hours with the engine off.	Energy Demand Reduction	Propulsion Upgrades_Land	2035	01	0216300A		$85,190
ARMY	Inland Petroleum Distribution System	Fuel Infrastructure	IPDS Fuel Unit COSIS	Supply Chain Resilience	Fuel Infrastructure	2020	02	0208031A	$14,304	$65,707
ARMY	800k Fuel System Supply Point	Fuel Infrastructure	800k FSSP COSIS	Supply Chain Resilience	Fuel Infrastructure	2020	02	0208031A	$557	$2,174
ARMY	HMMWV Hybrid Electric Vehicle (HEV)	HMMWV Hybrid Electric Vehicle (HEV) Christina Morris	HMMWV Hybrid Electric Vehicle (HEV) prototypes aimed to improve and provide new capabilities such as fuel demand reductions, reduced greenhouse gas emissions, increased automotive performance.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	05	0604642A		$182,010
ARMY	Joint Light Tactical Vehicle (JLTV) Hybrid Electric Vehicle (HEV)	JLTV HEV Mark Hildebrand	JLTV HEV prototypes aimed to improve and provide new capabilities such as fuel demand reductions, reduced greenhouse gas emissions, increased automotive performance.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	05	0605812A		$163,800
ARMY	Electric Light Reconnaissance Vehicle	electric Light Reconnaissance Vehicle Christina Morris	The electric Light Reconnaissance Vehicle (eLRV) platform through electrification will provide commanders a substantial competitive advantage in the Multi-Domain Operational (MDO) Environment against threat capabilities through reduction in	Energy Substitution and Diversification	Propulsion Upgrades_Land	2040	05	0604642A		$27,616

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
			acoustic and thermal signature, silent mobility, increased dash speed, extended range, increased reliability and reduction in Class (CL) III requirements. These attributes will enhance lethality and survivability of the mounted reconnaissance squad, platoon and troop.							
ARMY	HMMWV Hybrid Electric Vehicle (HEV)	HMMWV Hybrid Electric Vehicle (HEV) Michelle Marku	HMMWV Hybrid Electric Vehicle (HEV) production aimed to improve and provide new capabilities such as fuel demand reductions, reduced greenhouse gas emissions, increased automotive performance.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2035	01	0216300A		$210,683
ARMY	JLTV Hybrid Electric Vehicle (HEV)	JLTV Hybrid Electric Vehicle (HEV) Lauren Mento	JLTV Hybrid Electric Vehicle (HEV) production aimed to improve and provide new capabilities such as fuel demand reductions, reduced greenhouse gas emissions, increased automotive performance.	Energy Substitution and Diversification	Propulsion Upgrades_Land	2035	01	0216300A		$10,500
ARMY	Combat Service Support Equipment	Tactical Electric Power Recapitalization	The Deployable Power Generation and Distribution System (DPGDS) Power Unit (PU) provides mission critical power to the Nation's strategic missile defense assets; specifically, the Army/Navy Transportable Radar Surveillance (AN/TPY-2) X-Band Radar in Forward-Based Mode (FBM). These radars are critical in detecting, classifying and tracking ballistic missiles. The power unit is also used by the Army's only Prime Power Battalion to support its National Response Framework (NRF) mission, providing DoD support to civil defense and climate related disasters. The recapitalization effort leverages significant technology advancements over past 15 years and extends the economic useful life of these PUs by bringing the set back to zero hours and reduces emissions with a newer EPA Tier engine. The recapitalized MEP-PU-810 will come in two variants, the MEP-PU-810C, a towed trailer configuration, which will replace the legacy MEP-PU-810A model and the MEP-PU-810D, a 5th wheel configuration DoT approved for over the road use at 55 mph, which will replace the legacy MEP-PU-810B model. Throughout the DoD there are 165 'A' Models and 66 'B' Models supporting numerous missions, such as supporting the nation's Ballistic Missile Defense System. Capabilities: Once fielded, the MEP-PU-810C/D Power Unit (PU) will be at the center of the Army's Deployable Power Generation and Distribution System (DPGDS). Portable, multi-fuel engine driven generator rated to provide 800kW of electrical power at 4160 Volts Alternating Current (VAC) and provides the transition from tactical electric power to Shore/Host Nation	Supply Chain Resilience	Contingency Basing	2035	03	0216300A	$8,293	$37,016

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[21] ($K)
			power. Completely self-contained, integrated trailer. Capable of operating standalone or in a multi-unit Microgrid configuration equipped with modern, built-in controls and its own electrical distribution bus, High Altitude Electromagnetic Pulse (HEMP) hardened.							
								Army Total	$1,137,023	$8,715,086

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
NAVY	(U)NAVY WARFIGHTING EXPERIMENTS AND DEMONSTRATIONS	Naval Power Systems	NextSTEP [previously Energy Systems Technology Evaluation Program (ESTEP)] is a prototype viability assessment program at naval facilities that promotes adoption, scaling and deployment to the warfighter of defense and dual-use technologies	Energy Substitution and Diversification	Storage, Power Controls and Distribution	1319	03	0603758N	$7,282	$4,426
NAVY	Advanced Surface Machinery Sys	Integrated Power Systems (IPS)	Developing land-based test site for the integrated power and energy system for DDG(X). [PB23 Program Decision Memorandum (PDM) II - DDGX IPS]	Energy Substitution and Diversification	Storage, Power Controls and Distribution	1319	04	0603573N	$94,800	$0
NAVY	Advanced Surface Machinery Sys	Shipboard Energy Storage	****Not Tagged in PBIS as of 24 Feb 2023**** Large format lithium ion batteries. [FY 2023 Cong. Program Increase, C873]	Energy Substitution and Diversification	Storage, Power Controls and Distribution	1319	04	0603573N	$9,000	$0
NAVY	Advanced Surface Machinery Sys	Integrated Power and Energy Systems	Development of Next Generation Integrated Power and Energy System (NGIPES) technology aboard Navy Ships to enable current and future weapons and sensor systems.	Energy Substitution and Diversification	Storage, Power Controls and Distribution	1319	04	0603573N	$2,517	$13,355
NAVY	Advanced Surface Machinery Sys	Integrated Power and Energy Systems	Development of Next Generation Integrated Power and Energy System (NGIPES) technology aboard Navy Ships to enable current and future weapons and sensor systems.	Energy Substitution and Diversification	Platform Upgrades_Sea	1319	04	0603573N	$0	$230,124
NAVY	Air and Missile Defense Radar (AMDR) System	Efficient SPY-6 Transmit/Receive Integrated Multichip Modules (TRIMM)s (PDM II)	Swapping the High-Power Amplifiers with higher efficiency power amplifiers, reducing the overall DC power consumption on the ship. [FY 2023 PDM II - Climate DDGX IPS]	Energy Demand Reduction	Platform Upgrades_Sea	1319	05	0604522N	$2,000	$0
NAVY	Aircraft Energy	P-8 Engine Compressor Coating	To determine the erosion coating treatment suitability for the P-8 fleet, by performing a field services engineering analysis of the compressor treatments on the CFM 56 engine.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$5,350	$6,700

[22] FY2024 – FY2028

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
NAVY	Aircraft Energy	Low Speed Precision Control	To develop a system that will allow the helicopter that to land efficiently in high pilot workload environments, for example, carrier ops, brown out and white out ops. Thus reducing the fuel needed for these flight operations.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$3,400	$11,400
NAVY	Aircraft Energy	Navy Training Next	To develop an augmented virtual reality primary flight training system with corresponding curriculum. Making training more efficient and reducing the flight time needed by student pilots on the basics of flying to gain proficiency. This reduces the needed fuel to train pilots.	Energy Demand Reduction	Training and Education	1319	04	0603724N	$2,500	$4,300
NAVY	Aircraft Energy	H-53 Particle Inlet Separator	Develop an improved inlet particle separation system for H-53. This will enable the engine to power and efficiency longer, by eliminating degradation of turbine engine components in high particulate environments.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$1,700	$4,477
NAVY	Aircraft Energy	E2-D ILM	To determine the capabilities and requirements for developing an improved landing system for the E-2D aircraft. This system will reduce the workload of the pilot, increasing the efficiency of landing the E-2D on the Carrier.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$1,500	$500
NAVY	Aircraft Energy	EM Pusher (Jet Car Test Electrification)	To retrofit Jet Car Track Site sleds from turbine engines to energy efficient electric propulsion system, thus saving fuel.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$1,100	$1,750
NAVY	Aircraft Energy	F-18 G5 High Efficiency Generator	Evaluate and demonstrate alternative aircraft power generation/conversion technologies to provide more efficient power generation to meet legacy platform power deficiencies	Energy Substitution and Diversification	Platform Upgrades_Air	1319	04	0603724N	$860	$600
NAVY	Aircraft Energy	Common Affordable Safe Energy Storage (CASES)	Optimize aircraft battery performance, safety and cost through development of a common, scalable Li ion battery	Energy Substitution and Diversification	Platform Upgrades_Air	1319	04	0603724N	$800	$3,433
NAVY	Aircraft Energy	Aerial Refueling Drogue Stabilization	Develop, design and demonstrate Aerial Refueling Drogue Stabilization technology to reduce refueling drogue captures. Benefit is reducing fuel used by fleet during refueling operations that can be used for mission prosecution.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$800	$12,200
NAVY	Aircraft Energy	Integrated Thermal and Power Management Modelling	Development and validation of Integrated Power and thermal management models to develop integrated solutions in legacy and emerging platforms	Energy Demand Reduction	M&S, Studies, and Wargames	1319	04	0603724N	$738	$7,691
NAVY	Aircraft Energy	Aircraft Engine Foam Wash	Develop and test nucleated foam wash for Navy turbine engines. This technology will significantly increase cleaning capability resulting in much lower engine temperatures, which results in lower fuel consumption.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$650	$4,050
NAVY	Aircraft Energy	P-8 Finlets	Optimize design, installation and demonstrate of drag reducing finlets on P-8 test aircraft	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$600	$7,400

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
NAVY	Aircraft Energy	Engine Blade Scanning and Coating development	Develop, and demonstrate blade scanning technology and turbine engine coatings for compressors blades for naval aircraft to increase in compressor efficiency, resulting in lower fuel consumption.	Energy Demand Reduction	Propulsion Upgrades_Air	1319	04	0603724N	$600	$6,200
NAVY	Aircraft Energy	F-18 Engine Efficiency	Evaluate and demonstrate alternative aircraft power generation/conversion technologies to provide more efficient power generation to meet legacy platform power deficiencies	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$350	$4,300
NAVY	Aircraft Energy	Advanced Fuel Cells for UAS Applications (H2 Stalker)	Design, build, and test a drop-in ready hydrogen fuel cell power and propulsion (P&P) system for the VTOL Stalker to demonstrate improved operational performance.	Energy Substitution and Diversification	Propulsion Upgrades_Air	1319	04	0603724N	$280	$620
NAVY	Aircraft Energy	Aircraft Turbine Engine Recuperator (UAS)	Demonstrate using M250 engine utilizing an advanced recuperator design enabling 25 percent reduced specific fuel consumption (SFC). Reduced SFC would provide extended time on station improvement of 25 – 35 percent, critical to ISR mission.	Energy Demand Reduction	Propulsion Upgrades_Air	1319	04	0603724N	$150	$150
NAVY	Aircraft Energy	Linear Spring	A birdcage-like spring that buckles at a set force, so that the pressure regulator can remain wide open until the cutoff pressure is reached. Will be designed to be a drop-in replacement for the existing springs	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$150	$150
NAVY	Aircraft Energy	Copper Bird Test Facility	Build a physical test rig that will simulate entire EFG Electrical System for dual GEN operation, Power Management, GCU diagnostics of system of system phenomenon (systems connected to electrical system), address/test new components for obsolescence for G4 reliability, provide test data for flight clearances	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$120	$0
NAVY	Aircraft Energy	Achieving Full Electric Capability	Identify, obtain and accumulate existing electrical data sets that exist and collect relevant aircraft data to inform aircraft electrical system capability	Energy Substitution and Diversification	Platform Upgrades_Air	1319	04	0603724N	$100	$3,300
NAVY	Aircraft Energy	F-18 Formation Flying for Long Fires	Develop and demonstrate an automated formation flight capability to increase cruise efficiency. Precise position station-keeping through the use of novel sensors, coupled with optimal, gain-limited flight control inputs, will provide precise “wake surfing” to minimize fuel burn by 10-20 percent and extend aircraft range.	Energy Demand Reduction	Platform Upgrades_Air	1319	04	0603724N	$0	$5,250
NAVY	Aircraft Energy	Opportunity Studies	Provide seed funding to investigate potential aircraft Operational Energy solutions (Power and Thermal) and identify potential candidates to select for detailed projects. *	Energy Substitution and Diversification	Platform Upgrades_Air	1319	04	0603724N	$0	$20,108
NAVY	Battery Development and Safety	Advanced Batteries Supply Chain	This project area will generate analytics that characterize the Department's current and projected energy/advanced battery needs	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$5,000	$10,400

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
NAVY	Battery Development and Safety	Battery Safety Certification	This project area will accomplish improvements in the battery safety certification process increasing the rapid safe deployment of advanced battery systems to the DoN. Allowing more battery-based systems to be deployed more quickly will provide gains in fielding greener energy sources and increasing combat capabilities.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$1,440	$5,696
NAVY	Battery Development and Safety	Battery Hazard Reduction	This project area will reduce the hazard of fielded batteries. Allowing more battery-based systems to be deployed more quickly will provide gains in fielding more greener energy sources and increasing combat capabilities.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$1,420	$7,591
NAVY	Battery Development and Safety	Battery Commonality	This project area will accomplish development of battery commonality efforts. Battery commonality will save DoN money and time when fielding advanced weapon systems	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$1,035	$2,911
NAVY	Battery Development and Safety	Battery Technology Development	This project area will accomplish development, laboratory and Fleet testing to determine overall mission and cost effectiveness of improved storage technologies allowing for the transition to greener energy and power sources.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$960	$4,616
NAVY	Battery Development and Safety	Battery Standards	This project area will develop clear battery, battery system, and containment-oriented standards and requirements. Allowing more advanced battery systems to be deployed more quickly will provide gains in fielding greener energy sources and increasing combat capabilities.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$783	$1,895
NAVY	Battery Development and Safety	Advance Batteries Jumpstart Program	This project area will accomplish laboratory testing of COTS EV batteries, supporting future acquisition and integration of COTS batteries to field advanced weapon systems.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$0	$10,000
NAVY	Battery Development and Safety	Advanced Batteries R&D	This project area will accomplish development, laboratory, and Fleet testing of renewable energy generation and storage allowing more advanced battery systems to be deployed more quickly will provide gains in fielding greener energy sources and increasing combat capabilities.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	04	0603724N	$0	$4,000
NAVY	DEFENSE RESEARCH SCIENCES	Naval Power Materials Research	Energy storage and power generation materials basic research	Energy Substitution and Diversification	Materials and Design	1319	01	0601153N	$13,024	$5,315
NAVY	DEFENSE RESEARCH SCIENCES	Naval Power Systems	Advancing power and energy science through fundamental research in the areas of conductor and permanent magnet materials, energy conversion, combustion, and cyber physical system modeling. Advancing thermal science and technology through fundamental studies of multi-phase heat transfer, fluid	Energy Demand Reduction	Platform Upgrades_Sea	1319	01	0601153N	$10,247	$8,916

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			dynamics, and nanostructured materials to efficiently acquire, transport, and reject heat and enable higher power density electronic systems. Fulfill the power and energy needs of the Navy's next-generation weapons and platforms by improving (1) Education, (2) Reliability of power electronic devices, (3) Power density of power systems, and (4) Power Electronics Manufacturing costs.							
NAVY	DEFENSE RESEARCH SCIENCES	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Substitution and Diversification	Alternative Fuels Production	1319	01	0601153N	$5,782	$4,266
NAVY	DEFENSE RESEARCH SCIENCES	Sea Based Aviation Propulsion Basic Research	This Program provides long-term basic research that discovers new phenomena related power propulsion and thermal management, with the intent that they mature to provide transition opportunities for the associated applied research program. This Program also supports university research in these areas and the associated graduate student support to help build the number and quality of Scientists and Engineers with relevant skills to help further develop power and propulsion systems for future Sea Based Aviation platforms and weapon systems.	Energy Demand Reduction	Propulsion Upgrades_Air	1319	01	0601153N	$4,041	$2,855
NAVY	DEFENSE RESEARCH SCIENCES	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$3,574	$2,637
NAVY	DEFENSE RESEARCH SCIENCES	Naval Platform Operational Endurance and Climate Resiliency Science	Advancing research in novel materials and processes to reduce fossil fuel use and decrease emissions. Developing design tools to: explicitly target climate resilience to increase awareness of carbon footprint, or GHG emissions from platforms, fleets, etc.; and provide the capacity for predicting emissions from platforms, including unmanned platforms for power architecture design, performance, and performance prediction (to include emissions).	Energy Demand Reduction	Platform Upgrades_Sea	1319	01	0601153N	$1,455	$900
NAVY	DEFENSE RESEARCH SCIENCES	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination;	Energy Substitution	Alternative Power Sources	1319	01	0601153N	$1,270	$938

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	and Diversification						
NAVY	DEFENSE RESEARCH SCIENCES	Naval Biosciences - Microbial Fuel Cells	Microbial fuel cells (MFC) provide electricity harvested from specialized natural bacteria that use non-hazardous organic compounds as fuel, and then provide electrical current to an electrode. Can be used to sustainably power seafloor sensors/systems in place of batteries. Program focuses on study of fundamental mechanisms used for extracellular electron transport.	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$654	$482
NAVY	DEFENSE RESEARCH SCIENCES	ONRG International Research	Basic research with international principal investigators doing collaborative and cooperative research with the Naval research enterprise.	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$288	$180
NAVY	DEFENSE RESEARCH SCIENCES	Bioengineering and Life Sciences (Energy)	Basic research exploring biofabrication for generation of inorganic energy harvesting/conversion materials; bacterial-inorganic hybrid materials for fuel cells; nano-biomaterials for generating high intensity light sources; silk composites for energy harvesting and energy sources; and novel humidity responsive materials for harnessing energy for natural evaporation.	Energy Substitution and Diversification	Materials and Design	1319	01	0601153N	$160	$0
NAVY	DEFENSE RESEARCH SCIENCES	Bioengineering and Life Sciences (Energy)	Basic research exploring the fundamental mechanism of bacterial spores' water-responsive behaviors in order to develop practical applications using the evaporation energy harvesting technique, and next generation actuators.	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$160	$0
NAVY	DEFENSE RESEARCH SCIENCES	Naval Power Materials Research	Energy storage and power generation materials basic research	Energy Substitution and Diversification	Materials and Design	1319	01	0601153N	$0	$19,405
NAVY	DEFENSE RESEARCH SCIENCES	Naval Power Systems	Advancing power and energy science through fundamental research in the areas of conductor and permanent magnet materials, energy conversion, combustion, and cyber physical system modeling. Advancing thermal science and technology through fundamental studies of multi-phase heat transfer, fluid dynamics, and nanostructured materials to efficiently acquire, transport, and reject heat and enable higher power density electronic systems. Fulfill the power and energy needs of the Navy's next-generation weapons and platforms by improving (1) Education, (2) Reliability of power electronic devices, (3) Power	Energy Demand Reduction	Platform Upgrades_Sea	1319	01	0601153N	$0	$31,705

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			density of power systems, and (4) Power Electronics Manufacturing costs.							
NAVY	DEFENSE RESEARCH SCIENCES	Naval Platform Operational Endurance and Climate Resiliency Science	Advancing research in novel materials and processes to reduce fossil fuel use and decrease emissions. Developing design tools to: explicitly target climate resilience to increase awareness of carbon footprint, or GHG emissions from platforms, fleets, etc.; and provide the capacity for predicting emissions from platforms, including unmanned platforms for power architecture design, performance, and performance prediction (to include emissions).	Energy Demand Reduction	Platform Upgrades_Sea	1319	01	0601153N	$0	$2,393
NAVY	DEFENSE RESEARCH SCIENCES	Naval Biosciences - Microbial Fuel Cells	Microbial fuel cells (MFC) provide electricity harvested from specialized natural bacteria that use non-hazardous organic compounds as fuel, and then provide electrical current to an electrode. Can be used to sustainably power seafloor sensors/systems in place of batteries. Program focuses on study of fundamental mechanisms used for extracellular electron transport.	Energy Demand Reduction	Alternative Power Sources	1319	01	0601153N	$0	$1,735
NAVY	DEFENSE RESEARCH SCIENCES	Sea Based Aviation Propulsion Basic Research	This Program provides long-term basic research that discovers new phenomena related power propulsion and thermal management, with the intent that they mature to provide transition opportunities for the associated applied research program. This Program also supports university research in these areas and the associated graduate student support to help build the number and quality of Scientists and Engineers with relevant skills to help further develop power and propulsion systems for future Sea Based Aviation platforms and weapon systems.	Energy Demand Reduction	Propulsion Upgrades_Air	1319	01	0601153N	$0	$11,694
NAVY	DEFENSE RESEARCH SCIENCES	ONRG International Research	Basic research with international principal investigators doing collaborative and cooperative research with the Naval research enterprise.	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$0	$629
NAVY	DEFENSE RESEARCH SCIENCES	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$0	$10,207
NAVY	DEFENSE RESEARCH SCIENCES	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of	Energy Substitution and Diversification	Alternative Power Sources	1319	01	0601153N	$0	$3,647

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.							
NAVY	DEFENSE RESEARCH SCIENCES	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Substitution and Diversification	Alternative Fuels Production	1319	01	0601153N	$0	$16,739
NAVY	ENERGY CONSERVATION	Propulsion Systems - LM2500 Variable Stator Vane Optimization	Funds added by PDM II - Climate: This specific project aims to test and analyze the fuel consumption impacts of various efficiency upgrades to the LM2500 Gas Turbine Generators.	Supply Chain Resilience	Propulsion Upgrades_Sea	1319	04	0603724N	$3,571	$2,575
NAVY	ENERGY CONSERVATION	Power Generation and Storage - PA6B Electronic Fuel Injection	Funds added by PDM II - Climate: This specific project is to complete an Electronic Fuel Injection Upgrade for the Fairbanks Morse (FM) PA6B Diesel Engine. Upgrading the fuel delivery system to electronic fuel injection will improve fuel efficiency of the engine, reduces emissions and visible smoke, and reduce maintenance man hours and costs.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$1,894	$0
NAVY	ENERGY CONSERVATION	Energy Monitoring, Planning and Assessment - Global Energy Information System Development	Continue development of GENISYS, a suite of software systems providing critical near real time energy related information to Operational Forces and Navy engineers, creating opportunity for greater operational efficiencies, increased time on station, extended warfighter reach, improved fuel forecasting and logistics, and optimization of future platform designs and upgrades.	Enterprise-wide Energy Visibility	Metering and Monitoring	1319	04	0603724N	$1,343	$6,752
NAVY	ENERGY CONSERVATION	HVAC	This project will be utilized to accomplish prototype development, land and shipboard testing to determine cost effectiveness of improvements aimed at more efficient climate control of shipboard spaces.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$904	$2,422
NAVY	ENERGY CONSERVATION	Power Generation and Storage	This project area accomplishes development, laboratory and Fleet testing to determine overall mission and cost effectiveness of improved power generation and storage technologies.	Energy Substitution and Diversification	Platform Upgrades_Sea	1319	04	0603724N	$800	$5,761
NAVY	ENERGY CONSERVATION	Hull Hydrodynamics	This project area accomplishes development, modeling, laboratory and Fleet testing of ship modifications to propellers such as fouling release coatings and/or hull appendages to determine overall mission and cost effectiveness of these improvements.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$552	$4,937
NAVY	ENERGY CONSERVATION	Propulsion Systems	This project is utilized to identify requirements and perform land based and shipboard testing of ship propulsion system improvements on Gas Turbine,	Energy Demand Reduction	Propulsion Upgrades_Sea	1319	04	0603724N	$389	$6,979

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			Steam, and Diesel Engine systems to reduce overall fuel consumption and lower maintenance costs.							
NAVY	ENERGY CONSERVATION	Hull Husbandry	This project is utilized to identify and evaluate new underwater hull coating systems and underwater hull cleaning and maintenance techniques to reduce hydrodynamic drag on the hull and thereby increase fuel efficiency.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$365	$5,189
NAVY	ENERGY CONSERVATION	Electrical Systems	This project area is utilized to identify and perform land based and shipboard testing of ship electrical system improvements to reduce energy consumption.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$289	$2,874
NAVY	ENERGY CONSERVATION	Energy Monitoring, Planning and Assessment	This project area focuses on methods of capturing and displaying energy related data to shipboard personnel and navy engineers and planners as actionable information to employ energy conservation measures underway and in port as mission requirements permit.	Enterprise-wide Energy Visibility	Metering and Monitoring	1319	04	0603724N	$120	$1,913
NAVY	ENERGY CONSERVATION	Thermal Management	This project will be utilized to identify and evaluate potential uses for Thermal Management techniques designed to reduce or dissipate overall shipboard heat generation and reduce the shipboard electrical demand on HVAC systems.	Energy Demand Reduction	Platform Thermal Management	1319	04	0603724N	$120	$3,003
NAVY	ENERGY CONSERVATION	Auxiliary Systems	This project area is utilized to identify, test and evaluate new technologies for shipboard auxiliary systems aimed at reducing fuel consumption.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$120	$1,529
NAVY	ENERGY CONSERVATION	Hull Hydrodynamics - Sonar Dome Redesign Study	Study to explore a sonar dome redesign and its impact on reducing hull resistance for future surface combatants.	Energy Demand Reduction	Platform Upgrades_Sea	1319	04	0603724N	$0	$1,021
NAVY	ENERGY CONSERVATION	Thermal Management - Navy-MARAD Tech Demo - Advanced Absorption Chiller	Absorption chiller technology will be demonstrated on a MARAD vessel to validate design and expected improvements over traditional vapor compressor chillers. This effort will benefit the T-AKE class and could provide additional utility to any other large diesel-powered vessels as future weapon systems require more power and cooling capacity.	Energy Demand Reduction	Platform Thermal Management	1319	04	0603724N	$0	$2,500
NAVY	ENERGY CONSERVATION	Electrical Systems - Non-Intrusive Load Monitoring Network Integration and Technology Advancement	This project will further develop and mature non-intrusive sensor to monitor power usage, machinery health and operating conditions to enhance CBM and potential inclusion of cyber threat detection.	Energy Demand Reduction	Metering and Monitoring	1319	04	0603724N	$0	$1,830
NAVY	ENERGY CONSERVATION	Energy Monitoring, Planning and Assessment - Navy-MARAD Energy Technology Testing Coordination	Navy PM to coordinate Navy and MARAD project stakeholders and resources for Testing Energy Technologies at Sea	Enterprise-wide Energy Visibility	Platform Upgrades_Sea	1319	04	0603724N	$0	$1,250
NAVY	ENERGY CONSERVATION	Energy Monitoring, Planning and	Funding to support policy, training, education changes to improve energy culture. Includes a wargame in 24, and climate literacy.	Enterprise-wide Energy Visibility	Training and Education	1319	04	0603724N	$0	$10,000

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
		Assessment - Training Education and Support								
NAVY	ENERGY CONSERVATION	Energy Monitoring, Planning and Assessment - Enterprise-Wide Energy Visibility Development	Energy supportability assessment studies to document common and authoritative sources of Feet OE data for use in relevant OE analyses. The data identified in this study will be used to develop the Theatre Energy Model (TEM) which will be applied to current Energy Supportability Analysis (ESA) activities and will make considerations for the impacts of specific missions on energy usage, including the creation of mission-based burn rate (MBBR) fuel curves.	Enterprise-wide Energy Visibility	M&S, Studies, and Wargames	1319	04	0603724N	$0	$15,000
NAVY	ENERGY CONSERVATION	Energy Monitoring, Planning and Assessment - Emissions Monitoring on Self Defense Test Ship	Pilot project to conduct real-time monitoring of GHG emissions from the Self Defense Test Ship. Could be useful for optimizing engine performance and efficiency. Ship has the same engine type as the DDG-51 class.	Enterprise-wide Energy Visibility	Metering and Monitoring	1319	04	0603724N	$0	$1,300
NAVY	FORCE PROTECTION ADVANCED TECHNOLOGY	Naval Power Systems	Energy Magazine integrates with the ship's existing power system and eliminates the need for mission systems to each develop, build, and test their own unique energy storage systems, and supports zonal or whole ship UPS functionality. Modular building blocks to enable the energy infrastructure necessary to support DEW/EW system integration onto ships.	Supply Chain Resilience	Platform Upgrades_Sea	1319	03	0603123N	$0	$42,571
NAVY	FORCE PROTECTION ADVANCED TECHNOLOGY	Naval Power Systems	NextSTEP [previously Energy Systems Technology Evaluation Program (ESTEP)] is a prototype viability assessment program at naval facilities that promotes adoption, scaling and deployment to the warfighter of defense and dual-use technologies	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	03	0603123N	$0	$16,527
NAVY	FORCE PROTECTION APPLIED RESEARCH	Naval Platform Operational Endurance and Climate Resiliency Science	Advancing design tools focused on climate resilience and predicting emissions from platforms. Pursue technology development efforts addressed to impact climate remediation, including evaluation of Low Global Warming Potential (GWP) refrigerants, Subsea and Seabed Warfare (SSW) Energy Harvesting and Direct Air Capture and Blue Carbon fuel synthesis. This effort received funding via PBIS Issue 50402: "PDM I Climate - Green Tech carbon sequestration and capture FY24," $7M/year 23-27.	Energy Substitution and Diversification	Platform Upgrades_Sea	1319	02	0602123N	$15,506	$38,277
NAVY	FORCE PROTECTION APPLIED RESEARCH	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Substitution and Diversification	Alternative Power Sources	1319	02	0602123N	$3,054	$58

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
NAVY	FORCE PROTECTION APPLIED RESEARCH	Naval Power Systems	Pursue DDGX advanced propulsion and expand existing fuel efficiencies in the DDG-51 fleet.	Supply Chain Resilience	Platform Upgrades_Sea	1319	02	0602123N	$0	$389
NAVY	FORCE PROTECTION APPLIED RESEARCH	Naval Power Systems	SIOP- TR electrification studies, marine and green concrete studies, and worker QOL studies at each of the four (4) public shipyards	Supply Chain Resilience	Alternative Power Sources	1319	02	0602123N	$0	$8,511
NAVY	FORCE PROTECTION APPLIED RESEARCH	Naval Power Systems	Refueling and Support Package to Enable Communications and Situational Awareness (RASP-CASA): Hydrogen production development system to support unmanned systems fueled on and off ships (support ISR and small USV capabilities from a ship).	Supply Chain Resilience	Contingency Basing	1319	02	0602123N	$0	$996
NAVY	FORCE PROTECTION APPLIED RESEARCH	Sea Based Aviation Propulsion Applied Research	This Program provides medium-term, applied research to demonstrate advanced engine technologies applicable to engine components for naval aviation platforms in propulsion-related technology areas. The specific areas addressed in this program are: (1) Propulsion Cycles, Subsystems, and Engine-Airframe Integration (2) High Stage-Loading, Variable-Geometry, and Enhanced Durability Turbomachinery (3) Jet Noise Reduction for tactical aircraft (4) Hot Section Materials and Coatings, (5) Higher Power Density and Stability Combustion Systems, and (6) Small Propulsion Engine Technology for Autonomous Air Vehicles.	Energy Demand Reduction	Propulsion Upgrades_Air	1319	02	0602123N	$0	$14,380
NAVY	FORCE PROTECTION APPLIED RESEARCH	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Substitution and Diversification	Alternative Power Sources	1319	02	0602123N	$0	$9,665
NAVY	FORCE PROTECTION APPLIED RESEARCH	Propulsion Task Force Energy (TFE)	This Program, in partnership with the Variable Cycle Advance Technology (VCAT) program, has the objective to develop variable geometry and adaptive cycle gas turbine engine technology for next generation air dominance aircraft. The benefits of these technologies are anticipated to be reduced fuel consumption and hence greater operational range and reduced logistics tail, mostly by reducing the demand for deployed fuel and tanker aircraft support.	Energy Demand Reduction	Propulsion Upgrades_Air	1319	02	0602123N	$0	$34,551
NAVY	FUTURE NAVAL CAPABILITIES APPLIED RESEARCH	SW-FY21-02 Robust Combat Power Control (RCPC)	Develop Combat Power and Energy Control System to anticipate, align and configure shipboard resources based on system state and mission context.	Supply Chain Resilience	Platform Upgrades_Sea	1319	03	0603673N	$9,244	$0
NAVY	INFORMATION AUTOMATION	Global Energy Information System (GENISYS)	Fields and sustains GENISYS on the DDG-51 class and LPD-17 class with fleet wide implementation to follow. The GENISYS software suite increases operational	Enterprise-wide Energy Visibility	Metering and Monitoring	1804	01	0708020N	$2,263	$10,117

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			reach and endurance through the capture and utilization of shipboard power and energy data. The Electronic Logbook (eLogBook) application replaces engineering, deck and other paper logs with electronic versions. The Shipboard Energy Assessment System (SEAS) application digitizes and aggregates energy and logbook data to inform crew decision-making. The ashore application, Fleet Energy Conservation Dashboard (FECD), supports fleet fuel planning by collecting surface fleet power and energy data from a variety of sources to analyze and display energy production, consumption and efficiency of the fleet, ship class, ship, or ship system. [FY 2023 PDM II - Climate DDGX IPS PDM Load Sheet]							
NAVY	LITTORAL COMBAT SHIP (LCS)	Littoral Combat Ship (LCS) Stern Flaps	Complete engineering for stern flap installations on LCS. [FY 2022 PDM III-Operational Energy LOE 3 Add - Ship Mods]	Energy Demand Reduction	Platform Upgrades_Sea	1810	01	0204230N	$1,270	$0
NAVY	MARINE CORPS ADVANCED TECHNOLOGY DEMO (ATD)	Marine Corps Operational Energy: Energy Optimization and Logistic Burden Reduction	Advanced Technology Demonstration research to optimize energy usage and/or meet operational energy demand with renewable energy sources and reduce excess capacity or reduce logistic footprint/burden energy sources.	Supply Chain Resilience	Individual/Warfighter Power	1319	03	0603640M	$0	$25,393
NAVY	MARINE CORPS ADVANCED TECHNOLOGY DEMO (ATD)	Marine Air-Ground Task Force Sustainment (Hybrid Expeditionary Power (HEP))	Systems will be MV-22 transportable and capable of drawing upon multiple power sources (self-generated and scavenged) in order to provide electrical power to an onboard battery bank from which loads will be powered in real-time, or on which power will be stored for future use.	Supply Chain Resilience	Storage, Power Controls and Distribution	1319	03	0603640M	$0	$3,328
NAVY	MARINE CORPS LANDING FORCE TECHNOLOGY	Naval Power Systems (M)	Applied Research to increase energy efficiency in weapons systems, platforms, vehicles and equipment and extend tactical range/operational reach.	Supply Chain Resilience	Platform Upgrades_Sea	1319	02	0602131M	$0	$7,984
NAVY	Mobility Fuels	Rapid fuel analysis and impact assessment	Develop test methods, fuel-hardware interaction correlations and analytic tools to reduce operational impacts from field identified deficiencies	Supply Chain Resilience	Conventional Fuels Testing	1319	04	0603724N	$2,497	$12,454
NAVY	Mobility Fuels	Fuel Hardware Impact analysis	Conduct RDT&E necessary to allow operational and/or technical decision makers the ability to assess risk of fuel properties/chemistry on current and emerging operational or platform requirements	Supply Chain Resilience	Conventional Fuels Testing	1319	04	0603724N	$2,400	$15,113
NAVY	Mobility Fuels	Interoperability with Commercial and Allied forces (Worldwide)	Conduct RDT&E necessary to assure that Naval tactical forces (air, sea and ground) can operate seamlessly using allied and/or commercially procured fuels	Supply Chain Resilience	Alternative Fuels Certification and Testing	1319	04	0603724N	$1,400	$5,656
NAVY	Mobility Fuels	Deployed Sensor Development and Validation	Develop and validate technology to reduce time and resources necessary to provide fuel quality surveillance in forward deployed environments	Supply Chain Resilience	Metering and Monitoring	1319	04	0603724N	$1,042	$6,343
NAVY	Ocean Warfighting Environment Applied Research	Integrated Climate Weather and Ocean Decision Support	Long term Applied Research on improved integration of weather and ocean forecasts into ship routing, ship response and propulsion efficiency planning, and	Supply Chain Resilience	Alternative Power Sources	1319	02	0602435N	$0	$12,501

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			Refueling at Sea logistics planning, as well as prediction of hazardous and extreme weather events and trends for climate adaptation, resiliency, and mitigation.							
NAVY	Ocean Warfighting Environment Applied Research	Naval Power Systems	iPOWER power and energy simulation application and make it a CAC-protected cloud service accessible to users across the Marine Corps and Navy.	Supply Chain Resilience	Simulators Air	1319	02	0602435N	$0	$340
NAVY	Ocean Warfighting Environment Applied Research	Naval Power Systems	RASP-CASA: Hydrogen production development system to support unmanned systems fueled on and off ships (support ISR and small USV capabilities from a ship).	Supply Chain Resilience	Alternative Fuels Production	1319	02	0602435N	$0	$1,556
NAVY	OPLOG IPT Development	Seabased Petroleum Distribution System (SPDS)	Development of an offshore bulk fuel cache storage and over the shore transfer system. Replaces and improves on legacy OPDS systems.	Supply Chain Resilience	Mobile Fuel Distribution Assets	1319	04	0603564N	$15,066	$5,800
NAVY	SHIP PRELIMINARY DESIGN AND FEASIBILITY STUDIES	T-AKE Propeller Duct	Hydrodynamic analysis to include CFD and tow tank testing for the validation of the propeller duct for the T-AKE class.	Supply Chain Resilience	Propulsion Upgrades_Sea	1319	04	0603564N	$0	$250
NAVY	SHIP PRELIMINARY DESIGN AND FEASIBILITY STUDIES	Seabased Petroleum Distribution System (SPDS)	Development of an offshore bulk fuel cache storage and over the shore transfer system. Replaces and improves on legacy OPDS systems.	Supply Chain Resilience	Mobile Fuel Distribution Assets	1319	04	0603564N	$0	$0
NAVY	Shore Energy Technology	Shore Energy Technology	Energy RDT&E Project that will test, evaluate, and validate components as well as demonstrate cost-effective and technical viability of energy security, efficiency, resilience, reliability, and technologies. Specifically, this project aims to pursue three areas of development, testing and evaluation: (A) Modeling and possible prototype testing of new energy sources for use at Naval installations with potential for widespread applicability to energy security; (B) It will support demonstration and validation of advanced electric grid management systems, known as "Smart Grid" and "Micro Grid" technology, for use at Naval installations to enable improved energy security; (C) Demonstration and Validation of Alternative Energy, Energy Efficiency, and Resiliency and Smart Energy Management Technology.	Energy Substitution and Diversification	Alternative Power Sources	1319	04	0603724N	$1,946	$10,335
NAVY	SURFACE SUPPORT	Condition Assessment System	Supports Installation and Procurement of ICAS (Integrated Condition Assessment System) and eRM (Enterprise Remote Monitoring). These systems enable remote monitoring and real time health assessments of shipboard equipment.	Supply Chain Resilience	Platform Upgrades_Sea	1810	01	0204228N	$827	$4,368
NAVY	SURFACE SUPPORT	Light-Emitting Diode (LED) Lighting on DDG-51 Class Ships	Replace fluorescent lighting fixtures with energy efficient LED	Energy Demand Reduction	Materials and Design	1810	01	0204228N	$0	$3,500

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
NAVY	SURFACE SUPPORT	LED Lighting on DDG-51 Class Ships	Replace current fluorescent hangar bay lights with new LEDs	Energy Demand Reduction	Platform Upgrades_Sea	1810	01	0204228N	$0	$4,000
NAVY	UNDERSEA WARFARE APPLIED RESEARCH	Naval Power Undersea Weaponry (USW)	Applied research to develop component, subsystem and system technologies that are the critical building blocks for advanced high-energy-density and power-density propulsion systems, enabling increased endurance (days/weeks/months) and reliability in an air-independent environment. Approaches include modeling and simulation, fuel cells, engines, novel fuels/oxidizers and reactant storage/delivery systems.	Energy Demand Reduction	Propulsion Upgrades_Sea	1319	02	0602747N	$1,765	$0
NAVY	UNDERSEA WARFARE APPLIED RESEARCH	NRL - Energy	Long term Basic and Applied research into phenomena and mechanisms allowing for more efficient conversion of power; generation of power from solar illumination; from hydrogen conversion in fuel cells; storage of energy in improved battery technologies; augmentation of liquid fuels for greater energy density and exploitation of biological mechanisms for long-duration energy sources.	Energy Demand Reduction	Alternative Power Sources	1319	02	0602747N	$1,082	$3,528
NAVY	UNDERWAY REPLENISHMENT SHIPS	T-AO205 Seawater Pump VFD Retrofit	T-ALT development to install VFDs on the seawater circulating system to add variable capacity capability to reduce energy consumption during low load conditions.	Energy Demand Reduction	Platform Upgrades_Sea	1804	01	0204441N	$0	$500
NAVY	UNDERWAY REPLENISHMENT SHIPS	MSC Fleet Shipboard LED Lighting Upgrades	Replace legacy light fixtures with LEDs on MSC ships.	Energy Demand Reduction	Platform Upgrades_Sea	1804	01	0204441N	$0	$1,000
NAVY	UNMANNED CARRIER AVIATION (UCA)	Unmanned Aerial Refueler	Development of first CVN-based organic mission and recovery tanker. MQ-25 will extend the range and increase lethality of the CSG's CVW and will contribute to F/A-18EF shortfall by relieving tanker duties and returning a/c to the strike fighter role. MQ- 25 will also have a secondary ISR capability. IOC 4QFY24.	Supply Chain Resilience	Mobile Fuel Distribution Assets	1319	05	0605414N	$0	$1,185,171
NAVY	UNMANNED CARRIER AVIATION (UCA)	Unmanned Aerial Refueler Control System	Development of the MQ-25A mission control system Mission Control System.	Enterprise-wide Energy Visibility	Mobile Fuel Distribution Assets	1319	05	0605414N	$0	$70,009
NAVY	Warfighter Sustainment Applied Research	ONRG International Research	Early applied research with international principal investigators doing collaborative and cooperative research with the Naval research enterprise.	Enterprise-wide Energy Visibility	Alternative Power Sources	1319	02	0602236N		
NAVY	WARFIGHTER SUSTAINMENT APPLIED RESEARCH	Naval Power Systems	Microgrid Resilience on Naval Installations by Adoption of Shipboard Zonal Distribution - HIL (Hardware-in-the-Loop) and Pilot Hardware Demos.	Energy Substitution and Diversification	Platform Upgrades_Land	1319	02	0602236N	$0	$522
NAVY	WARFIGHTER SUSTAINMENT APPLIED RESEARCH	Naval Power Systems	Pilot and install small scale systems that produce water locally at tactical training facilities currently requiring significant logistics to provide externally	Energy Substitution and Diversification	Contingency Basing	1319	02	0602236N	$0	$6,347

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[22] ($K)
			sourced, and relatively small amounts of water at remote outpost location.							
								Navy Total	**$259,518**	**$2,166,410**

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[23] ($K)
USMC	Mobile Power Systems	Advance Power Sources	The Family of Mobile Power Equipment is a family-of-systems to continuously procure, update, and replenish approximately 19,000 items of Mobile Tactical Power Generation and Distribution Equipment to include the AMMPS system, and 22 different TAMCNs. The Family of Mobile Electric Power Equipment consists of skid and trailer mounted tactical generators ranging from 2 to 200 kilowatts, Mobile Electric Power Distribution Systems, Floodlight Sets, Load Banks and Electricians Tool Kits. This equipment is procured and fielded to provide electricity on the battlefield. Combat, combat support and combat service support units all require tactical power to operate weapons systems, C4I systems, medical and messing facilities, environmental control equipment, and water purification systems.	Supply Chain Resilience	Contingency Basing	1109	06	0206211M	$ 5,841	$112,894
USMC	Expeditionary Energy Office	Expeditionary Energy Office	E2O is currently focused on analyzing and developing Operational Energy solutions to support distributed operations, Expeditionary Advanced Base Operations, and the Marine Littoral Regiments. This is done through modelling and simulation and the Small Unit Power initiative as well as the modernization of bulk fuel assets.	Energy Substitution and Diversification	Individual/Warfighter Power	1319	07	0206313M	$ 14,125	$35,740
USMC	Family of Expeditionary Fuel Systems	Fuel Distribution	Development of an Autonomous littoral focused fuel distribution system to enable distributed operations and persistence of the Marine Littoral Regiment	Supply Chain Resilience	Metering and Monitoring	1109	06	0206315M	$0	$10,261
USMC	Medium Tactical Vehicle	Electrified MTT	Analyze, develop, prototype, engineer, and procure a Next Generation Medium Tactical Truck. The smaller and lighter system will use a hybrid electric drive to	Energy Substitution	Platform Upgrades_Land	1109	05	0206315M	$ 7,929	$38,178

[23] FY2024 – FY2028

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[23] ($K)
	Replacements (MTVR)		reduce fuel consumption, extend operational range, and provide an exportable power capability, use a modular and open systems architecture, incorporate scalable armor, facilitate advanced condition-based maintenance; all while improving or maintaining the same maneuverability and fordability as the current MTVR.	and Diversification						
USMC	Family of Shelters and Shelter Equipment	Shelters, Shelter Liners, and Lighting upgrades	R&D for future shelter systems and USMC lighting solution of the future.	Energy Substitution and Diversification	Contingency Basing	1319	07	0206623M	$ 177	$1,305
USMC	Family of Expeditionary Fuel Systems	Fuel Distribution	Development of an Autonomous littoral focused fuel distribution system to enable distributed operations and persistence of the Marine Littoral Regiment	Supply Chain Resilience	Metering and Monitoring	1319	07	0206623M	$0	$11,606
USMC	Mobile Power Equipment	Advance Power Sources	The Family of Mobile Power Equipment is a family-of-systems to continuously procure, update, and replenish approximately 19,000 items of Mobile Tactical Power Generation and Distribution Equipment to include the AMMPS system, and 22 different TAMCNs. The Family of Mobile Electric Power Equipment consists of skid and trailer mounted tactical generators ranging from 2 to 200 kilowatts, Mobile Electric Power Distribution Systems, Floodlight Sets, Load Banks and Electricians Tool Kits. This equipment is procured and fielded to provide electricity on the battlefield. Combat, combat support and combat service support units all require tactical power to operate weapons systems, C4I systems, medical and messing facilities, environmental control equipment, and water purification systems.	Supply Chain Resilience	Contingency Basing	1106	01	0206624M	$ 892	$4,731
USMC	Medium Tactical Vehicle Replacements (MTVR)	Electrified MTT	Analyze, develop, prototype, engineer, and procure a Next Generation Medium Tactical Truck. The smaller and lighter system will use a hybrid electric drive to reduce fuel consumption, extend operational range, and provide an exportable power capability, use a modular and open systems architecture, incorporate scalable armor, facilitate advanced condition-based maintenance; all while improving or maintaining the same maneuverability and fordability as the current MTVR.	Energy Substitution and Diversification	Platform Upgrades_Land	1319	07	0206624M	$14	$200
USMC	Mobile Power Equipment	Advance Power Sources	The Family of Mobile Power Equipment is a family-of-systems to continuously procure, update, and replenish approximately 19,000 items of Mobile Tactical Power Generation and Distribution Equipment to include the AMMPS system, and 22 different TAMCNs. The Family of Mobile Electric Power Equipment consists of skid and trailer mounted	Energy Substitution and Diversification	Alternative Power Sources	1319	07	0206624M	$4,836	$30,444

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[23] ($K)
			tactical generators ranging from 2 to 200 kilowatts, Mobile Electric Power Distribution Systems, Floodlight Sets, Load Banks and Electricians Tool Kits. This equipment is procured and fielded to provide electricity on the battlefield. Combat, combat support and combat service support units all require tactical power to operate weapons systems, C4I systems, medical and messing facilities, environmental control equipment, and water purification systems.							
USMC	Mobile Power Equipment	Advance Power Sources	The Family of Mobile Power Equipment is a family-of-systems to continuously procure, update, and replenish approximately 19,000 items of Mobile Tactical Power Generation and Distribution Equipment to include the AMMPS system, and 22 different TAMCNs. The Family of Mobile Electric Power Equipment consists of skid and trailer mounted tactical generators ranging from 2 to 200 kilowatts, Mobile Electric Power Distribution Systems, Floodlight Sets, Load Banks and Electricians Tool Kits. This equipment is procured and fielded to provide electricity on the battlefield. Combat, combat support and combat service support units all require tactical power to operate weapons systems, C4I systems, medical and messing facilities, environmental control equipment, and water purification systems.	Supply Chain Resilience	Contingency Basing	1109	06	0502511M	$0	$2,005
USMC	Mobile Power Equipment	Advance Power Sources	The Family of Mobile Power Equipment is a family-of-systems to continuously procure, update, and replenish approximately 19,000 items of Mobile Tactical Power Generation and Distribution Equipment to include the AMMPS system, and 22 different TAMCNs. The Family of Mobile Electric Power Equipment consists of skid and trailer mounted tactical generators ranging from 2 to 200 kilowatts, Mobile Electric Power Distribution Systems, Floodlight Sets, Load Banks and Electricians Tool Kits. This equipment is procured and fielded to provide electricity on the battlefield. Combat, combat support and combat service support units all require tactical power to operate weapons systems, C4I systems, medical and messing facilities, environmental control equipment, and water purification systems.	Supply Chain Resilience	Contingency Basing	1109	06	0502514M	$0	$24,317
USMC	Applied Research	Energy Efficiency and Demand Reduction	Applied Research to increase energy efficiency in weapons systems, platforms, vehicles and equipment and extend tactical range/operational reach.	Energy Demand Reduction	Individual/Warfighter Power	1319	02	0602131M	$ 9,539	$10,824
USMC	Expeditionary Energy Office	Expeditionary Energy Office	This funding will accommodate the incremental science and technology efforts that will inform the final requirements for the Medium Tactical Truck (MTT),	Energy Substitution	Platform Upgrades_Land	1319	04	0603635M	$0	$6,000

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[23] ($K)
			which is the replacement for the Medium Tactical Vehicle Replacement (MTVR). Specifically, it will support an effort through the Office of Naval Research (ONR) spanning two fiscal years that includes designing/building/fabricating (FY24), and testing (FY25) the MTT advanced technology demonstrator.	and Diversification						
USMC	Marine Corps Warfighting Lab/Futures Directorate (MCWL/FD)	Hybrid Expeditionary Power	Combining proven technologies in a novel way, program seeks to provide an ULTV towable, V-22/CH-53/C130 transportable, Mobile Hybrid Power source that can use multiple fuel types to provide quiet sustained power.	Energy Substitution and Diversification	Individual/Warfighter Power	1319	03	0603640M	$760	$4,476
USMC	Advanced Technology Demo	Energy Optimization and Logistic Burden Reduction	Advanced Technology Demonstration research to optimize energy usage and/or meet operational energy demand with renewable energy sources and reduce excess capacity or reduce logistic footprint/burden energy sources.	Energy Demand Reduction	Individual/Warfighter Power	1319	03	0603640M	$5,538	$34,458
USMC	Medium Tactical Vehicle Replacements (MTVR)	Electrified MTT	Analyze, develop, prototype, engineer, and procure a Next Generation Medium Tactical Truck. The smaller and lighter system will use a hybrid electric drive to reduce fuel consumption, extend operational range, and provide an exportable power capability, use a modular and open systems architecture, incorporate scalable armor, facilitate advanced condition-based maintenance; all while improving or maintaining the same maneuverability and fordability as the current MTVR.	Energy Substitution and Diversification	Platform Upgrades_Land	1106	01	0702808M	$197	$5,888
								USMC Total	**$49,848**	**$333,327**

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[24] ($K)
OUSD(A&S)	Office Under Secretary of Defense	Operational Energy	Identify and reduce logistics and operational risk	Supply Chain Resilience	Policy and Oversight	0100	04	0903399D8Z	10,027	141,598
OUSD(A&S)	Office Under Secretary of Defense	Operational Energy Capability Improvement Fund (OECIF)	Advanced technologies in operational energy across warfighting platforms and domains	Energy Substitution and Diversification	Storage, Power Controls and Distribution	0400	03	0604055D8Z	199,170	906,306
OUSD(A&S)	Office Under Secretary of Defense	Operational Energy Prototyping	Operational energy prototype and demonstration	Energy Substitution and Diversification	Storage, Power Controls and Distribution	0400	04	0604555D8Z	39,479	289,355

[24] FY2024 – FY2028

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[24] ($K)
								OUSD(A&S) Total	**248,676**	**1,337,259**

ORG	OE Program Title	OE Initiative Title	OE Project Description	OE Strategy Objectives	OE Activity Classification	Treasury Code	BA Code	Program Element	FY2023 ($K)	FYDP[25] ($K)
DLA	Battery Network (BATTNET)	BATTNET	Battery Network (BATTNET) is one element of DLA's Improving Industrial Base Manufacturing Processes' strategic focus area and will improve the supply and reduce the cost of procured batteries used in fielded weapon systems, such as communication radios and armored vehicles. Batteries exhibit dynamic challenges for military logistics. BATTNET relies on a community of battery supply chain members, engineering support activities, researchers, and users to conduct research and development on sustainment issues or risks, and bridge technical solutions for specific groups of batteries.	Energy Substitution and Diversification	Alternative Power Sources	0400	03	0603680s	$ 2,823	$ 28,496
DLA	Energy Readiness Program (ERP)	ERP	ERP addresses current and future issues connected to areas encompassing the Class III Bulk (Petroleum, Oils and Lubrication) fuel supply system in order to maintain and improve current warfighter product requirements.	Supply Chain Resilience	Alternative Fuels Certification and Testing	0400	03	0603712S	$2,311	$12,258
								DLA Total	**$5,134**	**$40,754**

[25] FY2024 – FY2028

Appendix P – Report on Installation Resilience Projects

10 U.S.C. 2815 Fiscal Year 2023 Energy Resilience and Conservation Investment Program Annual Report												
FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2021	Air Force	Fort Smith Air National Guard	AR	Photovoltaic (PV) Arrays and Battery Storage	Construct photovoltaic generation and battery storage to provide energy resiliency to Fort Smith Air National Guard Base. This scope includes locating roof mounted PV arrays on-base property, providing inverter system and performance monitoring equipment, and installing meters compatible and linked to local energy management system for instantaneous and historic performance tracking.	Planning and Design	$2,600	$4,706	By having stored energy generated and replenished by the solar array working in conjunction with the existing stand-by generator, the building will be able to continue functioning for several days longer than if powered by the generator alone and/or allow for stored diesel fuel to be diverted to an unexpected critical mission function while this building operates off the battery array. The project will enable a smoother transition and also extend the number of days the mission can continue without power.	This project will cover 100 percent of the critical load for a minimum duration of 10 hours, and significantly longer depending on the available sunlight. The diesel generator will provide secondary supplemental electricity during inclement weather when renewable generation is limited.	All components of this project are tied to the critical load. The 188th Wing features three primary mission sets: Remotely Piloted Aircraft (MQ-9 Reaper); Intelligence, Surveillance, and Reconnaissance (Distributed Ground Station-Arkansas); and Targeting (Space-Focused). This project would significantly reduce the disruption risks to critical mission operations.	Outages have occurred in the past affecting mission readiness, and this is a major step to improving the mission resiliency. Energy security and resilience is depended on backup generators, which does obtain total longer-term mission effectiveness during a major incident. The proposed solar arrays and proposed battery storage with the existing generators would diversify the Air National Guard's electric supply and contribute to the DoD's energy resiliency, energy assurance and energy security.
2021	Army	Military Ocean Terminal Concord	CA	Military Ocean Terminal Concord Microgrid	Construct a microgrid system capable of islanding Military Ocean Terminal Concord's (MOTCO) entire electrical distribution system, starting at the inland primary substation (the point of common coupling with Pacific Gas and Electric Company) and extending to the tidal area where on-loading and off-loading missions are conducted.	Construction	$29,000	$35,317	This project is critical for the mission assurance of MOTCO as the only strategic seaport for shipping ammunition to the Pacific region, and the only port capable of safely storing and moving ammunition, with the largest ammunition shipping capacity, on the West Coast. This project provides for the continuity of operations and mission essential functions of the entire installation.	This project enhances mission assurance by constructing a secure microgrid including an islanding and distribution control system. Pier modernization will increase the throughput velocity at MOTCO, better allowing the port to surge capabilities as needed to meet U.S. Pacific Command requirements.	As the only strategic seaport for shipping ammunition to the Pacific region, MOTCO remains the only port capable of safely storing and moving ammunition. Three critical loads must be sustained to conduct uninterrupted shipping ammunition missions: ship-to- shore gantry cranes; illumination at night to conduct 24/7missions; and power to the Network Enterprise Center (NEC). During islanded operation, the microgrid system will support all of MOTCO's critical loads.	Currently, diesel generators dedicated to the cranes and NEC provide 48-72 hours of backup power, and mobile lighting diesel generators provide very limited backup lighting services. During a sustained grid outage, generators would require regular refilling with fuel procured off-base. Installing an islandable microgrid at MOTCO will greatly improve the energy security and resilience of this strategic seaport.
2021	Army	Fort Liberty (formally Fort Bragg)	NC	Security Operations Training Facility (SOTF) Chilled Water Upgrade	Install a common chilled water secondary loop which can be independently fed by either the North or South energy plants at the SOTF.	Completed	$6,100	$7,414	This system is completely internal to the SOTF. This project supports mission critical operations by providing resilient cooling to the facility control and data centers in the SOTF.	This project enhances mission assurance by providing a redundant water loop to remove heat from critical mission systems. Without a redundant water loop, the facility has a significant risk for not being able to perform mission critical operations. This unique mission does not have a continuity of operations (COOP) site, therefore mitigating this risk would directly and positively impact mission assurance.	The project provides the necessary cooling required to maintain mission critical operations within SOTF. The SOTF, located at Fort Liberty, is a component of the United States Special Operations Command (USSOCOM) through the U.S. Army Special Operations Command (USASOCOM). It is charged with several critical missions including the study, development, training, and execution of special operations missions worldwide.	The chilled water loop was the single point of failure for heat removal. Failure in one section causes the entire facility cooling system to fail. This project provides the water needed to remove energy (heat) from the facility.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2021	DIA	Joint Base Anacostia-Bolling	DC	Industrial Controls System Modernization	Construct a modernized Industrial Controls System (ICS) that consists of a network distribution, replacement and modernization of the facility industrial control system, modernizing the facility operating technologies to provide consistent, secure, updated platforms and components to comply with the latest ICS.	Construction	$19,092	$15,978	Work will replace the legacy ICS network by providing a new ICS to include a network distribution to provide communication, system interfaces, real-time monitoring, and control of the system enhancing installation resilience.	Replacement of the ICS for the Facility Related Control System (FRCS) at DIA will ensure security of the mission and supporting activities. Upgrading the legacy control systems will enhance productivity by reducing consumption, increasing efficiency of the energy consuming legacy operating system, and integrating cost effective and mission compatible technologies. Additionally, the ICS and associated security measures will monitor and control the energy systems throughout the facility, allowing the mission essential activities to have priority.	Every mission operation (critical and non-critical) is located within the HQ facility. The success and outcome of this project will directly and indirectly remediate the risks. These operations include Information, Analysis, Operations, Science and Technology, and Joint Operations.	The current ICS system is an organically connected and distributed ICS network that has grown over the history of the facility on an as-needed basis. The system is functional, interconnected, and integrated into the current building management system but it does not have the standard supervisory control and data acquisition (SCADA) security processes and procedures in place to either identify or react to cybersecurity situations. This project implements cybersecurity requirements.
2021	Navy	Naval Support Activity (NSA) South Potomac	MD	Chemical Biological Incident Response Force (CBIRF) / Indian Head Explosive Ordnance Disposal Technical Division Headquarters (IHEODTD) / Housing Potable Water	Replace potable water (PW) lines and reroute fire protection lines at the Naval Support Facility (NSF) Indian Head; both critical to Installation Tenant missions and base support including the barracks for the Chemical Biological Incident Response Force (CBIRF) Marine Detachment.	Construction	$18,460	$25,700	For safety, due to the nature of the materials being produced and lack of redundant distribution, production of NSWC critical mission energetics material must be stopped when potable water services are unavailable or at significant risk for unavailability. This project will rehabilitate the distribution system and improve redundancy for reliability and safe operations at the installation.	Currently, with the 60 percent+ potable water distribution system loss averages across Indian Head, draw from the potable water wells exceed our permits from the State of Maryland and increase the strain on the existing aquifers that serve Indian Head and are shared with the surrounding business/ residential community. This project will reduce potable water usage in the project area by half, improve capacity for mission expansion, and preserve water service for the surrounding community.	The project area supports CBIRF barracks and operating headquarters; NSF Indian Head core functions of security, fire department, command headquarters, Child Development Center and accompanied housing. This project nearly eliminates critical mission disruptions from potable water failures in the CBIRF, IHEODTD, and Housing areas of Indian Head. Currently there are disruptions every month that jeopardize readiness and production schedules and cause unplanned work stoppages.	The rate of potable water system repairs at the installation is approximately 100 repairs per year and it is assumed this rate will double every subsequent 10 years. The existing potable water system layout does not provide adequate redundancy to provide reliable service to the Base. Replacement of the potable water system is critical in maintaining high water quality standards with efficient operation and reduced maintenance costs through upgraded equipment and materials.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2021	Navy	NSA Bethesda	MD	NSA Bethesda-16 Replace Chillers 3 through 9	Demolish and replace seven existing chillers to provide adequate facility temperature and humidity regulation for maintaining life safety standards within critical patient care facilities at the NSA Bethesda).	Completed	$13,840	$13,840	Reliable chilled water production and distribution is required for adequate facility temperature and humidity regulation to maintain life safety standards within critical facilities at NSA Bethesda whose critical mission is patient care. NSA Bethesda's 109 facilities include emergency rooms and patient rooms at the Walter Reed National Military Medical Center (WRNMMC); treatment facilities; barracks supporting the Wounded Warrior Program; laboratories and testing facilities for the Armed Forces Radiobiology Research Institute; and classrooms and nurseries at the Child Development Center. This project alone will increase chiller availability/ capacity by at least 17 percent and will reduce the Plant's electrical demand by over 20 percent.	During the 2017 cooling season, the chilled water demand was not met. This led to multiple unplanned chilled water interruptions, impacting mission success. All chillers proposed for replacement are at or near their useful lives and some are 5 years beyond, making them prone to failure and lengthy repairs. This project will provide adequate chilled water capacity with redundancy under current loads.	Critical missions are medical/emergency care, research and medical education. The medical center has already experienced times during which minimum facility needs could not be met by the chiller plant due to equipment failures. This project provides sufficient reliable chillers that Walter Reed Military Medical Center may fully function at all times, including known additional future facilities and services.	The current requirement during peak demand requires 7-8 chillers to be in operation. The age and condition of the current chillers has resulted in a maximum of six chillers available for operation at one time. This project provides sufficient reliable chillers to satisfy peak demand.
2021	Navy	NSA Monterey	CA	Cogeneration Plant at B236	Install a combined heat and power (CHP) plant at the NSA Monterey to generate electricity from natural gas and provide backup power to critical Navy computer centers; increasing overall energy security and resiliency at the base.	Planning and Design	$10,540	$15,810	In addition to increased energy resiliency, shifting a portion of the NSA Monterey electrical consumption from grid-source electricity to power produced on-base at the CHP will save the Navy on annual commodity purchase costs.	Greatly enhanced energy security and energy resiliency for NSA Monterey will result from the ability of the CHP plant to operate on natural gas in times of electrical grid failure, and also to be able to operate in resiliency mode to provide direct power to mission-critical Navy computer centers and to other select facilities at the base.	NSA Monterey provides primary support to the Naval Postgraduate School, which is the largest producer of advanced graduate degrees for the DoD, with thousands of students from all services and over 50 countries. Research is a critical element of this program. This project directly remediates disruption risks to the data center and sensitive compartmented information facility.	The data center in B302 has no backup generator at present and risks going down in the event of a Pacific Gas and Electric (PG&E) grid failure.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2021	Navy	NSA Naples	Italy	Smart Grid	Construct a Smart Grid Industrial Controls System (ICS) to provide end-to-end monitoring and control of electric, lighting, HVAC, water, and wastewater utilities, from the point of generation, or interconnection with the local utility company, down to the end-use equipment inside facilities.	Planning and Design	$3,490	$11,385	Multiple dedicated networks will be consolidated to the Smart Grid ICS using a common fiber optic network that will improve communications effectiveness and cyber security, including allowing IA (Information Assurance) compliance. Project includes renovation of existing facility space at NSA Naples to construct the regional Facility Energy Operation Center (FEOC) providing future centralized monitoring capabilities to the Europe, Africa and Southwest Asia (EURAFSWA) area of responsibility.	The project increases the cyber-resiliency of the system and the availability, reliability and quality of the attached systems improving mission assurance.	The project will help remediate mission risk by increasing cyber-resiliency of the systems and shall provide better availability of monitoring, reliability, and quality assurance of the ICS systems. Project will support mission critical functions by monitoring in real time the systems and energy use of facilities such as the command, control, computer, and communication (C4I) building and administrative headquarters for the installation and region.	NSA Naples current functional stand-alone intelligent facility systems are undermanned, not standardized, not fully interconnected, and not maximizing efficiency. This impacts mission. The new smart grid will eliminate these issues.
2021	Navy	Surface Combat Systems Center Wallops Island	VA	Generation and Distribution Resiliency Improvements	Replace an existing electrical substation and underground electrical lines, that support a critical Aegis V-10 Combat Weapons Facility, with new electrical back-up generation for use during extended electrical outages.	Construction	$9,100	$6,311	This project provides full electrical generator backup for the entire Aegis facility to insure mission operations with no anticipated interruptions. Feeder and substation replacement will improve reliability for the entire critical load.	Replacement of the old electrical feeder and substation along with the installation of the new generator and associated fuel tank will all work in concert to ensure the critical loads at the Aegis complex have resilient electrical power and remediate risk of mission disruption.	The Surface Combat Systems Center (SCSC) Aegis Combat Weapons Facility (V-10, V-20, V-21; 108,000 sf) is an Aegis Combat Weapons System engineering, testing, evaluation and team training center. SCSC performs Aegis combat weapons system engineering testing and training and cooperates in live fleet exercises. Project will dramatically improve system reliability and resiliency. 100 percent of the critical Aegis load will be affected and improved by this project.	Served by single rural electrical cooperative circuit and is vulnerable to outages which cause mission disruption. Mission interruptions and tactical equipment damage due to power outages will decrease, saving more than $2M/yr. This project provides continuous power for the Aegis facility and remediates mission risks from power outages including equipment damage and lost operational time.
2021	USMC	Marine Corps Air Ground Combat Center Twentynine Palms	CA	Install 10 MW Battery Energy Storage for Various Buildings	Install high density batteries, grid-tied bi-directional Smart-Grid Inverters and control systems to provide battery storage and power normalization of existing PV arrays.	Planning and Design	$11,646	$34,611	This battery energy storage system (BESS) project remediates risk based on the current state of availability/reliability and the improvement expected to meet mission requirements of the critical missions identified. The BESS System will greatly improve the stability and robustness of the microgrid by providing 2-4 hours of battery energy as well normalizing power generation fluctuations during heavy load periods. This significantly reduces brownouts and downtime and will allow islanding capabilities to be extended during load variabilities.	The BESS project will provide energy to either individual facilities served by the PV/BESSs for each building or provide power to the installation utility grid. Mission assurance is achieved by providing not only back-up power to critical systems during an electrical outage event, but also ensures stabilization of key resources, islanding capabilities, and micro grid resiliency, allowing uninterrupted mission critical activities and training.	The project will help ensure MCAGCC readiness and uninterrupted training activities by integrating the 10 MW of BESS into the base grid which already has 24 MW of traditional cogeneration assets and 9.149 MW of solar PV systems.	The current base-wide reliability is adequate under normal conditions but is significantly affected by any Loss of Power (LOP) event due to slow response time of cogeneration facility and black start sequences. Currently the base relies on both Southern California Edison's grid supplied electricity, on-site cogeneration electrical production, and facility level generators; however, not all mission critical activities are supported during extended outages.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Air Force	Cavalier Air Force Station	ND	PCARS Emergency Power Plant Fuel Storage	Construct a 648,000-gallon diesel fuel storage farm with accompanying ancillary structure and equipment for exclusive use by the Emergency Power Plant that directly supports the Perimeter Acquisition Radar Attack Characterization System (PARCS AN/FPQ-16).	Planning and Design	$24,150	$25,700	A properly sized and configured fuel farm to provide an adequate fuel storage supply for emergency power in support of the PARCS mission. Strategic refueling capability from a secure source is extremely limited, as the nearest supporting military installation, Grand Forks AFB, is 90 miles away through mostly isolated two-lane farmland roads, and may not have sufficient fuel storage capacity of its own on base to support Cavalier ASN in addition to its own primary mission.	Continual operation is critical to the U.S. Space Force (USSF) and DoD space tracking programs. The Fuel Storage System has sufficient capacity to ensure islanding capability for the PARCS AN/FPQ-16 for a duration of time consistent with DoD regulations.	This project provides the required fuel storage to provide energy to the 10th Space Warning Squadron of the 21st Space Wing. This mission supports NORAD by monitoring and tracking potential missile launches against North America with the GE AN/FPQ-16 Enhanced Perimeter Acquisition Radar Attack Characterization System (PARCS). The PARCS also monitors and tracks over half of all earth-orbiting objects to enable space situational awareness and space control.	Cavalier AFS currently has only 45,000 gal of on-site fuel storage for the PARCS AN/FPQ- 16 system. This project adds additional fuel storage tanks as part of an overall requirement to provide hardened emergency power in support of the PARCS mission. his project is for exclusive use by the Emergency Power Plant that directly supports the PARCS AN/FPQ-16 system, which constitutes 100 percent of the critical load on the site.
2022	Air Force	MacDill AFB	FL	Transmission and Switching Stations	Upgrade existing switchgear near the Tanker Way Gate substation feeder and construct new underground electrical distribution lines and switching stations to provide redundancy along with resiliency to the base's existing electrical distribution system.	Construction	$22,000	$40,513	The Base would suffer a major power outage with loss of either the Tampa Electric Company's (TECO) Substation at the MacDill Gate, the feeder from the TECO Substation or the MacDill Gate. The new underground distribution line and switching station add redundancy to the base's existing distribution system and will back-feed power to the eastern and southern regions of the base in the event of the loss of either single point of failure.	This project eliminates the electrical outages on installed distribution lines due to lightning strikes and high winds. It also prevents a single failure of the northern commercial power feeds from either the MacDill or Interbay substations if the power is disrupted. It enhances mission assurance by enabling the primary distribution system to operate independent from all commercial feeders.	Critical mission partners include the 6th Air Refueling wing (6 ARW), HQ USCENTCOM, HQ USSOCOM, U.S. Special Operations Command Central (SOCCENT), Joint Communications Support Element (JCSE) and the Marine Corps Forces Central Command (MARCENT). These critical mission functions have critical facilities with high energy intensity. The combined energy consumption for critical HQ USCENTCOM and HQ USSOCOM buildings exceeds the energy needs of all other critical facilities on base.	The local utility (TECO) provides the installation's primary power feed and any disturbance on the primary feed causes overloading on the line and mass outages on the eastern and southeastern portions of the installation. This project will provide redundancy to the Base utility power and allow the base to use either source of power (primary or redundant feed).

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Air Force	Memphis Air National Guard	TN	PV Arrays and Battery Storage	Construct PV generation and battery energy storage (BES) to provide resiliency power to facilities on-base, and install meters linked to local energy management systems for instantaneous and historic performance tracking.	Construction	$4,780	$17,760	By having stored energy generated and replenished by the solar array working in conjunction with the existing stand-by generator, the building will be able to continue functioning for several days longer than if powered by the generator alone and/or allow for stored diesel fuel to be diverted to an unexpected critical mission function while this building operates off the battery array.	Outages have occurred in the past affecting mission readiness, and this is a major step to improving the mission resiliency. All components of this project are tied to the critical load.	Memphis ANBG is part of the National Disaster Medical System. The purpose was to create a nationwide medical response system that could supplement state and local medical resources during an emergency or disaster, and to provide backup medical support to the military/VA medical care systems during an overseas conflict. This project will cover 100 percent of the critical load for a minimum duration of 10 hours, and significantly longer depending on the available sunlight. The diesel generator will provide supplemental electricity during inclement weather when renewable generation is limited.	The only resiliency energy systems Memphis Air National Guard Base has are individual building generators and one small roofed mounted PV system. The performance of the PV has been exceptional and that it is reasonable to continue PV array application on other roofs and ground base systems. The proposed solar arrays and proposed battery storage with the existing generators would diversify the ANG's electric supply and contribute to the DOD's energy resiliency, energy assurance and energy security.
2022	Air Force	Mountain Home Air Force Base	ID	Water Treatment Plant and Pump Station	Construct a treatment plant for processing surface water to drinking water standards, including connections to existing distribution systems.	Construction	$33,800	$102,501	The base will retain its existing potable water distribution system. New distribution paths will connect the existing water districts together to allow for primary distribution from the new treatment plant source, as well as distribution paths from the retained wells beyond their current districts to any other point on the installation as a backup source.	The old wells that exist in areas without PFOS/PFOA contamination will be retained and act as a backup to the primary treatment plant source in case of an emergency, natural disaster, or other disruption of supply from the primary treatment plant.	MHAFB is the home of the 366th Fighter Wing, the mission of the 366th FW is to develop and deploy combat-ready airmen; take care of gunfighters; and protect and enhance national resources. An active Idaho Air National Guard unit, the 266th Range Squadron (RANS) controls and maintains emitter sites within the 7,412-square-mile operational training range located in southern Idaho. The resiliency aspects of potable water support all facets of the Mission at MHAFB. There is currently no tolerance for the absence of this resource.	Wide-spread water shortages are common. MHAFB supplies 100 percent of the potable water demand for the installation and housing areas from three on-base wells that pump groundwater from the Mountain Home Aquifer in Elmore County, Idaho. However, these wells can only supply 50 percent of the daily installation demand. The USAF seeks to resolve the water resilience impact of the regional aquifer drop by constructing a water treatment plant and utilize surface water provided by the Snake River pipeline.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Air Force	Springfield-Beckley Municipal Airport	OH	Base-Wide Microgrid with Natural Gas Generator, Photovoltaic, and Battery Storage	Reconfigure existing electrical utility services and replace electrical distribution equipment, providing a base-wide electrical loop with on-site renewable PV generation, natural gas generator, battery energy storage (BES), and micro-grid controls that provides resilient power for essential missions.	Planning and Design	$4,700	$17,500	The base-wide micro-grid will have the capability to monitor incoming utility power and detect poor power quality or outages, allowing it to automatically disconnect from the utility and operate on emergency power. Multiple levels of surge suppression will absorb any disturbances within the micro- grid system to prevent damage and allow continued operations.	The micro-grid control system generator/ PV/BES will constantly monitor and react automatically as needed to minimize or eliminate outages and downtime. The controls system will provide a real time display of how the system is operating and any equipment that requires maintenance or repair. The micro-grid controls system will allow the base to seamlessly operate independent of utility power 24/7 using multiple distributed generation sources resulting in little or no interruption to critical missions.	This project will ensure energy resilience to 24/7 unmanned aircraft (MQ-9 Reaper) operations and intelligence missions in buildings 146 and 150 and provide a base-wide micro-grid control system to have resilient power for supporting facilities which include critical communications for these missions.	The base currently has nine RPIE generators, six with 24-hour diesel fuel tanks and three piped for natural gas. The base currently has 5,000 gallons of bulk diesel storage, authorized storage quantity is 8,000 gallons based on usage. This does not meet 7 days' worth of diesel fuel available per Air Force and DoD requirements. This project provides the mission with additional resilience to provide diverse non-diesel generation, and ultimately maintain uninterrupted operation.
2022	Army	Fort Moore (Formally Fort Benning)	GA	Construct 4.8MW Generation and Microgrid	Construct a natural gas power generation plant and BESS, incorporate an existing PV generation asset, and install microgrid controls to enable self-sufficient islanding capability to the distribution system during commercial grid outages.	Construction	$17,000	$56,100	The microgrid will provide Fort Moore a distribution level control system capable of isolating from the larger power grid into a self-sufficient grid with continuous power to support mission critical facilities. The benefits of deploying a microgrid system include the reduction of impacts to critical missions during power outages by maintaining power, as well as increase energy resilience and surety.	Compared to traditional back-up power, the microgrid will provide operational reliability, maintenance sustainability, safety, and intelligent management to more loads utilizing both new and existing systems. This project enhances mission assurance by remaining on standby to support power delivery to mission critical facilities during power outage events.	Fort Moore (formally Fort Benning) is a Power Projection Platform and possesses the capability to deploy combat- ready forces by air, rail, and highway. It is also the residence of various military training schools, 75th Ranger Regiment, Military Advisor Training Academy (MATA) and Security Forces Assistance Brigade (SFAB) and other tenant units. Several critical facilities will be served by this project including multiple HQ facilities, barracks, a fueling station, and dining facilities. This project would ensure backup generation is available for critical missions during power disruptions.	Currently, Fort Moore has four transmission feeds coming onto the installation at three of its seven substations. The electric system does not have the capability to strategically reduce loads and adequately distribute electric energy to installation loads. In the event of a short-term grid outage, there are numerous small back-up diesel generators at various utilities and buildings. This project will reduce the potential for electrical interruptions or blackouts, providing for improved energy security.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Army	Fort Novosel (formally Fort Rucker)	AL	Construct 10mw Generation and Microgrid	Construct a natural gas power plant to provide blackstart capabilities and the ability to supplement existing PV generation; and install a microgrid control system.	Planning and Design	$24,000	$38,400	The benefits of deploying the microgrid system include the reduction of impacts to Critical Mission operations during utility power outages by maintaining power to critical assets, as well as dramatically increase energy resilience and surety. With the addition of the reciprocating internal combustion engine(RICE) generators and supplemental PV power, Fort Novosel will be able to meet its normal average load during daylight hours and support its Critical Mission Load 24/7.	Compared to traditional backup power, the microgrid will provide operational reliability, maintenance sustainability, safety, and intelligent management to more loads utilizing both new and existing generation assets.	Fort Novosel is the largest training helicopter installation in the world with over 600 aircraft conducting operations from five base-fields and seventeen stage-fields. Project supports the U.S. Army Aviation Center of Excellence, the largest helicopter training installation in the world. Installation cannot currently meet sustainment needs during an extended outage. Project increases energy resilience by enabling power generation capability for critical missions and decreases risks associated with a single distribution source.	Currently Fort Novosel has a single connection to the electrical grid at its transmission substation. Mission critical facilities and operations are served with numerous building attached back-up diesel generators. In an emergency situation, with the commercial electric power grid down, there are insufficient staff to purchase, arrange transport, fuel and maintain the many back-up generators. Project will provide operational reliability, maintenance sustainability, safety, and intelligent management to more loads utilizing both new and existing generation assets as compared to traditional backup power.
2022	Army	Camp Grayling	MI	Construct 650 kW Gas-Fired Micro-Turbine Generation System	Install a gas fired micro-turbine power generation system, underground electrical distribution, transformers, SCADA upgrades and controls, switches, relays, compressed natural gas system, and enlarged underground gas line to provide emergency backup during power outages.	Planning and Design	$5,700	$9,120	This project will build redundancy to an existing single source electrical and gas systems. Only the air traffic control tower has back-up power generation.	This project provides critical mission assurance of the Michigan ARNG due to the projected increased demand on the local infrastructure, increasing operational tempo, increasing cost of electricity and increased potential for electrical outages.	The airfield operations require dependable sources of electricity and natural gas. On the Grayling Army Airfield there are 27 facilities that support the following functional areas: Airfield operations, Crash Rescue Building, Barracks, Dining Facility, Classrooms, Readiness Center, Entry Control Point, and multiple logistical buildings. Most of these facilities are mission critical and currently rely on a single electrical circuit and an undersized natural gas feed for operational use. The project establishes electrical resiliency through redundancy from on-site power generation.	This project is critical to the state of Michigan due to the current situation of total reliance on commercial power resulting in a single point failure of all Grayling Army Airfield operations. This project eliminates the single point failure and electrical grid outages which disrupt the operation of critical facilities supporting aviation operations and training.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Army	Camp Shelby	MS	10 MW Generation Plant and Feeder Level Microgrid System	Construct a natural gas generation plant connected to the installation's primary substation and install a micro-grid control system.	Planning and Design	$30,000	$27,000	The generation plant will provide power to substation feeders during an emergency scenario when the electric grid is compromised. The micro-grid control system will provide smart switching capabilities at the feeder level providing critical power to the entire installation.	The energy security and resilience the microgrid provides will significantly mitigate known vulnerabilities and will ensure mission critical functions can be performed during utility outages. This project will directly and positively impact mission assurance of the ARNG by ensuring continuity of critical operations and mission command during training, planning, alert, assembly, preparation and deployments of soldiers in support of federal overseas contingency operations or DSCA operations during emergencies.	Camp Shelby is a Power Projection Platform, Mobilization Force Generation Installation, Joint Forces Training Center (JFTC) for Active Components of the Army, Navy, Marine Corps, and Air Force, Army Reserves, and National Guardsmen. Critical missions and facilities at Camp Shelby are spread throughout the installation and providing emergency power to the entire installation was more cost effective and efficient than providing emergency power to sections of the distribution grid.	This project will alleviate the current limitations. The critical facilities and operations are served with numerous small back-up diesel generators. In an emergency situation, with the commercial electric power grid down, there is insufficient personnel refuel, maintain, and operate the many back-up generators spread out over the installation. Additionally, the fuel storage supply on the installation is limited to 72 hours maximum and incapable of sustained emergency operations for any extended period of time.
2022	Army	Camp Arifjan	Kuwait	Install Microgrid Controller, 1.25 MW Solar PV, and 1.5 MWh Battery	Construct a secure microgrid, including a modern smart grid control system, roof-mounted PV solar arrays and BESS.	Construction	$15,000	$13,047	The integrated microgrid system surrounding the power plant in Zone 4 at Camp Arifjan will greatly improve energy security and significantly increase the ability to ensure resiliency and reliability for the continuity of operations of mission essential functions located on base.	The microgrid will be installed on a stand-alone network with the possibility of future interconnection to the private utilities monitoring and control system (UMCS). The energy security and resilience of the microgrid that this project provides, will significantly mitigate known vulnerabilities and will ensure improved mission critical functions can be performed during normal operations and fuel supply constraints.	Camp Arifjan is an Army forward logistics base in Kuwait which accommodates elements of the DoD and international community. Essential missions served by this project include air-conditioned pre-positioned equipment storage. Developing an integrated microgrid system surrounding the power plant in Zone 4 at Camp Arifjan will greatly improve energy security and significantly increase the ability to ensure the continuity of operations and mission essential functions located on base.	The Zone 4 power plant's (Z4PP) energy is produced by six 1 megawatt (MW) diesel distributed generator system, manufactured in 2015, with a load bank that maintains the optimum loading on the generators while maintaining the spinning reserve. No other alternative generation is on site that adds resiliency or redundancy to supply clean power. This project provides the resiliency and redundancy with clean power.
2022	Army	Fort Drum	NY	Wellfield Expansion Resiliency Project	Construct five new on-site groundwater well production areas with pumps and enclosures to provide a resilient water supply on post and improve supply reliability and quality.	Planning and Design	$27,000	$43,200	This project will allow Fort Drum to pivot toa sustained water supply to the installation. The requirement is to construct a new water well production field to increase the production and supply to Fort Drum to reduce and remove the need to acquire water from outside sources. The water production area would be constructed away from contaminated areas making the Fort Drum water supply self- sufficient for the future years.	The new production site will offer reliability and sustainability for the future with needed production of the new wells. This project directly remediates the disruption risks to critical mission operations on Fort Drum. In terms of water reliability with this project, all the elevated water towers on Fort Drum would be at full capacity. In terms of water quality with this project, there would be lower levels of contamination such as per- and polyfluoroalkyl substances (PFAS) and disinfectant bi-products in the drinking water.	Fort Drum's primary purpose is to support force readiness. Installation tenants support force readiness through the delivery of integrated and synchronized services such as medical, logistical, maintenance, and other base operations. This required water infrastructure project will support installation daily critical water demand for 35,000 customers to include soldiers and their families and civilians on Fort Drum and fire protection sprinkler demand for 22 million square feet of buildings.	In the past, due to lack of resiliency, when all of the water wells were off-line, Fort Drum was obliged to purchase water from the municipal water supplier Development Authority of the North Country (DANC). The sole use of the DANC contaminated water caused the Environmental Protection Agency (EPA) to issue an Executive Order against Fort Drum. Fort Drum can solely use water from DANC for only 7 days.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Army	Fort Allen	Puerto Rico	Microgrid Control System, 690 Kilowatt (kW) PV, 275 kW Gen, 570 Kilowatt Hour (kWh) BESS	Construct a microgrid system that includes a smart grid control system, carport structure(s) and/or ground-mounted (depending on specific siting) PV solar array, BESS, diesel generator, and fuel tank.	Planning and Design	$12,190	$20,480	Following Hurricane Maria, Juana Diaz experienced an electrical outage lasting over four months and a water outage of two weeks. Extreme weather conditions such as these, along with other major events such as the recent significant earthquakes, have tested the current system resiliency and have identified system wide weaknesses. While the generators are dependent on fuel deliveries, a microgrid system with PV array and BESS will greatly enhance the system's ability to remain online and fully functional following such major events.	Implementing a microgrid at Juana Diaz will greatly improve the energy security and resilience of Juana Diaz and will significantly increase the ability to ensure the continuity of operations and mission essential functions 1st MSC and 81st Readiness Division for southern Puerto Rico.	Juana Diaz (LTC Hernan G. Pasquera Army Reserve Center (ARC), PR010) is one of three (3) sites selected by the Army Reserve (AR) HQ identifying it as a critical facility to the AR missions based on its location on the island of Puerto Rico. The ARC is part of the 81st Readiness Division and supports a total of six (6) facilities with approximately 110,000 square feet of mission critical operations. The project will provide energy resilience and security by nearly eliminating the need for commercial utility power/energy demands.	Juana Diaz (LTC Hernan G. Pasquera ARC, PR010) currently uses power from the local electric utility company, Puerto Rico Electric Power Authority (PREPA). PREPA's power distribution system and energy supply are unreliable, which adversely impacts the energy security and operations of Juana Diaz. This project will provide energy resiliency and security for critical operations.
2022	Army	Aguadilla Ramey Unit School	Puerto Rico	Microgrid Control System, 460 kW PV, 275 kW Generator, 660 kWh BESS	Construct a secure integrated feeder level microgrid including a smart grid control system, carport structure(s) and/or ground-mounted (depending on specific siting) PV solar array BESS, diesel generator, and fuel tank.	Planning and Design	$10,120	$16,960	A microgrid at Ramey will improve the energy security and resilience and will increase the ability to ensure the continuity of operations and mission essential functions of both the 1st MSC and 81st Readiness Division for western Puerto Rico. This project will directly and positively impact mission assurance of the 81st Readiness Division by ensuring continuity of operations and mission command during planning, alert, assembly, preparation, and deployments of Soldiers in support of Federal or State missions.	This project is critical for mission assurance of Ramey due to the increased demand on existing infrastructure, increasing operational tempo, increasing cost of electricity, and increased occurrences of electrical outages. The microgrid and added distributed generation that this project provides will significantly mitigate known vulnerabilities and will ensure mission critical functions can be performed during utility outages.	Ramey (PR010) is one of three (3) sites selected by the Army Reserve (AR) HQ identifying it as a critical facility to the AR missions based on its location on the island of Puerto Rico. This ARC is part of the 81st Readiness Division and supports a total of six (6) facilities with approximately 73,000 square feet of mission critical operations.	Ramey currently uses power from the local electric utility company, PREPA. PREPA's power distribution system and energy supply are unreliable, which adversely impacts the energy security and operations of Ramey. This project not only will provide energy resiliency and security for critical operations, but it will also save energy costs for the facility.
2022	Army	Fort Stewart	GA	Construct a 10 MW Generation Plant, with Microgrid Controls	Construct a gas-fired generation plant that incorporates up to 10 MW from an existing PV generation asset and installs microgrid controls to enable self-sufficient islanding during a commercial grid outage.	Construction	$22,000	$48,750	The installation has a utility owned grid-facing solar PV asset. The PV does not support the installation during times of power outages. This project implements the December 2018 Energy Resilience Readiness Exercises report recommendation that the installation of centralized generation at a circuit level to increase the number of redundant backups to critical facilities.	The microgrid will provide a distribution level control system capable of isolating from the larger power grid into a self-sufficient grid with continuous power to support mission critical facilities. The benefits of deploying a microgrid system include the reduction of impacts to critical missions during power outages by maintaining power, as well as increase energy resilience and surety.	The project will support mission critical activities for the following: Headquarters for Armored Brigade, Headquarters for Sustainment Brigade, critical mission training complex, providing power to potable water well pumps, power to the Fort Stewart Fuel Facility, and providing power to the installation data center.	Mission critical facilities and operations are served with numerous building attached back-up diesel generators. In an emergency situation, with the commercial electric power grid down, there are insufficient staff to purchase, arrange transport, fuel and maintain the many back- up generators. The microgrid will be installed on a stand-alone network should a disruption occur with the primary electrical system.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Army	Camp Shelby	MS	Electrical Distribution Infrastructure Undergrounding Hardening Project	Construct a hardened, underground electrical distribution system with individual back-up generation at nine (9) mission critical facilities.	Planning and Design	$11,155	$14,500	On-site power generation, hardening (under grounding) of the electrical distribution system, and back-up generators are the three main components of Camp Shelby's energy resilience strategy. On-site generation will provide power and resilience at the installation level when there is a commercial grid outage. Backup generation will provide on-site energy and resilience at the building level when the primary commercial energy grid and on-site generation are off-line.	Hardening of the electrical distribution infrastructure at Camp Shelby will greatly improve the energy security and resilience of the installation; it will significantly increase the ability to ensure the continuity of operations and mission essential functions of the Army and ARNG.	Critical facilities served by this project include Range Control HQ, Garrison HQ, Fire Department, medical clinic, Soldiers Readiness Processing, Medical Supply, and Camp Shelby Training Site, and Medical Logistics Command. Activities critical to the core operations of the installation are located in these facilities and would require power during an extended electrical grid outage.	A disruptive risk that Camp Shelby faces is a major weather event such as the 2005 Hurricane Katrina, tornado or flooding. The main benefit of hardening the electrical distribution system will reduce the number of disruptions to Critical Mission operations through weather and other all-hazard events.
2022	Army	Fort Liberty (formally Fort Bragg)	NC	10 MW Microgrid Utilizing Existing and New Generators	Install a microgrid system capable of islanding the Joint Special Operations Command (JSOC) compound from the Fort Liberty distribution system.	Construction	$19,464	$25,809	The proposed microgrid will greatly improve the energy security and resilience of the JSOC compound and will increase the ability to ensure mission assurance during disruptions to electrical grid service.	The project will provide the capability to form a campus microgrid utilizing on-site generation to support 100 percent of the load at the JSOC compound. The JSOC, located on Fort Liberty, is a component command of the USSOCOM. It is charged with several critical missions including the study, development, training, and execution of special operations missions worldwide.	The project will provide power to critical missions within JSOC during an electrical outage. Critical facilities include command center, two data facilities, headquarters facility, three critical unit facilities, two critical logistics facilities. This project will support 100 percent of the JSOC compound load, including all critical loads.	Mission critical facilities are supported by Uninterruptable Power Supply (UPS) lasting 5-15 minutes and building level diesel backup generators. The diesel fuel storage supply on the installation is limited to 72 hours maximum and incapable of sustained emergency operations for any extended period of time. The islandable capability provided by this microgrid project, including its smart controls, on-site fuel storage, new generation, and integration of existing generation will significantly mitigate known vulnerabilities as well as potential threat vectors.
2022	Army	Fort Liberty (formally Fort Bragg)	NC	Emergency Water System	Construct an emergency water system capable of supplying potable water to the Fort Liberty water distribution system; includes reverse osmosis technology and hydroturbine power generation.	Planning and Design	$7,705	$9,123	This system would be completely internal to Fort Liberty and under Fort Liberty control, whereas current supplies are not. It will be powered by a hydroturbine at the dam on the Little River. Hydroturbine will supply power to the emergency water system when in operation. Otherwise, power will be supplied to the Fort Liberty electrical grid.	The emergency water system pulls surface water from the Little River which runs through the northern portion of the installation. Using a package plant with reverse osmosis capabilities, potable water can be supplied to the distribution system which includes 3.8 million gallons of storage. If required, numerous gate valves across the water distribution system can isolate water to direct it to the critical missions.	Fort Liberty is the home of the Airborne and Special Operations Forces. It is a Power Projection Platform, mobilization force generation installation for U.S. Army Special Operations Command (USASOC) and one of 92 military components designated as a critical defense installation. The emergency water system will provide water to critical missions in the event off-post commodity providers can no longer supply Fort Liberty.	Fort Liberty currently has two sources of potable water that meet at a single point of failure. The emergency water system could support critical missions during a routine outage and in the case of a black sky event where providers are not able to supply Fort Liberty.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	DIA	Joint Base Anacostia-Bolling	DC	PV Carports	Construct carport structures and PV system at nine locations on Joint Base Anacostia-Bolling, with the capability of providing excess power to the commercial grid.	Planning and Design	$28,500	$40,650	This project will be able to reduce the Defense Intelligence Analysis Center's overall energy intensity footprint and serve as a renewable energy project at DIA. Utilizing the solar carport as a means of powering the DIA facility will also provide increase reliability for operations and 24/7 mission.	Power resiliency is required to meet and maintain mission critical posture. The existing generators are 11 years old and are not very efficient and will be reaching life expectancy. In house power generation for critical loads will be provided through solar power production. If power is lost, the solar load will continue providing power eliminating disruptions.	DIA provides critical DoD element needed to enable the warfighter and needed to support national security. This project would directly remediate disruption to the DIA campus, which is deemed as Defense Critical Asset (DCA).	DIA has very little tolerance to power outages. The generators on site have a limitation on how long they can run due to the fuel limit. The current emergency generators are inefficient providing a limited amount of resiliency.
2022	DIA	Joint Base Anacostia-Bolling	DC	DIA HQ Cooling Towers and Cond Pumps	Replace four (4) cooling towers and six (6) condenser water pumps at DIA headquarters building 6000.	Planning and Design	$1,450	$0	The current cooling towers are used at near 100 percent capacity, thus when they fail, disrupt data center cooling, creature comfort and even emergency generator cooling. This project is necessary to enhance facility resilience.	Power resiliency is required to meet and maintain mission critical posture. The project will reduce cooling failure for data centers increasing mission assurance for critical data center operations.	This would directly remediate disruption to the DIA campus which is deemed as DCA. Existing equipment is at end-of-life (~ 35 years old) and provides cooling to Building 6000, that includes mission critical areas. If equipment were to fail prior to replacement the system could not support the demand and the critical missions would be significantly impacted.	This project will replace four towers that are 2k/gal/min with a delta 10 deg F. The current old towers produce at approximately 0.0263kW/TON while the newer towers produce at approximately 0.0154kW/TON. The newer types of towers are nearly 52 percent more efficient than the older tower.
2022	Navy	Naval Air Facility Atsugi	Japan	Smart Grid for Utility and Facility Controls	Install an Industrial Controls System (ICS) to provide end-to-end electrical, steam, water, and wastewater utilities and building energy management at Naval Air Facility (NAF) Atsugi.	Construction	$3,810	$4,870	This project will directly allow for the separate monitoring and control of installation energy management and utility systems from the point of generation, or interconnection with the local utility company, down to the end-use equipment inside the buildings in order to directly remediate disruption risks to electrical power and critical mission operations.	The project will provide an ICS Work Desk that will interface, communicate, and control HVAC systems within existing facilities. This interface will allow system operators to actively monitor and control building HVAC systems in order to maximize efficiency without sacrificing facility mission support. This will provide the communications backbone that allows future facilities and existing buildings to be integrated into the ICS system.	This project enhances mission assurance of communication centers (NIPR, SIPR, DISA) and airfield operations (fixed wing and helicopter), which remain operational at all times. It directly supports mission critical functions related to Installation Electrical, Steam, Potable Water and Wastewater Utility Systems, and Facility Control Systems and addresses known vulnerabilities to Command and Control of Installation Utility Systems.	NAF Atsugi has intelligent facility systems that are functional, but do not cover all applicable systems, are not fully interconnected, are not maximizing efficiency, and periodically impact the facility mission. Given that the facility systems are not covering all applicable systems, the cost saving functions of the systems are no longer being used to their maximum potential. This project integrates the different operation systems (DDC, SCADA, advanced meter infrastructure (AMI)) into a common ICS system, establishing a common database, developing analytics and dashboards, and developing a standard graphical interface.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Navy	Kings Bay Naval Submarine Base (NSB)	GA	Electrical Transmission and Distribution	Replace thirteen (13) circuit breakers with breakers having integrated current transformers for relaying and metering; Replace thirty-four (34) medium voltage Keaney SF6 gas switches with vacuum bottle pad mount switches (integrate into base SCADA); Replace medium voltage switchgear, shore power unit substations, low-voltage shore power switchgear, breakers, and feeder cables in three Refit Wharf buildings 5061, 5092, and 5116 (integrate into base SCADA).	Construction	$19,314	$14,274	The degradation of the utility infrastructure (transmission, distribution, generation) over the last decade has left the installation vulnerable to frequent facility level outages. Critical infrastructure is beyond a sustainable condition for mission reliability. This project greatly improves distribution reliability and transmission resiliency, directly enhancing the readiness of our critical mission operations.	This project improves the reliability of critical electrical infrastructure that is aged and obsolete, and positively impacts mission readiness by providing each tenant secure and reliable power.	This project directly supports NSB Kings Bay's tenants that include TRIDENT Refit Facility (TRF), TRIDENT Training Facility (TTF), United States Coast Guard (USCG) Marine Force Protection Unit (MFPU), and Strategic Weapons Facility Atlantic (SWFLANT) guided missile submarines (SSGNs) and ballistic missile submarines (SSBNs).	SUBASE has 99.9 percent reliability from our local utility provider. However, local distribution degradation has increased the rate of facility level outages. Due to the outdated relays and monitoring equipment, operators must coordinate circuit wide outages to locate failure and perform maintenance. The goal of this project is to improve the reliability and resiliency scores thus improving the installations overall score.
2022	Navy	Polaris Point Submarine Base	Guam	Inner Apra Harbor Resiliency Upgrades (Phase I)	Construct a 12MW generator plant and BESS at Polaris Point to correct adverse, grid-wide frequency drops and voltage sags, and to ensure operational and emergency continuity at Naval Base Guam during an energy disruption event.	Planning and Design	$38,300	$53,464	Guam serves as a vital strategic presence in the Western Pacific. The forward deployed presence of submarines in Guam allows for increased mission days and are the first to respond when needed. Polaris Point is important as U.S operating forces from the Seventh Fleet and other Fleet Forces use Naval Base Guam as a pivot point when transiting to the Persian Gulf or further into the Asia Pacific AOR. This project will close the energy mission gap to provide installation energy resilience.	The existing power distribution system lacks the current configuration to provide clean reliable power to Polaris Point. The addition of onsite generation would eliminate the power spikes, sags, and undersized/ unreliable backup power from Orote power plant.	The additional backup power generation will primarily serve loads for the Ohio-class submarines and supporting shore facilities. The BESS will provide power quality by correcting grid-wide frequency drops and voltage sags, which adversely affect the mission	At Polaris Point NBG, grid-wide frequency drops, and voltage spikes/sags are adversely affecting the mission. In addition, the existing power plant generators at Orote substation are undersized and incapable of supplying reliable backup power to Polaris Point during utility power outages. This project addresses known frequency deviations throughout the island of Guam that cause damage to sensitive mission critical electronics on submarines in the engineering and navigation

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	Navy	Naval Air Weapons Station China Lake	CA	Solar Energy Storage System	Construct an energy storage system (ESS) to capture excess solar energy from an existing PV array supplying power to the installation via a solar power purchase agreement (PPA), which is currently being curtailed due to excessive generation.	Planning and Design	$8,950	$19,490	The intent of this project is to install the energy storage system at the installation's incoming Main Substation with the ability to distribute energy to most installation facilities. This provides the ability to serve critical loads during daylight hours and limited nighttime operations.	All missions supported involve long planned, high cost, critical aircraft and weapons system test and evaluation procedures critical to the Navy and our national defense. This project eliminates interruption of these tests that could have significant cost and schedule impacts to weapons procurement programs rippling throughout DoD and potentially delaying deployment of critical weapons to the fleet.	Critical missions supported by this project include operations focused on air weapons development and testing, airfield and flight operations, RDAT&E laboratory work, explosives handling, and range testing. Aircraft platform RDT&E facilities that will be more resilient include the F/A – 18, E-2D, AV-8B and EA - 18G. In addition, this project will add resiliency to the Navy's premier open-air range electronic warfare threat environment used to test all next generation jamming devices and aircraft platform electronic warfare systems across all DoD services.	The risk that this project partially remediates is critical load loss. Resiliency enhanced by this project includes increasing the ability to produce electrical energy locally to feed critical facilities during grid outages. This project provides on-base distributed generation that could be available to offset a portion of critical facility loads that do not currently have sufficient backup generation to support mission requirements.
2022	NGA	National Geospatial-Intelligence Agency Campus East	VA	Electrical System Redundancy	Reconfigure an existing electrical transmission network by installing switching equipment, transformers, cabling, duct banks, and manholes to provide a second power feed to mission critical facilities.	Construction	$5,299	$4,608	This project increases the reliability of the electrical power transmission system by increasing operational availability of electrical power service to critical functions. The branch of the electrical power transmission network that serves the Main Gate, the Visitor Control Center, the Remote Inspection Facility, and the Parking Garage is currently subject to periodic outages for electrical equipment maintenance. This project eliminates the power outages for periodic maintenance for these critical functions.	This project eliminates power outages and improves NGA's security posture by maintaining power to NGA's Main Gate, VCC, RIF and Parking Garage. This project provides continuous electrical power service to critical operational functions and facilities during scheduled and unscheduled power outages essential for continuity of operations at NCE.	This project provides a second power feed to mission critical campus facilities, including the primary campus Access Control Point (ACP), the Remote Inspection Facility (RIF), the Visitor Control Center (VCC), and life safety and security devices in the Parking Garage (PG). Completion of this project improves the resiliency of the campus electrical system to maintain electrical power transmission to the end users during periodic equipment maintenance activities.	During periodic electrical equipment maintenance activities, the electrical power service to NCE Main Gate, the Visitor Control Center, the Remote Inspection Facility, and the parking garage is disconnected. This project remediates electrical power outages to critical functions in order to perform periodic maintenance.
2022	USMC	Marine Corps Air Station Miramar	CA	Additional landfill gas (LFG) Power Meter Station	Install a high voltage meter station to integrate a new LFG-powered generator, switchgear, electrical infrastructure and demand response controls to achieve high reliability with renewable resources.	Planning and Design	$4,054	$4,662	This project ensures the installation mission by providing supplemental renewable power resources to the microgrid. It addresses known vulnerabilities in the diversity of power resources to support microgrid	Provide critical infrastructure to align with City of San Diego Intergovernmental Services Agreement (IGSA) landfill power delivery. It provides energy resilience by installing a high voltage electric feeder to allow LFG power to support the microgrid. Also, it displaces the use of fossil generation from diesel and natural gas that is currently being used to generate microgrid power.	This project directly supports the mission critical functions associated with the 3rd Marine Air Wing at the Station and ensures the Station can provide national and state Emergency Operations Center requirements.	There is currently no ability to connect the new LFG power generator to the base electric grid via microgrid. This project improves the electric infrastructure and controls systems to enhance microgrid resilience with optimization of power quality and supply.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2022	WHS	Pentagon	VA	Recommissioning of HVAC Systems, Part B	Recommission HVAC systems to improve energy efficiency and extend the useful life of the existing equipment and mechanical systems.	Construction	$2,600	$2,600	This project will allow WHS to continue recommissioning work in 9.4 million square foot of real property in the National Capital Region. This will enable WHS facilities to improve their energy resilience by reducing energy waste in mission critical systems. By reducing energy required to operate the facilities, critical backup generation resources, such as fuel storage, becomes more efficient as less fuel is needed to sustain the mission.	This project supports the Pentagon's mission by reducing overall building electric load, requiring less generation capacity to backup critical functions.	This project supports mission critical areas in the Pentagon's command centers and data centers, making systems more efficient and reliable.	Recommissioning extends equipment life by returning equipment to optimal operating functionality and allows operation and maintenance staff to focus time on mission critical issues rather than routine tenant calls. The anticipated energy savings from this project will be immediate, and the impact to the utility bills will likely be recognized in the same year the project is implemented.
2023	Air Force	Patrick SFB	FL	Underground Electric Distribution System	Replace an existing overhead electric distribution line and transformers with an underground electric distribution system and pad-mounted transformers.	Construction	$8,400	$14,149	Reliable and resilient electrical distribution infrastructure is required at Cape Canaveral Space Force Station (CCSFS) supporting critical launch functions. Additionally, this will provide reliable launch infrastructure, launch teams, and seamless partnership with launch and satellite programs.	Project directly remediates disruption risk by providing new undergrounded electrical circuits in the vicinity of critical loads and outside of biologically sensitive areas. The area is susceptible to hurricane and storm damage, as exhibited by past tropical storm activity impacting the county (e.g., Charley, Frances, and Jeanne in 2004, Matthew in 2016, Irma in 2017). This project would eliminate vulnerability to critical loads supporting Falcon 9 launches, Navy missions, and port operations.	A scrubbed launch due to an electrical issue will result in complete mission failure. Depending on the payload and proposed orbit, these payloads have a very tight window to launch. Some launch windows only come once or twice a year, so a scrubbed launch could result in major delays to national security or manned spaceflight missions.	The CCSFS installation location is less than one mile from a saltw a t e r source, causing all metal components to deteriorate at an increased rate. Current infrastructure maintenance is impeded due to lack of spares based on the age of the electrical line. This project would decrease downtime for repair and unanticipated costs by using current spares, as opposed to antiquated or phased-out spares.
2023	Air Force	Patrick SFB	FL	Water Distribution Loop	Install 41,505 linear feet of new high-density polyethylene (HDPE) potable water mains, with variable frequency drive (VFD) booster pumps and isolation valves, to provide looped water systems to the Integrate-Transfer-Launch (ITL) and north areas on CCSFS.	Planning & Design	$7,300	$19,530	Most of the facilities in the ITL and North Cape areas are launch critical. A failure of the existing mains would directly impact launch capability.	Space Launch Delta 45's mission is to provide the activities and resources for safety of flight, range instrumentation, infrastructure, and scheduling required to support and assure space and ballistic launches, and other operations. This project would decrease the likelihood of infrastructure-driven launch mission delays and scrubbed launch missions. This project will eliminate the need to flush 20.9 million gallons (MGal) per year	Project directly remediates disruption risk by providing a redundant potable water supply to critical facilities in the ITL and North Areas of CCSFS. Lack of potable water supply has prompted immediate installation response to provide potable water from filled deluge tanks and water towers in order to provide water for cooling tower and chiller systems of critical facilities. This project would result in the potable water system being able to function while having an unscheduled outage in one path of the potable water pathway.	One of the critical requirements for successful rocket launches is a reliable water distribution infrastructure. The multiple stakeholders at this installation all have high water demand. CCSFS must flush 1.5 to 3 million gallons of water every month to maintain drinking water quality throughout the Installation. Re-circulating and re-chlorinating potable water in the ITL and North Cape areas would significantly reduce flushing, thereby reducing energy and water usage.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	Army	U.S. Army Reserve Center, Conroe	TX	Power Generation and Microgrid	Construct a carport mounted PV solar array with solar panels, mounts, inverters, transformers, controls and communication, AC/DC wiring with conduit, low voltage switchgear, and security measures.	Planning and Design	$9,600	$19,300	The ARC at Conroe supports the Army Reserve UH-60 Black Hawk medium lift transport aviation mission. The microgrid system will include master microgrid controller, and an automated distribution switch to disconnect the Army Reserve Center from local electrical utility grid during grid outages.	This project will directly and positively impact mission assurance of the 63rd Readiness Division by providing about 96 percent of its average annual energy consumption with on-site renewable energy generation, thereby ensuring continuity of operations and mission command during planning, alert, assembly, preparation, and deployments of Soldiers in support of federal or state missions during emergencies.	The ARC is a critical facility and supports multiple facilities with approximately 110,000 square feet of mission critical operations for the 63rd Readiness Division. All buildings on the Conroe site are deemed critical to support mission needs. This project will directly and positively impact mission assurance of the 63rd Readiness Division by providing about 96 percent of its average annual energy consumption with on-site renewable energy generation.	The Conroe ARC currently does not have on-site energy generation capability; it is fully reliant on the local utility grid. This reliance presents an energy security vulnerability to Conroe ARC's critical missions. This project provides about 96 percent of its average annual energy consumption with on-site renewable energy generation reducing sole dependance on the local grid.
2023	Army	Camp Arifjan	Kuwait	Power Generation and Microgrid	Install a grid tie from Zone 4 Power Plant (PP) to power critical facilities, a rooftop mounted solar PV array, BESS, automatic transfer switches (ATS), and Microgrid controls.	Planning and Design	$26,850	$35,047	The U.S. Army-owned central generation plant is providing redundant and resilient power to these critical facilities. Both compounds are powered by aged and failing diesel generators which are dependent on truck delivered diesel fuel. This project will install new dual feed substations at both TIP-C and MCF Signal Compounds connecting them to Host Nation power and Zone 4 to provide energy resilience.	Camp Arifjan requires resilient power supply architecture for 24/7 operations. This new system will provide resiliency benefits. It will provide redundant grid-tie feeds to the critical facilities. With a new dedicated feed between the two most critical compounds outlined in the detailed project description, islanding for any possible lengthy HN grid outages will be much more simplified and faster than utilizing existing distribution feeders/ substations/Utility Distribution Systems.	Critical loads at Camp Arifjan are required for 24/7 operations, and this system would cover a major portion of these loads with improved resilient power supply architecture for two critical facility compound areas: (1) The TIP-C compound and (2) The MCF signal compound.	Most essential missions are on the Host Nation power grid, receive Assistance-in-Kind power, and have conventional diesel genset back-up power. Two major facilities with large electric loads, TIP-C and MCF compounds, are not currently connected to the Host Nation grid. Currently, both TIP-C and MCF are served primary power and backup power by diesel generators. This project would connect these facilities to the Host-Nation power grid.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	Army	Fort Riley	KS	Power Generation and Microgrid	Construct a microgrid connected to the electrical distribution system with modular natural gas generation.	Planning and Design	$25,780	$28,900	Fort Riley supports training, assembly, and deployment missions associated with the U.S. Army Forces Command units located on the installation, particularly, the 1st Infantry Division. Currently, the division must rely on its portable warfighting equipment and power generation capability to maintain installation operations should there be a prolonged loss of power. Once operational, the microgrid will provide Fort Riley the ability to island up to 35 percent of its critical load from the external power grid.	Developing a microgrid at Fort Riley will greatly improve the energy security and resilience of the installation and will significantly increase the ability to ensure the continuity of operations and mission essential functions of Fort Riley as a Power Projection Platform (PPP), Mobilization Force Generation Installation Contingency Inactive, and Regional Collective Training Capability installation.	Of the 129 critical facilities, 57 have no backup power. There are 60 generators (35 natural gas, 22 diesel, and 3 propane) serving one or more critical facilities. This project provides emergency power and will enhance the protection and resilience of mission critical facilities and capabilities located on feeders connected to three (3) electrical substations. Facilities served include First Infantry Division Headquarters, Four (4) Brigade Headquarters Buildings, and five (5) training and training support centers. Project will allow these facilities to operate under emergency in the event of a power outage.	There is an inadequate number of backup generators and onsite fuel storage to sustain critical facilities across the installation. This microgrid project will mitigate that vulnerability and greatly improve the installation's ability to support the Army's mission
2023	Army	Fort Stewart - Hunter Army Airfield	GA	Power Generation and Microgrid	Construct a natural gas generation plant, with existing Reciprocating Internal Combustion Engines (RICE) generators, connected to the installation's distribution switching station, owned by Canoochee Electric Membership Cooperative, and to the installation's natural gas infrastructure. In addition to the generating equipment, the plant will consist of generators, a newly installed microgrid control system, and a paralleling switchgear	Planning and Design	$25,400	$37,800	This project provides energy resilience in times of grid outage to allow the critical missions to continue operation at Hunter Army Airfield. It supports mission critical activities by providing emergency power for Headquarters for 3rd Combat Aviation Brigade, Bulk Fuel (JP-8) Storage and fueling, DAAG/Ready Building, Ammo Supply Points, and shelter locations for families. Sectionalization at the circuit level to serve only critical loads would be cost prohibitive, therefore the project is designed to power the entire installation.	This Project will provide a microgrid control system that will enhance the protection and resilience of critical assets and capabilities by providing smart switching capabilities at the feeder level, providing critical power to meet the needs of the Hunter AAF. This project will support 100 percent of critical missions and critical facilities at Hunter Army Airfield.	This project installs an 8 MW microgrid to provide redundant backup to support 100 percent of mission critical facilities and cover nominal base load. The benefits of constructing a second electric supply point to Hunter Army Airfield include supplying the Airfield with generation for 100 percent of mission critical facilities.	In an emergency situation, if the commercial electric power grid is down, there are insufficient staff to purchase, arrange transport, fuel and maintain the many back- up generators. This project enhances mission assurance by providing standby power to support energy delivery to mission critical facilities during power outage events. This project provides energy resilience in times of grid outage to allow the critical missions to continue operation at Hunter Army Airfield.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	Army	Fort Cavazos (formally Fort Hood)	TX	Power Generation and Microgrid	Construct a microgrid system that includes natural gas fueled reciprocating internal combustion engines (RICE), a solar PV parking canopy array, BESS, automated switches required to isolate, island and blackstart the system, and microgrid controls.	Planning and Design	$31,500	$40,500	Fort Cavazos is a mobilization and demobilization station for Army Reserve and National Guard units, a Power Projection Platform (PPP) and a primary active mobilization force generation installation. It is also the Army's premier Regional Collective Training Center, and the only post in the United States capable of stationing and training two armored divisions. This project will support 14 percent of the total installation critical load.	The proposed microgrid will power multiple mission critical facilities with an estimated critical load of 2.12 MW at the Phantom Warrior Central campus. The project addresses Critical Mission Sustainment (CMS) and reduces risk by providing electricity to critical LRC loads during power disruptions. PV and battery storage reduce fuel supply risk.	The microgrid will provide resiliency for multiple critical missions at Fort Cavazos for a minimum of 14 days through backup power generators, battery storage and new power generation. The microgrid will enable the continuity of mission command for critical commanders and staff in III Corps HQ, Garrison HQ, NEC, Directorate of Plans, Training, Mobilization and Security, Directorate of Emergency Services, and the Blood Donor Center.	There are existing deficiencies in generator coverage and the current cost of contingency power, to include the cost and resources needed to refuel diesel systems, without this project. The installation would be at risk from electrical interruptions without improved, longer-term energy resilience. This project reduces these vulnerabilities.
2023	DIA	Redstone Arsenal	AL	Missile and Space Intelligence Center (MSIC) Backup Power Generation	Install UPS and seven (7) generators for the MSIC facility, including power transformers, switchgear, distribution panels, power, control and communication cabling, control hardware, and building management system programming.	Planning and Design	$10,700	$0	This project provides redundancy of service, transmission and generation to MSIC to decrease the possibility of significant mission downtime and equipment damage due to loss or power.	There are various facilities that perform essential mission functions that are critical to the DOD and Intelligence Community and have zero tolerance for power interruptions. Mission availability for this facility is 99.99 percent; there is no tolerance for downtime. These facilities contain numerous laboratories and scientific computing systems. Power failure to these facilities during operations risk damage to high value equipment and components, corruption of data results essential to DIA/MSIC's mission and potential delays in MSIC's ability to respond to high priority intelligence requirements.	DIA's MSIC facility is plagued by frequent short duration outages and intermittent extended outages both of which have significant impact on mission critical systems and functions. The existing power infrastructure supports only 30 percent of these mission critical facilities peak demands. There is currently a single 1 MW generator that, given its age, can only provide 75 percent of its rated capacity. This falls far short of the 2.31 MW peak demand required to support the critical loads of the (10) prioritized MSIC facilities.	The MSIC complex has experienced an average of 29 outages annually, for the last 5 years. Utility reliability is notoriously low as evidenced by the frequent power outages annually. The project includes providing procurement, installation, testing, and commissioning of all electrical, control and civil work required to provide an additional generation capacity, UPS backup power and the ability to independently accomplish load bank testing on the seven generators and UPS without power interruption.
2023	DoDEA	Kadena Air Base	Japan	Lighting Upgrades	Replace lighting at the Kadena High School with, dimmable LED luminaires that have a high efficiency LED light driver that deliver energy savings and low maintenance.	Planning and Design	$780	$837	If the lighting is not upgraded, the substandard environment will continue to hamper the educational program for students. The required maintenance and repair of expired and failing systems will continue to strain maintenance capabilities and budgets. If not funded, the energy savings of 125,000 kwh per year will not be realized.	This project is an energy conservation project in accordance with 10 U.S.C. 2914.	This project is an energy conservation project IAW 10 U.S.C. 2914.	The existing lighting installed at Kadena High School is in poor condition. The lighting uses outdated T-8 fluorescent technology which requires significant efforts to keep it operating. The existing lighting levels are substandard.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	Navy	Naval Air Station (NAS) Jacksonville	FL	Facility Energy Operations Center Renovation	Renovate portions of the Facility Energy Operations Center (FEOC) to provide required operational workspace. Includes relocation of interior walls to provide private offices, open offices, breakroom, bathrooms, janitorial and communications rooms; renovation of interior finishes; reconfiguration of HVAC, electrical, plumbing, fire suppression, and telecommunication utilities.	Planning and Design	$2,400	$2,925	The FEOC improves mission assurance and readiness by providing a comprehensive picture of the operational condition of regional energy infrastructure. The Smart Grid will integrate data from installation building control systems, automated utility meters (AMI), and utility plant control systems (SCADA) through the cyber security platform enclave to a centralized data collection point with data analytics software, the FEOC. Through metering and control system integration, the FEOC will empower the Navy's ability to proactively manage energy, minimize equipment downtime, and effectively reduce costs and energy consumption.	The FEOC improves mission assurance and readiness by providing an actionable, comprehensive picture of the operational condition of regional energy infrastructure. The plan for Smart Grid is to integrate data from installation building control systems, AMI, and utility plant control systems (SCADA) through the cyber security platform enclave to a centralized data collection point with data analytics software, the FEOC.	This project supports all 17 Commander, Navy Region Southeast's installations which include CNIC, NAVAIR, NAVSEA, TRIDENT Refit Facility, TRIDENT Training Facility, Strategic Weapons Facility Atlantic, and other non-Navy or Non-DoD (e.g., USCG and U.S. Customs and Border Protection) critical missions and operations.	The planned FEOC is necessary to realize Full Operation Capability of the Smart Grid Initiative at FEC SE. Currently the ICS Monitoring Station and staff occupy one office in Building 103, as the staffing and systems to be monitored grow the current space will not be able to accommodate the operation. This project provides the necessary space to support the ICS.
2023	Navy	NAVBASE Ventura / Point Mugu	CA	Ground Mounted Solar PV	Construct a fix-tilt, ground-mounted system solar PV system at San Nicolas Island, Naval Base Ventura County that interfaces with the installation power grid.	Construction	$13,360	$22,712	This project provides a second renewable energy source and a third distributed energy resource for the island electrical system. It will integrate with existing electrical generation assets to improve reliability and resilience and serve to significantly reduce diesel generator operating time and extend on-island fuel storage capacity.	This project supports multiple critical and non-critical missions on board San Nicolas Island. SNI is located approximately 60 miles southwest of Naval Base Ventura County Point Mugu Site and lays within the 36,000 square- mile Naval Air Systems Command Point Mugu Sea Range.	In an event of a power failure, critical weapons test schedules and timelines will be at risk. Mission critical exercises, testing, and training at SNI is of the utmost importance to avoid safety mishaps. It is crucial that continuous power be provided during critical mission events, for example live tracking of target and missile launches that are constantly monitored between SNI, Point Mugu, China Lake, and Vandenberg AFB.	The solar PV system remediates the risk of critical load failure. This will provide supplemental power during low, or no wind power conditions, complementing the existing wind turbine electrical production and further extending the length of time that the island grid can provide power without diesel generator operation. It eliminates the risk of fuel delivery and disruption and the dependence on any one source of power.
2023	Navy	SUBASE Kings Bay	GA	SCADA Modernization	Upgrade Sub-base Kings Bay's existing Integrated Digital Network infrastructure by repairing passive and active infrastructure including centralized server architecture, main/intermediate/edge distribution nodes, and upgrading software that improves control, alarming, and data logging functionality.	Planning and Design	$11,200	$15,325	A disruption in energy distribution will obviously affect the customers on that circuit but due to the interdependency of critical missions at SUBASE a single facility outage can create a cascading mission impact. Proposed improvements directly enhance the SUBASE mission through improved SCADA network functionality /redundancy and removes latency issues in preparation for the COLUMBIA class submarine program.	SUBASE mission requires the delivery of reliable and quality power to support the TRIDENT missile production program and provide complete support to the Atlantic submarine fleet who call Kings Bay homeport. Unplanned electrical outages cause immediate stoppage of production and lead to future downtime for scheduled repairs.	The Water Front Utility Plant provides critical compressed air and chilled water to the pier side nuclear submarines. The nation's strategic nuclear deterrent mission relies on each SUBASE tenant's ability to perform at a high rate of efficiency and protocol. This includes the TRIDENT missile production program and TRIDENT Refit Facility transition from OHIO Class to COLUMBIA class submarines.	In current outage situations, operators must restore power through switching to redundant circuits where available, isolate the circuit, locate the failure, dispatch repairs, and ultimately restore power through the normal power circuits. The improvements scoped in this project aim to perform these actions concurrently from the SCADA control building ultimately minimizing disruption risks to near zero. When this downtime affects our strategic deterrent tenants, work stoppages occur, and the labor cost is estimated at $17K per each hour of lost

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	Navy	NAVBASE Guam	Guam	Electrical Distribution System	Construct up to 16 MW of backup generation (4 MW at Orote Power Plant, 12 MW at Polaris Point) and all ancillary equipment (including any necessary modifications to pre-planned/existing facilities generation will be located inside). Upgrades of control system and all ancillary equipment (including any necessary modifications to pre-planned/existing facilities generation will be located inside).	Construction	$34,360	$51,560	NAVBASE Guam lacks sufficient backup generation and distribution capabilities to reliably power critical facilities and pier loads of Ohio Class submarines during utility outages. Distribution power system modifications and microgrid capabilities are required to create a loop network configuration between Polaris Point and Orote Power Plant to improve resilience and allow seamless transfer to backup generation during utility outages.	This project provides backup power and mission resiliency for the piers in line, and addresses known frequency deviations throughout the island of Guam that cause damage to sensitive mission critical electronics on submarines in the engineering and navigation spaces.	This project enhances mission assurance for the Ohio Class submarine, support facilities and port operations (access to backup generation). The lack of adequate back up power and the low power quality has a detrimental effect on the sensitive electronics throughout the Ship and instrumentation for the reactor plant.	The current amount of backup power and ability of the antiquated electrical power distribution configuration is insufficient for both critical mission loads, and pier loads (OHIO class submarines). This project directly remediates that deficiency.
2023	Navy	Camp Lemonnier	Djibouti	Enhanced Energy Security and Control Systems	Install and integrate secured FRCS, and designs and builds several energy conservation measures to reduce fuel consumption and operation and maintenance costs at the Prime Power Plant on Camp Lemonnier. The system will encompass electric, water, and wastewater utilities, as well as building energy management to provide reliability, resiliency, and efficiency by integration and communication of control systems.	Planning and Design	$24,000	$34,320	This project provides resilience and reliability in the form of cyber secure FRCS, with the ability to monitor and respond to outages and power quality issues quicker, ensuring less interruption in mission.	This project reduces interruptions to mission and tenants/customers by shortening response time to power outages and quality issues and reduces base energy demand. It provides monitoring and control of energy/utility systems from the point of generation down to the end-use equipment.	Camp Lemonnier, Djibouti is a forward operating site (FOS) and the primary base of operations for U.S. Africa Command in the Horn of Africa. Camp Lemonnier presently has some intelligent facility systems that are functional, but are undermanned, not standardized, not programmable, not fully interconnected, are not maximizing efficiency, and periodically breakdown which impacts facility missions.	Camp Lemonnier generates its own power using diesel fuel-fired generators. There is no supplementary connection to an off-base electric distribution grid. The camp purifies underground water reserves for all water supply needs through the process of reverse osmosis, which requires significant energy use.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	Navy	NAVSUPPORT Hampton Roads	VA	Primary Distribution Substation	Replace a substation, four (4) transformers, and switchgear to create redundant unit pairs capable of carrying full electrical loads.	Construction	$19,000	$11,400	The Maritime Operations Center provides critical service and command operations to the Atlantic Fleet working operationally out of Building NH-95. These substations are required to provide redundancy backup electrical service to critical loads for the Center and other operational support of the missions in NH-95.	The Maritime Operations Center provides critical service and command operations to the Atlantic Fleet working operationally out of NH95. These substations are required to provide redundancy backup electrical service to critical loads for the Center and other operational support of the missions in NH-95. This creates a "true" redundancy of this project's unit pairs so that if one unit goes down or is brought down for maintenance the other unit in the pair carries the full load.	The Maritime Operations Center provides critical service and command operations to the Atlantic Fleet working operationally out of Building NH-95. These substations are required to provide redundancy backup electrical service to critical loads for the Center and other operational support of the missions in NH-95	Replacement parts are proving to be difficult, if not impossible to find, and repair time is increasing because of equipment wear and age fatigue. The wear on the equipment places a danger to all maintenance personnel because of the inherent arc flash hazards associated with equipment that no longer performs as specified by the manufacturer.
2023	Navy	Naval Support Activity Hampton Roads	VA	Backup Power Generation	Renovate the Joint Use Intelligence Center (NH139); replace the emergency generator and paralleling controls, two existing UPS system back-ups and associated panels; and provide new mounting pads.	Planning and Design	$3,400	$4,164	NH139 emergency generator, the paralleling controls, and portions of the UPS back-up system are past their useful life requiring upgrade/ replacement. Building NH139 lacks a communication link to NH95 to alert facility technicians, who are located at NH95, of any system issues or failure.	Operations occurring at Building NH139 service the Atlantic Fleet for critical missions. This project will provide adequate facilities and service and repair and upgrade to NH139 to address failing or faltering systems.	This project improves energy resilience. NH139 services the Atlantic Fleet with facility space and supports critical missions of the tenant agency.	This project eliminates the vulnerabilities to NH139. NH139 suffers from an aging control system, lack of UPS reliability, failing generator, and failing programmable logic controllers (PLCs) that are aging and past their useful life. NH139 also lacks an alarm system communication link to NH95 for technicians in the event of issues or failure.
2023	Navy	JBPHH / Joint Base Pearl Harbor-Hickam	HI	Primary Electrical Distribution	Replace a 4.16KV distribution system to improve electrical power resiliency and reliability for Naval Computer and Telecommunications Area Master Station Pacific, Wahiawa, JBPHH.	Planning and Design	$25,000	$39,345	Naval Facilities Engineering Command (NAVFAC) Hawaii Utilities is required to provide adequate, reliable and uninterrupted electrical power throughout JBPHH, including Naval Computer and Telecommunications Area Master Station Pacific, Wahiawa, JBPHH.	This project provides end of service life replacement of deteriorated conductors and associated antiquated transformers and switchgear for reliability and resiliency metrics.	This project provides reliability, resilience and power security for C5ISR OPS – COMMAND AND CONTROL and MMDS C5IRS OPS - NETWORK OPERATIONS/ TELECOMMUNICATIONS critical missions and multiple facilities.	Reliability is impacted negatively at current mission requirement. Old, deteriorating and failing electrical infrastructure and conductors are what causes extended electrical outages to the mission.
2023	NGA	NGA Campus East Springfield, Ft. Belvoir	VA	Chilled Water Redundancy	Reconfigure existing chill water cooling network that serves the NGA Campus East technology/data center by installing additional overhead supply and return piping with all necessary insulation, metering and valve equipment, ensuring that cooling capability is maintained to end users during unexpected equipment maintenance or failure activities.	Construction	$1,100	$1,650	Completion of this project improves the resiliency of the campus's data center cooling system by creating a parallel, multiple point of failure system to maintain capabilities to the end users during potential unexpected equipment maintenance activities or critical failure events.	Continuous data center equipment cooling service in support of critical operational functions during scheduled and unscheduled maintenance activities or system failure events.	Current cooling configuration is (N), with any single point of critical failure or needed maintenance crippling cooling potential to the data center. This project increases operational availability and eliminates catastrophic data center operational downtime of critical missions.	During potential critical maintenance or equipment failure events, cooling capacity to the data/technology center at NGA Campus East would be partially or completely interrupted. Currently no viable temporary mitigation measures are implemented to keep these critical operations functioning during the duration of the maintenance/failure activity.

FY	Military Service/ Defense Agency	Location	State / Country	Title	Project Scope	Current Project Status	Authorized Amount ($000)	Current Working Estimate/ Award Amount ($000)	How This Project Enhances Installation Resilience	How This Project Enhances Mission Assurance	How The Project Supports Critical Mission Functions	How This Project Addresses Known Vulnerabilities
2023	NSA	Fort Meade	MD	Reclaimed Water Infrastructure Expansion	Expand a reclaimed water system for cooling tower makeup water on West Campus and Central Campus at NSA Washington.	Planning and Design	$23,310	$26,700	The existing potable water system used for makeup water at the cooling towers on West and Central Campus is the only source available. If this system is disrupted, the campus will rely on the water storage of the water towers suppling the campus. This storage will not provide enough water to maintain mission for prolonged amounts of time.	Building cooling relies on cooling towers which require a constant source of makeup water. At the campus facilities, potable water is the only source of makeup water. If the potable water system fails or is shut down, cooling can no longer be provided and could force the shutdown of mission operations to prevent overheating.	The cooling systems at West Campus and Central Campus facilities, including critical facilities, rely upon water to provide critical cooling required for continuing their missions. The reclaimed water will provide a redundant utility for mission-critical operations. Using reclaimed water in the cooling towers will also reduce water costs and drawdown from the capacity-constrained wells that serve Ft. Meade and NSA Washington.	The potable water system is the single source of water that is required for the cooling towers which are part of the process to provide the essential cooling required for mission operations. If the potable water system fails or is shut down, cooling can no longer be provided and could force the shutdown of mission operations to prevent overheating.
2023	USMC	Marine Corps Mountain Warfare Training Center Bridgeport	CA	Microgrid and Backup Power	Construct a cyber secure microgrid to enable synchronous operation of solar PV array, BESS, generators, an energy operations center, and microgrid controls.	Planning and Design	$25,560	$41,634	This level of energy assurance is critical because the base is in an unforgiving, remote, and arduous mountain environment where annual temperatures range from -20 to +90°F, wildfires impact energy supply, and snow accumulation reaches 8 feet. The base's mission, as well as the health, safety, and productivity of personnel conducting critical training, education, and RDT&E missions, is dependent on a reliable energy supply.	This project contributes to mission readiness by providing Bridgeport with a reliable, resilient, efficient, and cybersecure microgrid that enables islanding and continuity of operations for several weeks.	The mission of the Mountain Warfare Training Center is to "conduct service-level Marine Air-Ground Task Force integrated exercises and supporting formal schools, develops warfighting doctrine, supports RDT&E for specialized equipment for use in mountain warfare operations, and maintains installation infrastructure and services in order to facilitate increased USMC readiness." With temperatures that drop below negative 20°F, all activities are critically reliant upon consistent access to energy.	This project will significantly reduce the risk of catastrophic failure during the winter months as it reduces the dependency on the existing 1 MW generator. The project will allow the system to be operated in multiple modes, depending on the generation and storage assets that are available at the time, and still support the entire critical load.

Appendix Q – Points of Contact

INSTALLATION ENERGY	
Name	Email
Paul Pergolizzi	paul.j.pergolizzi.civ@mail.mil
Andrew Knox	andrew.d.knox4.civ@mail.mil
Alex Lackner (CTR)	alexander.w.lackner.ctr@mail.mil

OPERATIONAL ENERGY	
Name	Email
Oliver Fritz	oliver.h.fritz.civ@mail.mil
Camila Lopez Silva (CTR)	camila.a.lopezsilva.ctr@mail.mil